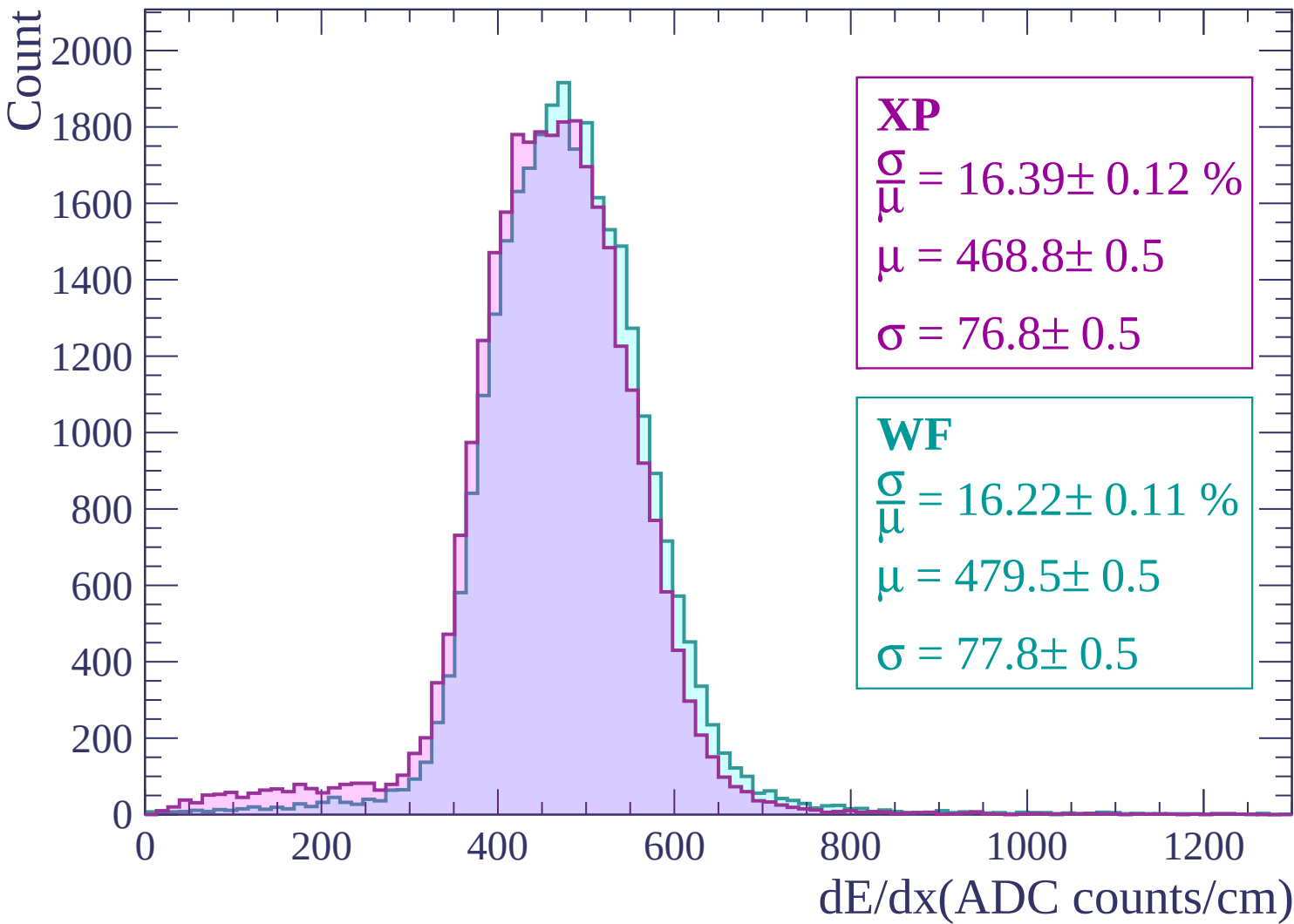
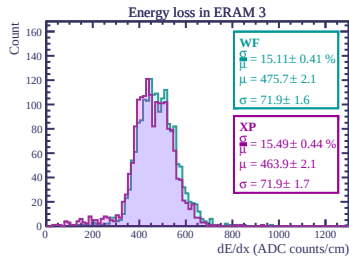
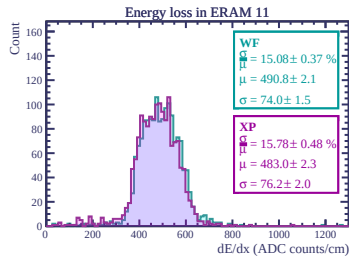
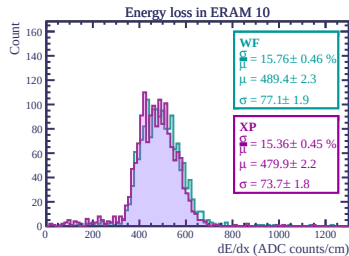
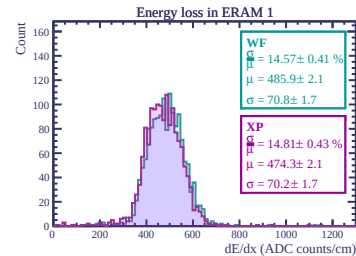
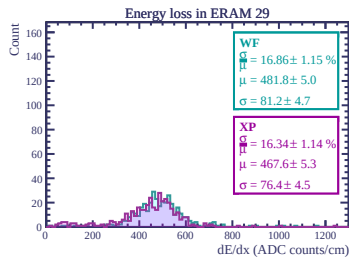
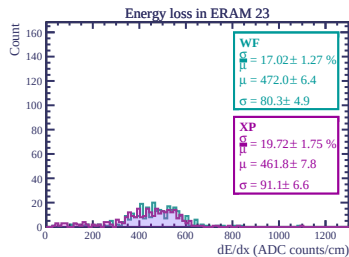
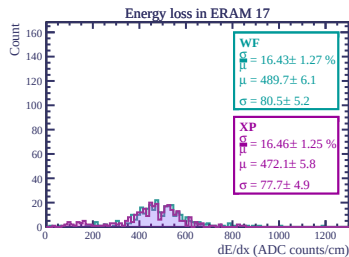
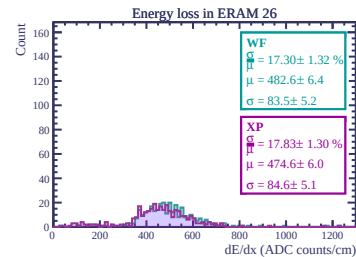
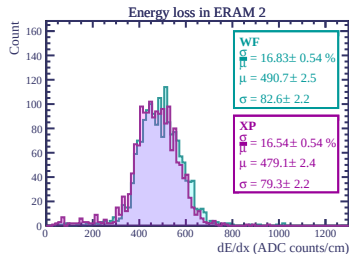
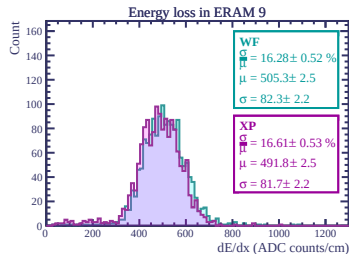
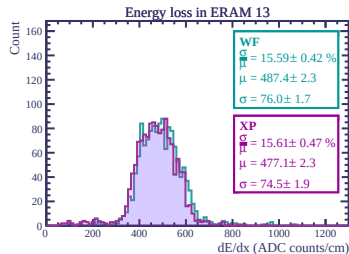
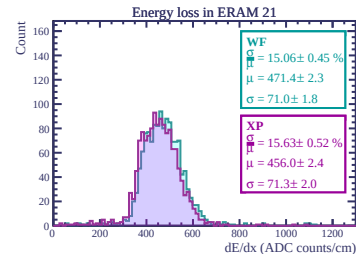
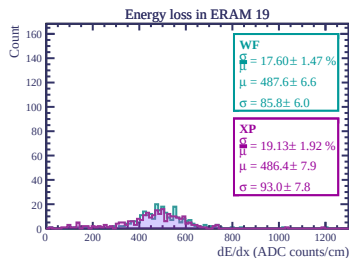
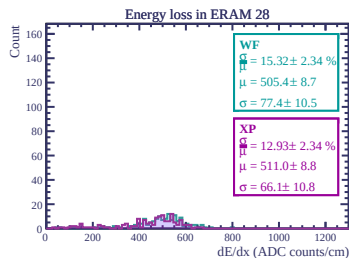
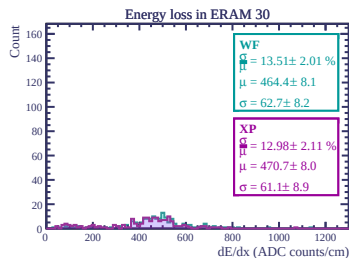
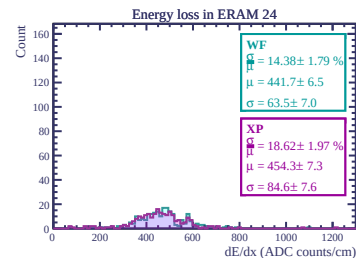
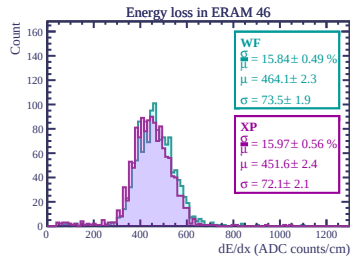
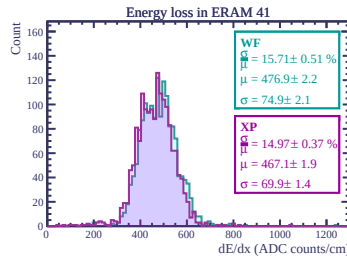
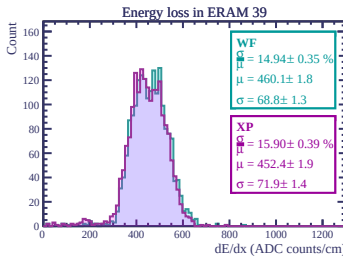
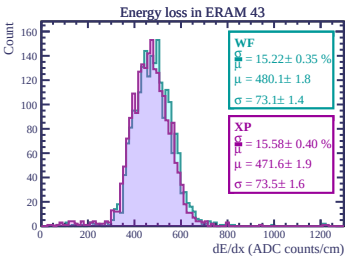
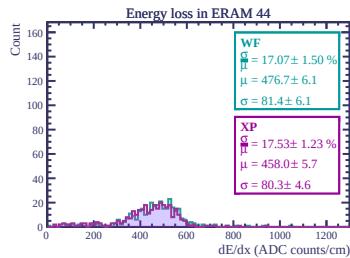
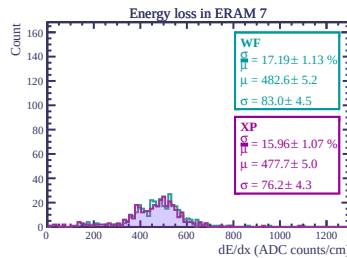
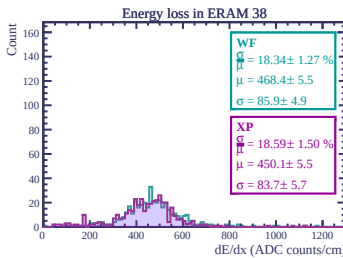
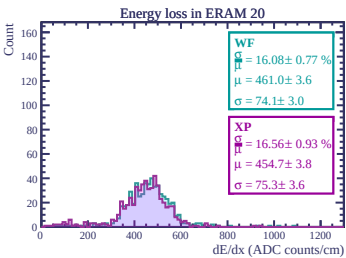
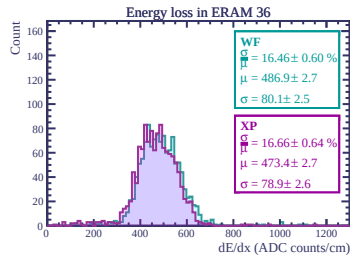
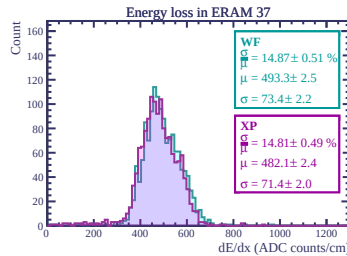
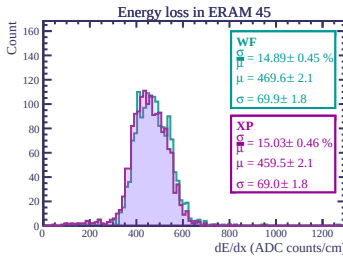
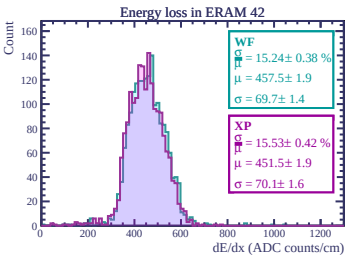
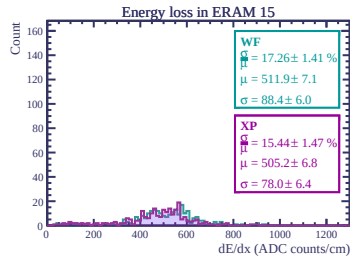
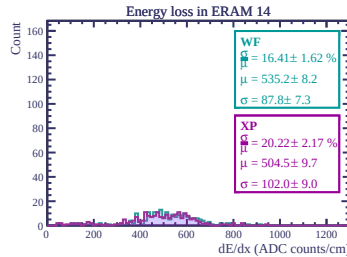
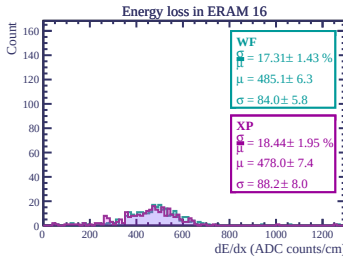
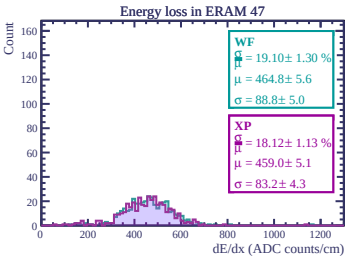
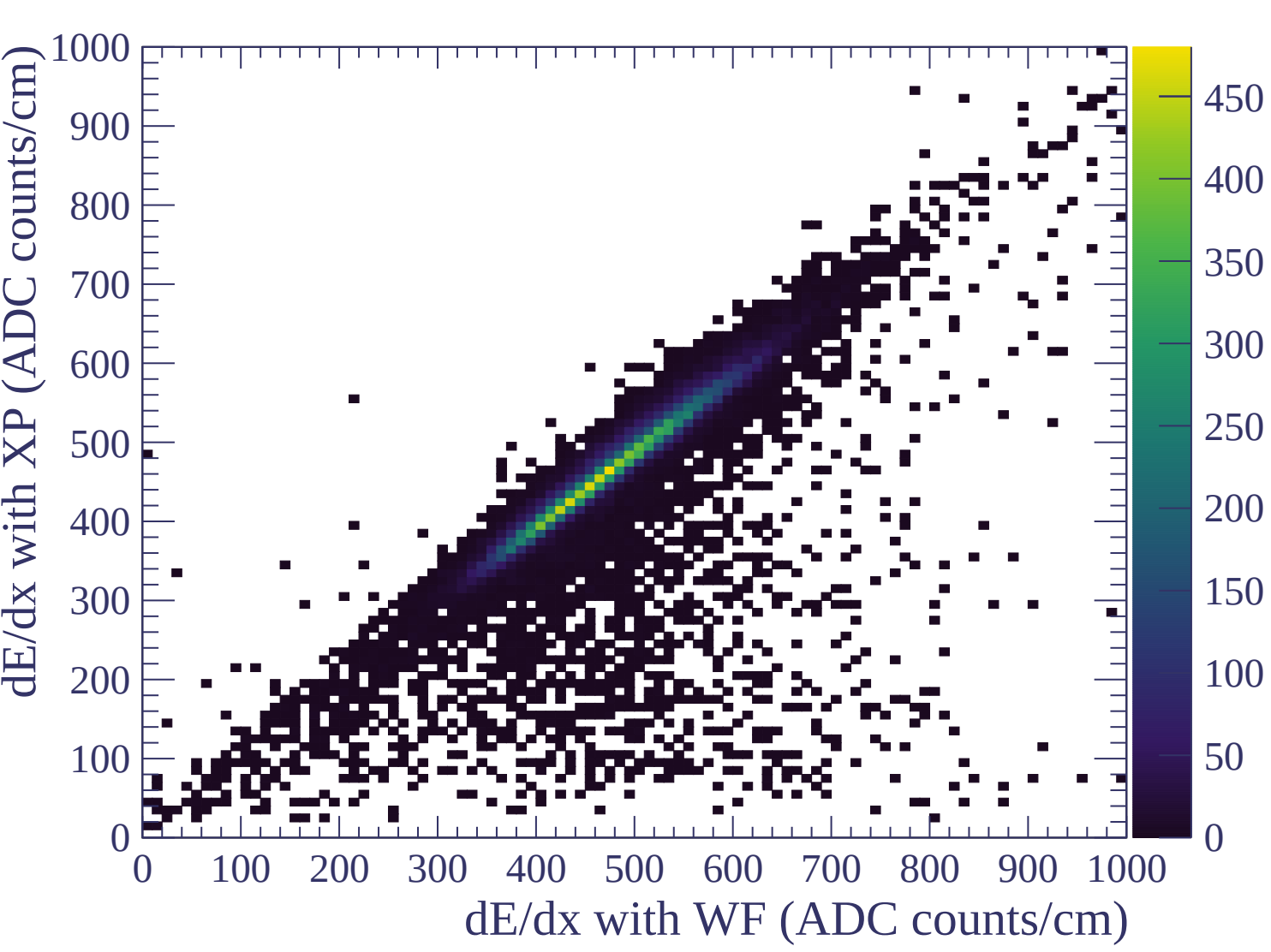
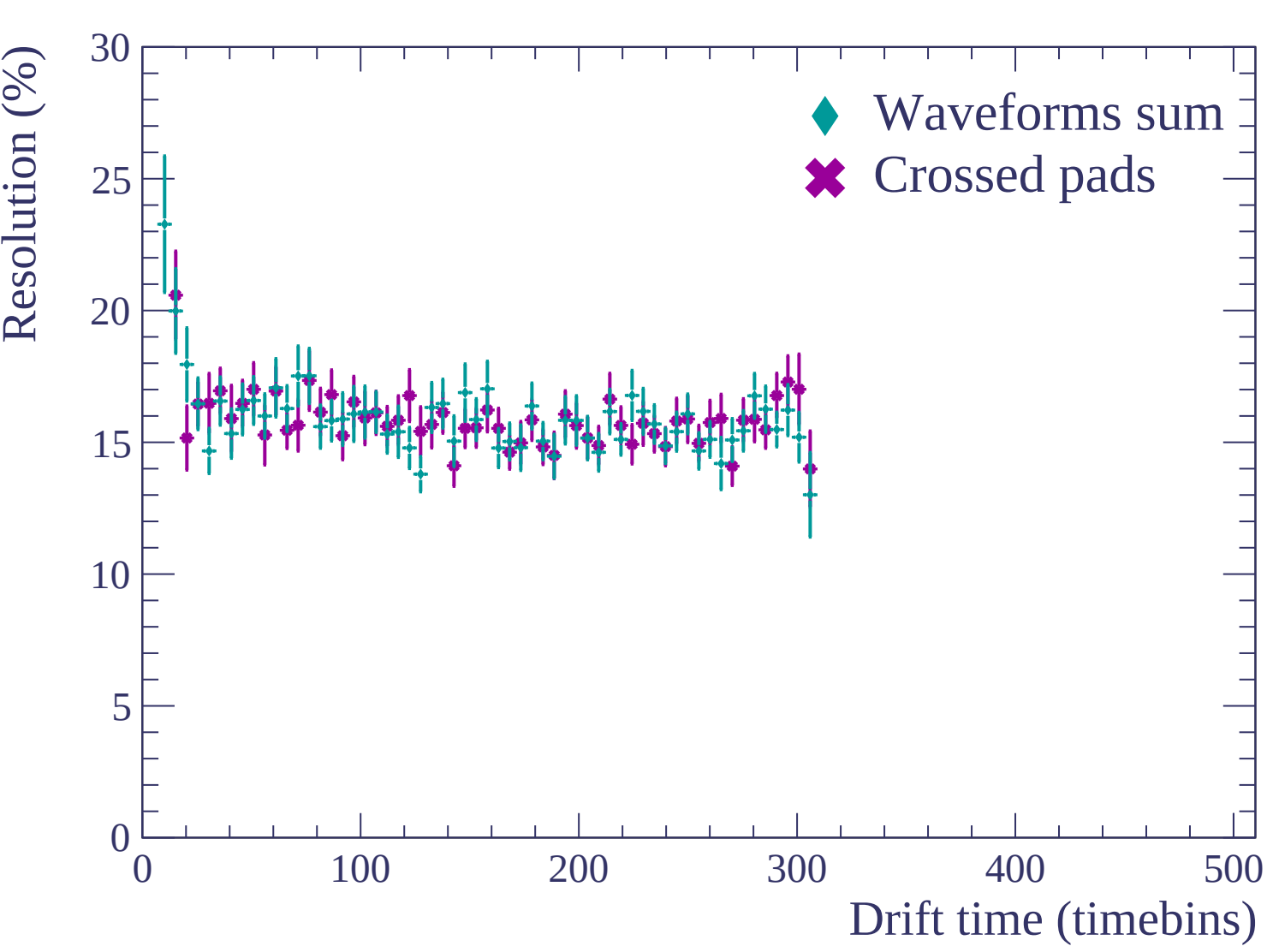


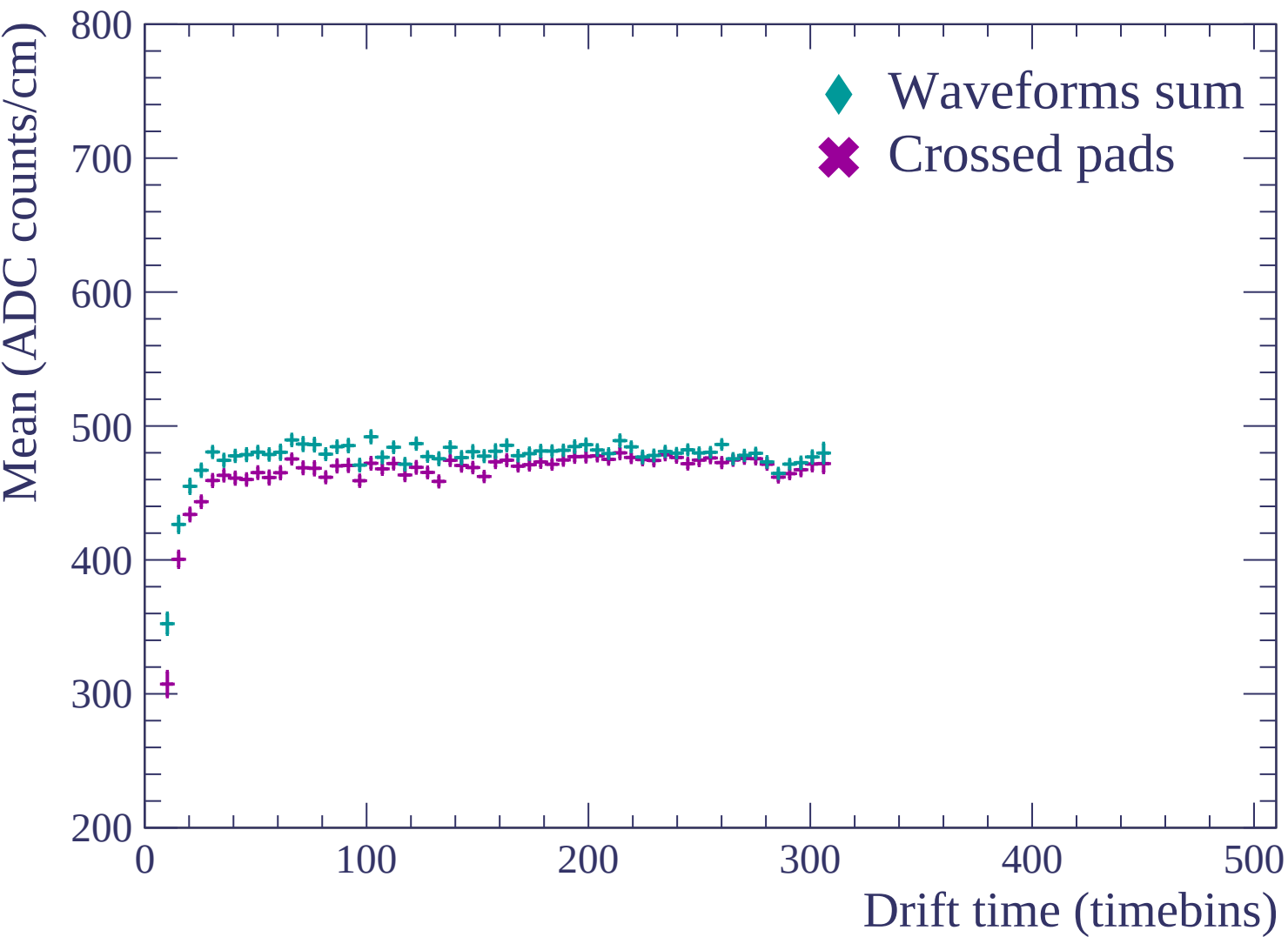


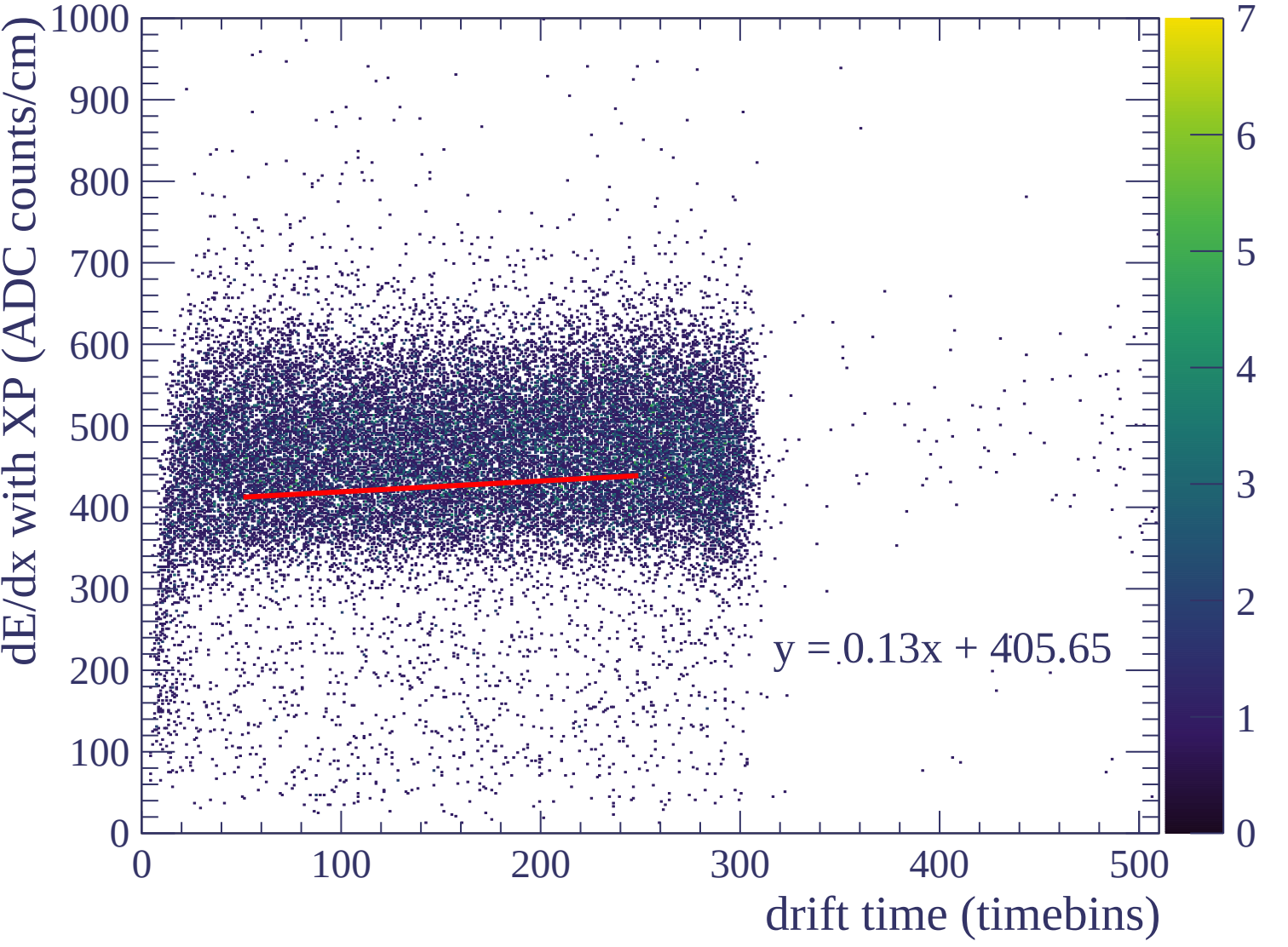


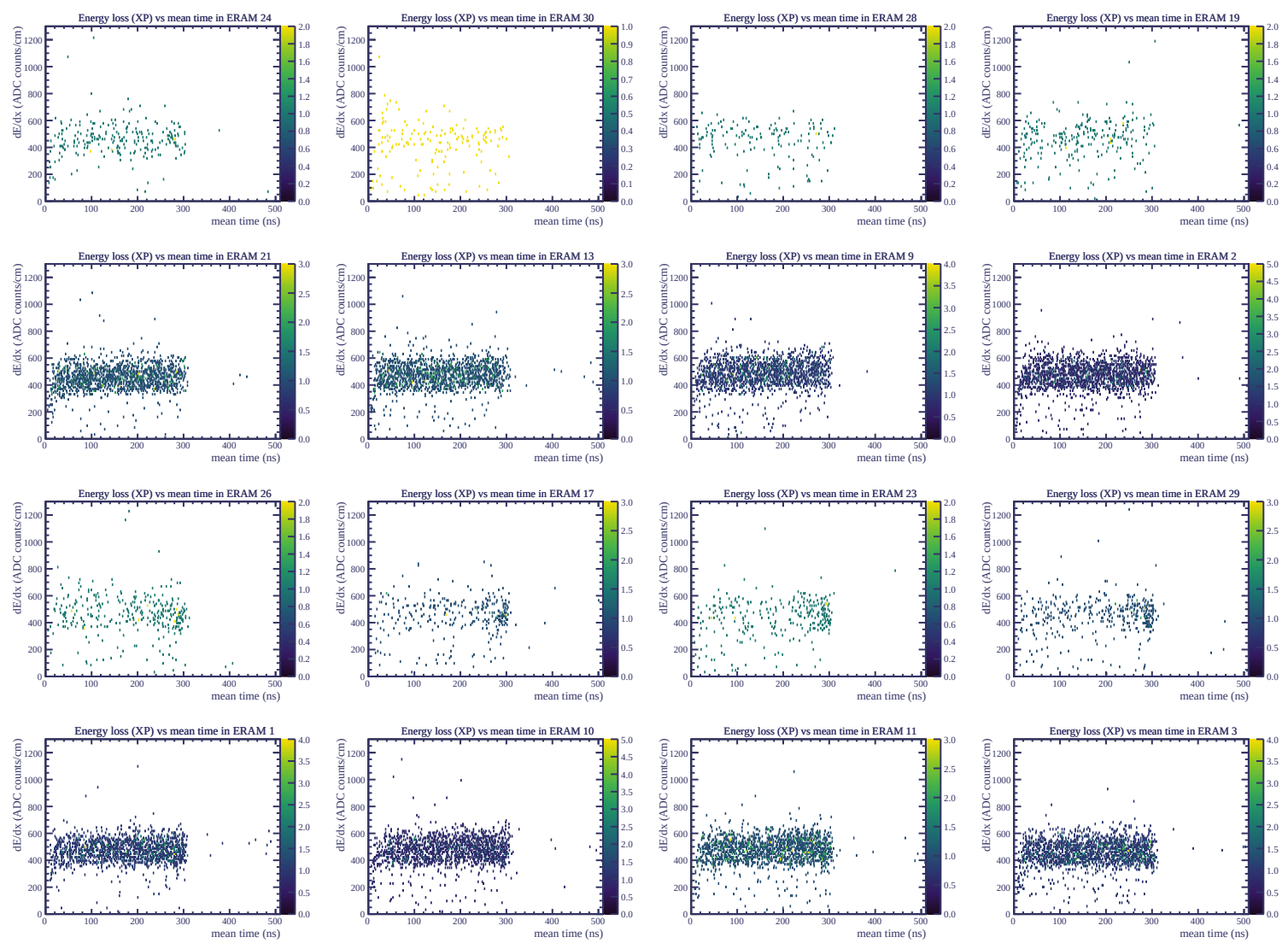


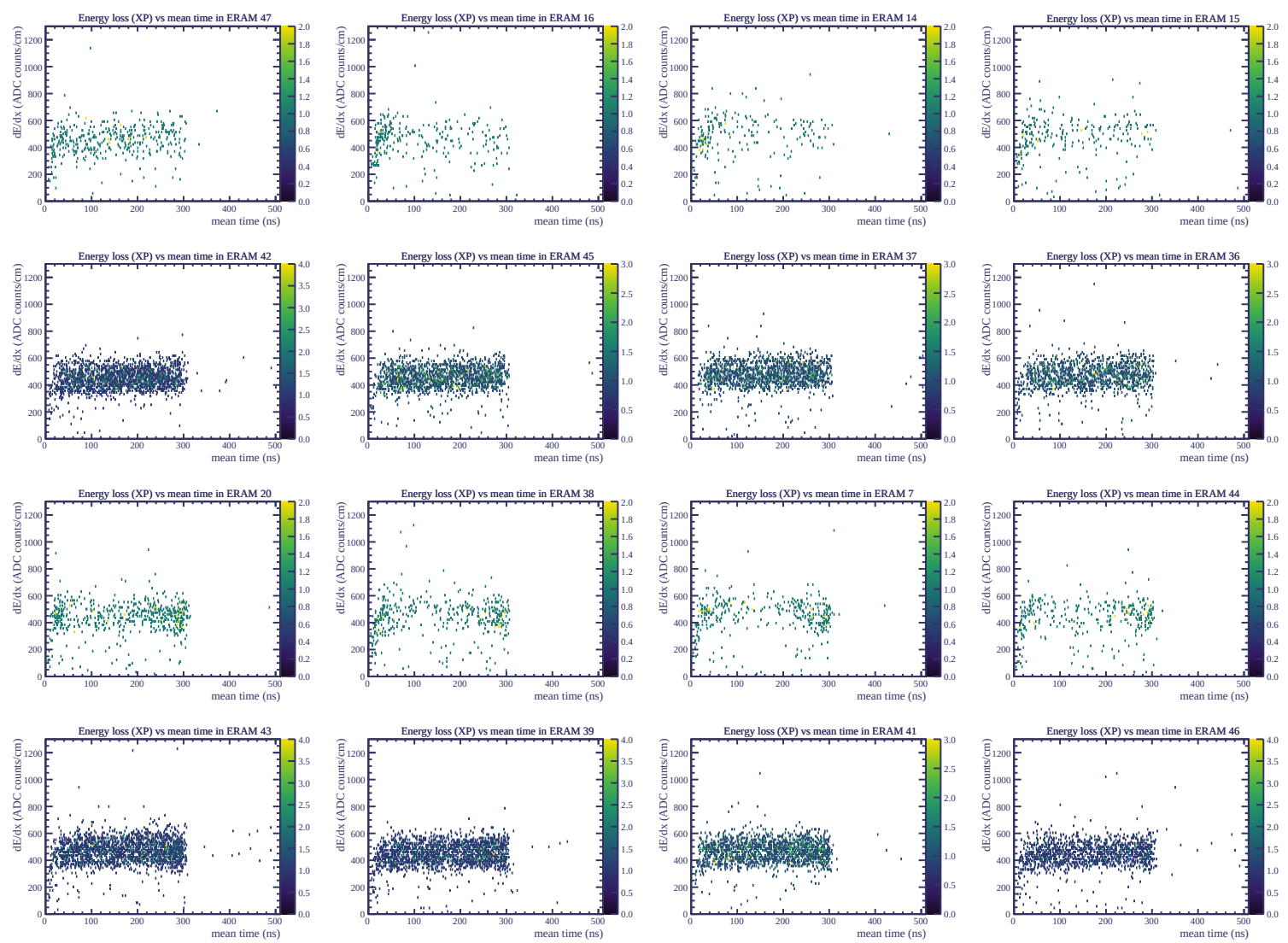


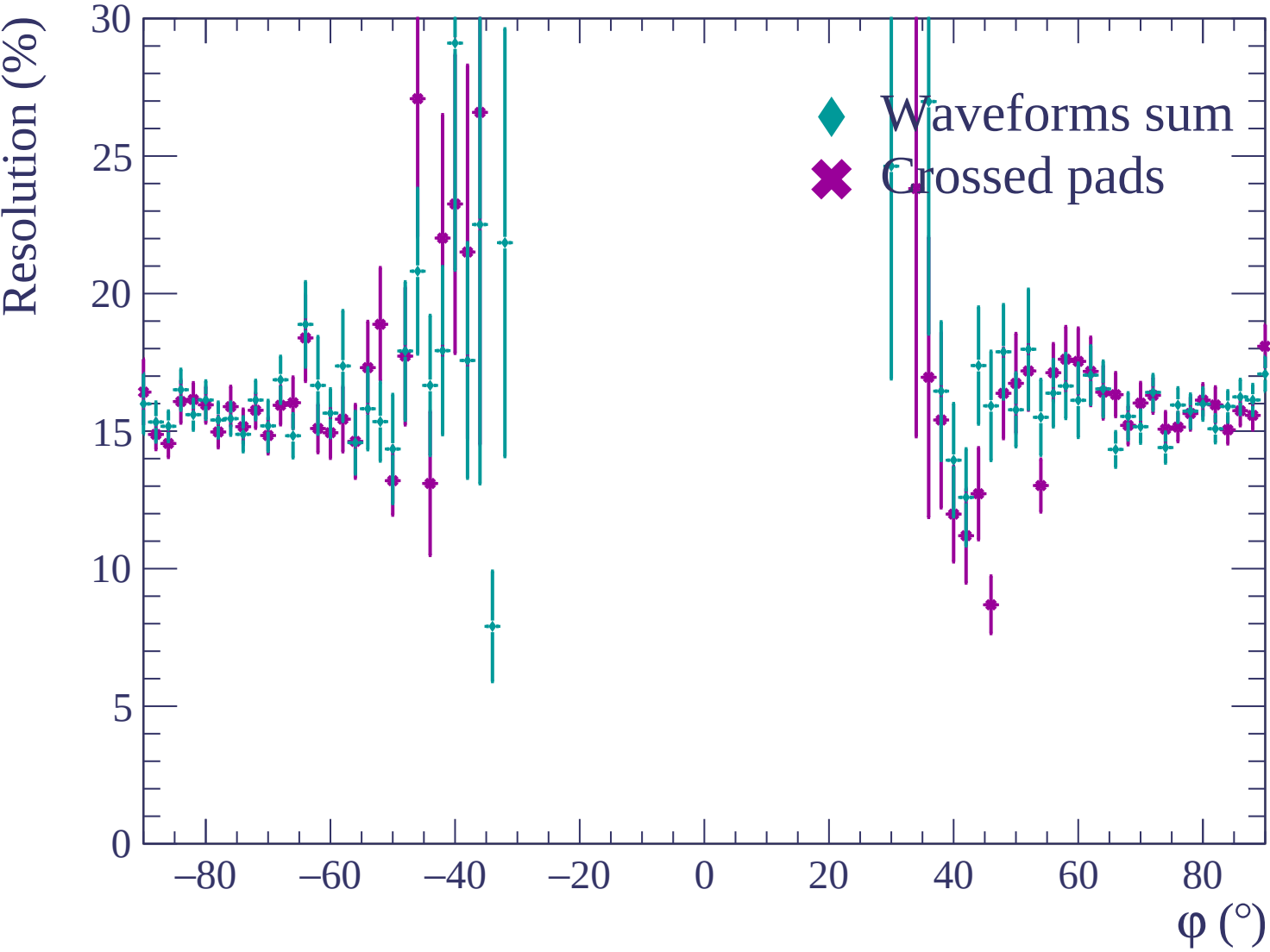


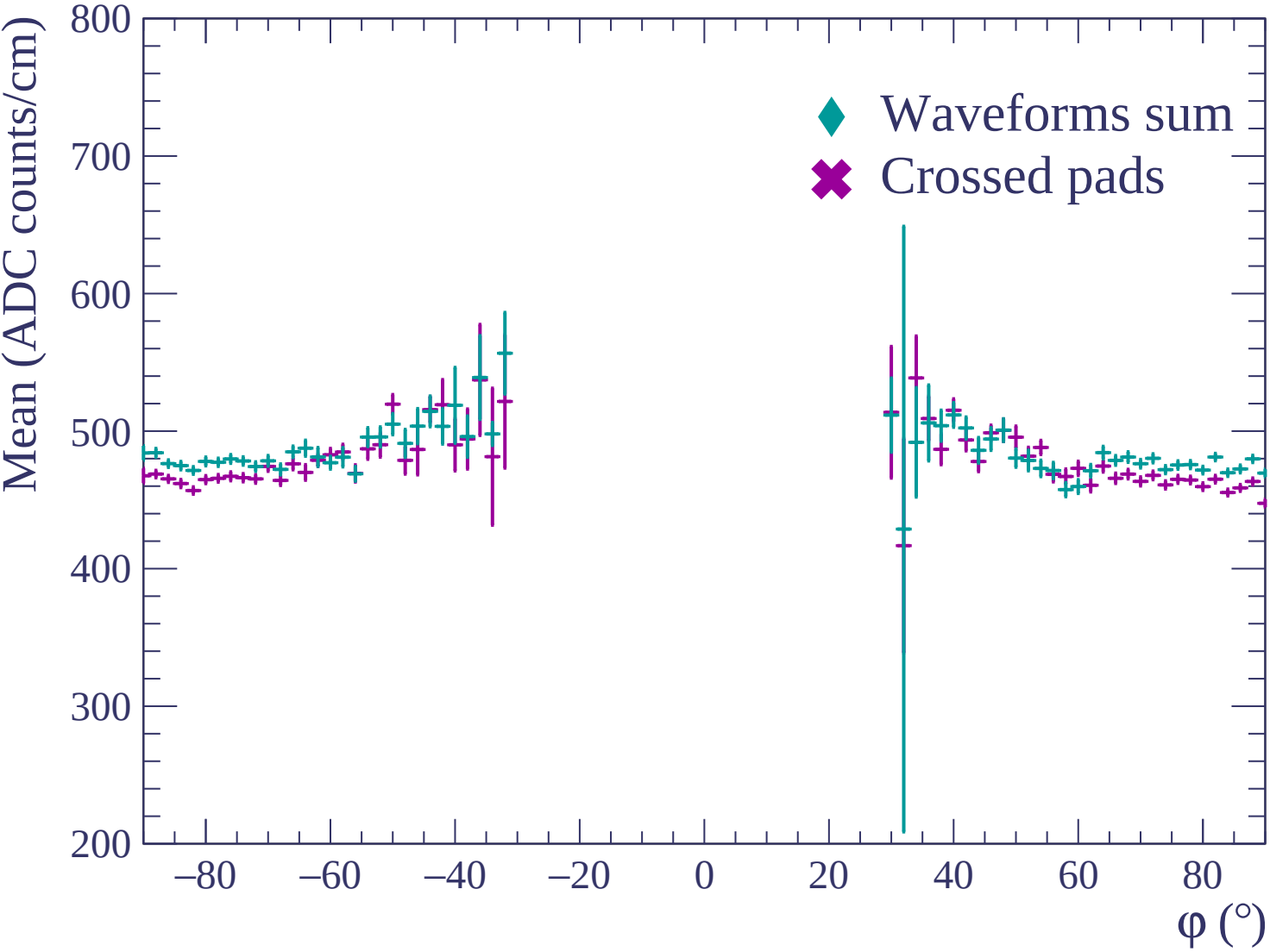


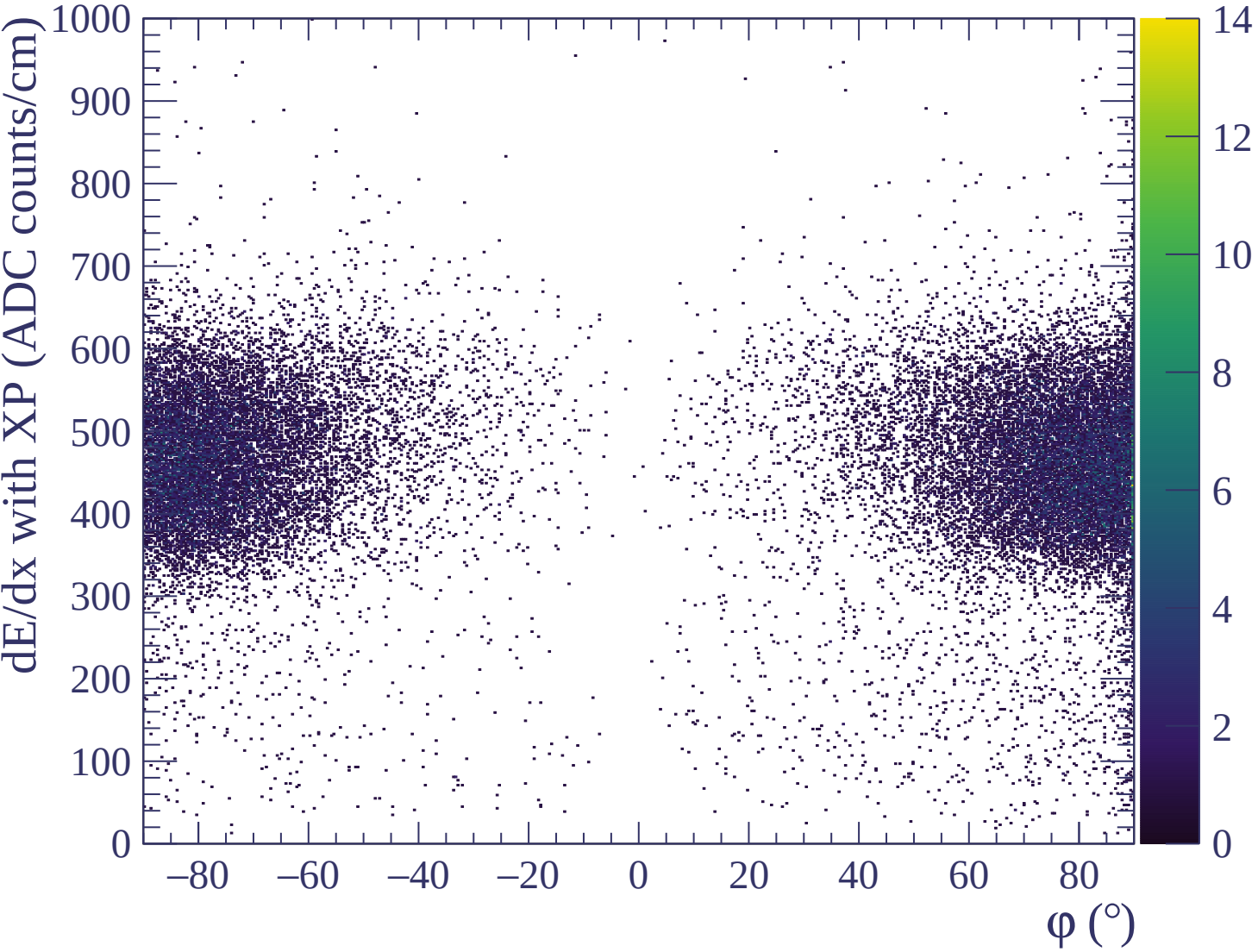


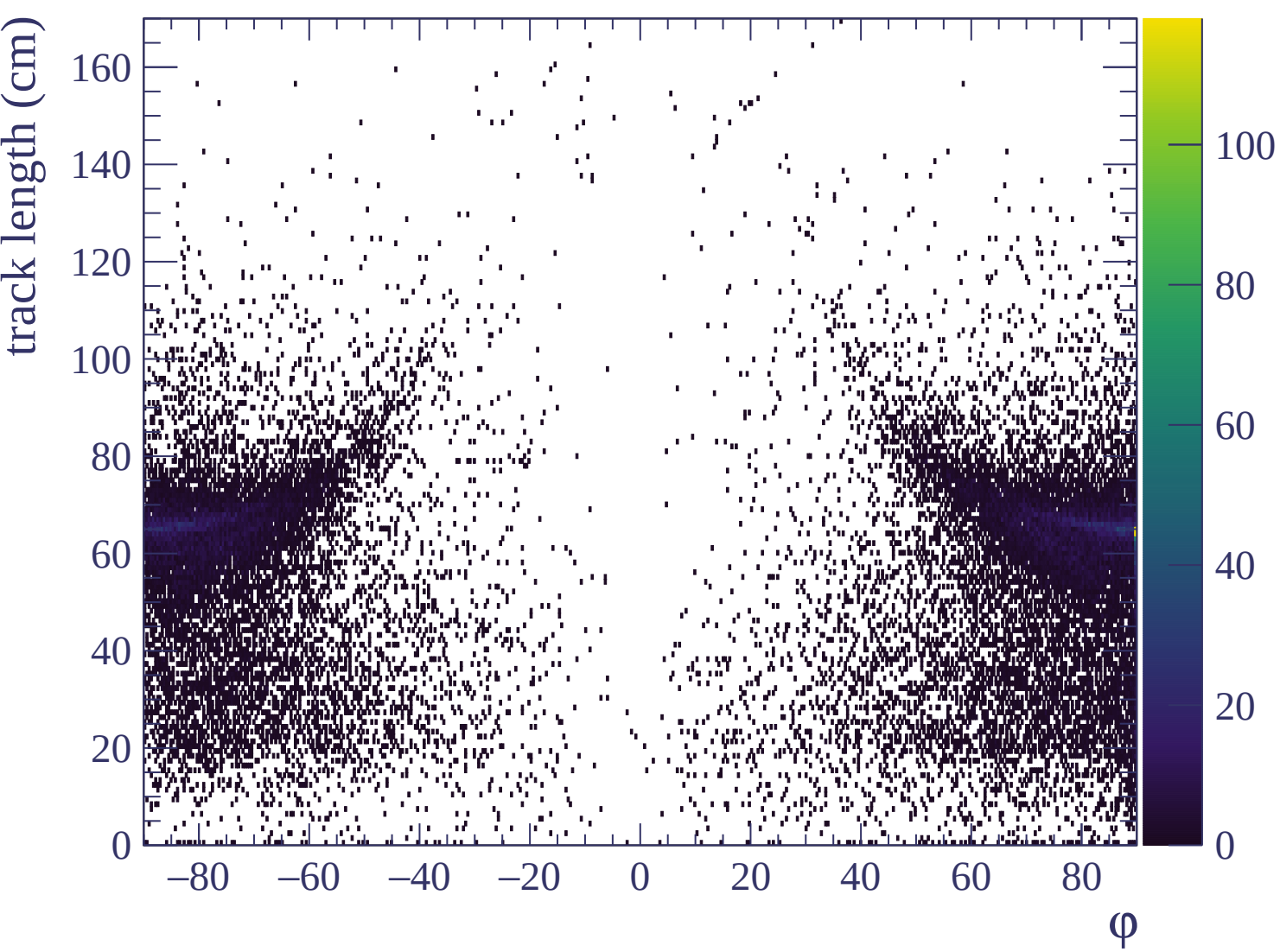


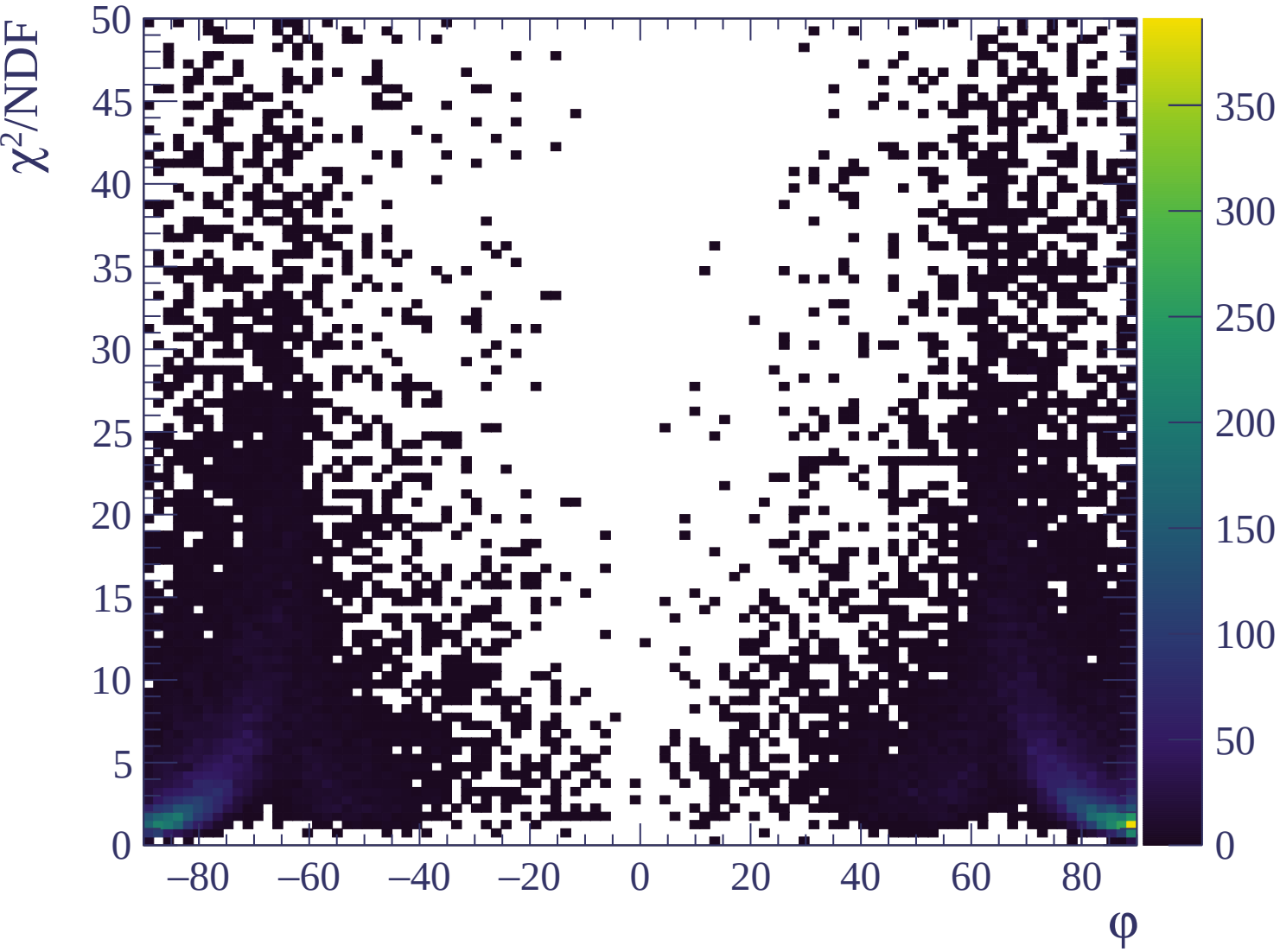


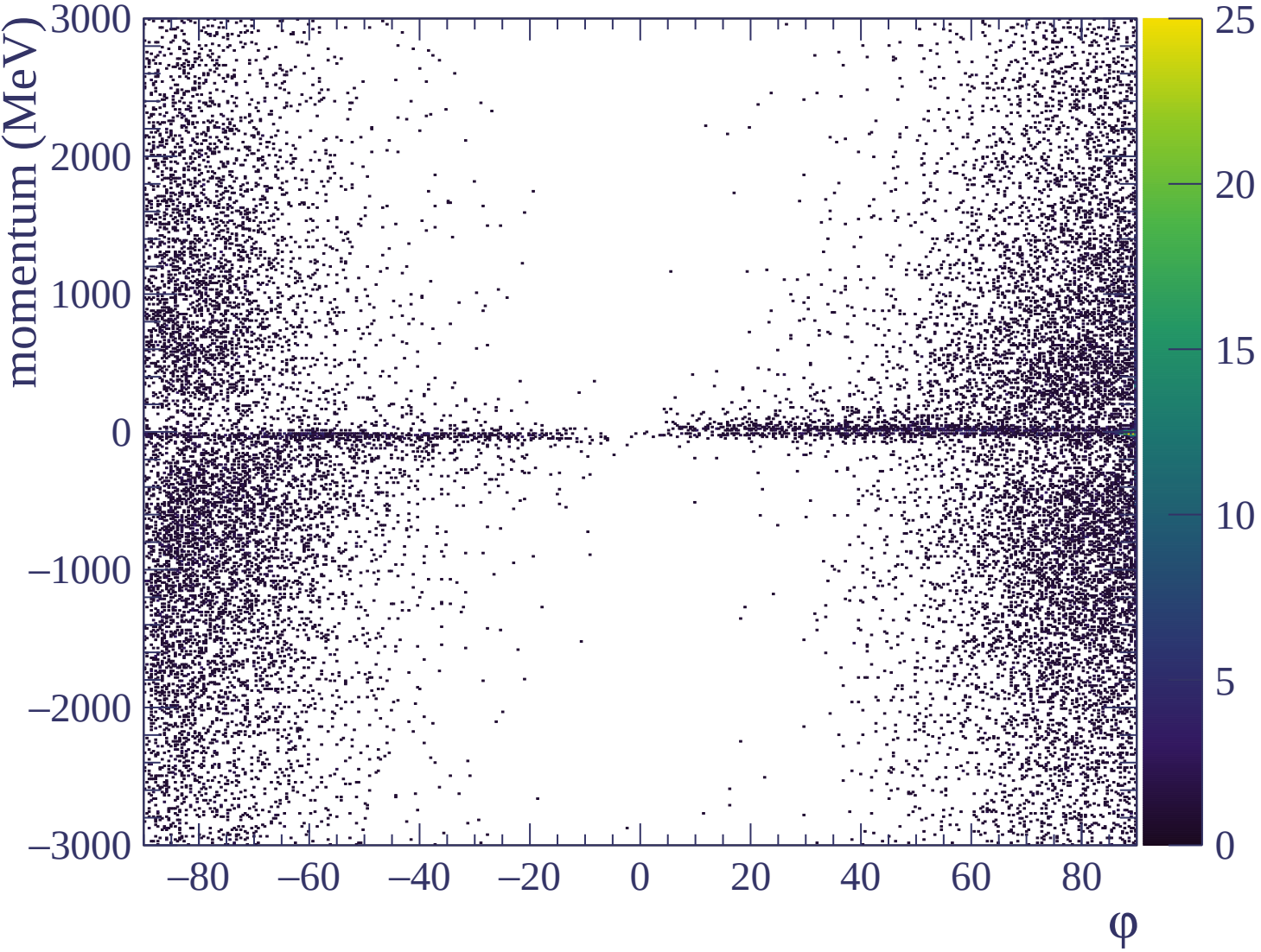


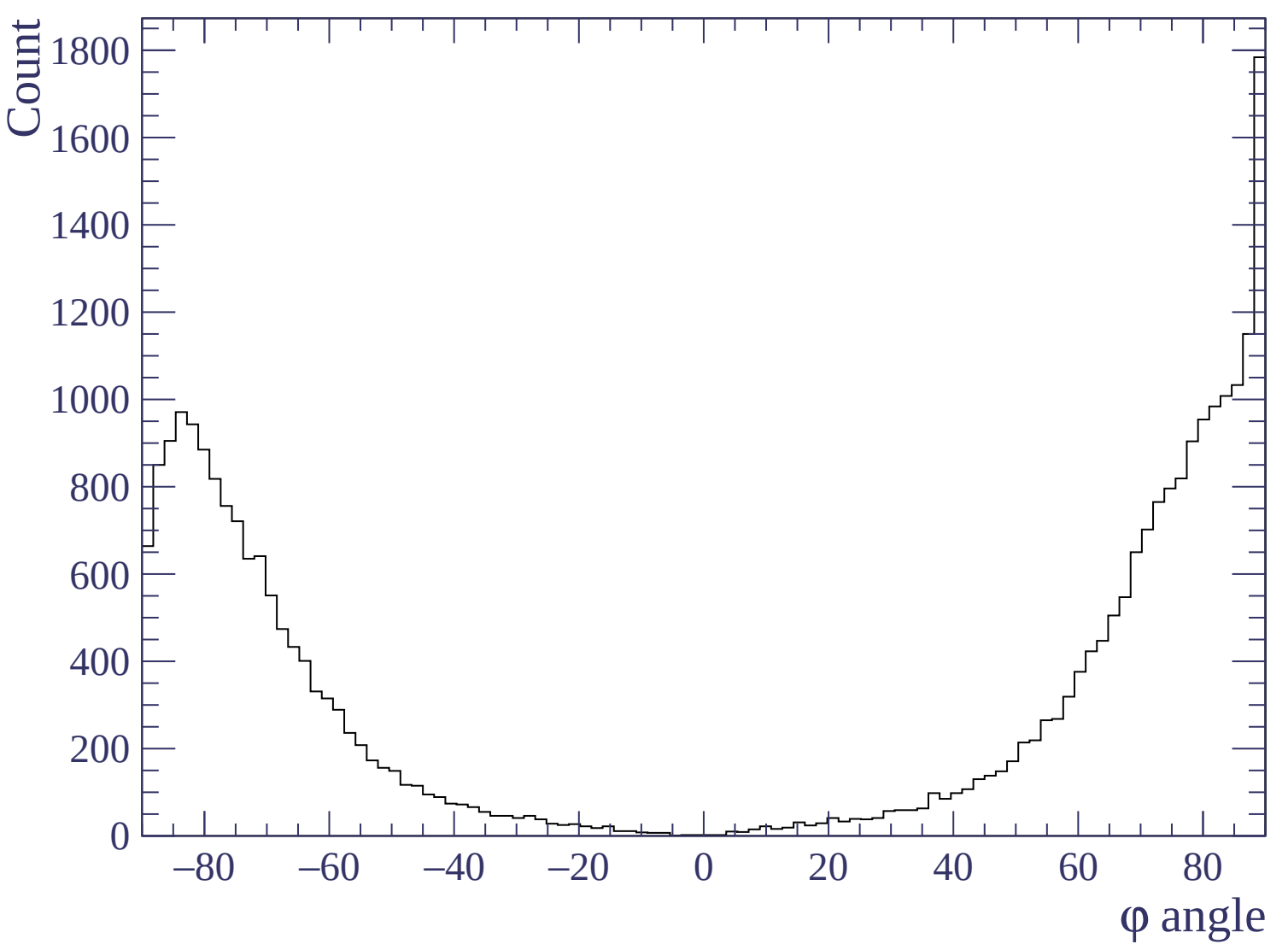


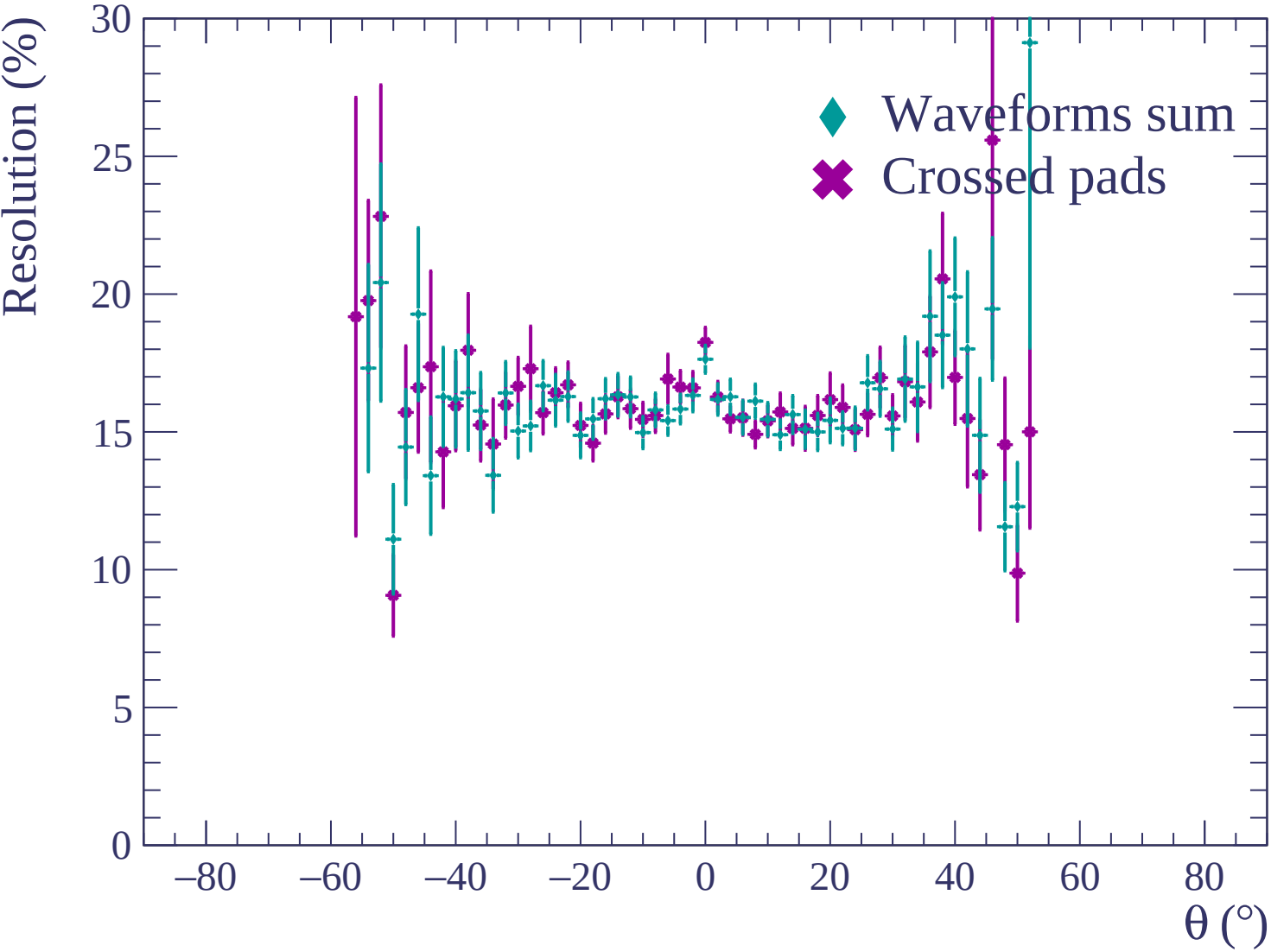


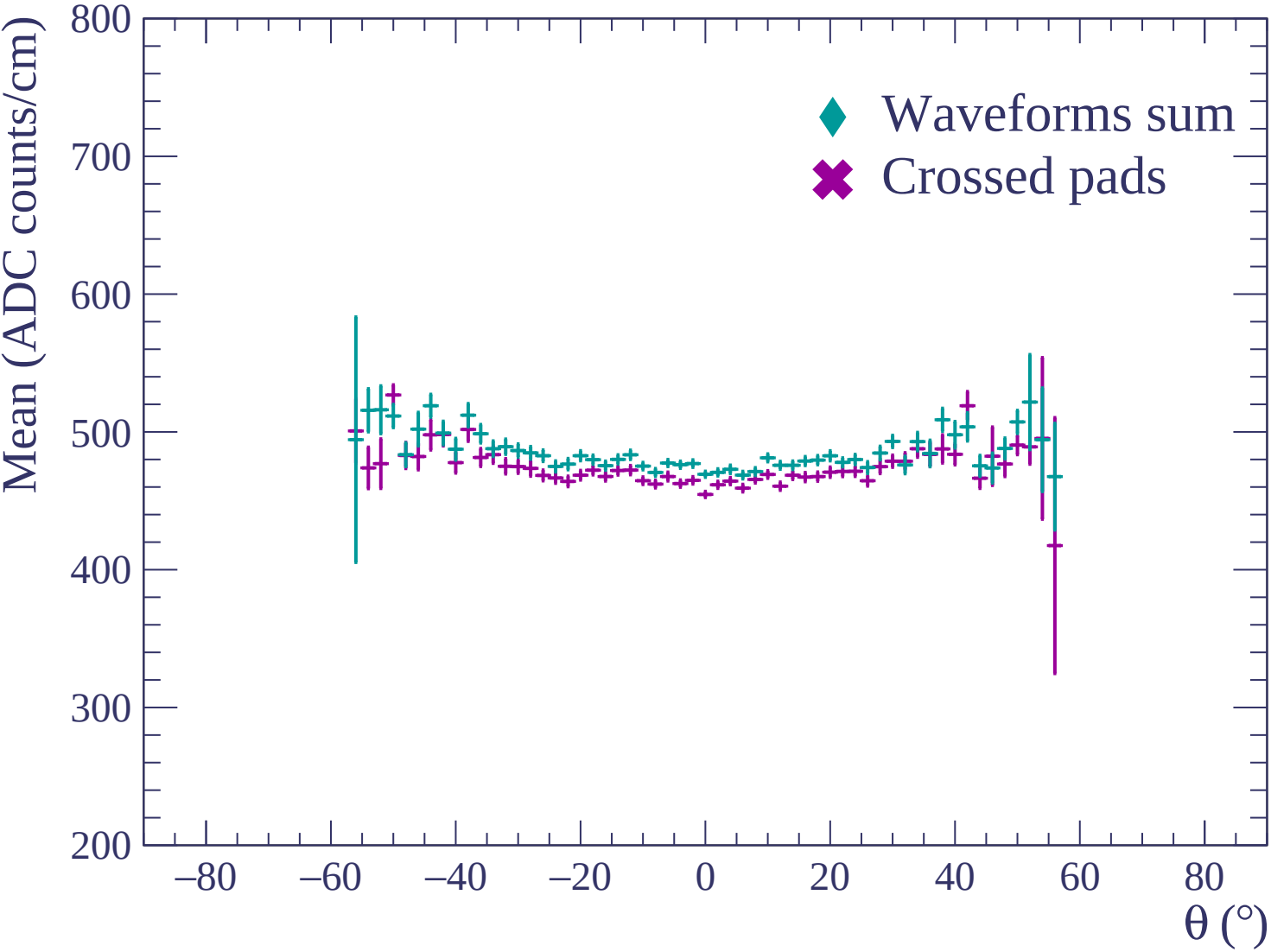


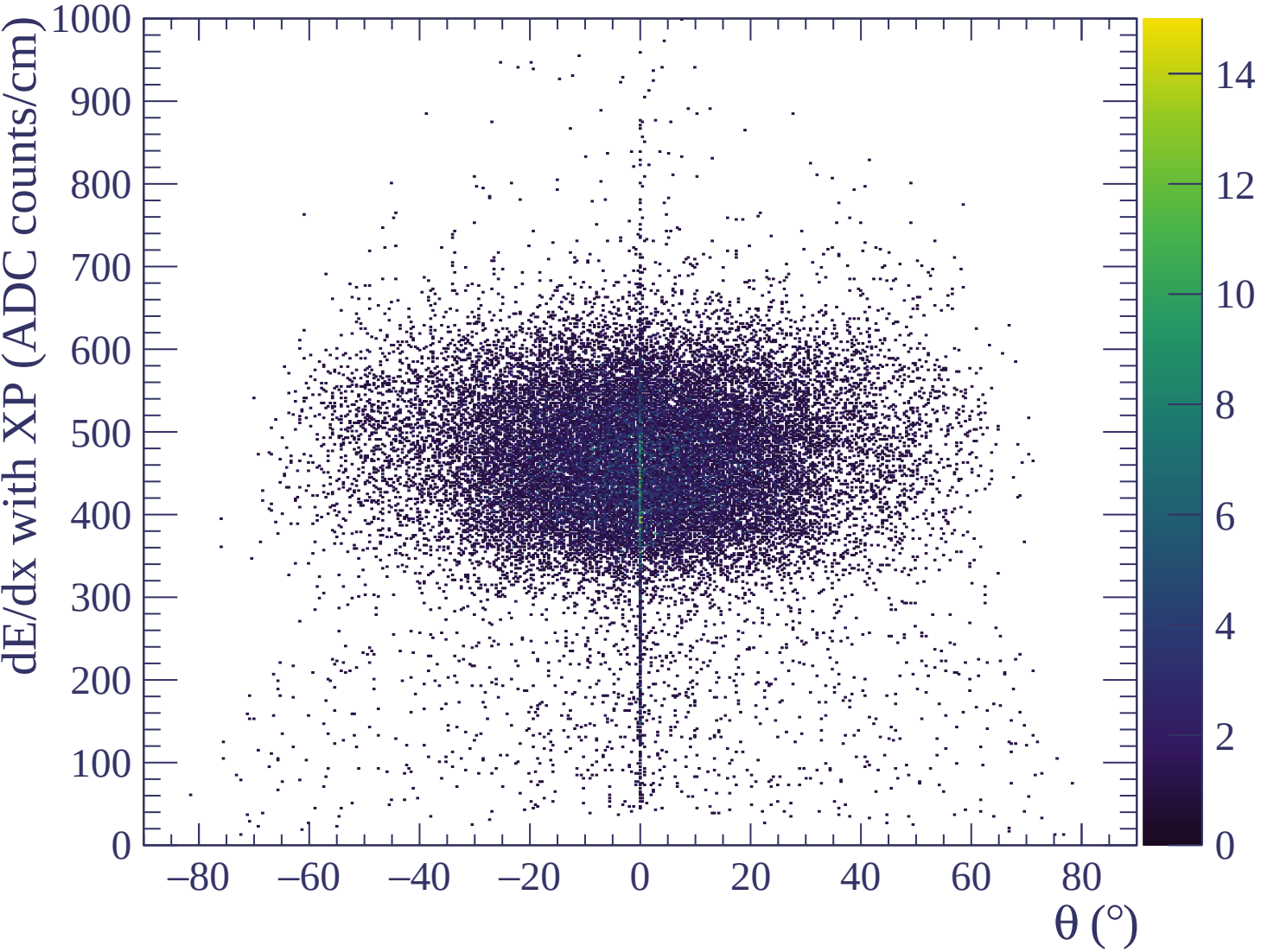


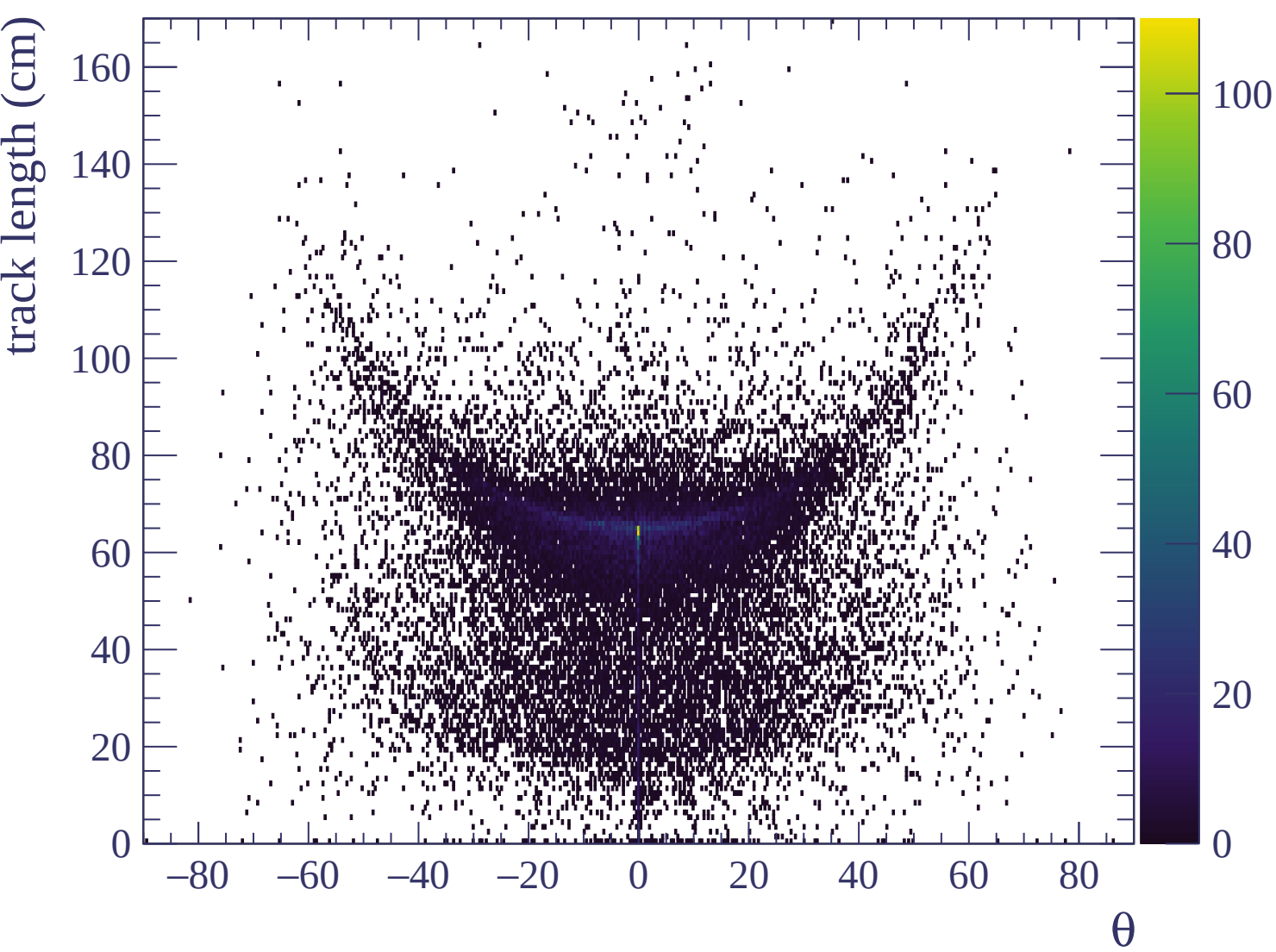


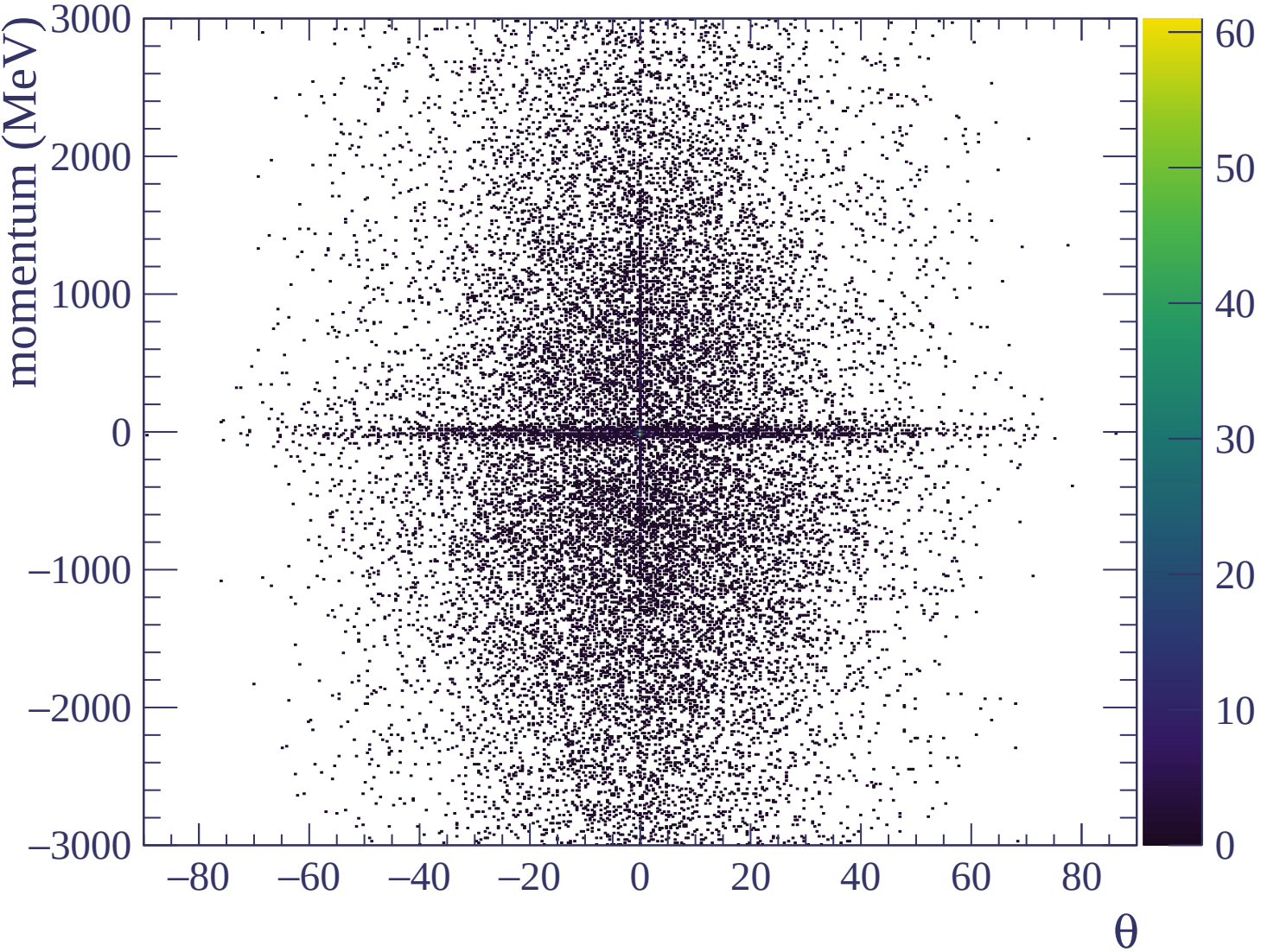


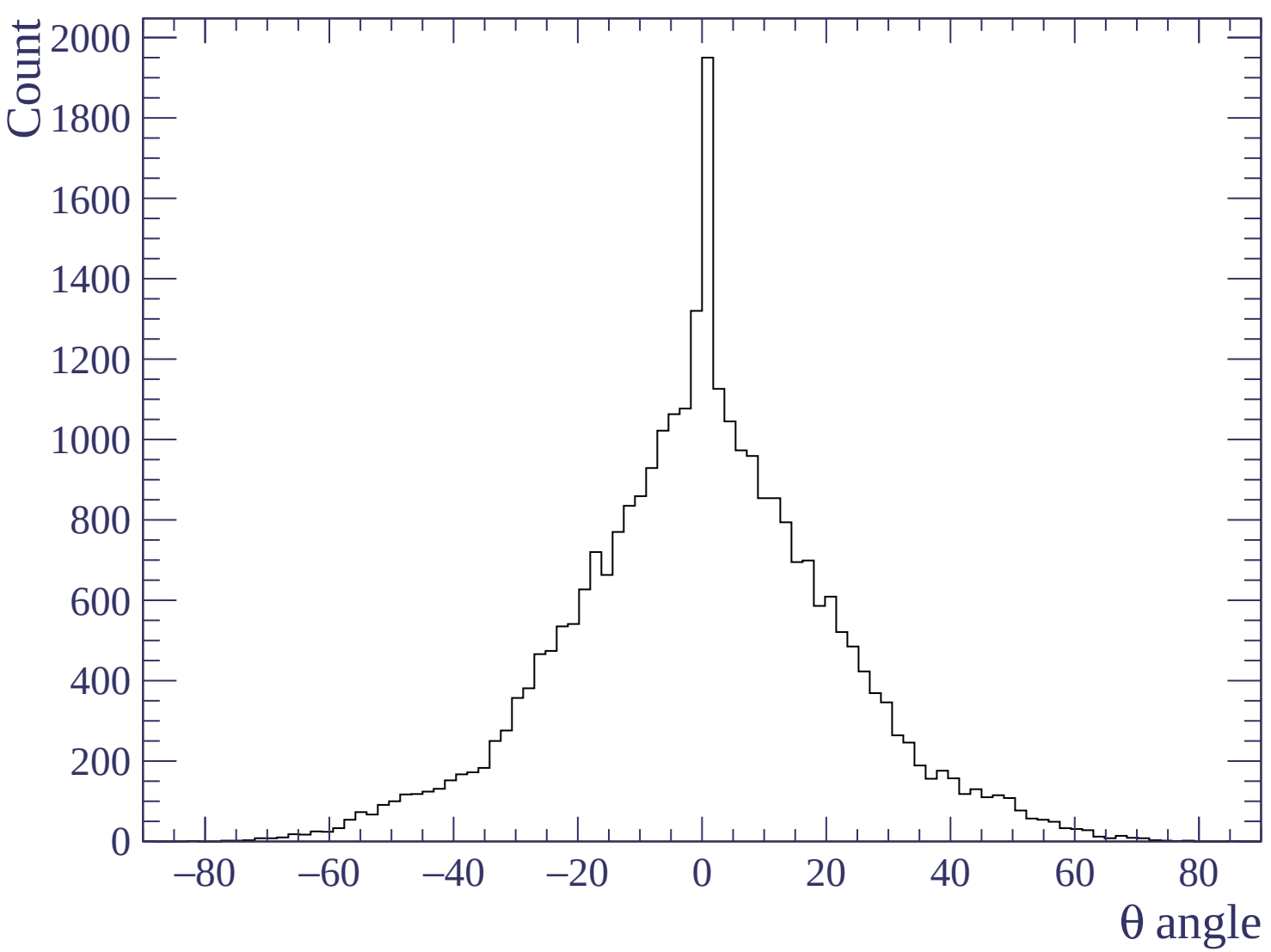


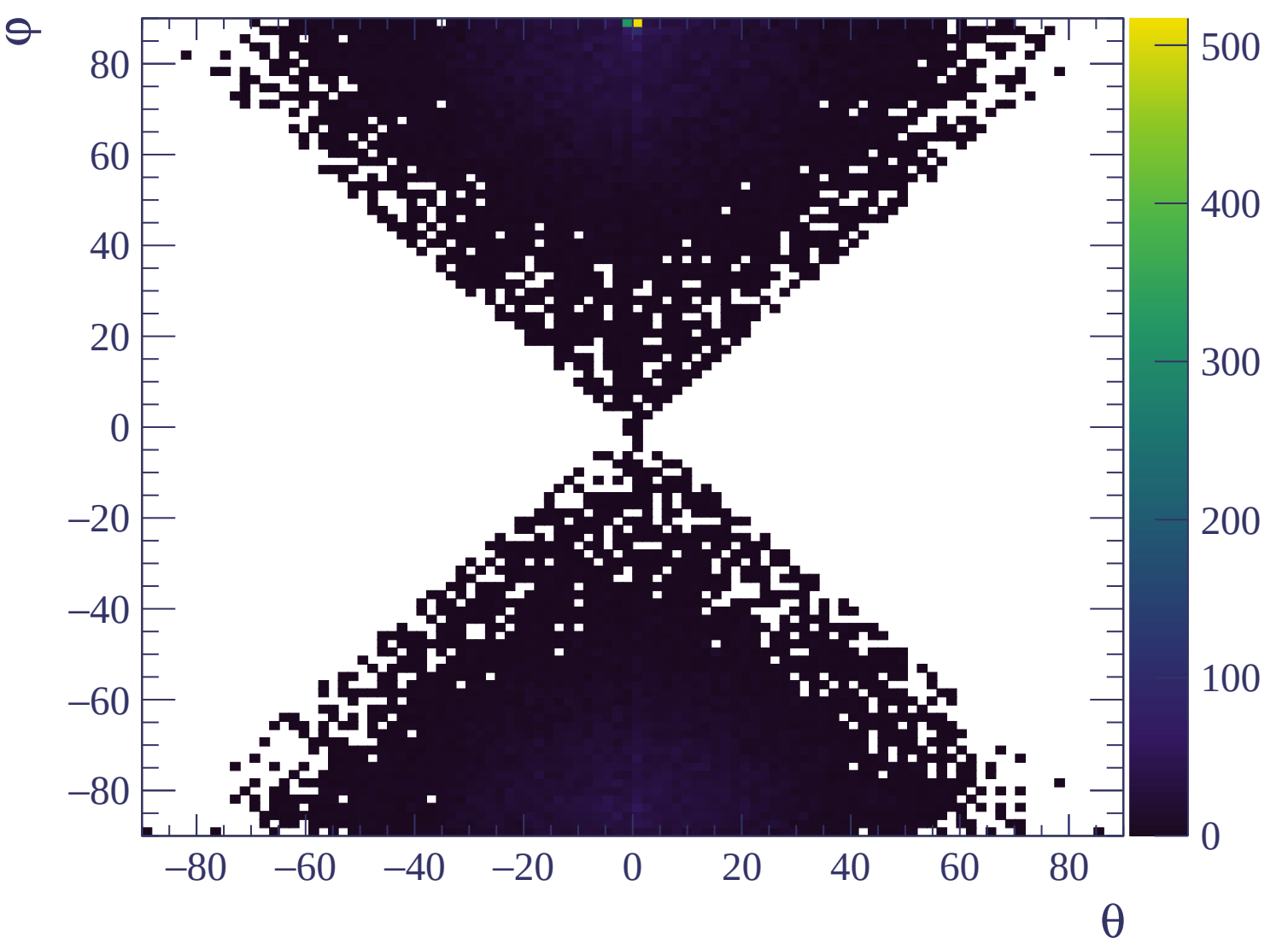


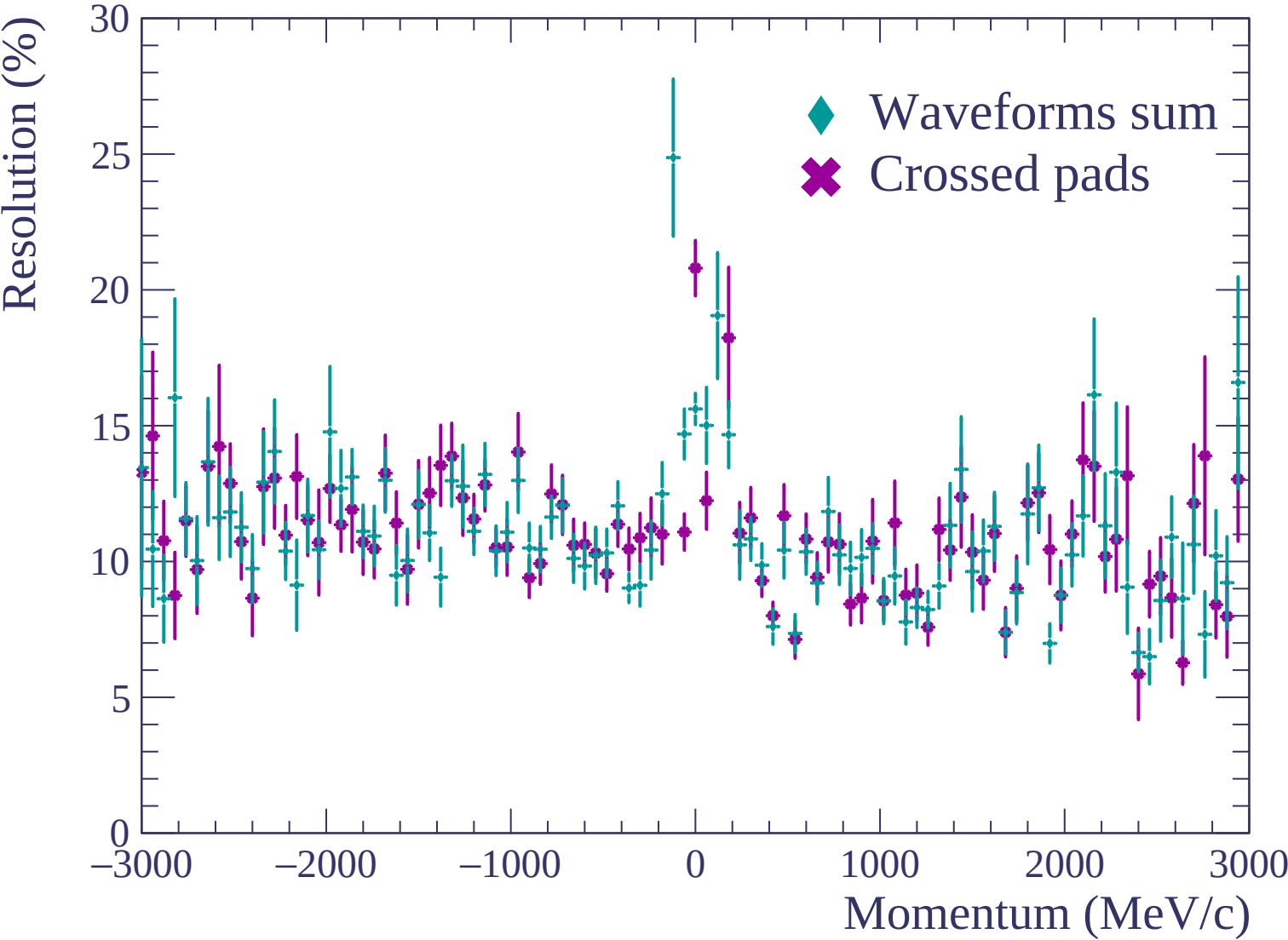


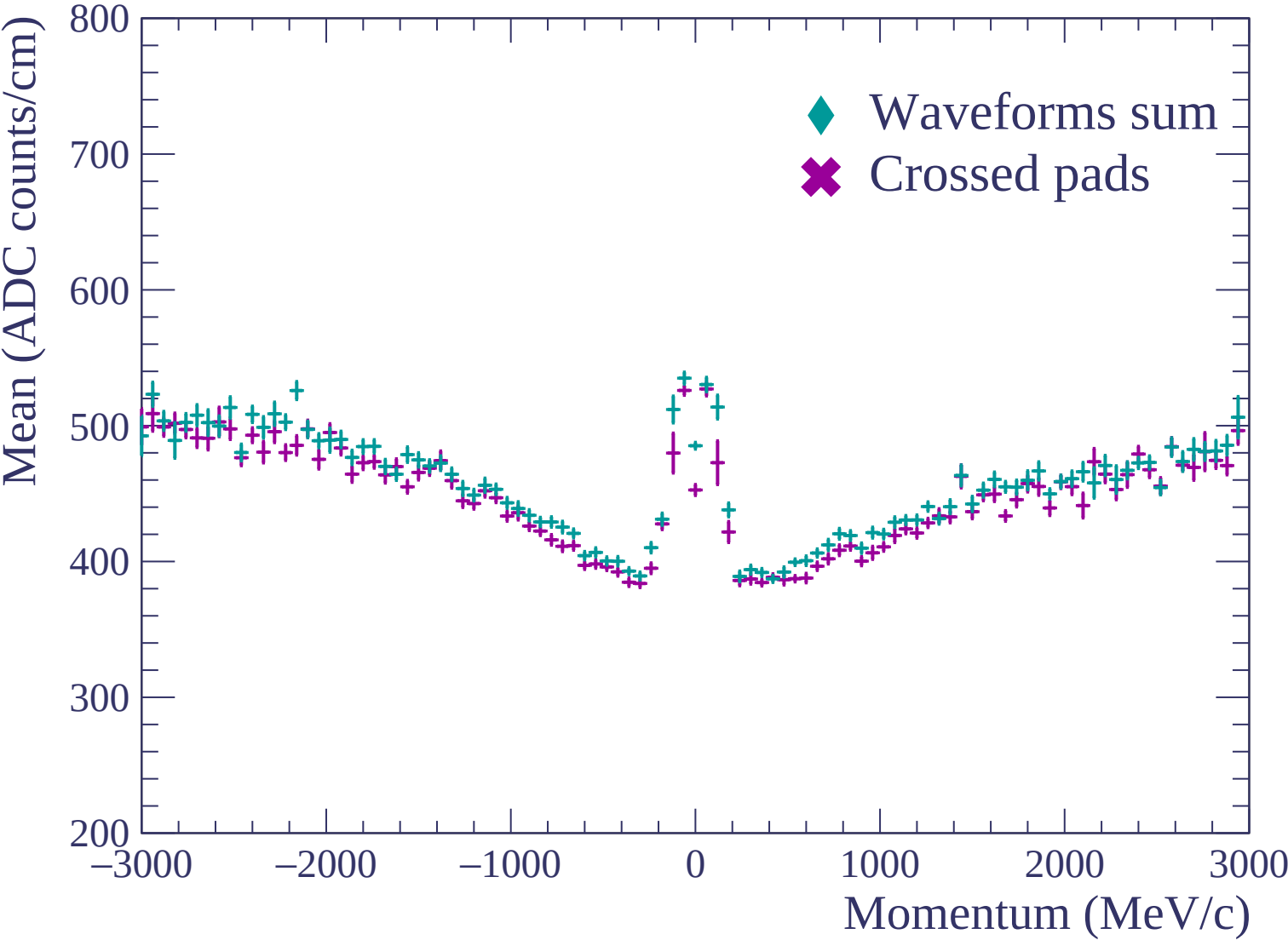


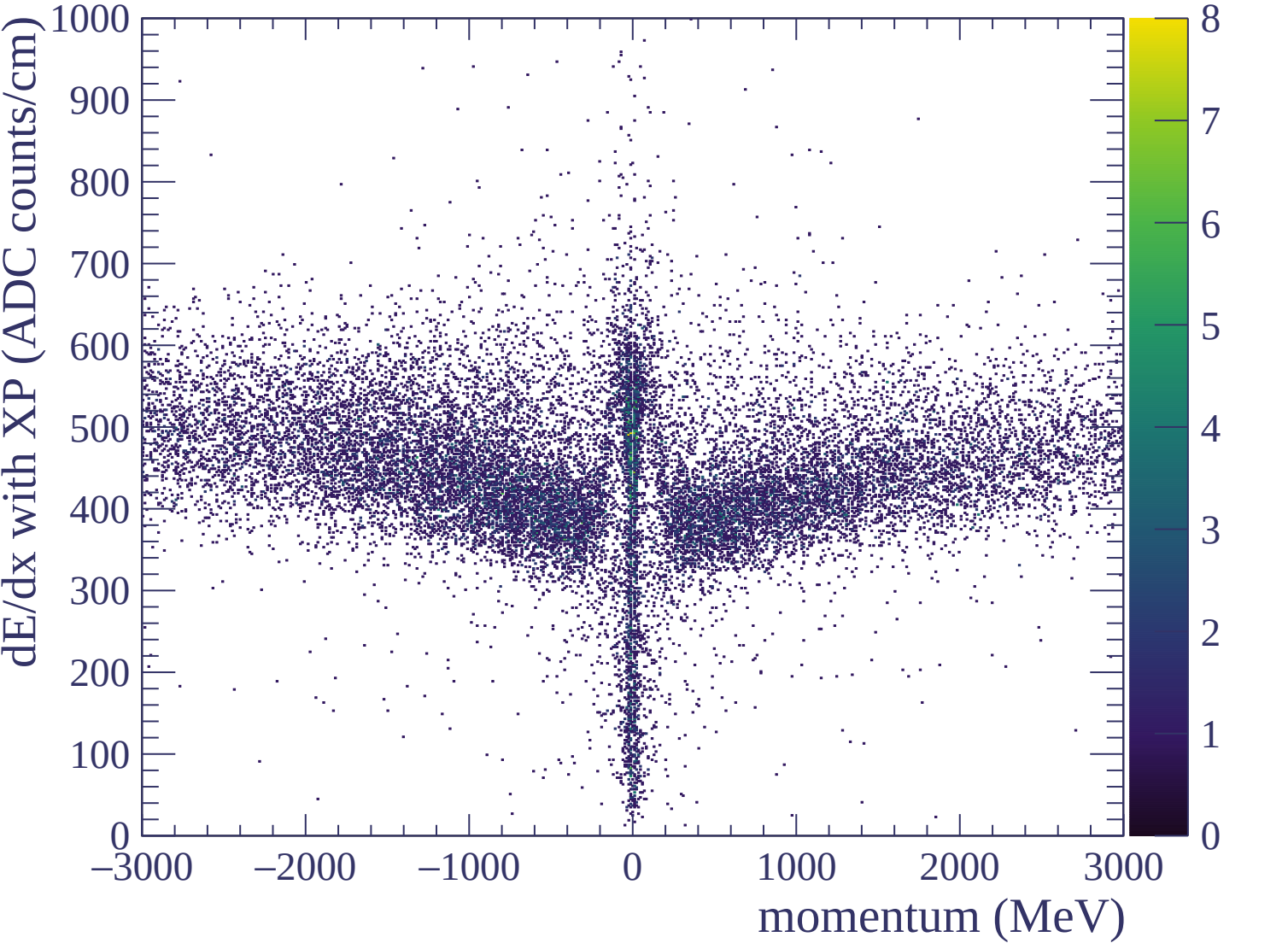


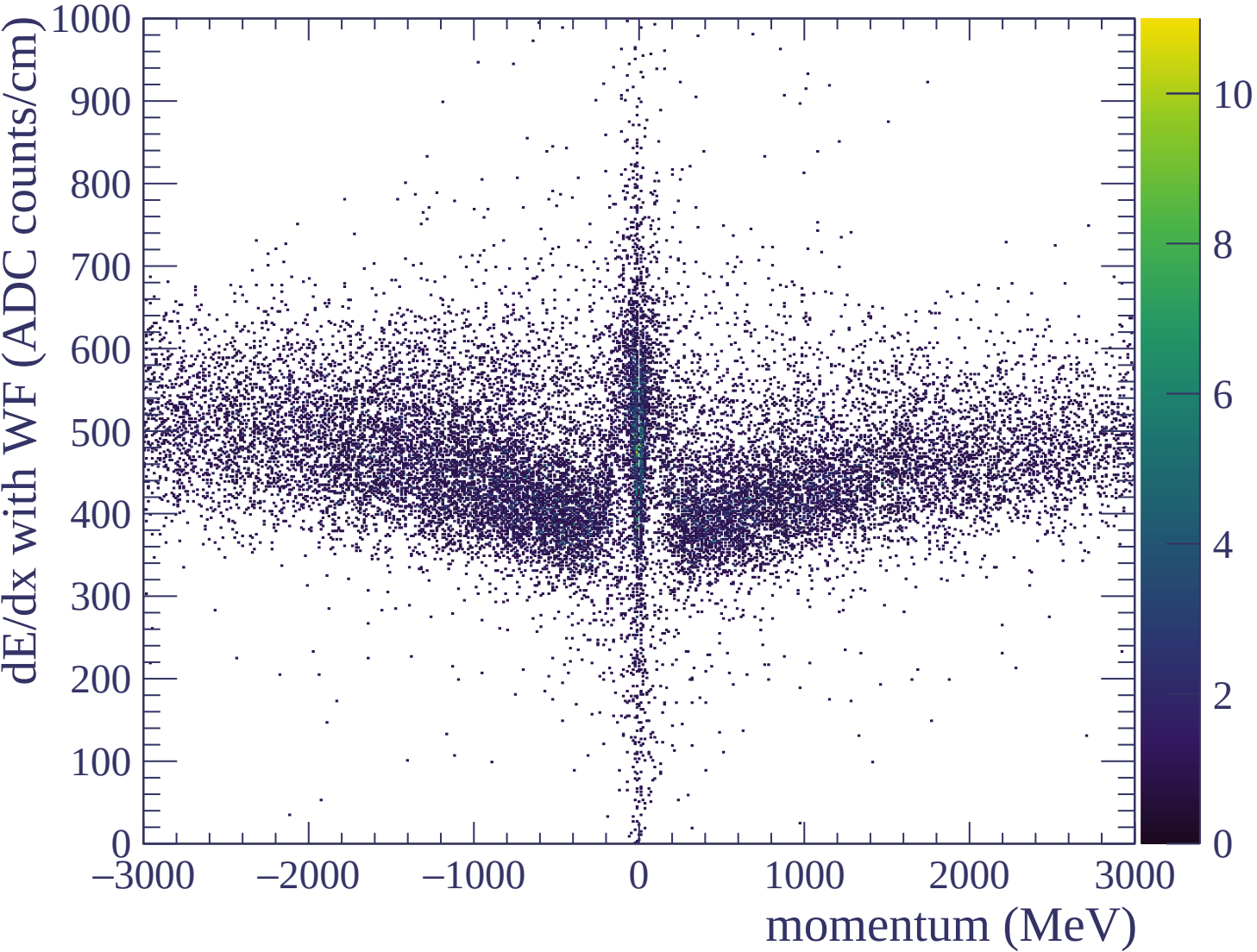


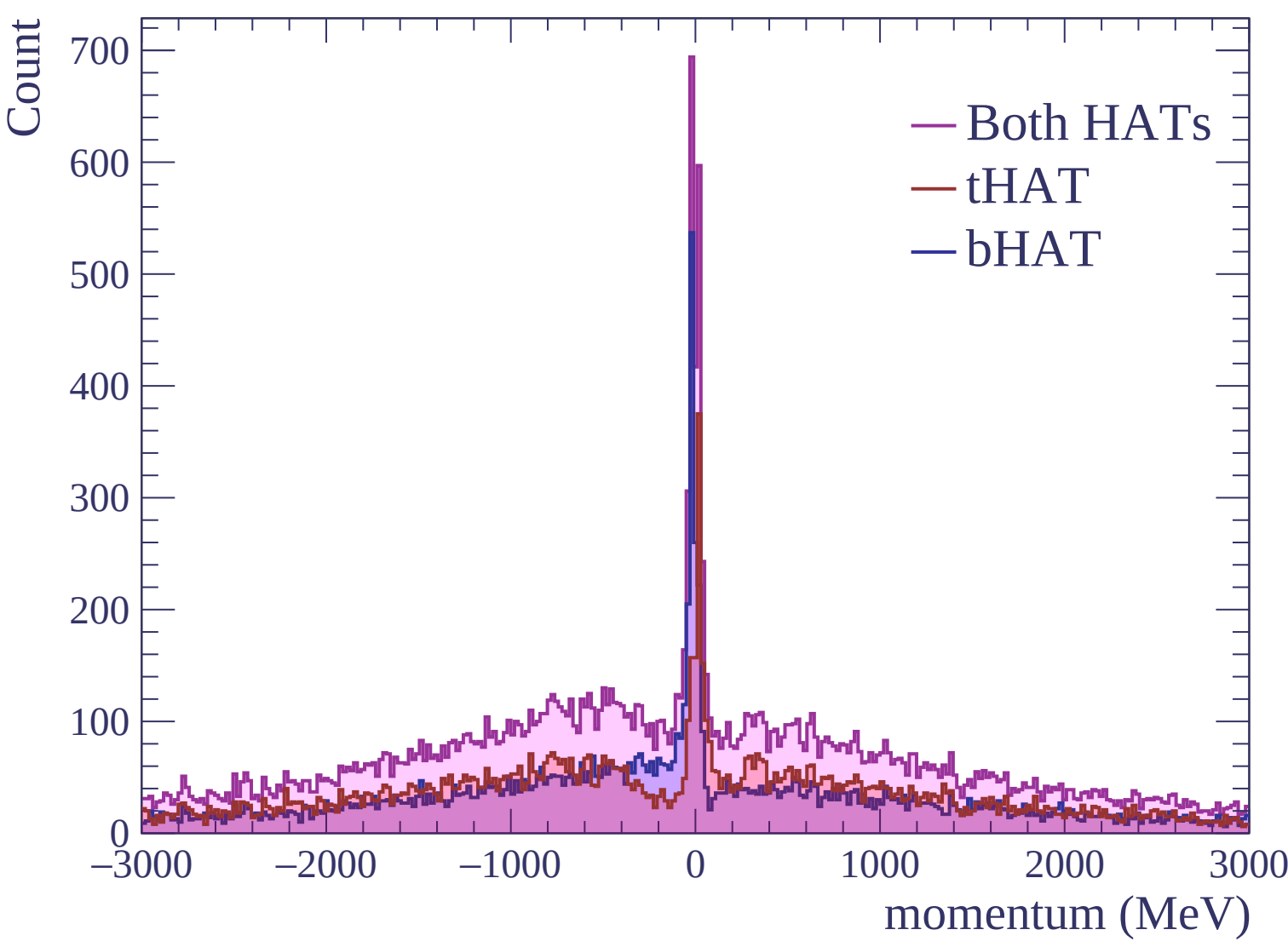


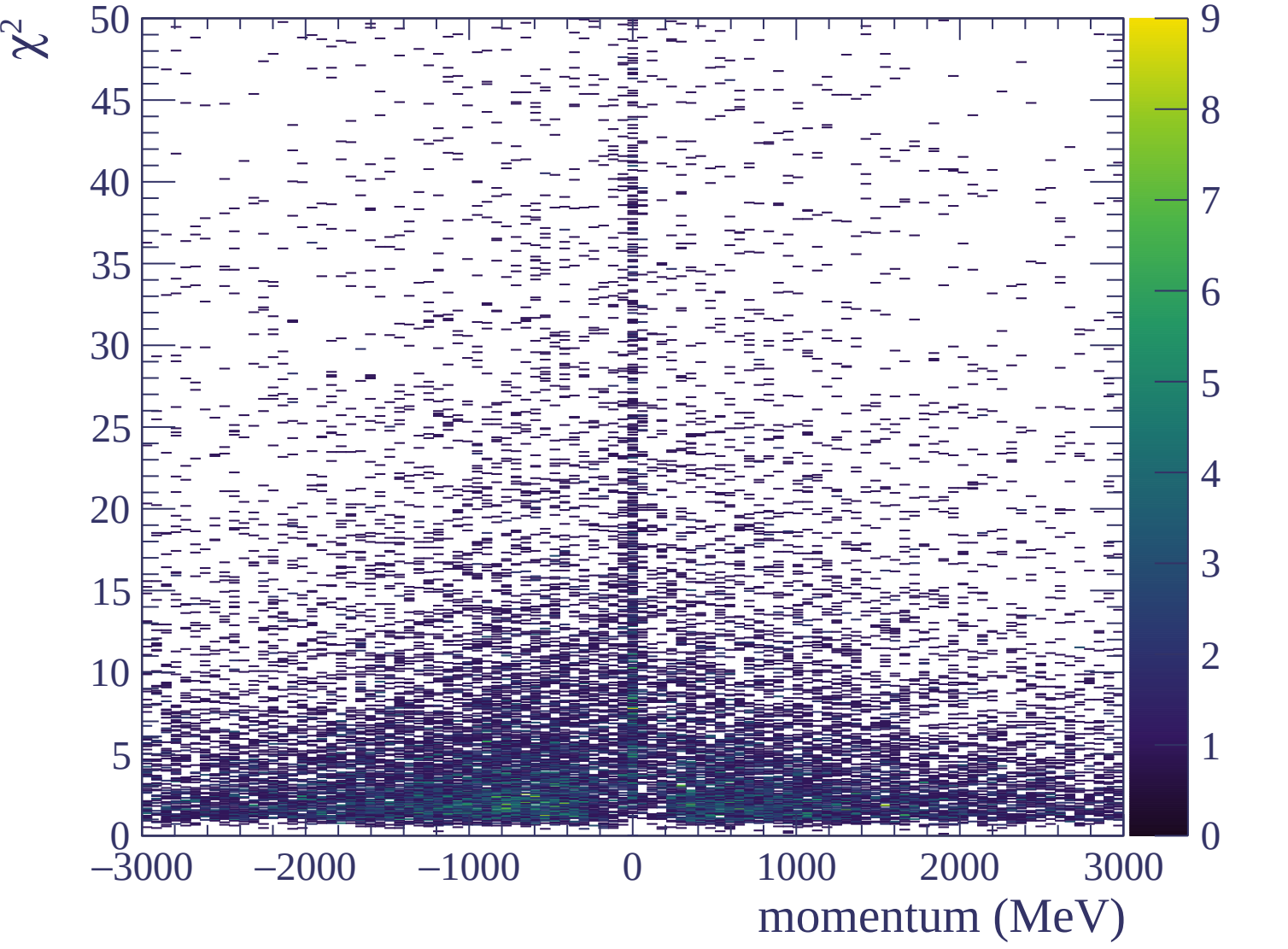


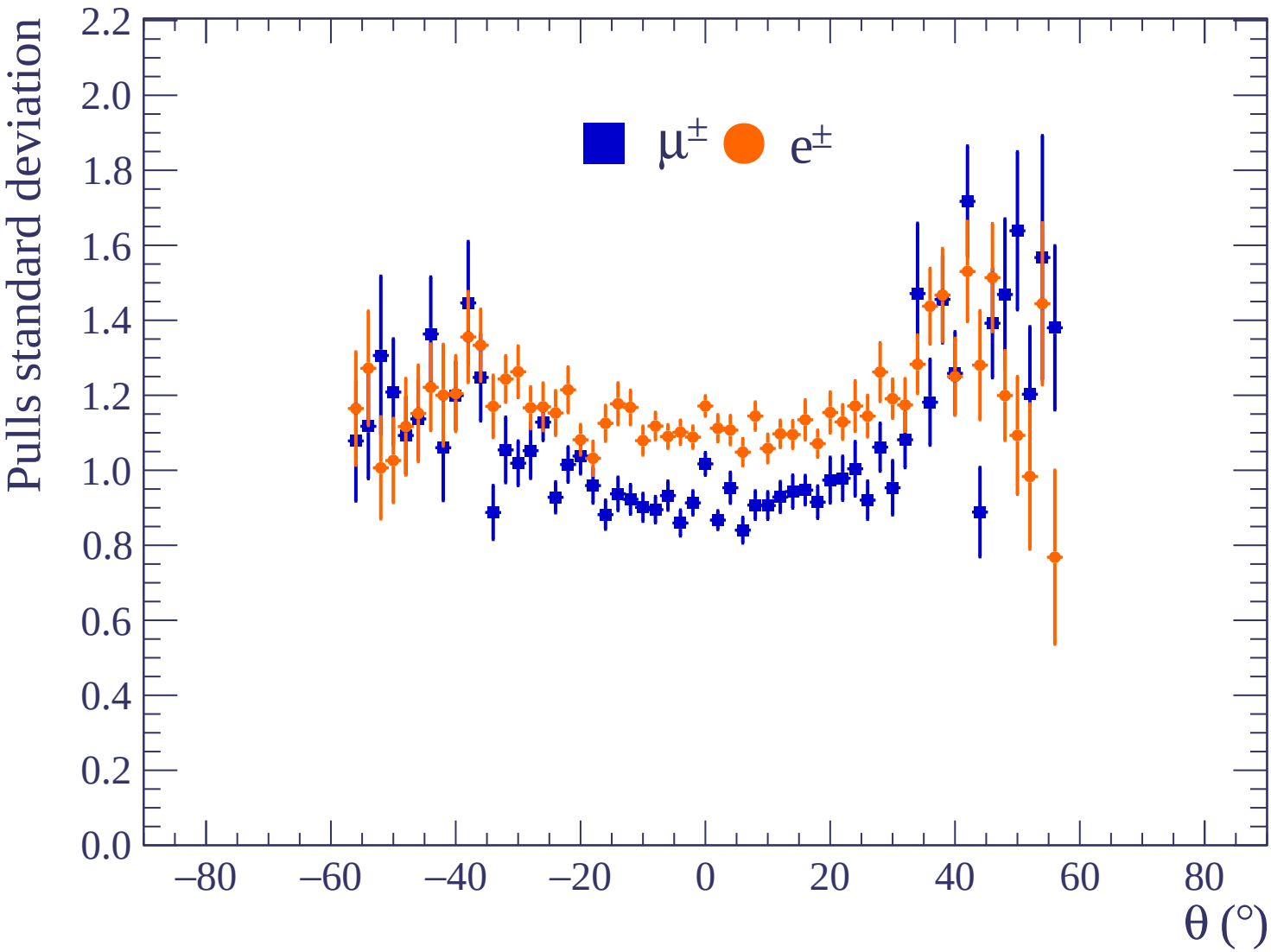


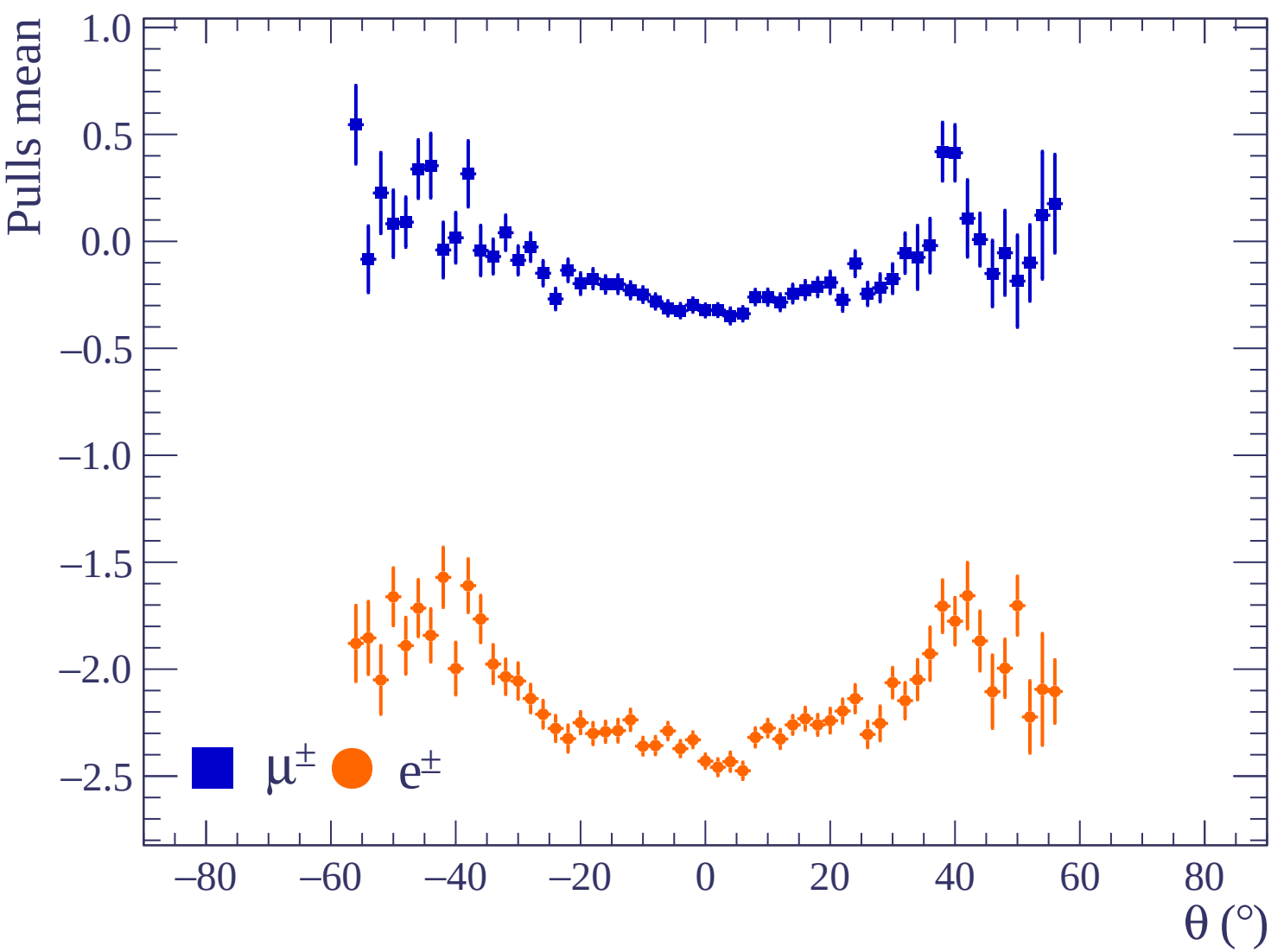


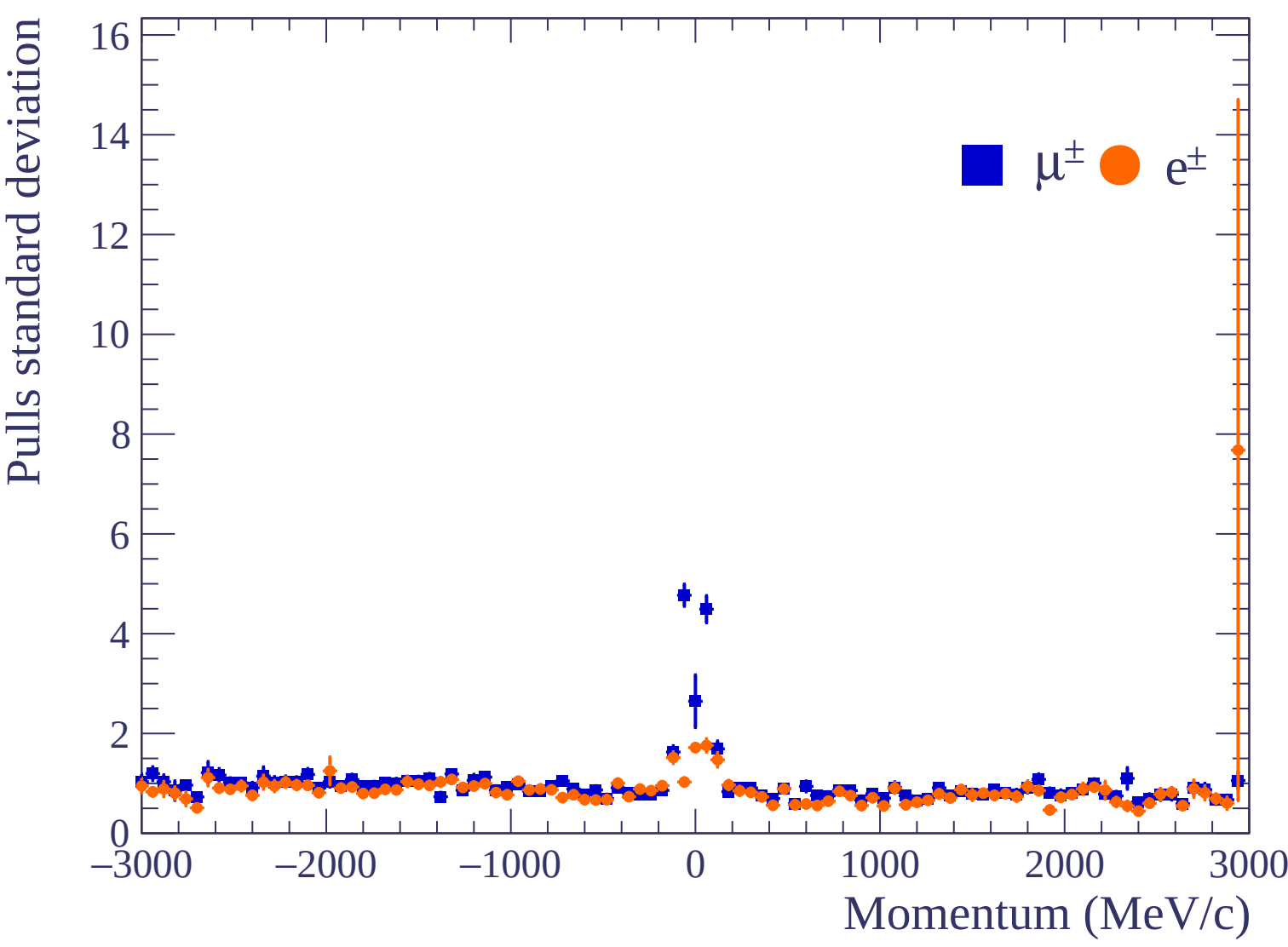


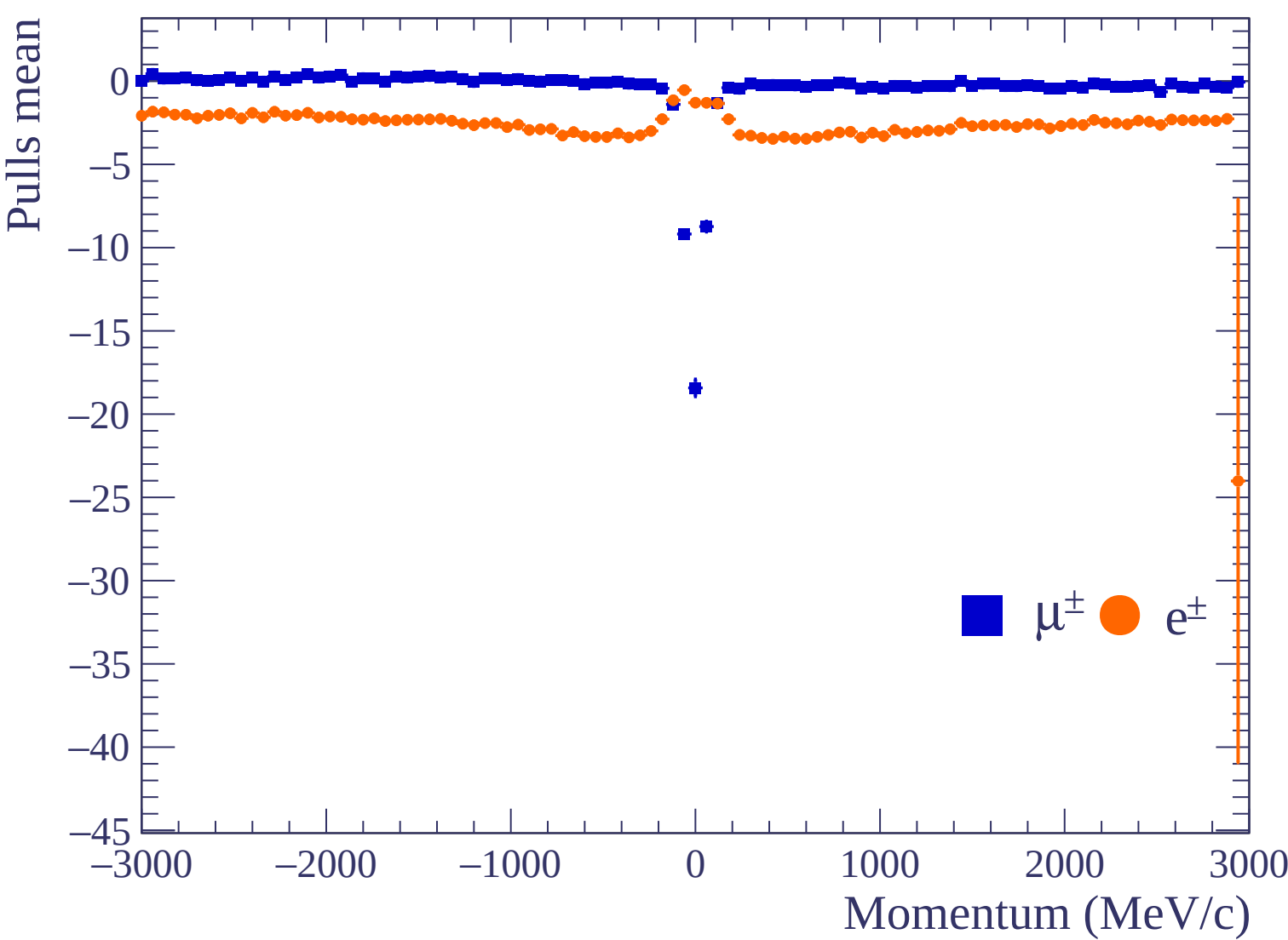


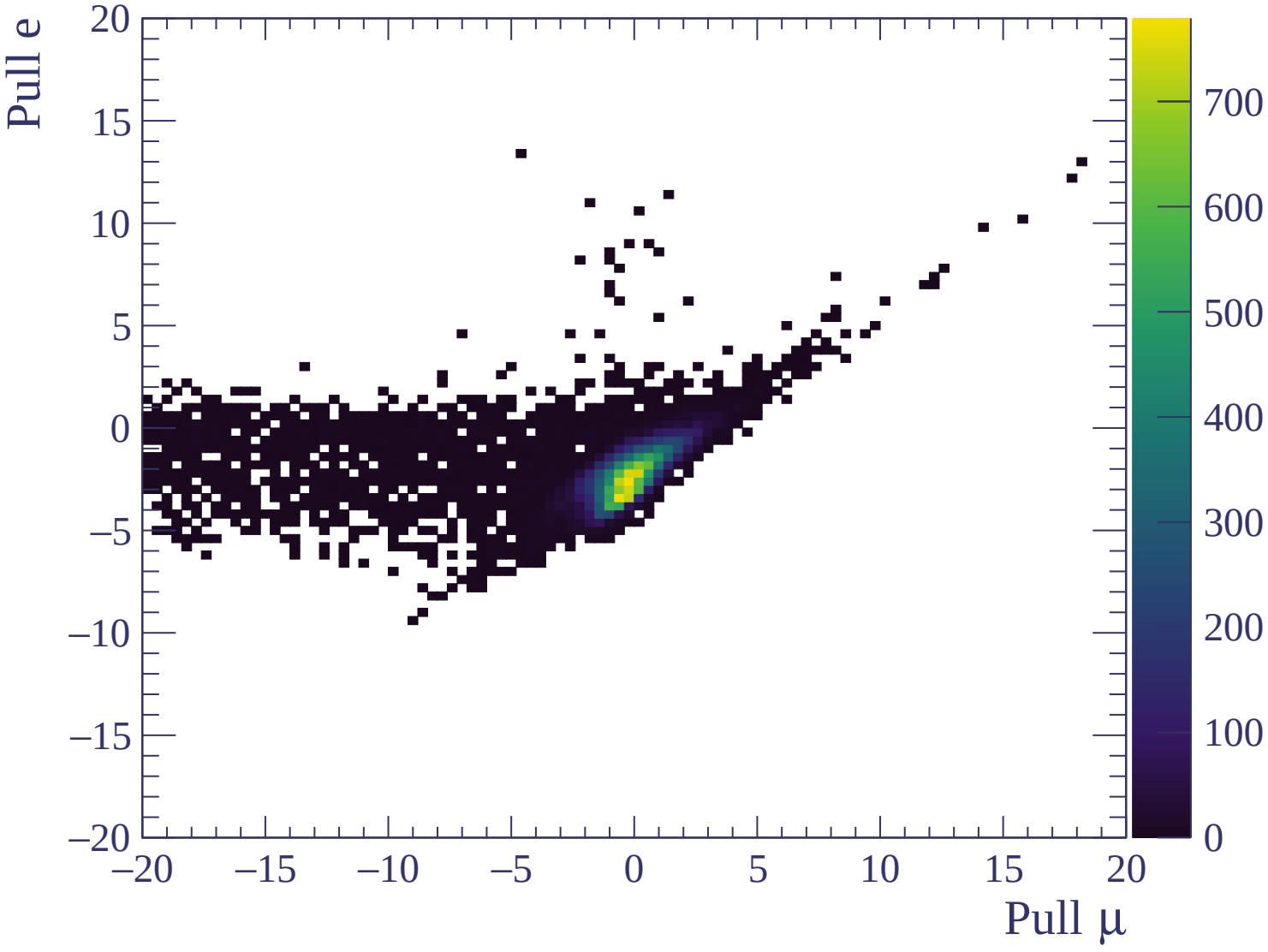


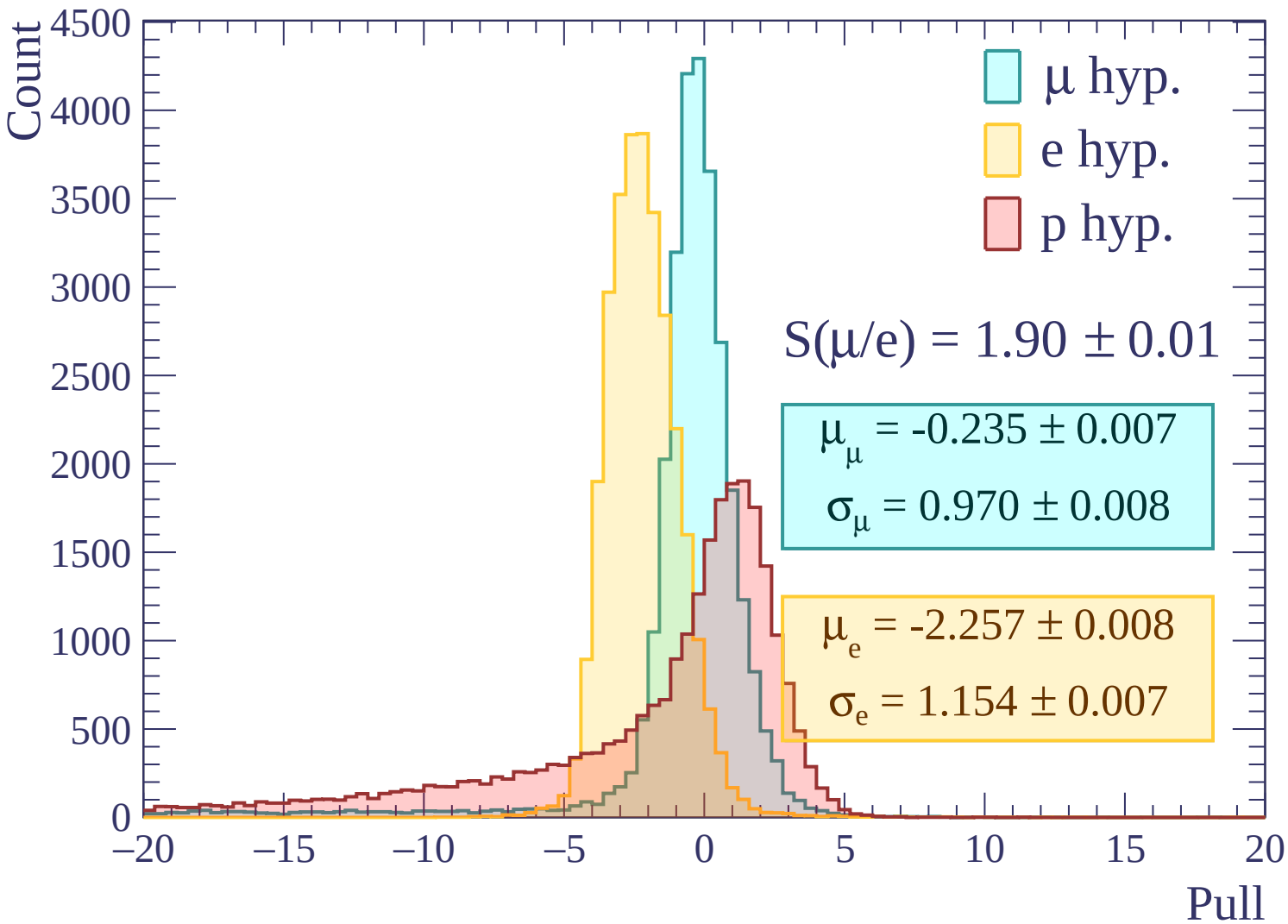


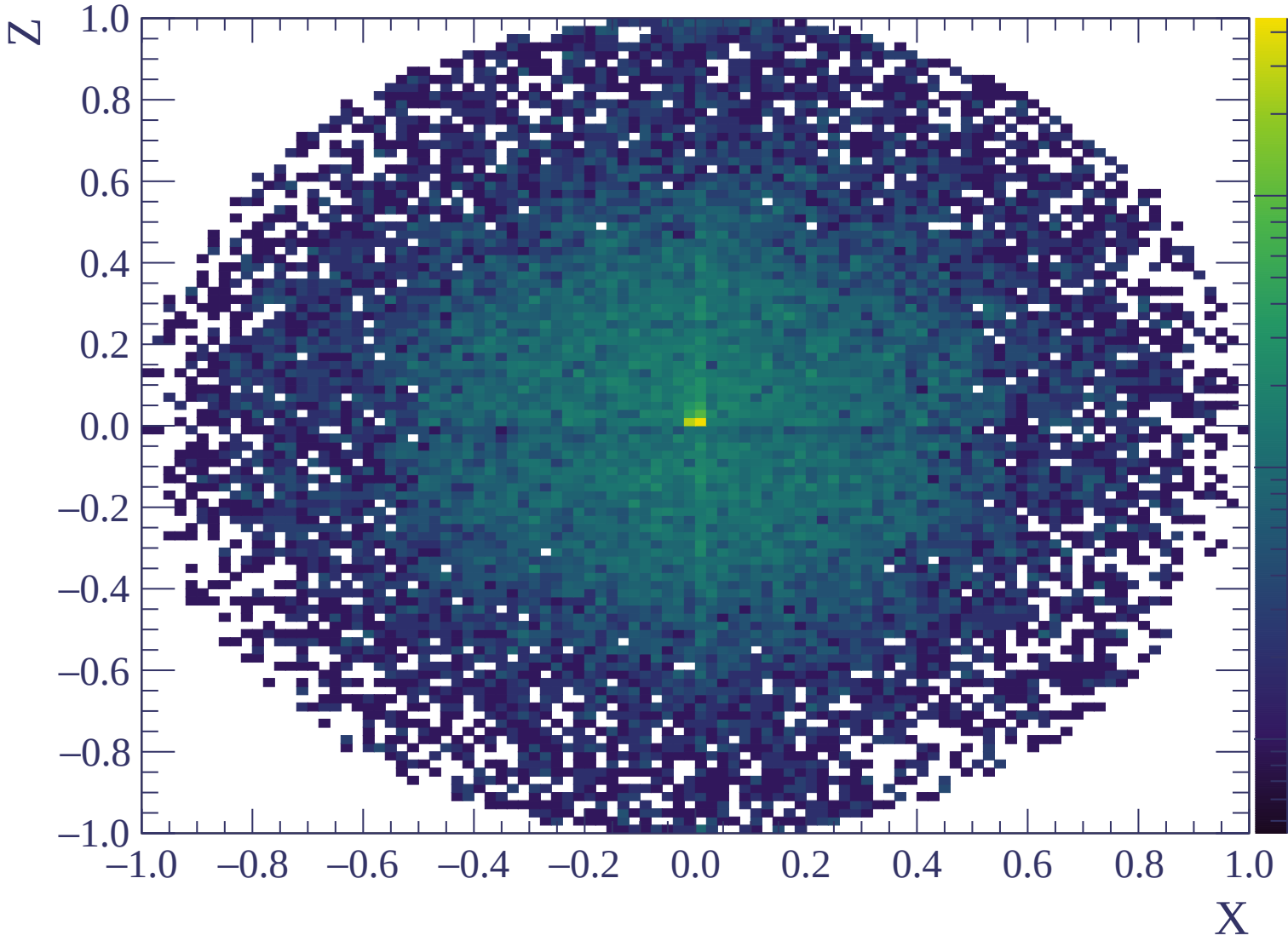


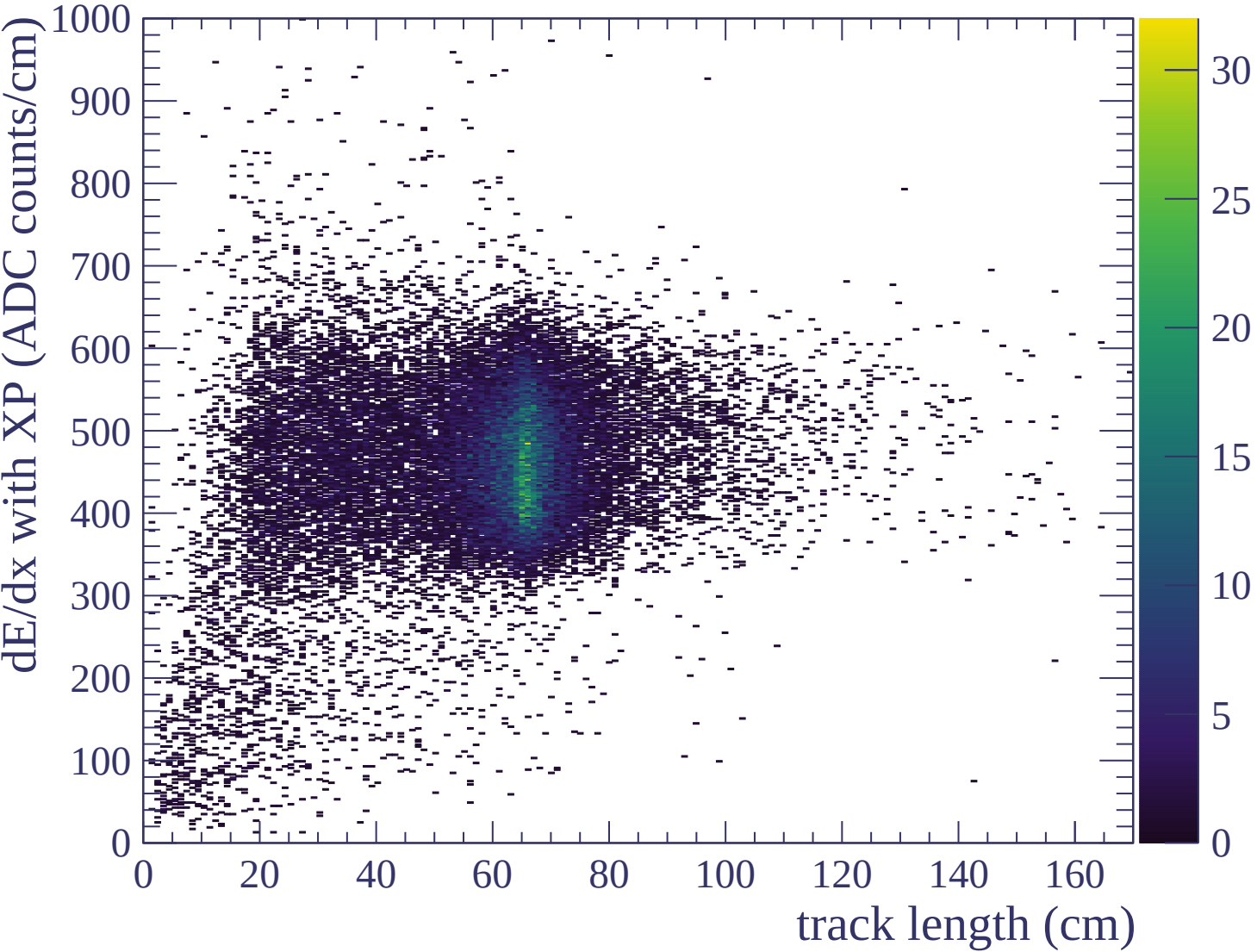


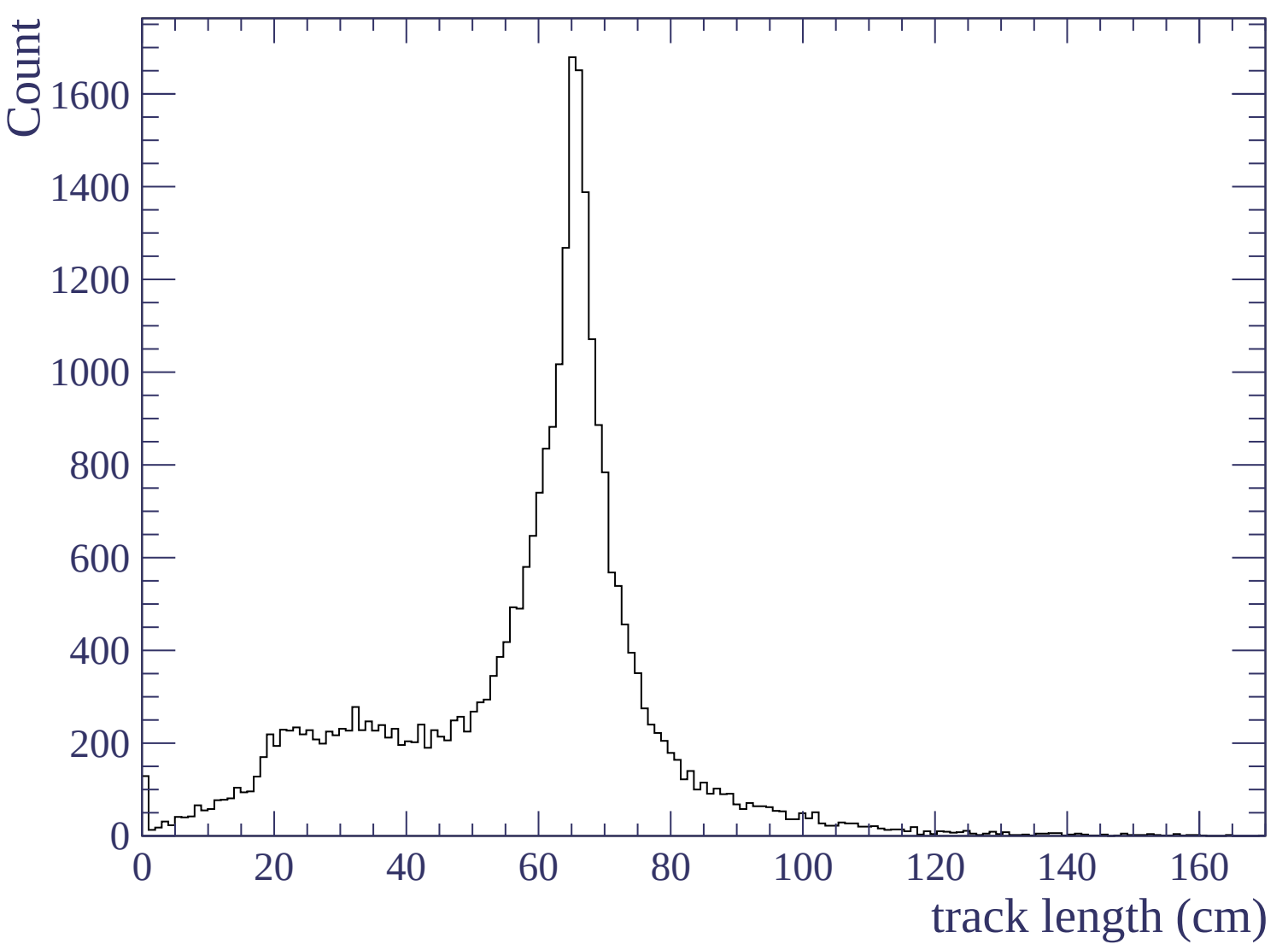


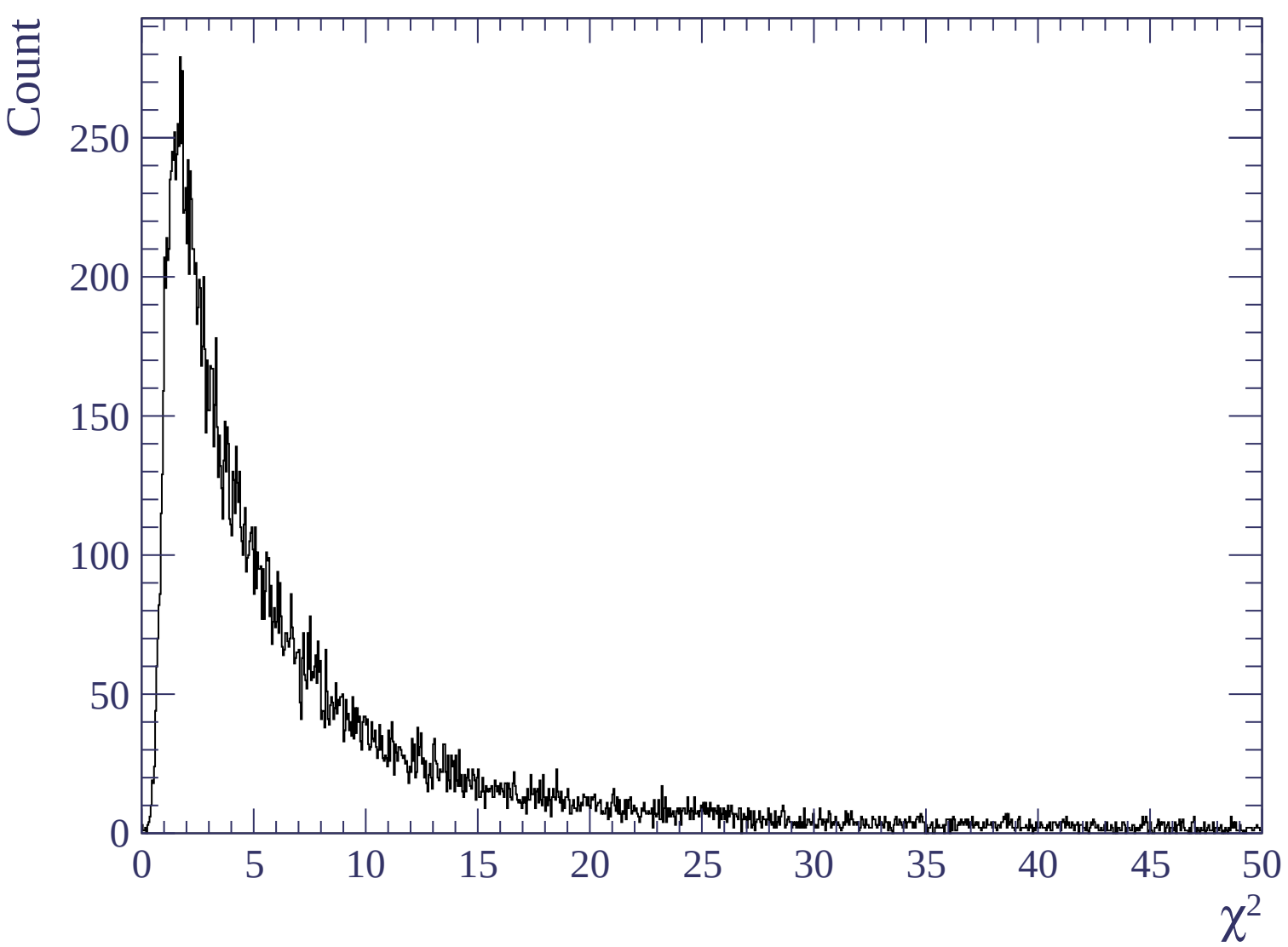


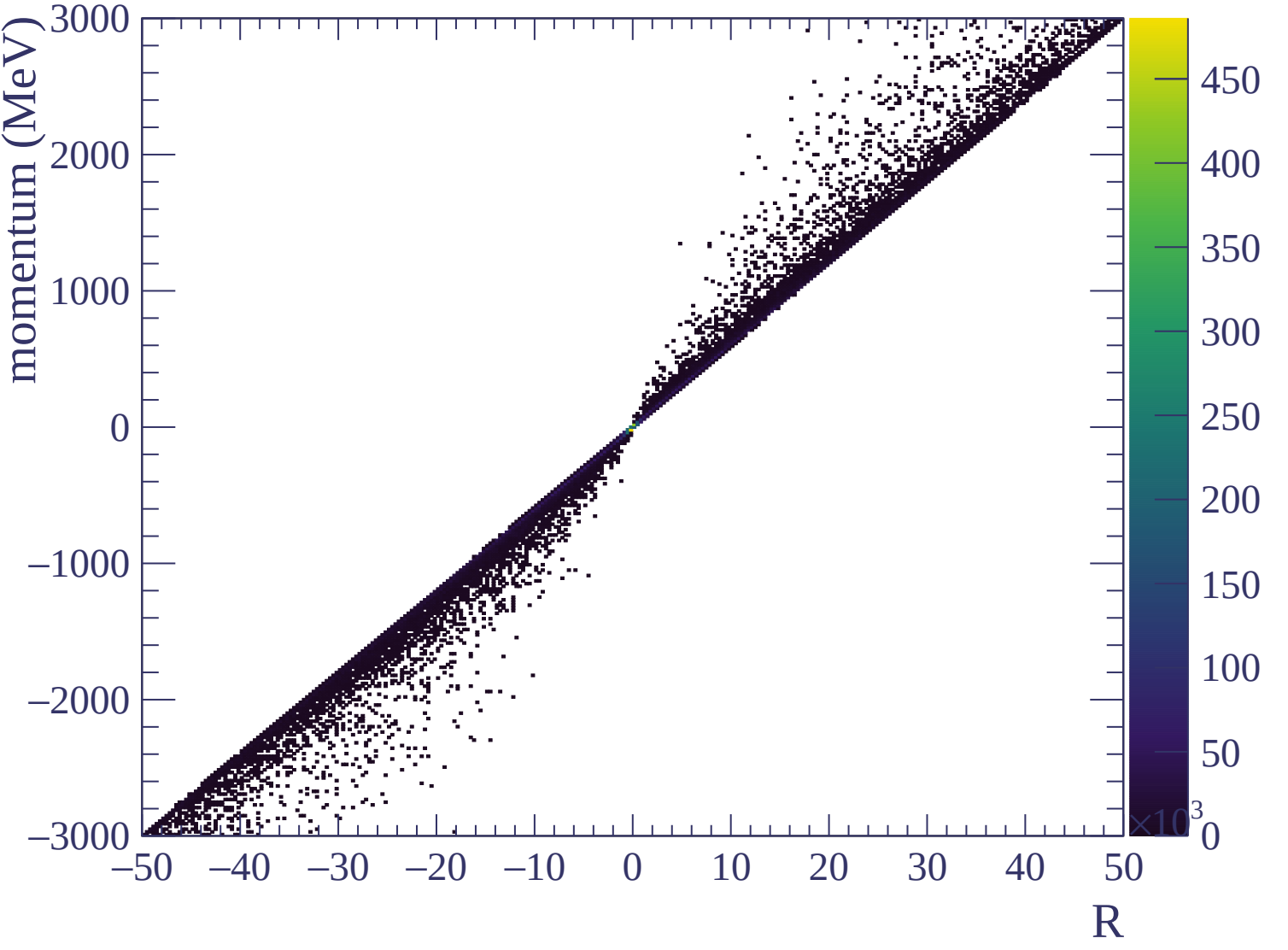


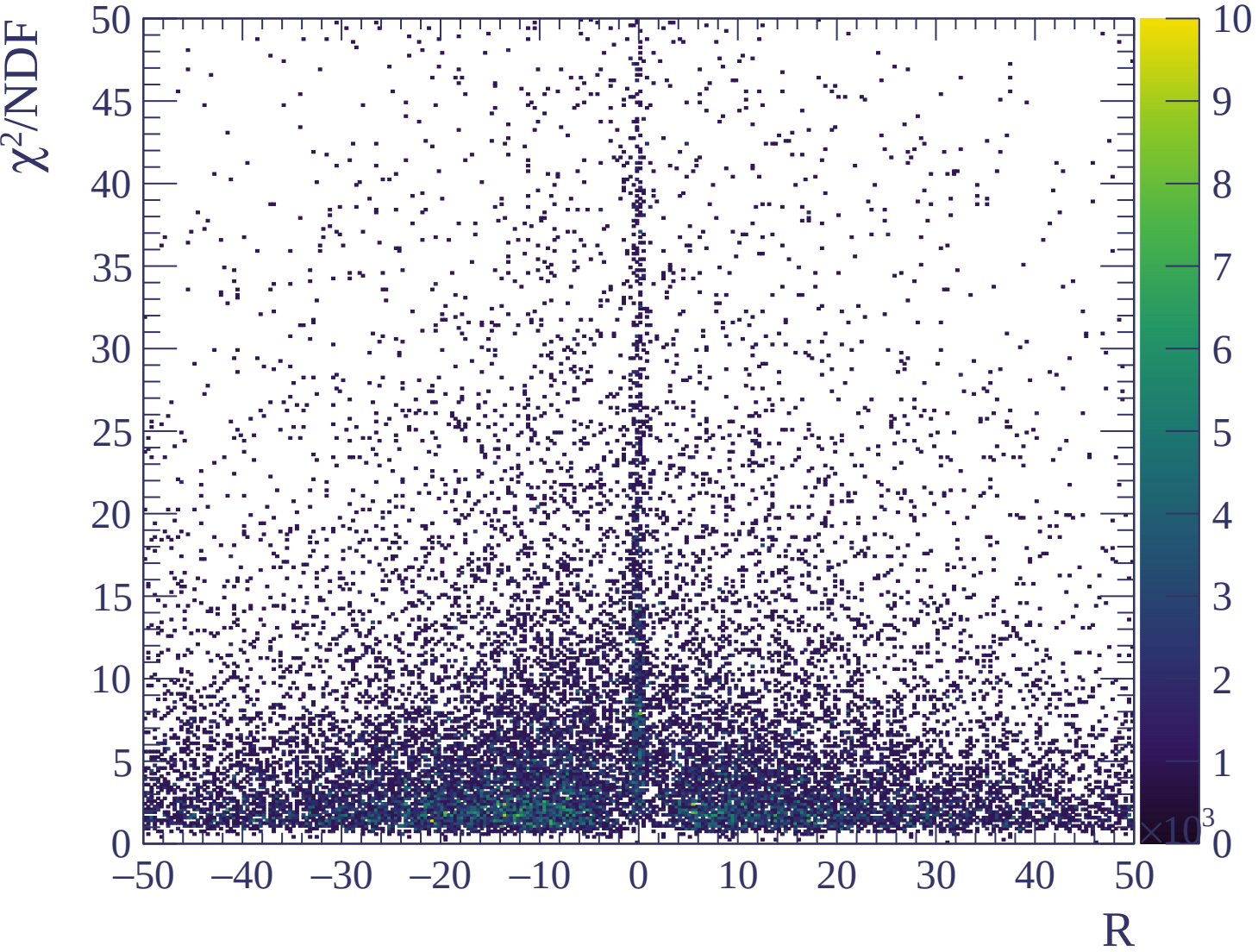






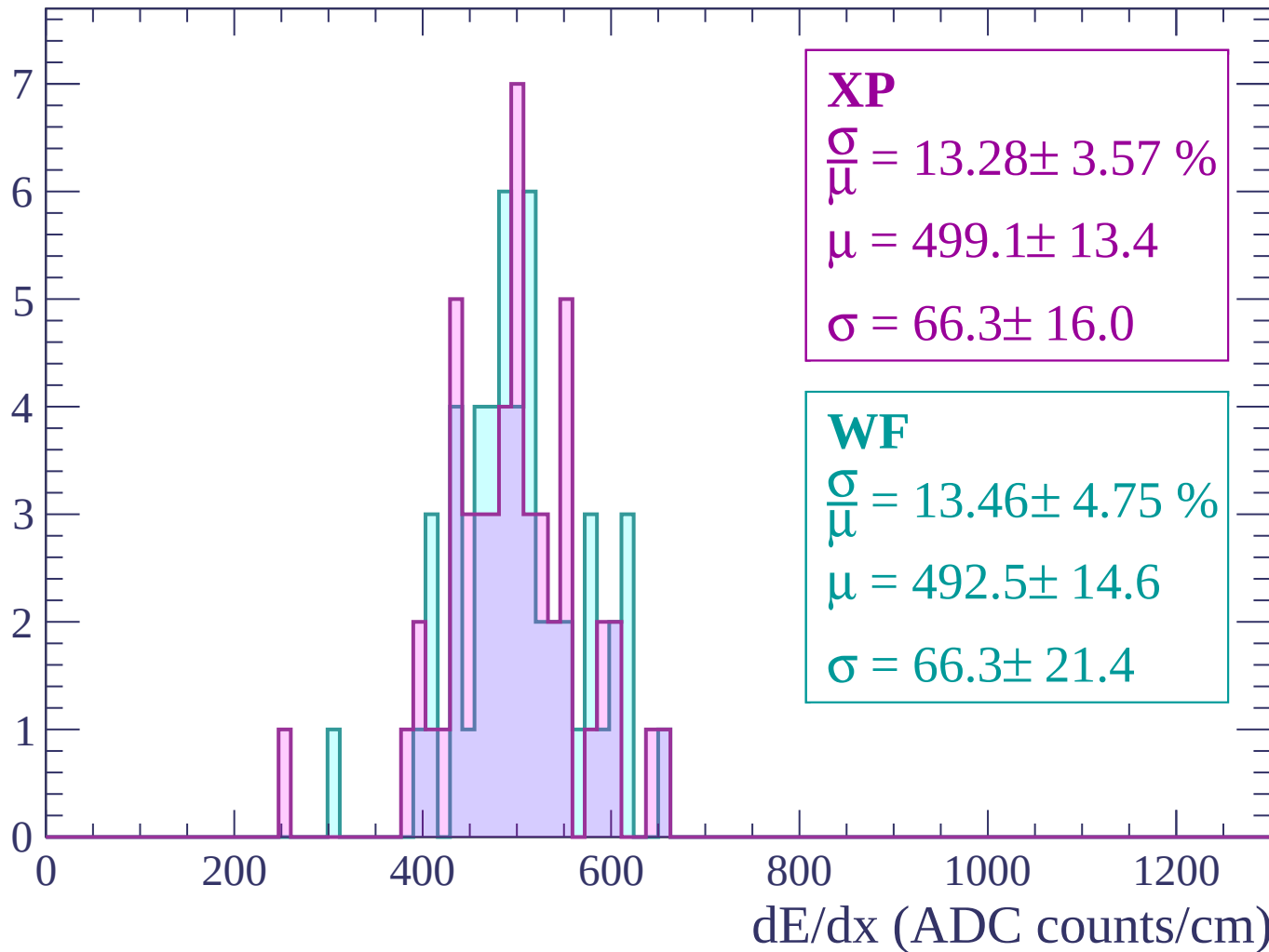






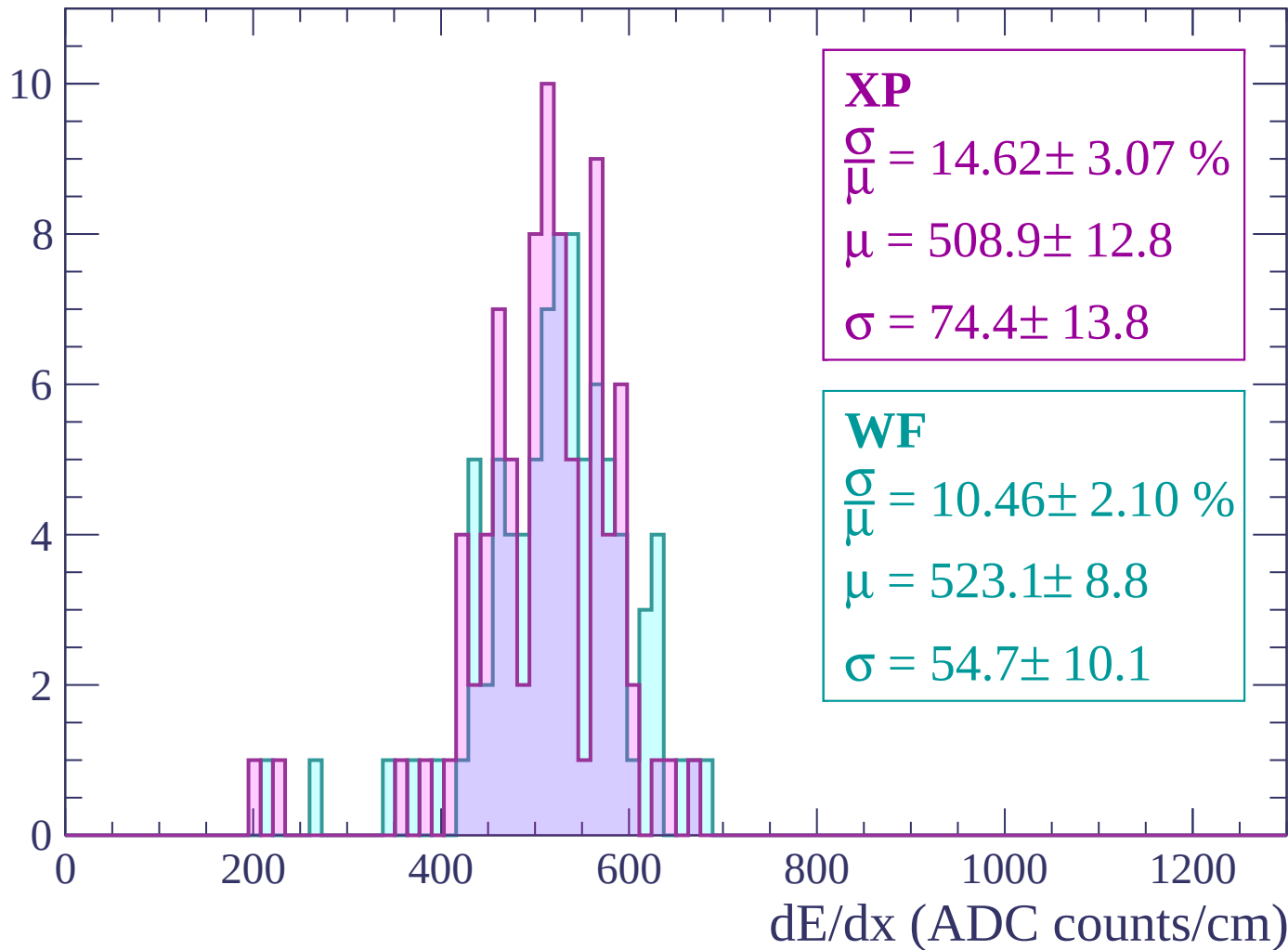
Energy loss | -3000 < p < -2940

Count



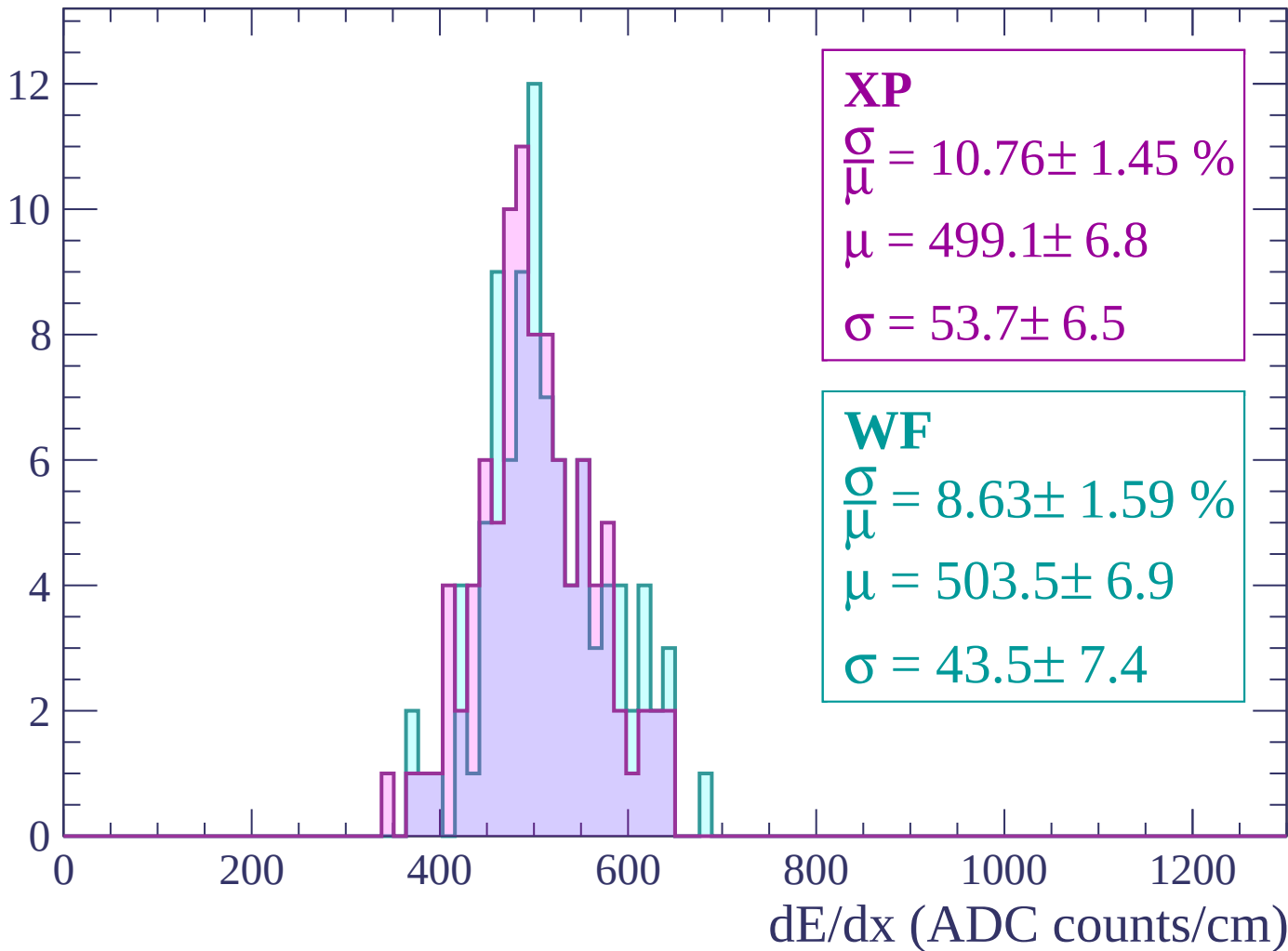
Energy loss | -2940 < p < -2880

Count



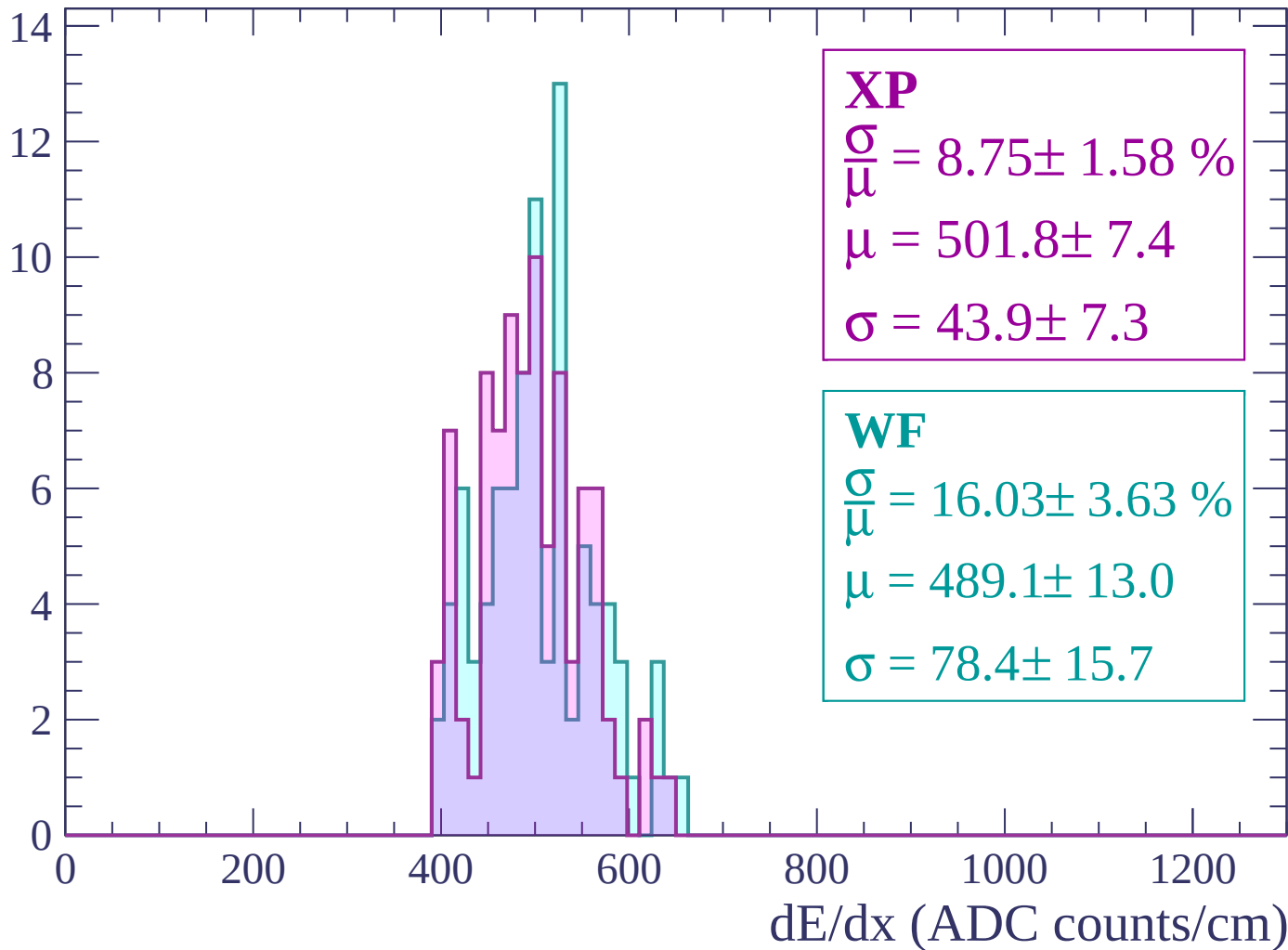
Energy loss | -2880 < p < -2820

Count



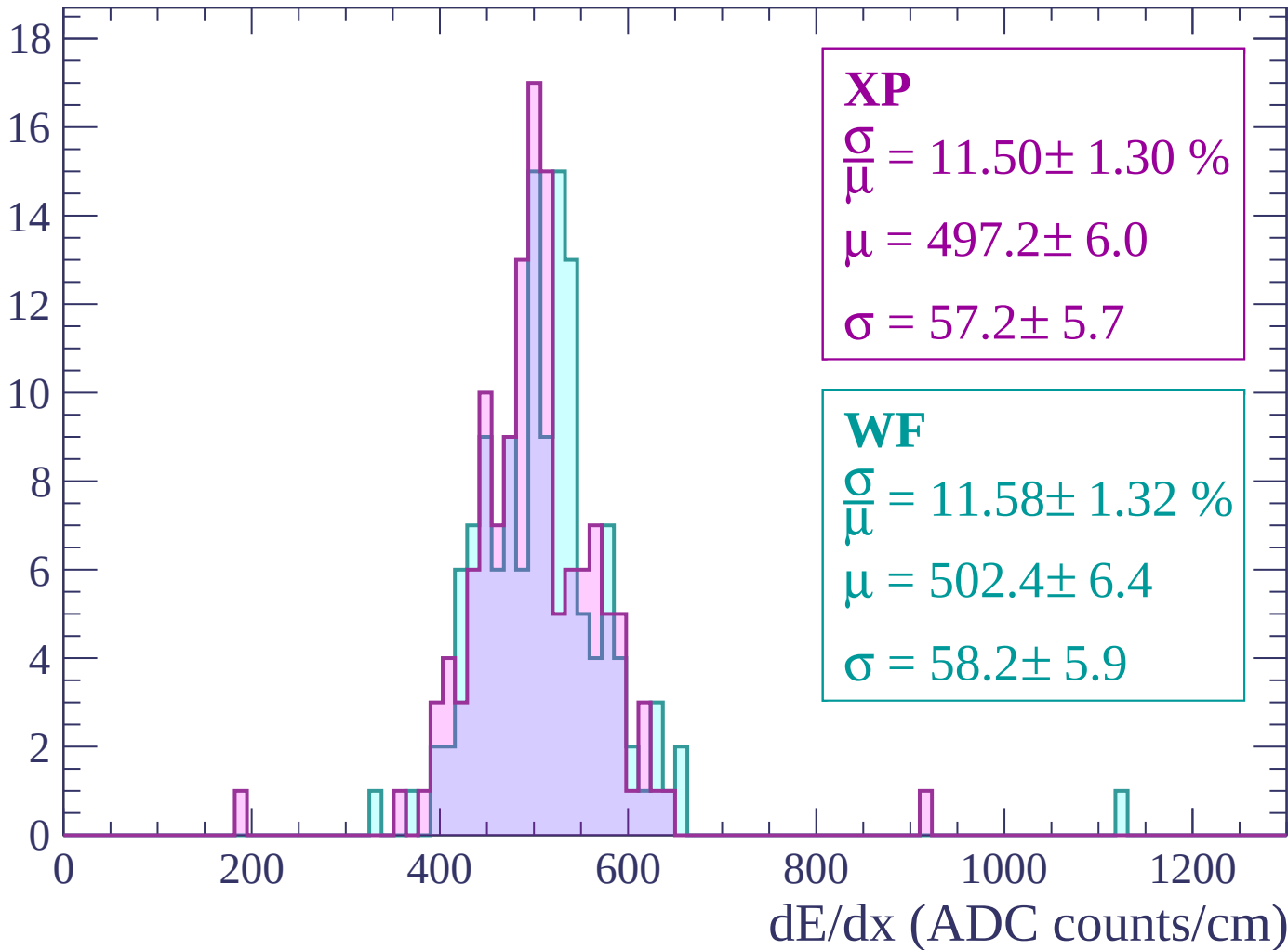
Energy loss | -2820 < p < -2760

Count



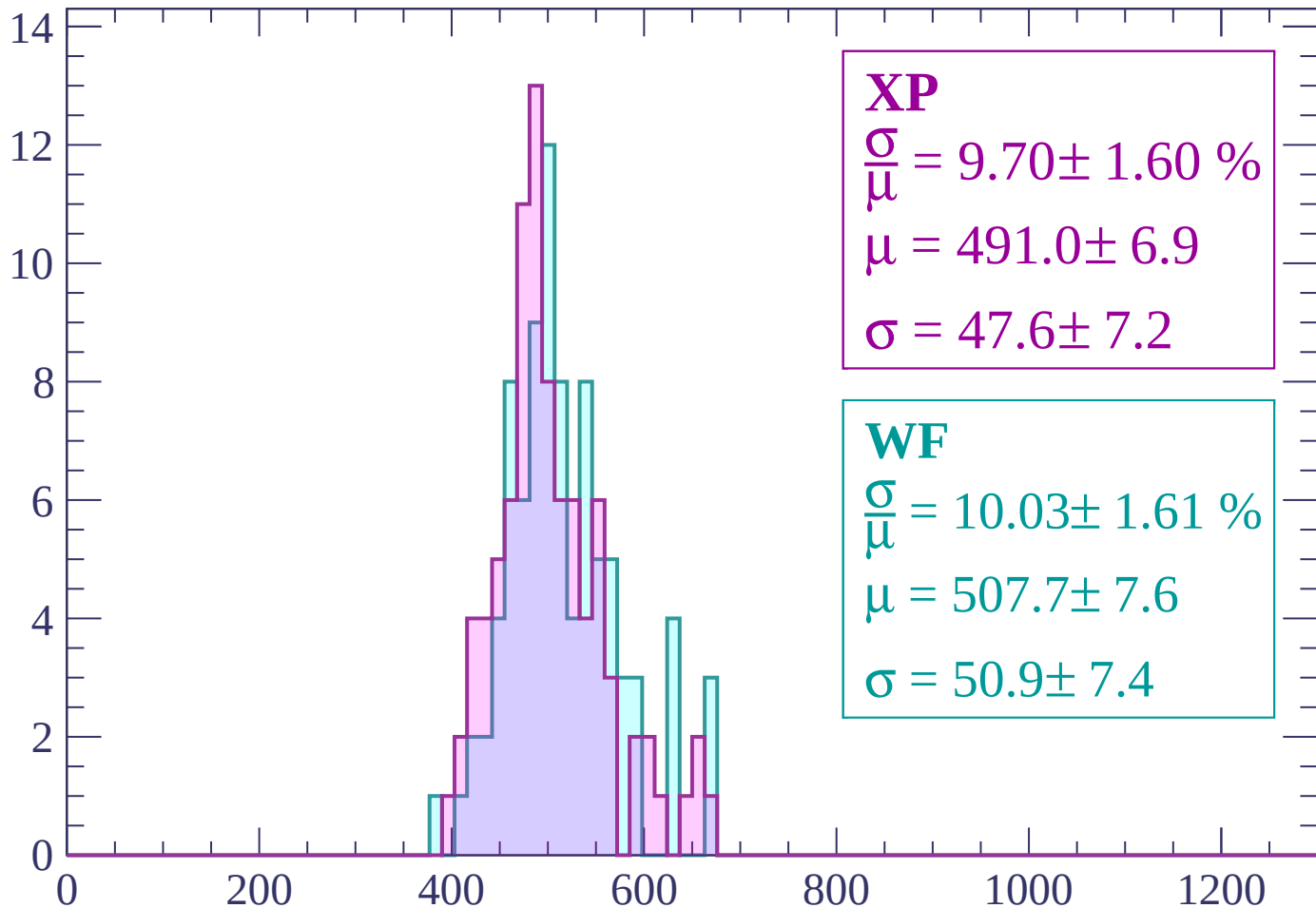
Energy loss | -2760 < p < -2700

Count



Energy loss | -2700 < p < -2640

Count



XP

$$\frac{\sigma}{\mu} = 9.70 \pm 1.60 \%$$

$$\mu = 491.0 \pm 6.9$$

$$\sigma = 47.6 \pm 7.2$$

WF

$$\frac{\sigma}{\mu} = 10.03 \pm 1.61 \%$$

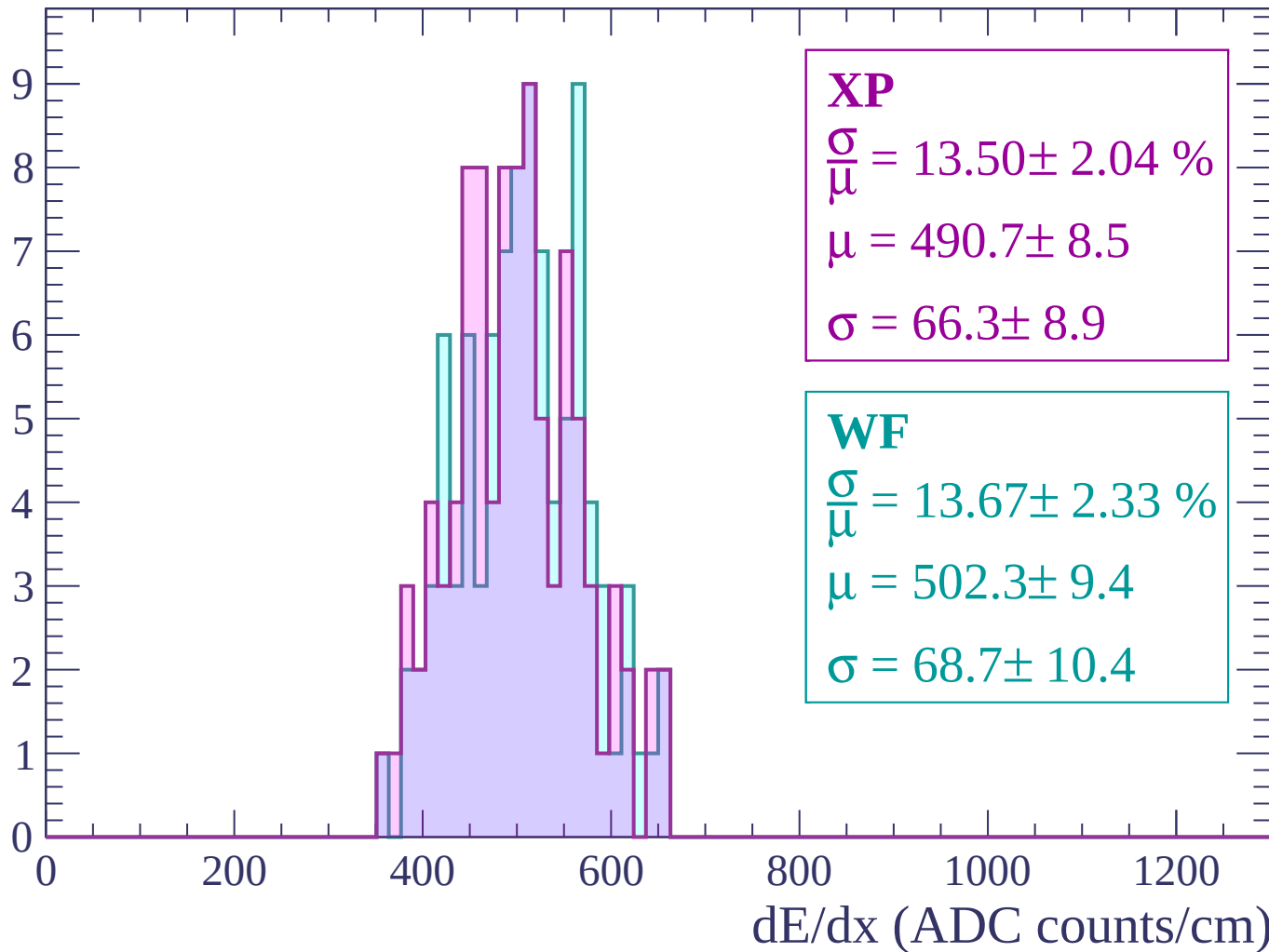
$$\mu = 507.7 \pm 7.6$$

$$\sigma = 50.9 \pm 7.4$$

dE/dx (ADC counts/cm)

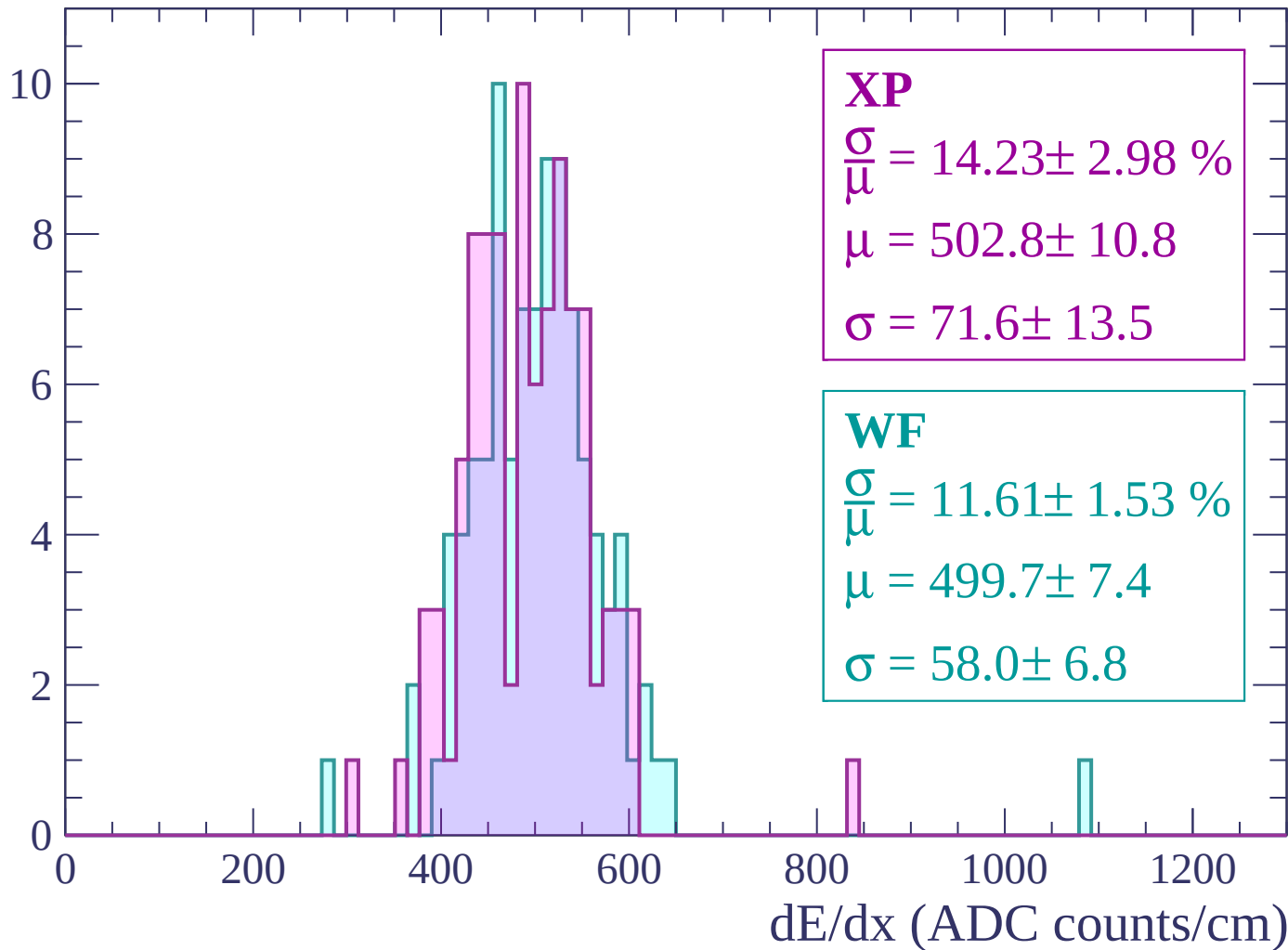
Energy loss | $-2640 < p < -2580$

Count



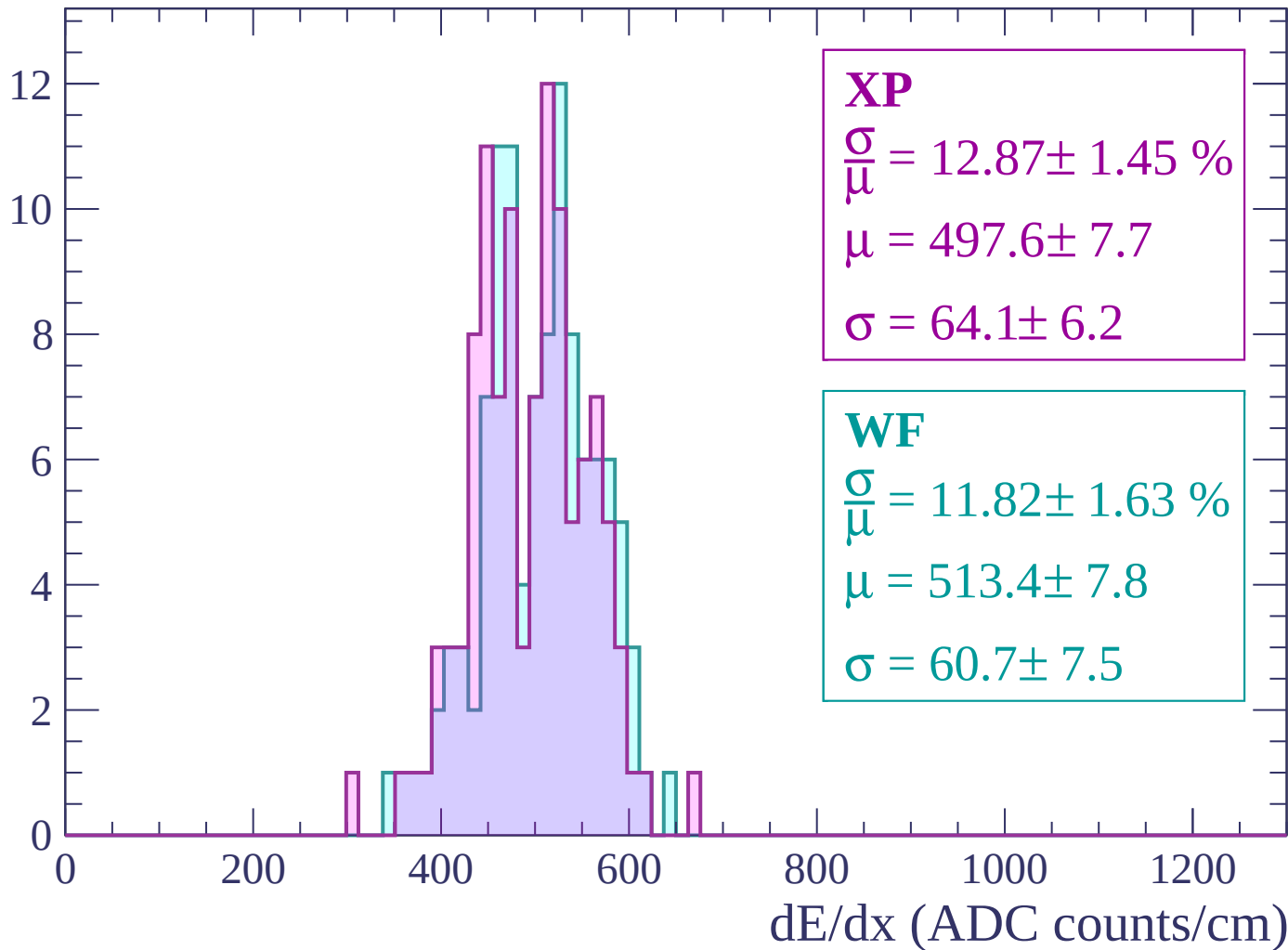
Energy loss | -2580 < p < -2520

Count



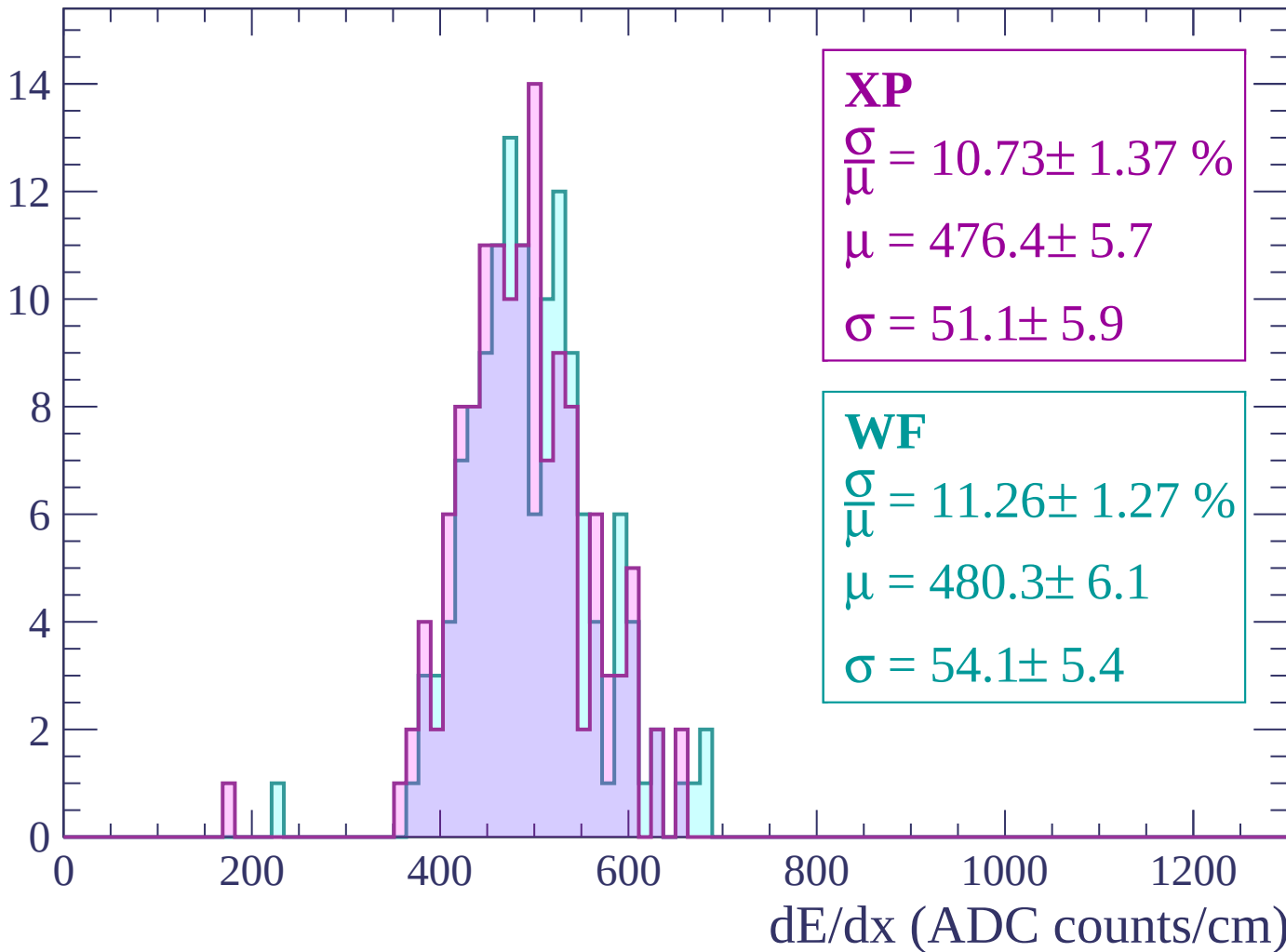
Energy loss | -2520 < p < -2460

Count



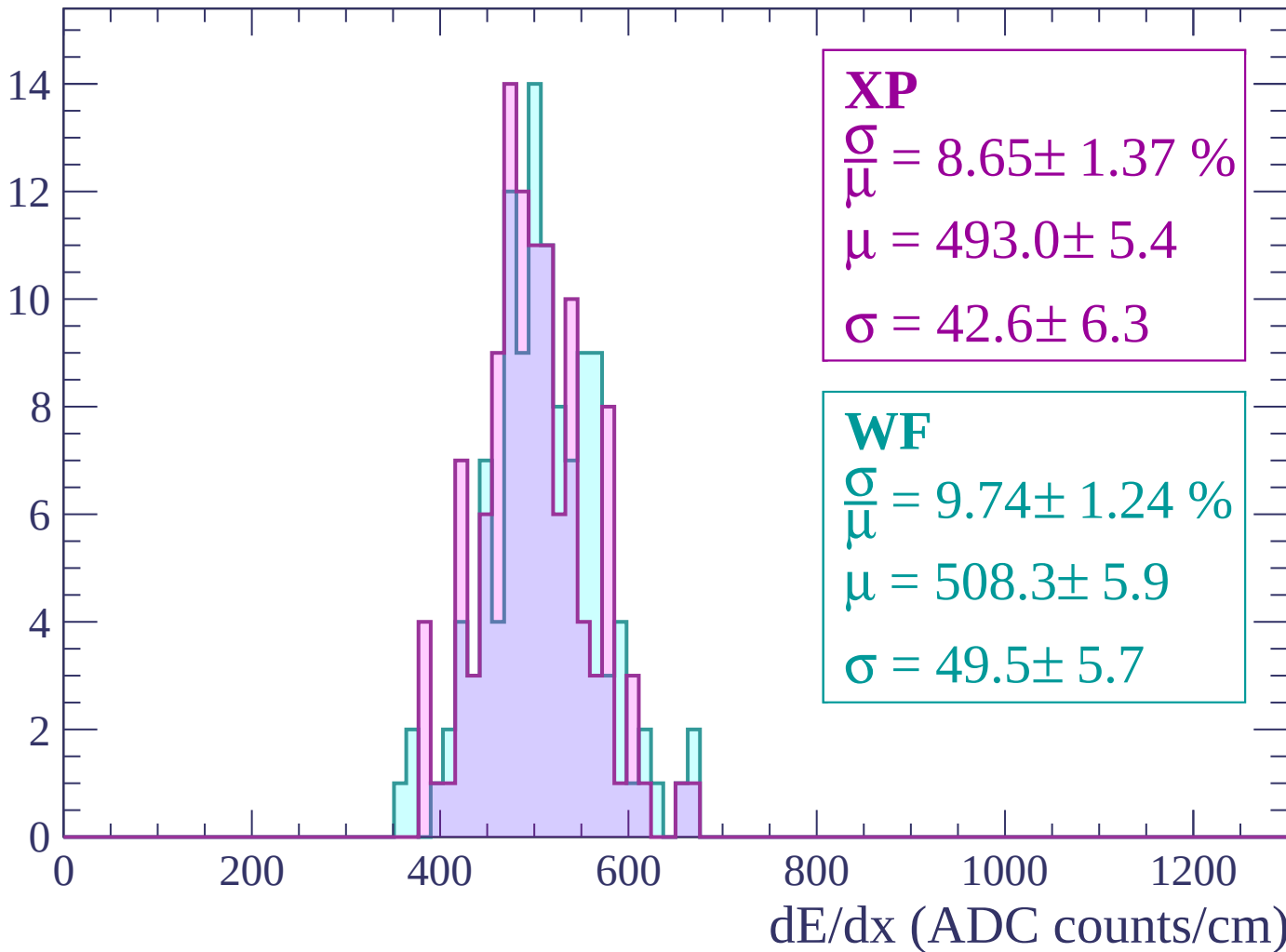
Energy loss | -2460 < p < -2400

Count



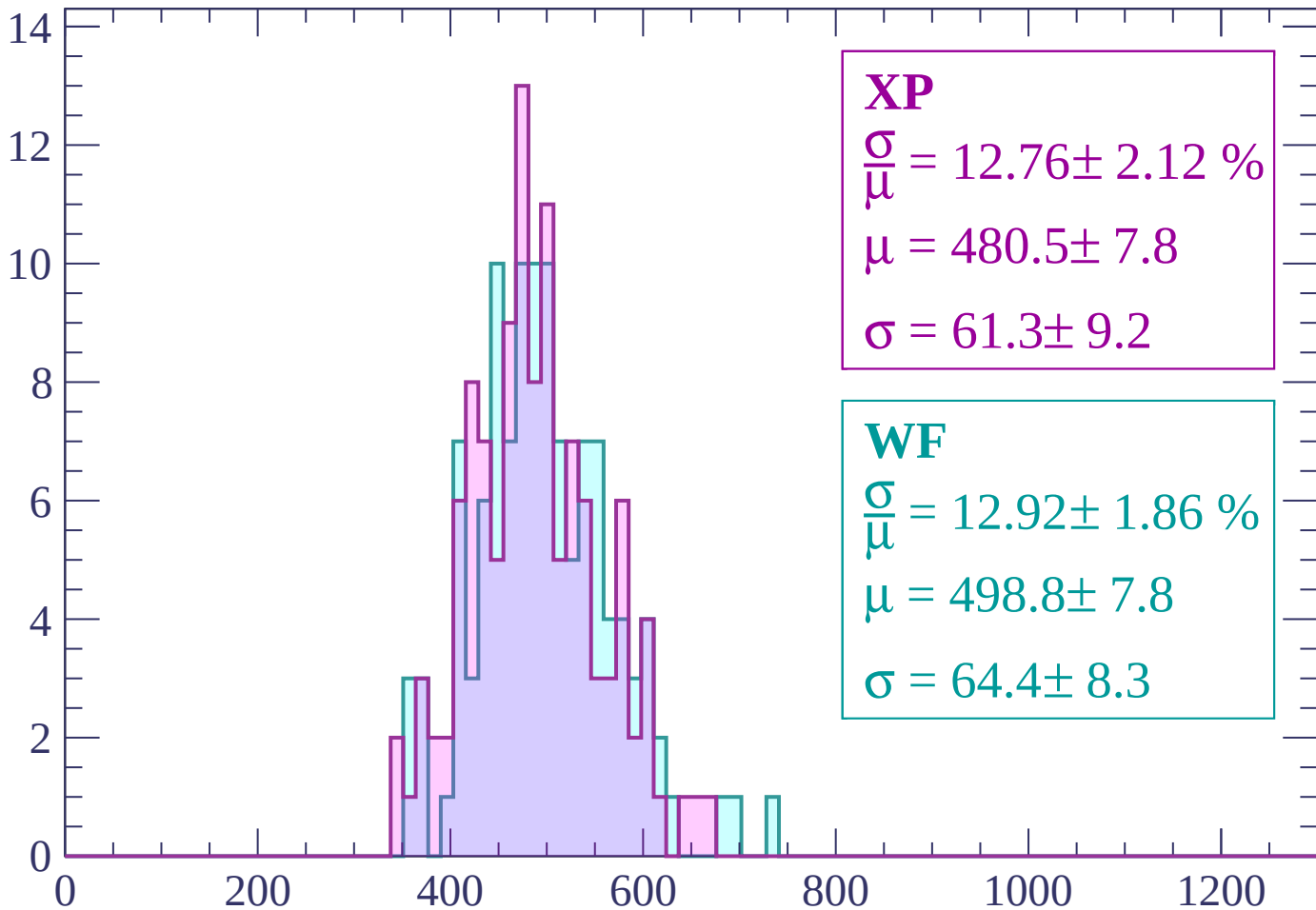
Energy loss | $-2400 < p < -2340$

Count



Energy loss | $-2340 < p < -2280$

Count



XP

$$\frac{\sigma}{\mu} = 12.76 \pm 2.12 \%$$

$$\mu = 480.5 \pm 7.8$$

$$\sigma = 61.3 \pm 9.2$$

WF

$$\frac{\sigma}{\mu} = 12.92 \pm 1.86 \%$$

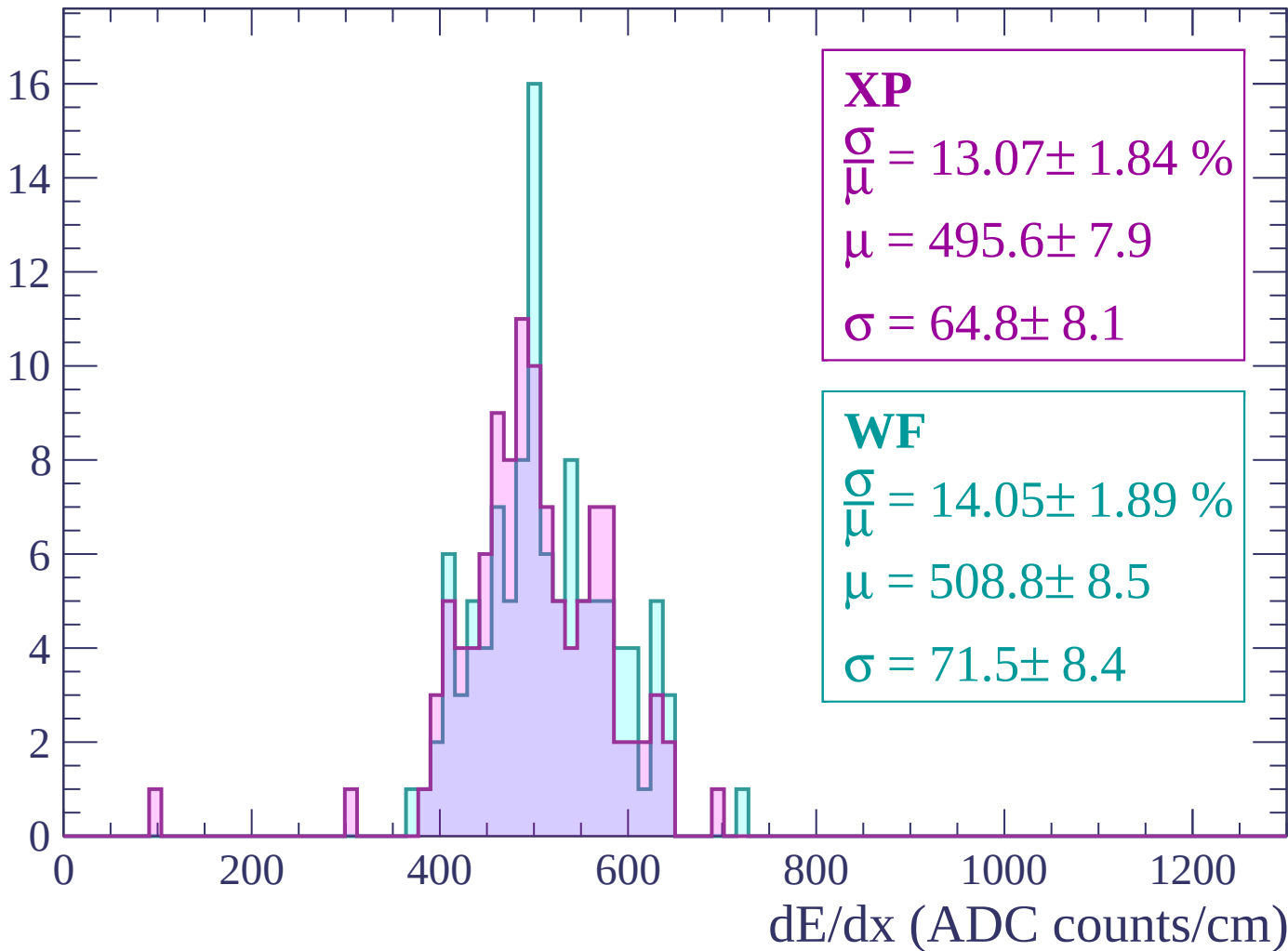
$$\mu = 498.8 \pm 7.8$$

$$\sigma = 64.4 \pm 8.3$$

dE/dx (ADC counts/cm)

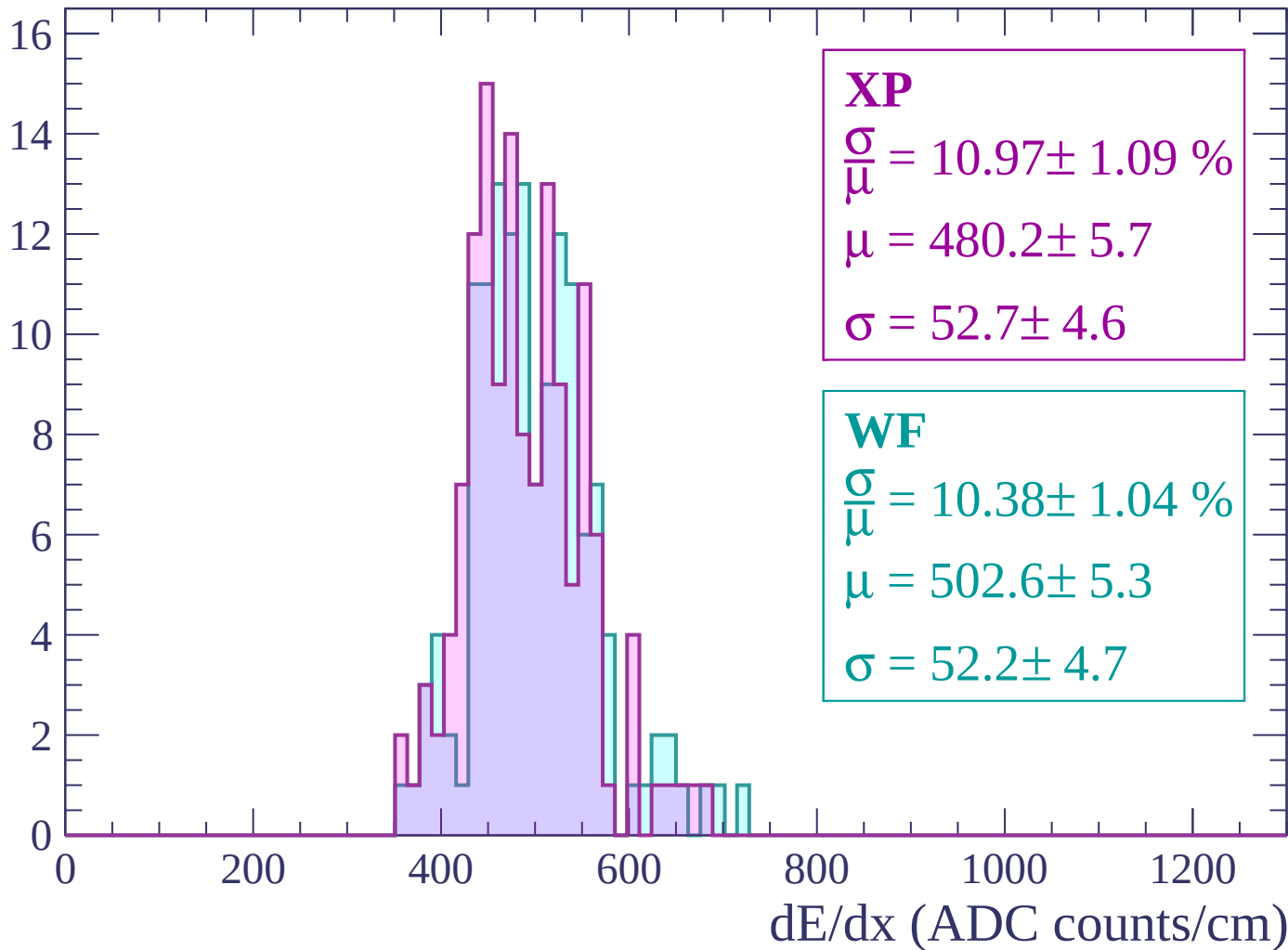
Energy loss | $-2280 < p < -2220$

Count



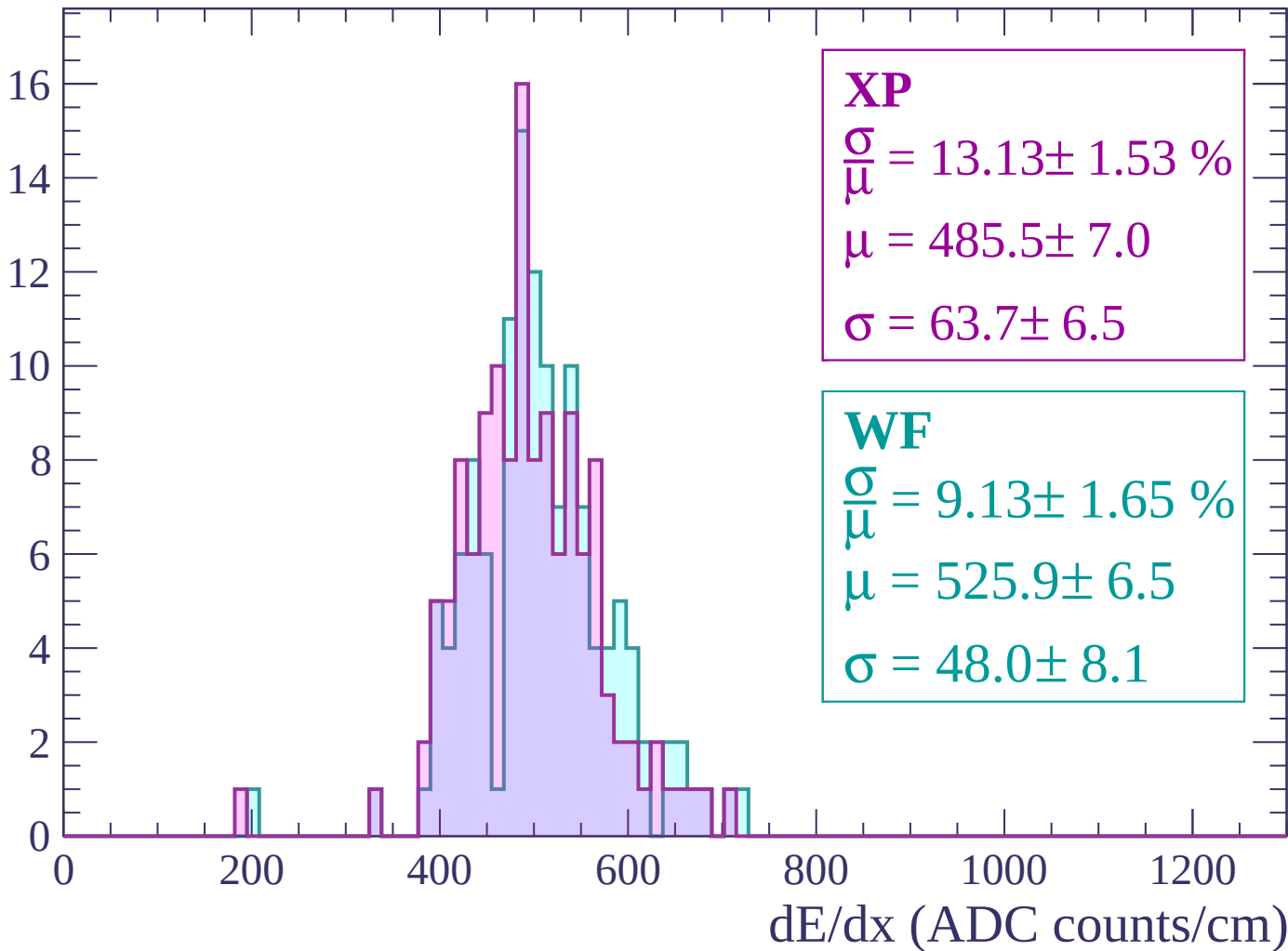
Energy loss | -2220 < p < -2160

Count



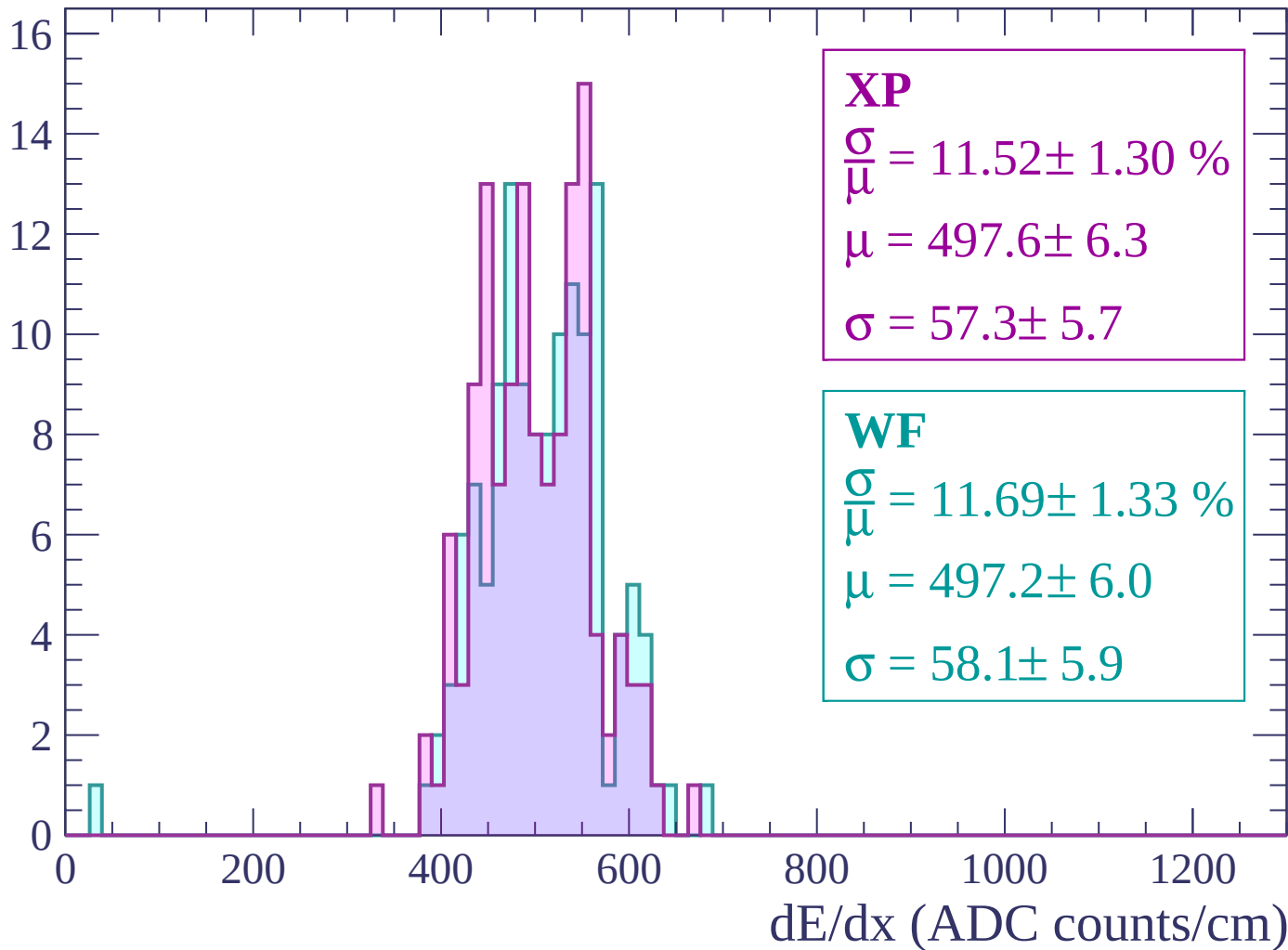
Energy loss | -2160 < p < -2100

Count



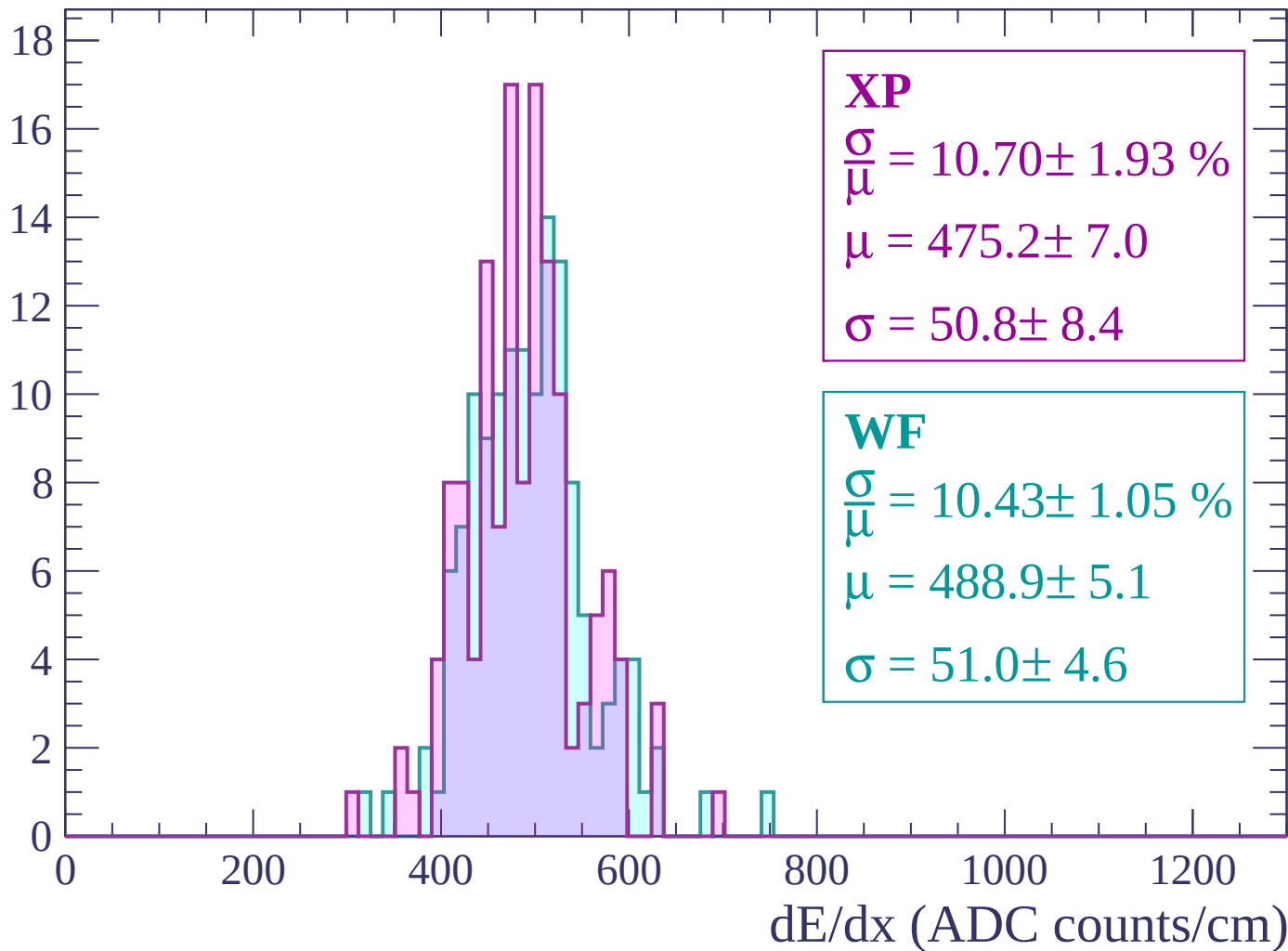
Energy loss | -2100 < p < -2040

Count



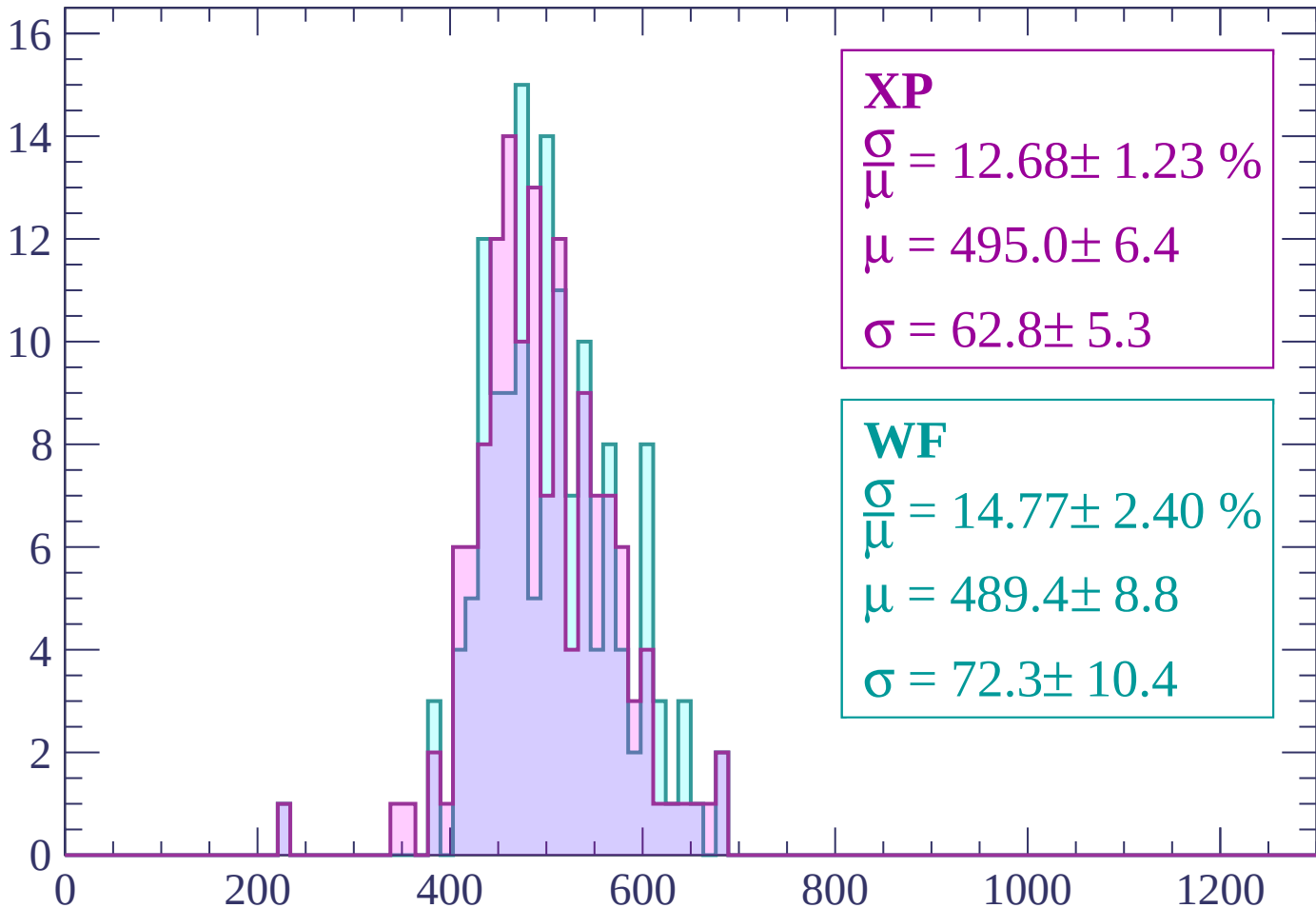
Energy loss | -2040 < p < -1980

Count



Energy loss | -1980 < p < -1920

Count



XP

$$\frac{\sigma}{\mu} = 12.68 \pm 1.23 \%$$

$$\mu = 495.0 \pm 6.4$$

$$\sigma = 62.8 \pm 5.3$$

WF

$$\frac{\sigma}{\mu} = 14.77 \pm 2.40 \%$$

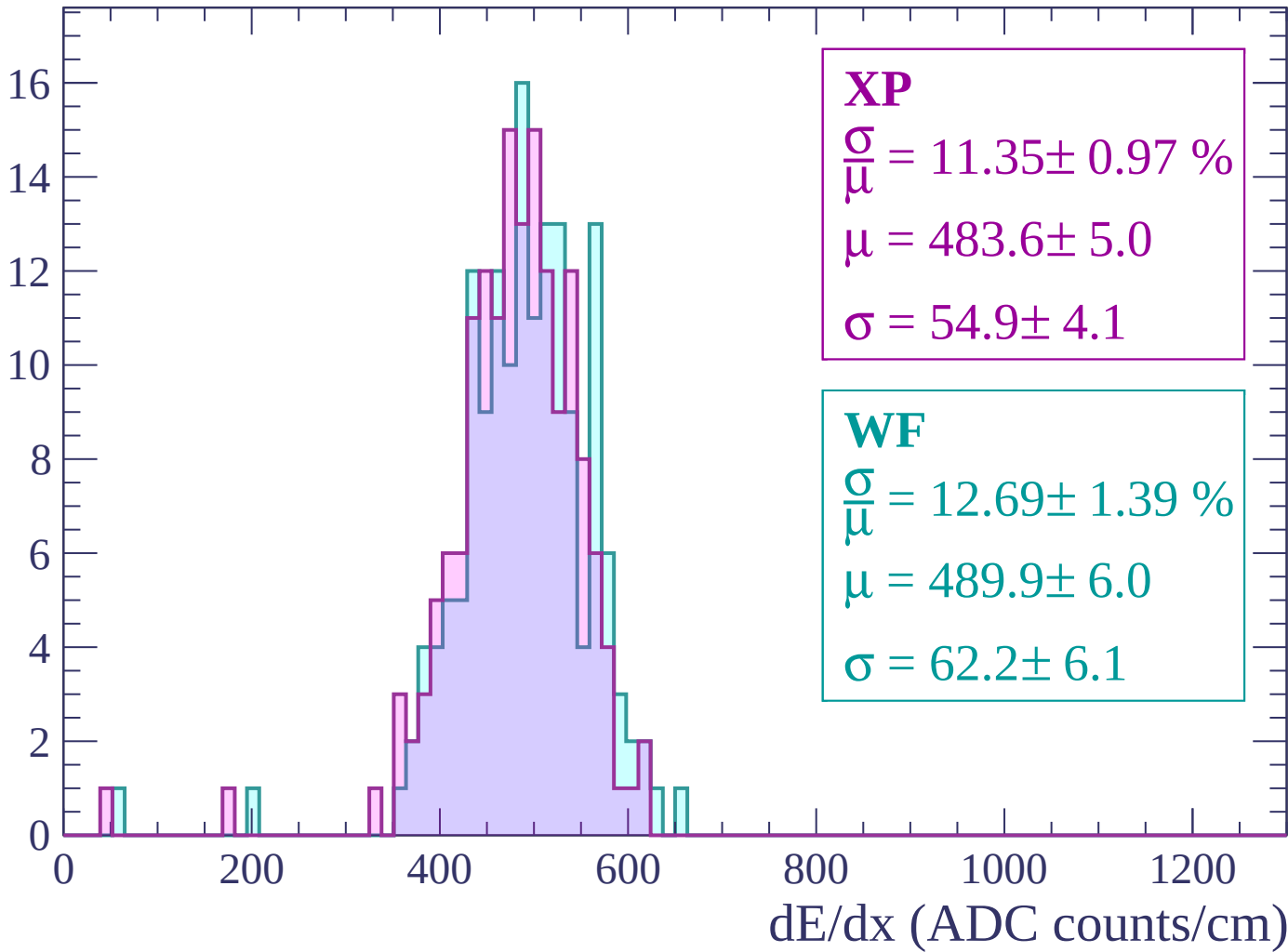
$$\mu = 489.4 \pm 8.8$$

$$\sigma = 72.3 \pm 10.4$$

dE/dx (ADC counts/cm)

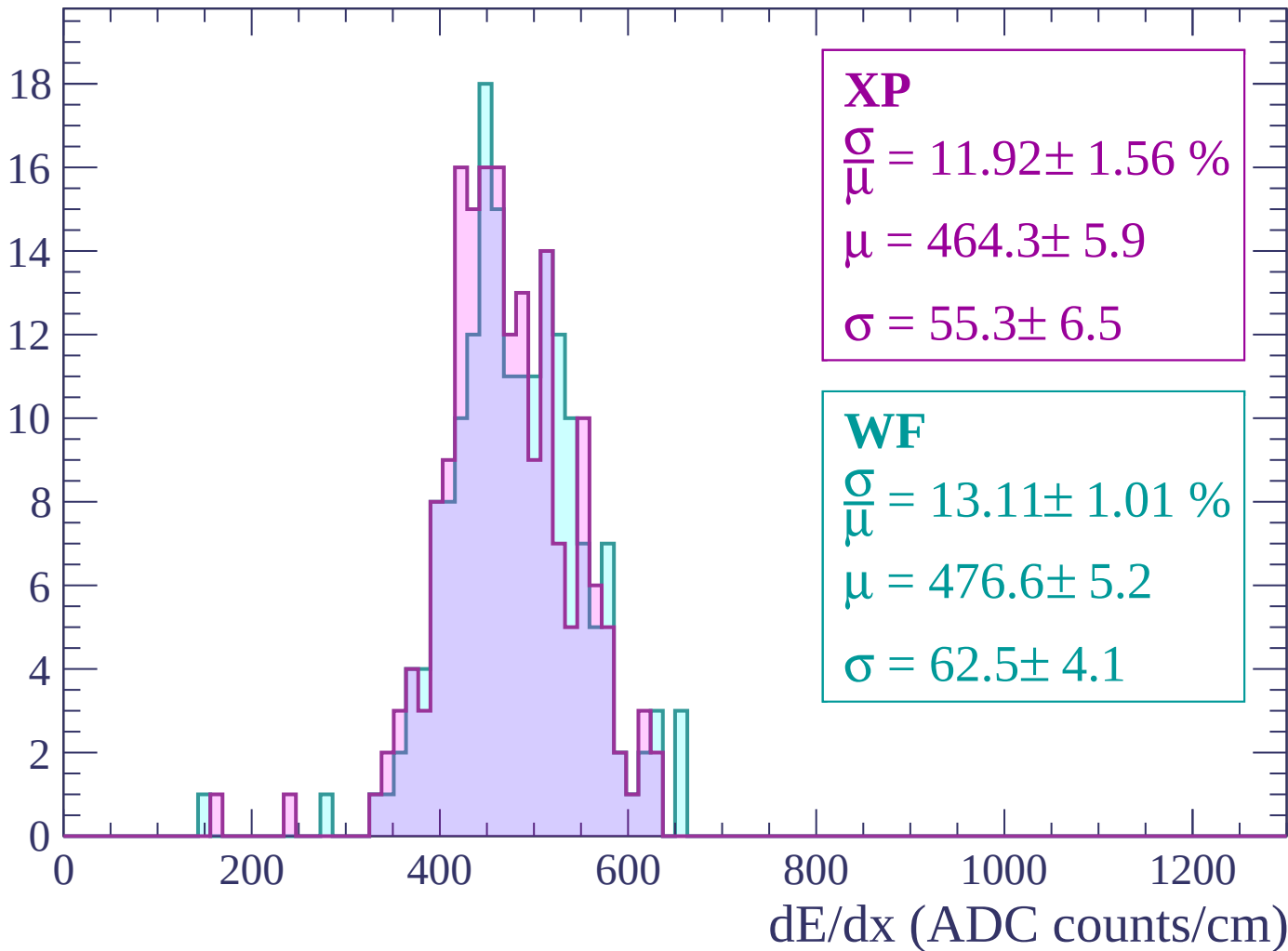
Energy loss | -1920 < p < -1860

Count



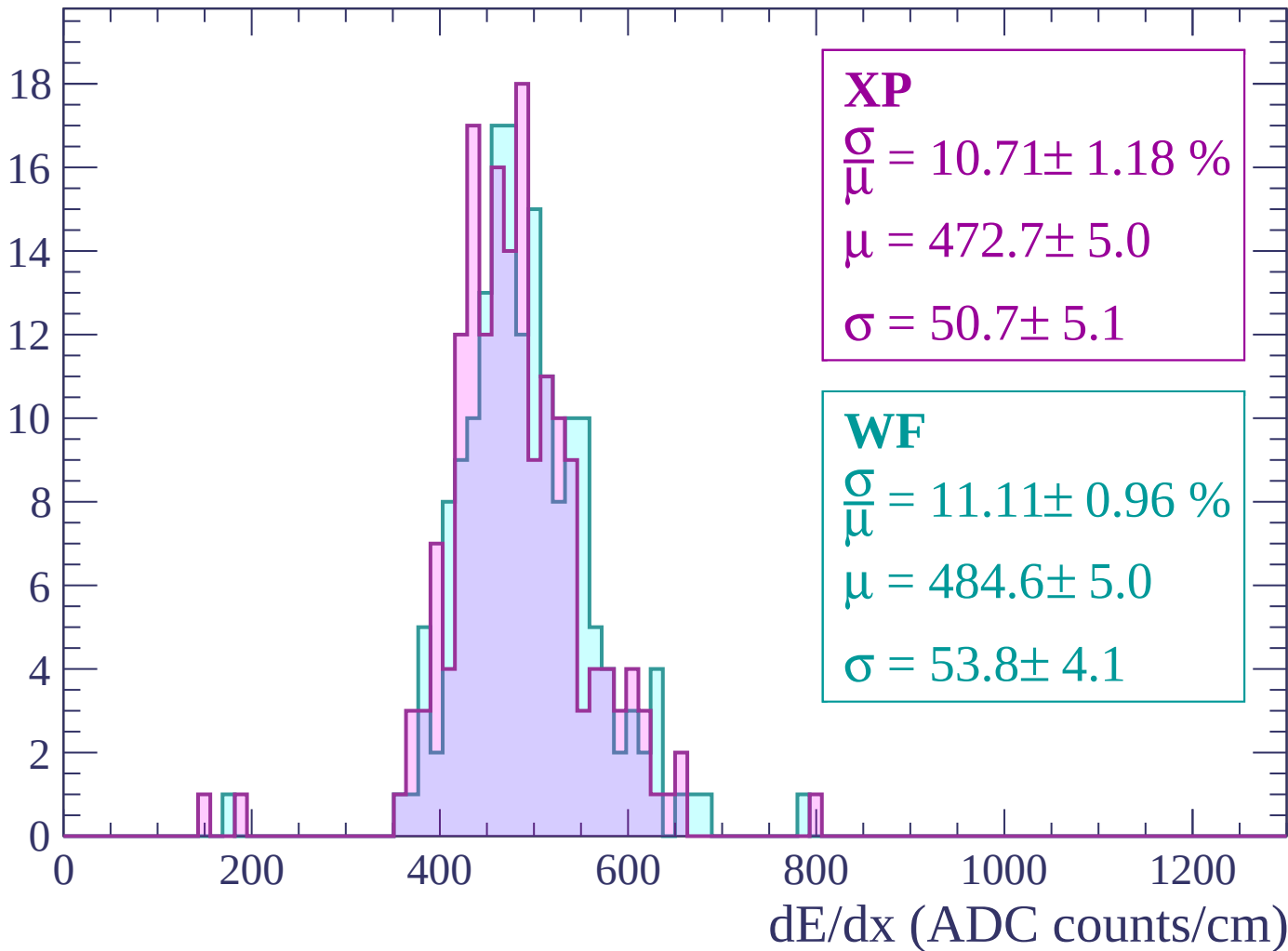
Energy loss | -1860 < p < -1800

Count



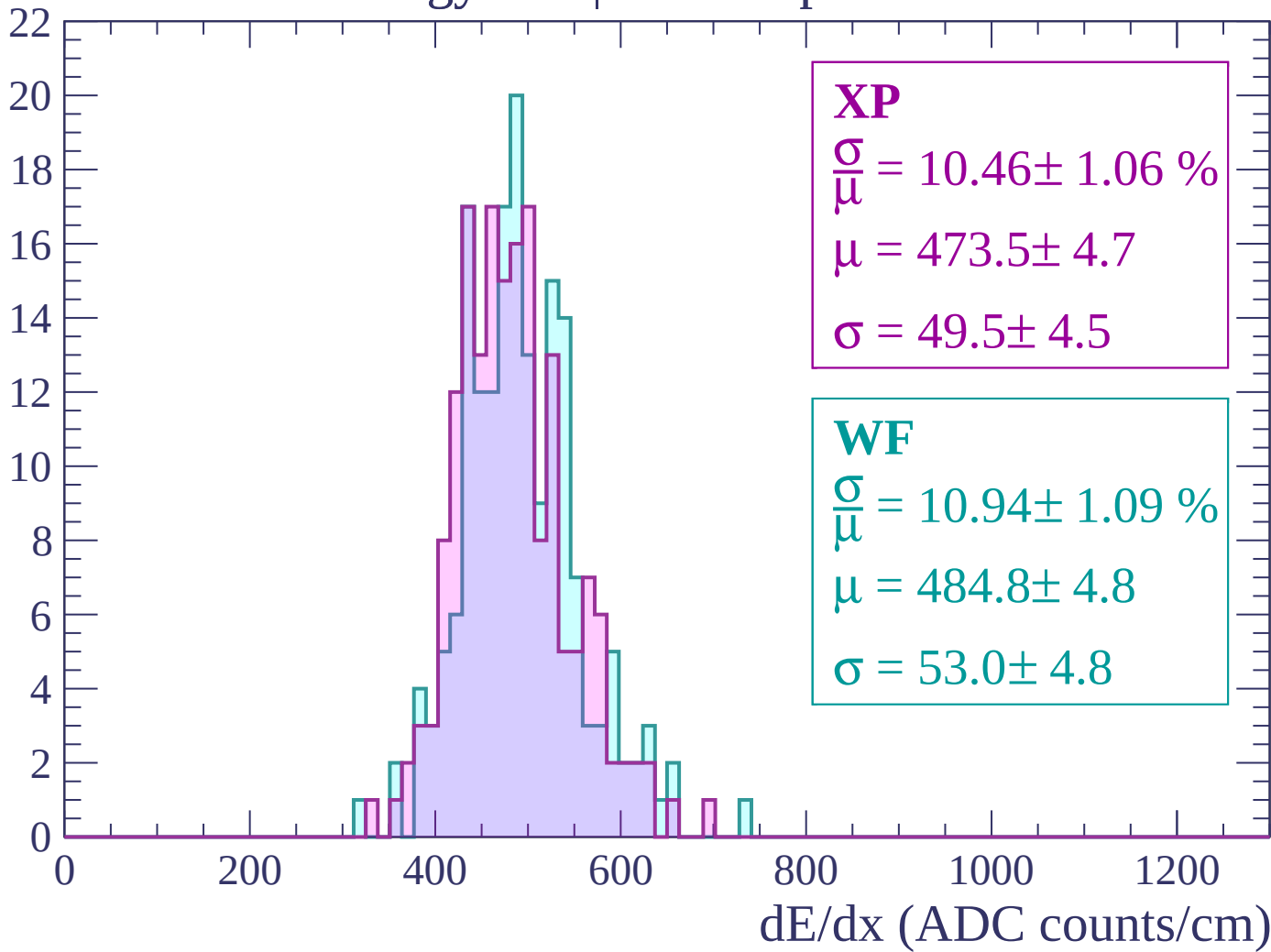
Energy loss | -1800 < p < -1740

Count



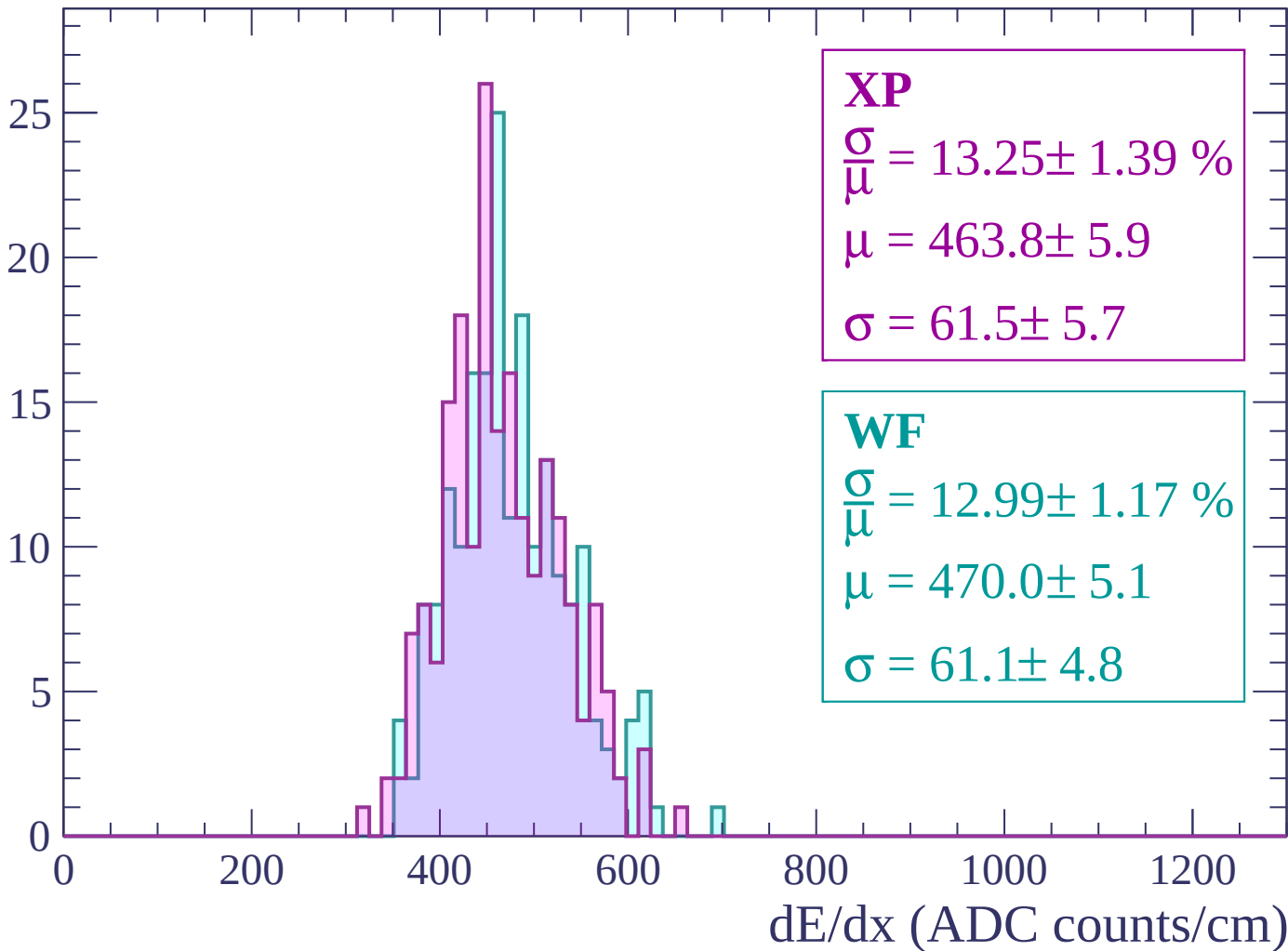
Energy loss | -1740 < p < -1680

Count



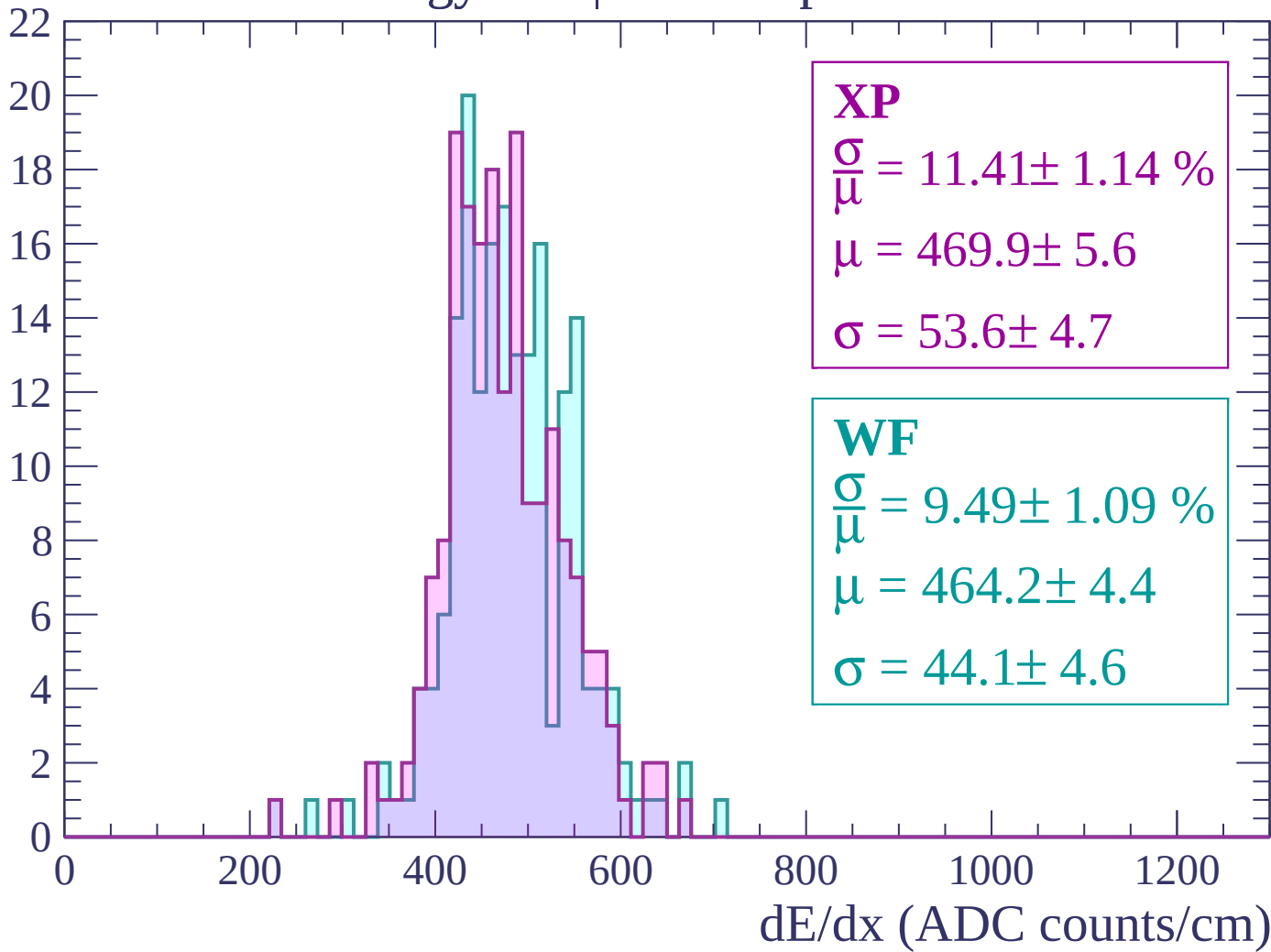
Energy loss | -1680 < p < -1620

Count



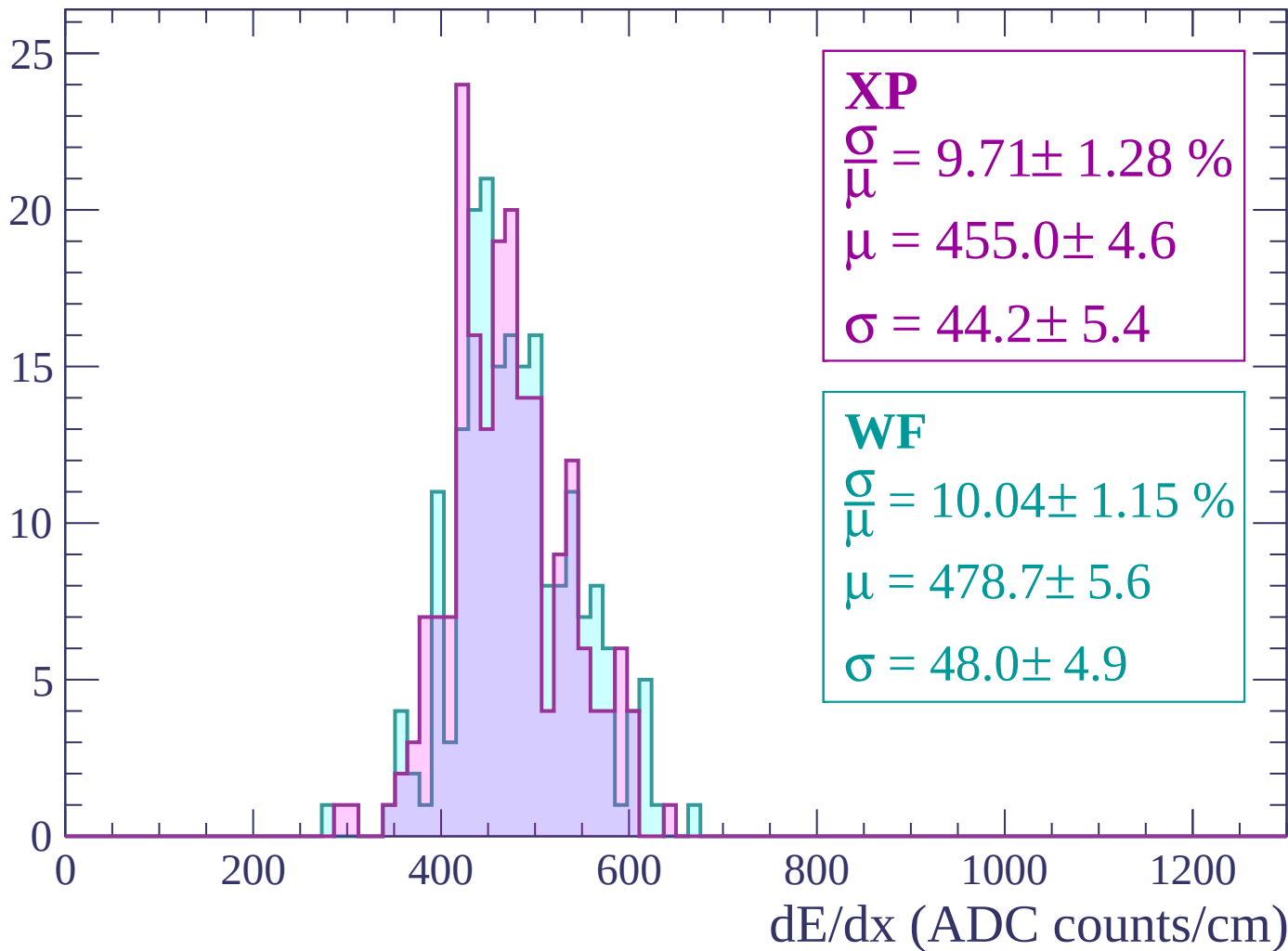
Energy loss | -1620 < p < -1560

Count



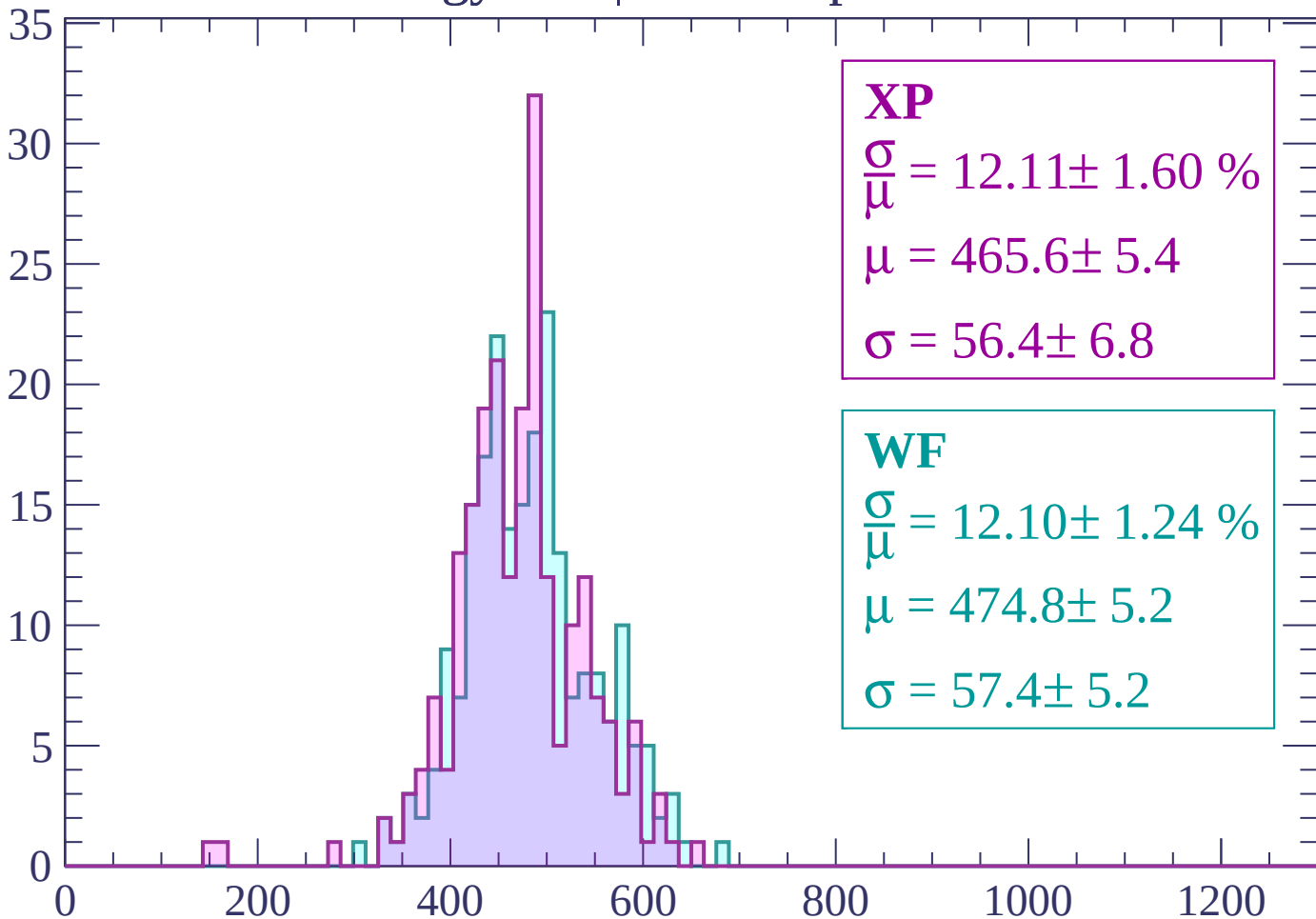
Energy loss | -1560 < p < -1500

Count



Energy loss | -1500 < p < -1440

Count



XP

$$\frac{\sigma}{\mu} = 12.11 \pm 1.60 \%$$

$$\mu = 465.6 \pm 5.4$$

$$\sigma = 56.4 \pm 6.8$$

WF

$$\frac{\sigma}{\mu} = 12.10 \pm 1.24 \%$$

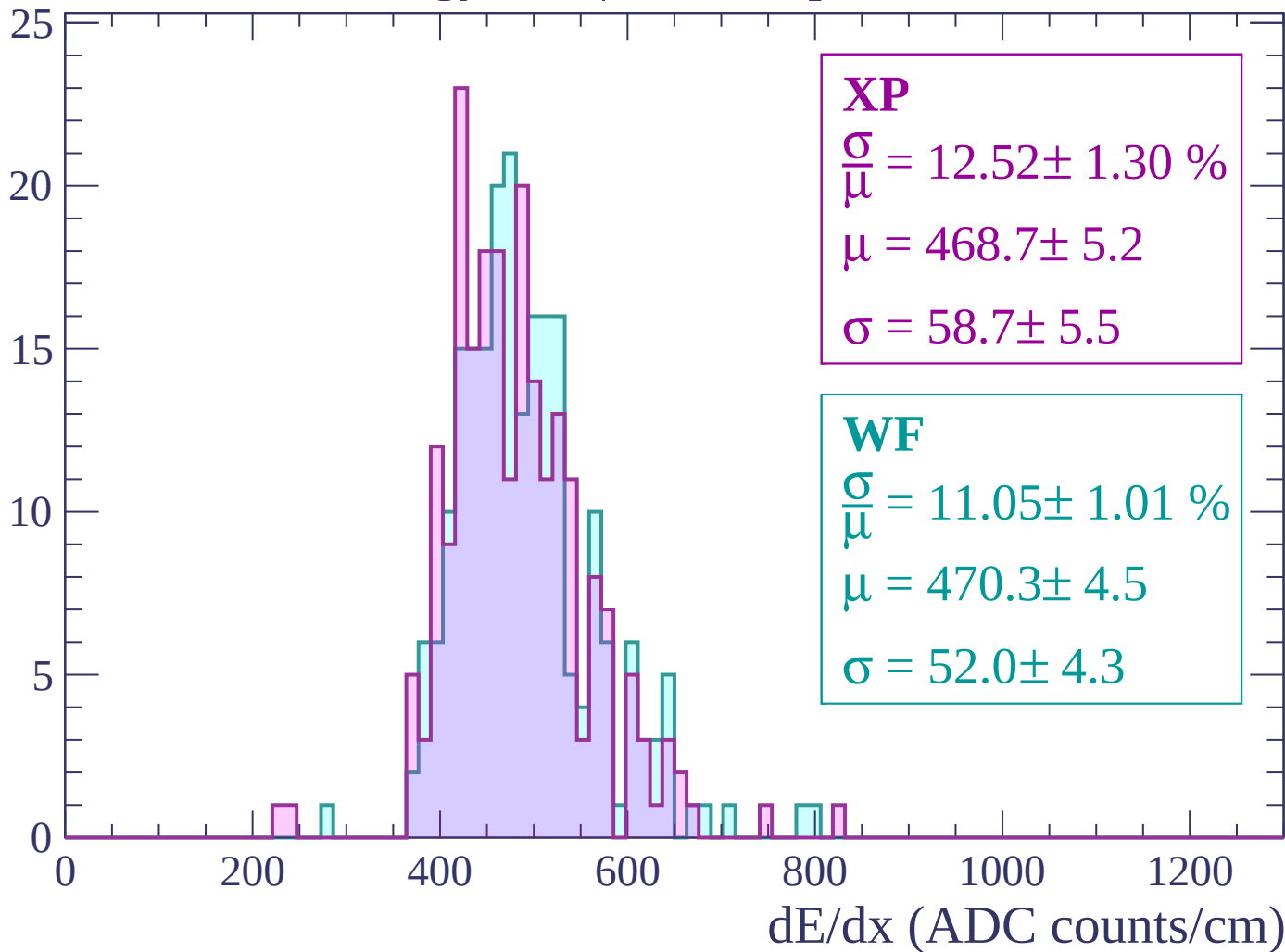
$$\mu = 474.8 \pm 5.2$$

$$\sigma = 57.4 \pm 5.2$$

dE/dx (ADC counts/cm)

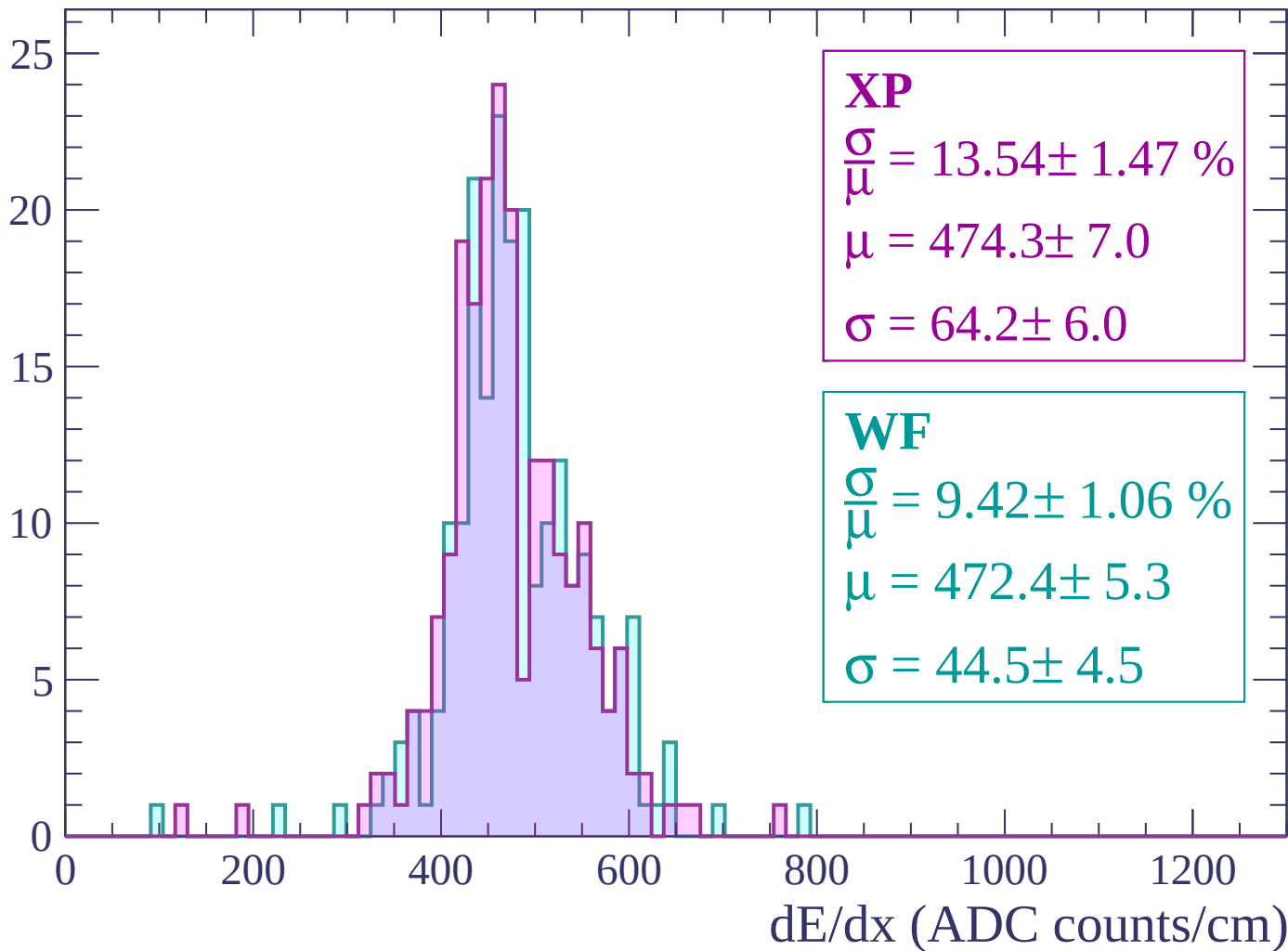
Energy loss | -1440 < p < -1380

Count



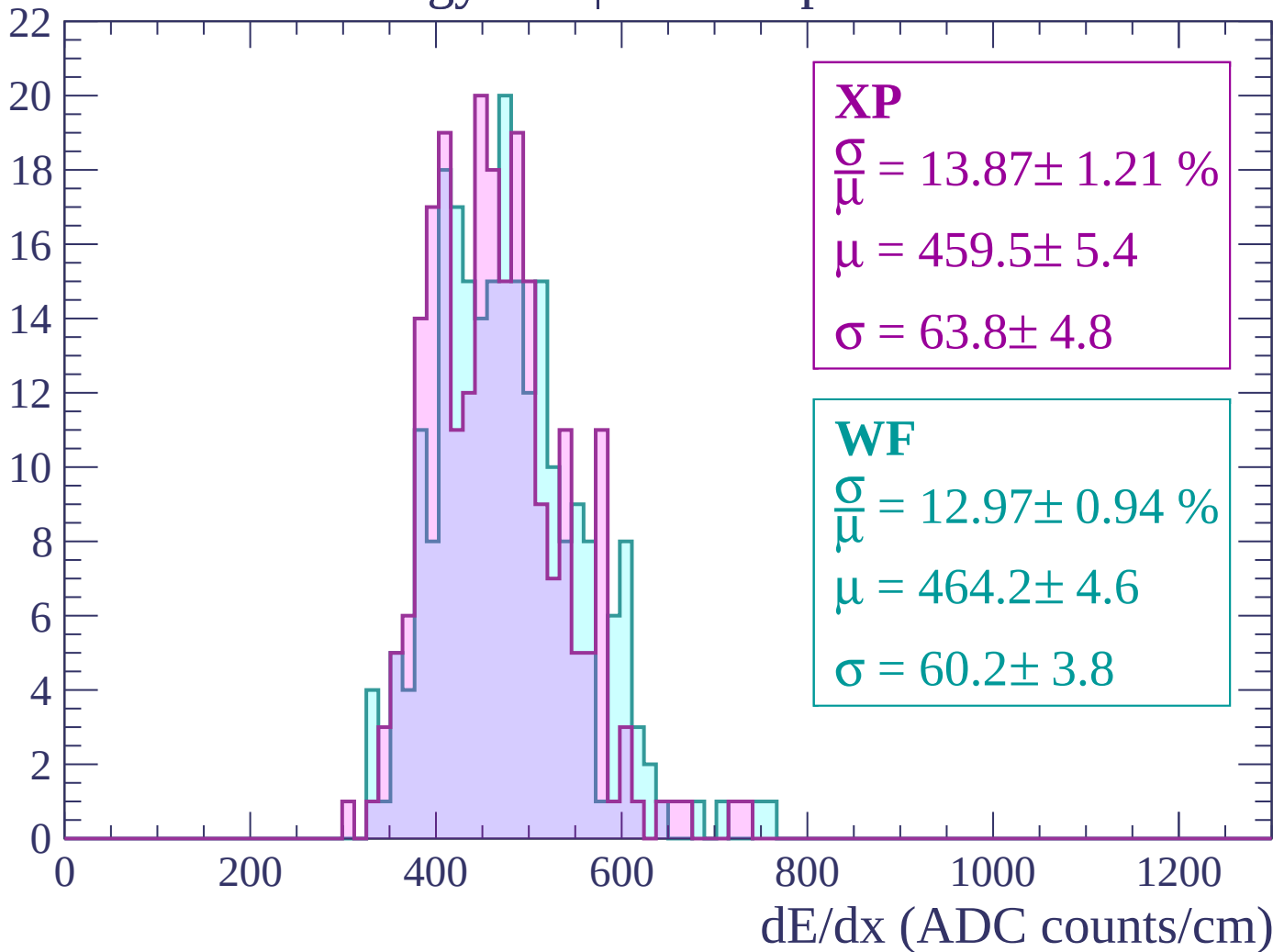
Energy loss | -1380 < p < -1320

Count



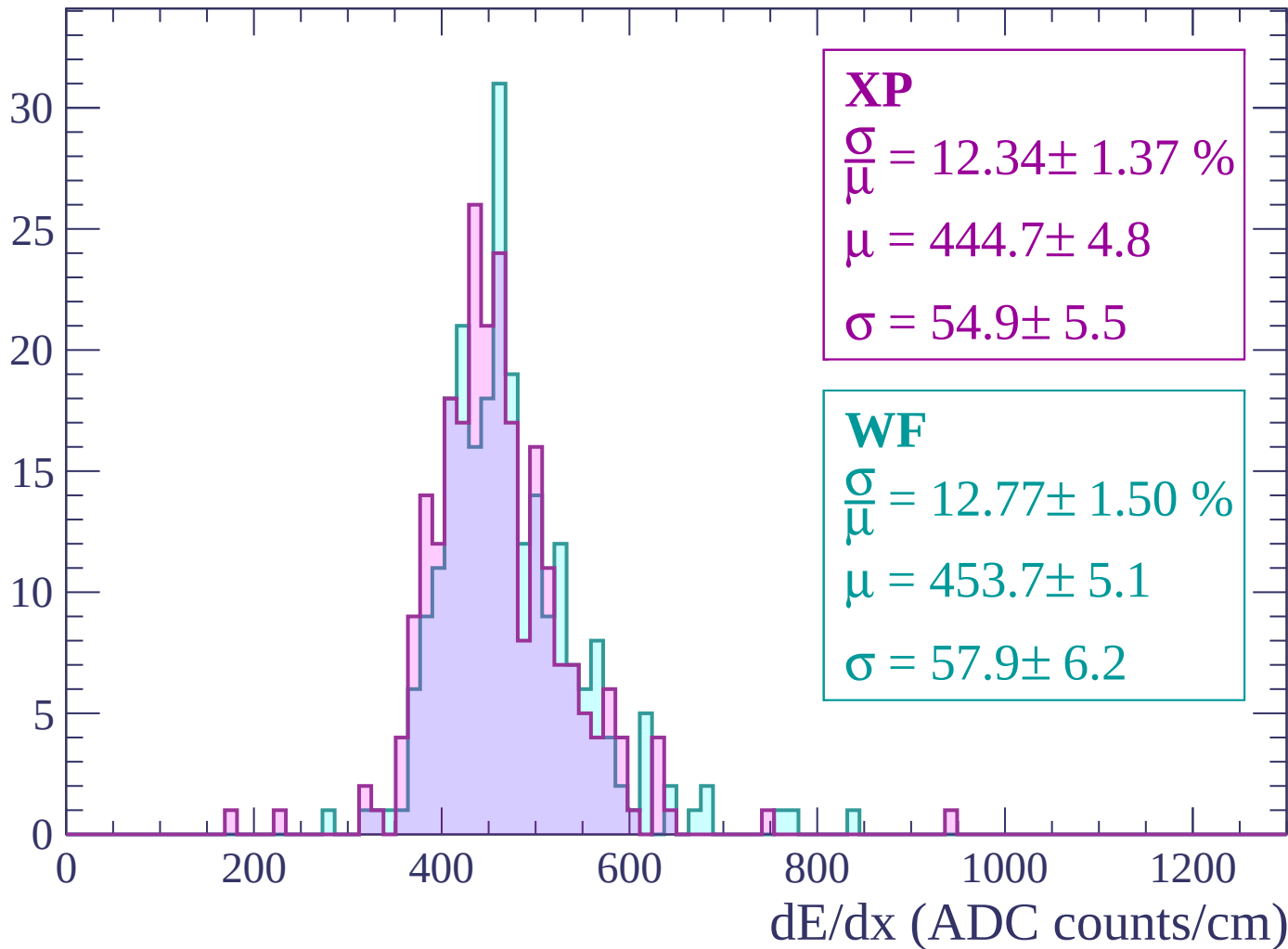
Energy loss | -1320 < p < -1260

Count



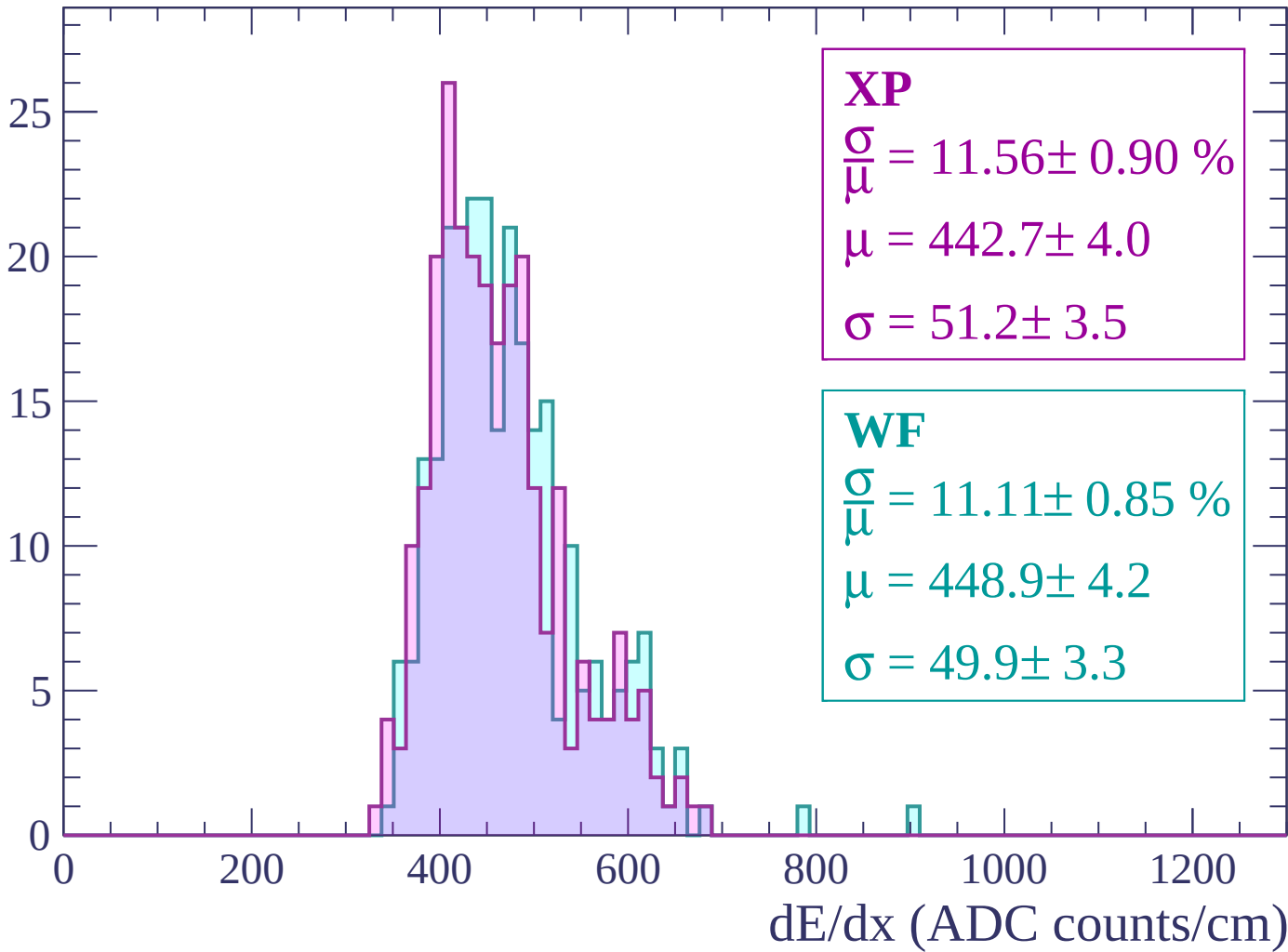
Energy loss | -1260 < p < -1200

Count



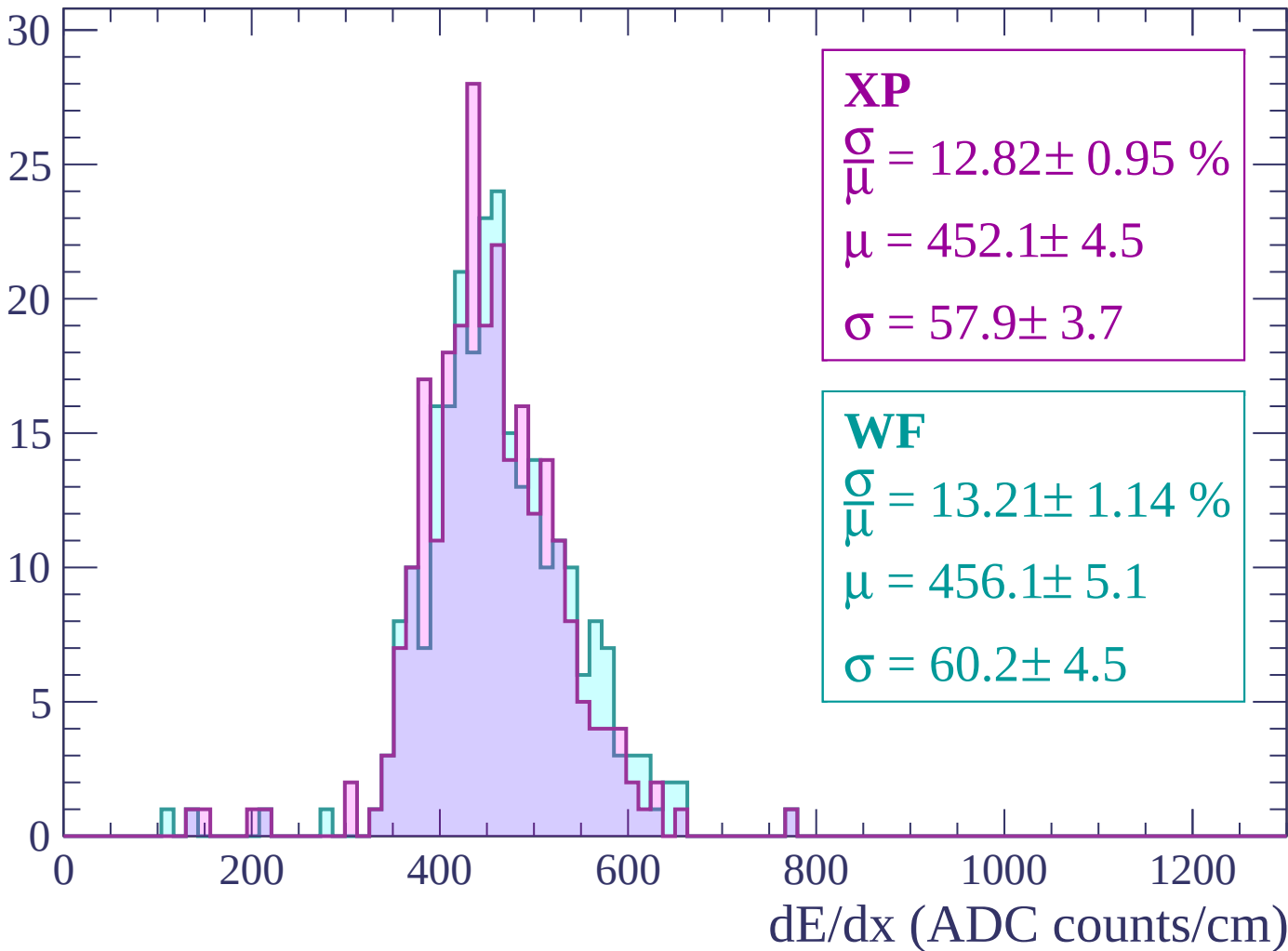
Energy loss | -1200 < p < -1140

Count



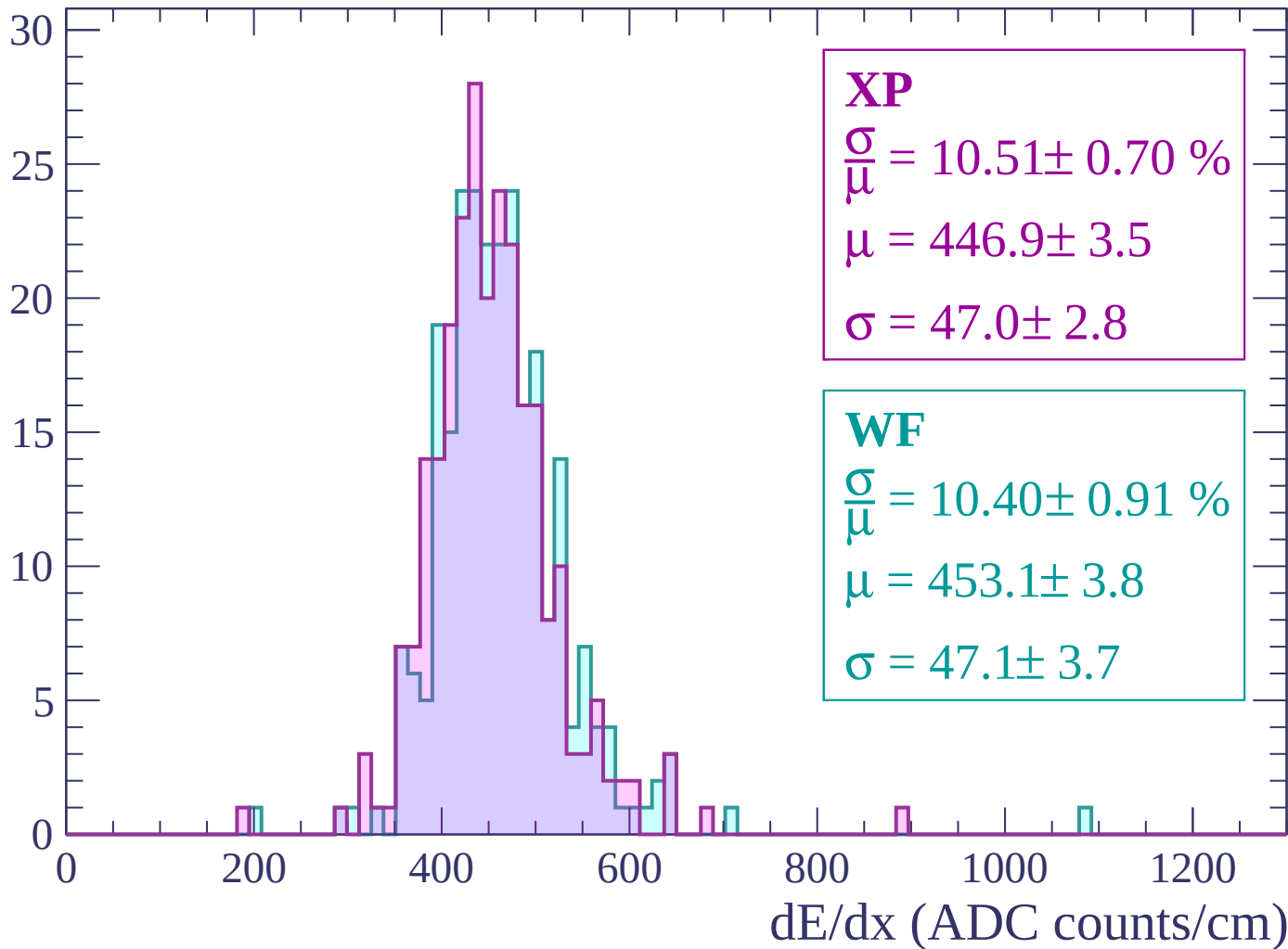
Energy loss | -1140 < p < -1080

Count



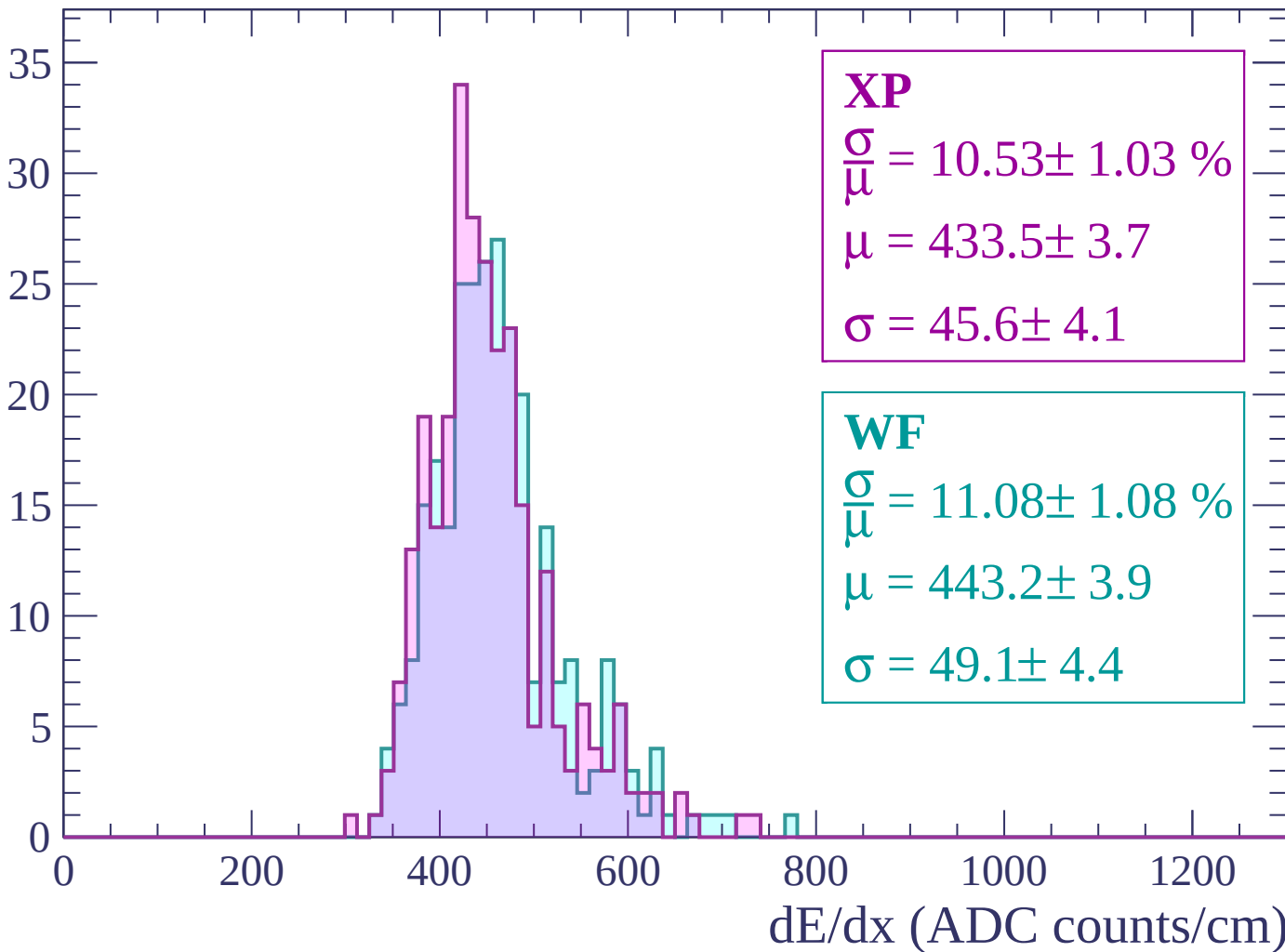
Energy loss | -1080 < p < -1020

Count



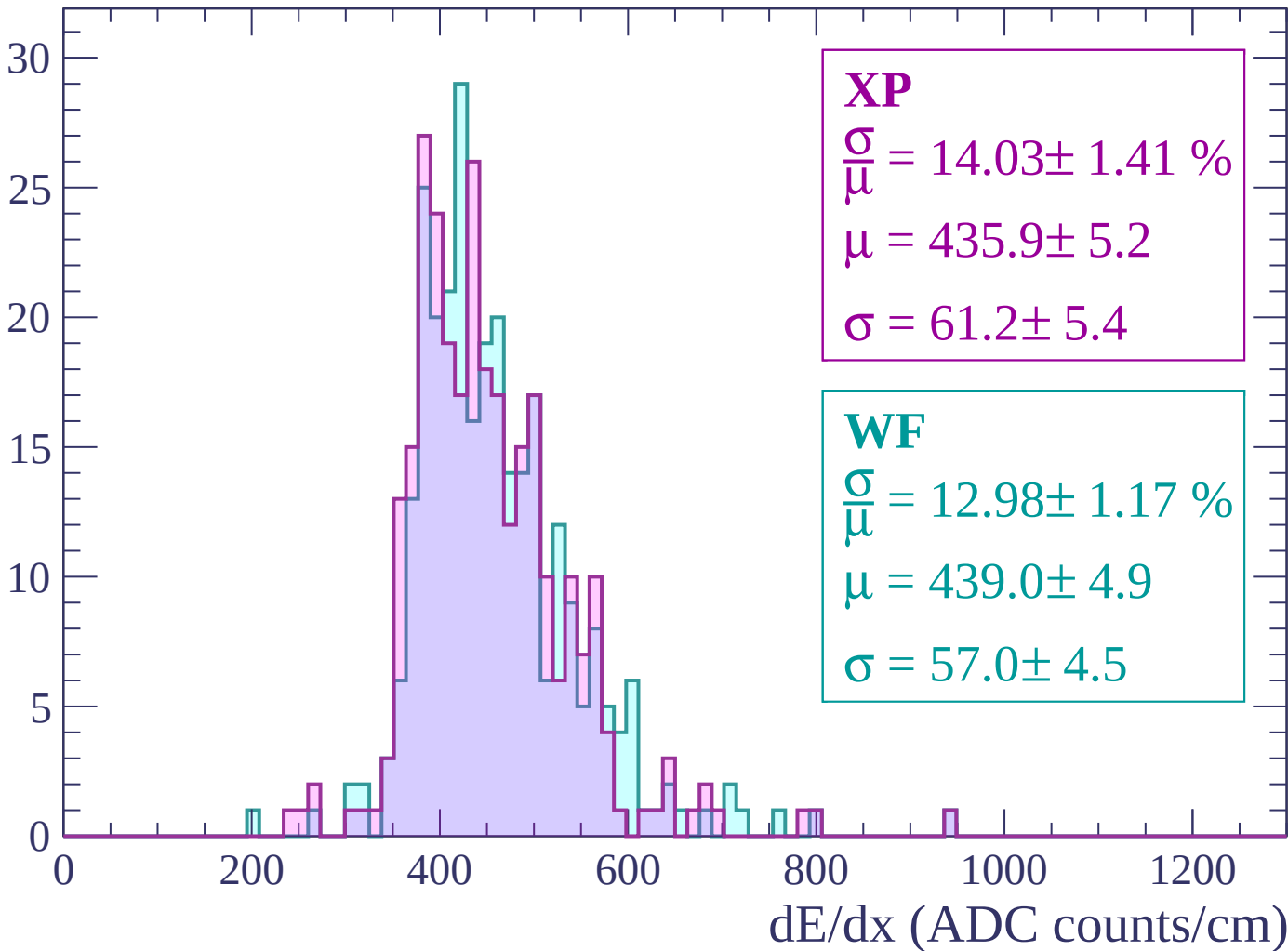
Energy loss | $-1020 < p < -960$

Count



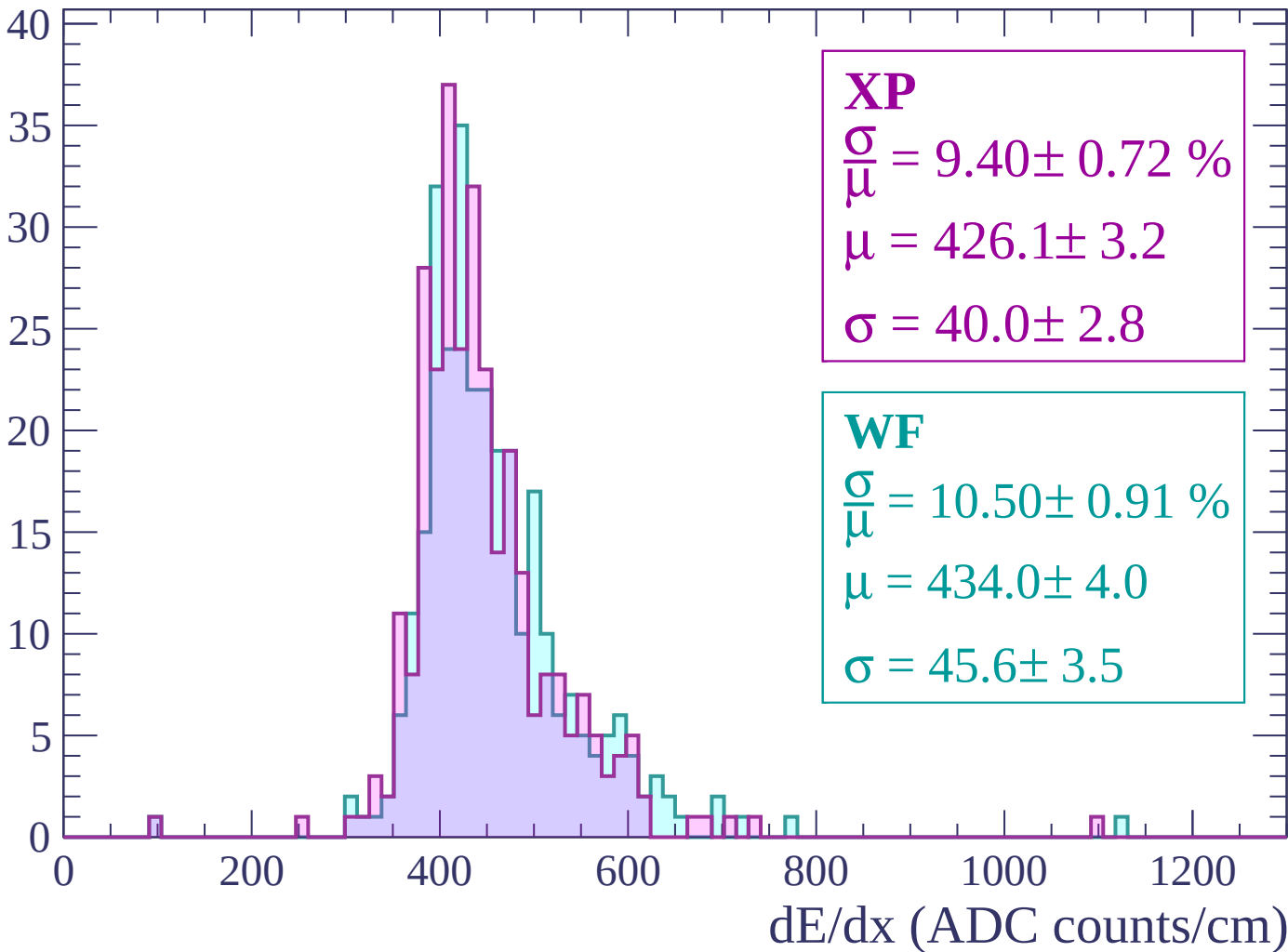
Energy loss | $-960 < p < -900$

Count



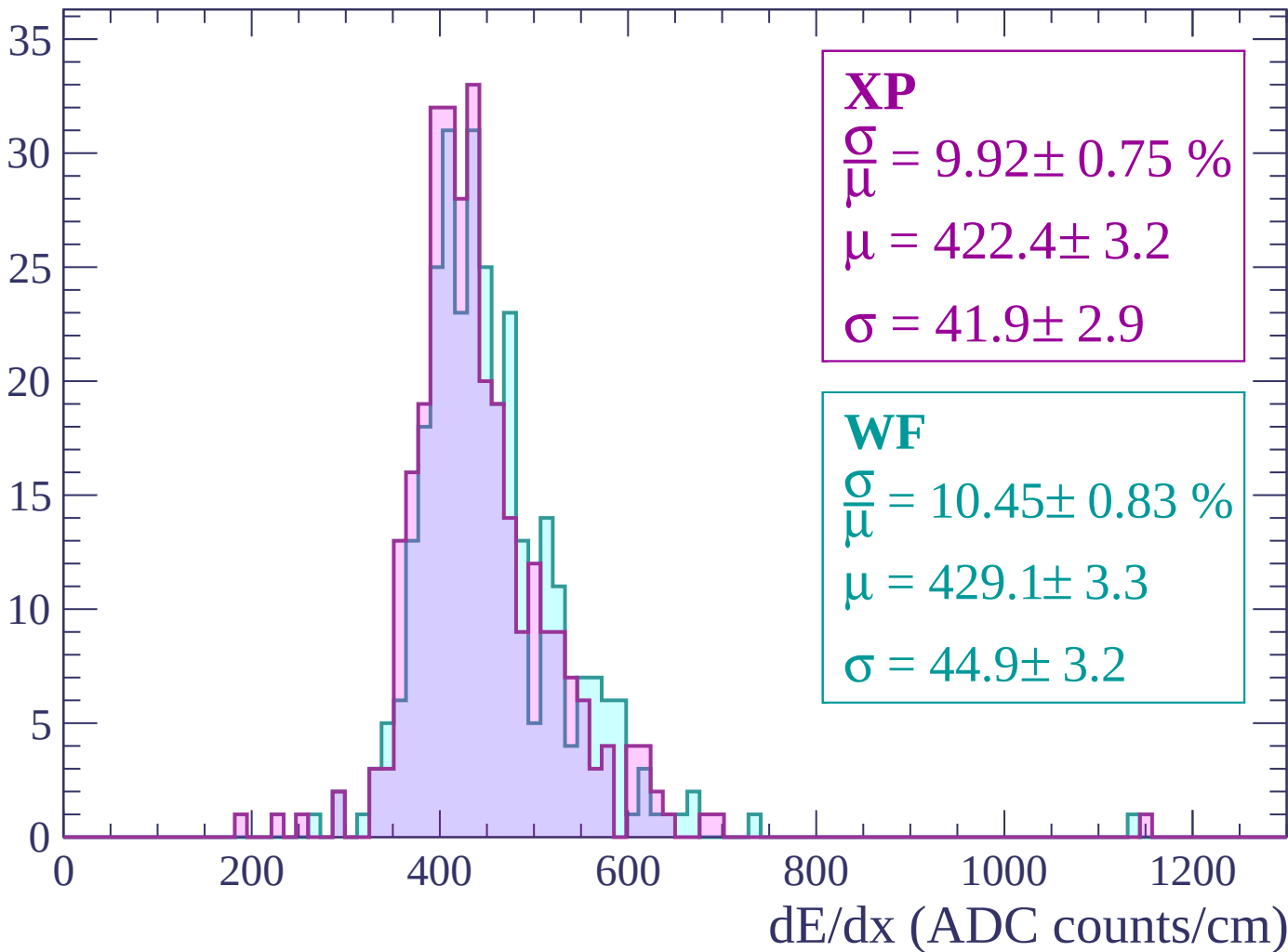
Energy loss | $-900 < p < -840$

Count



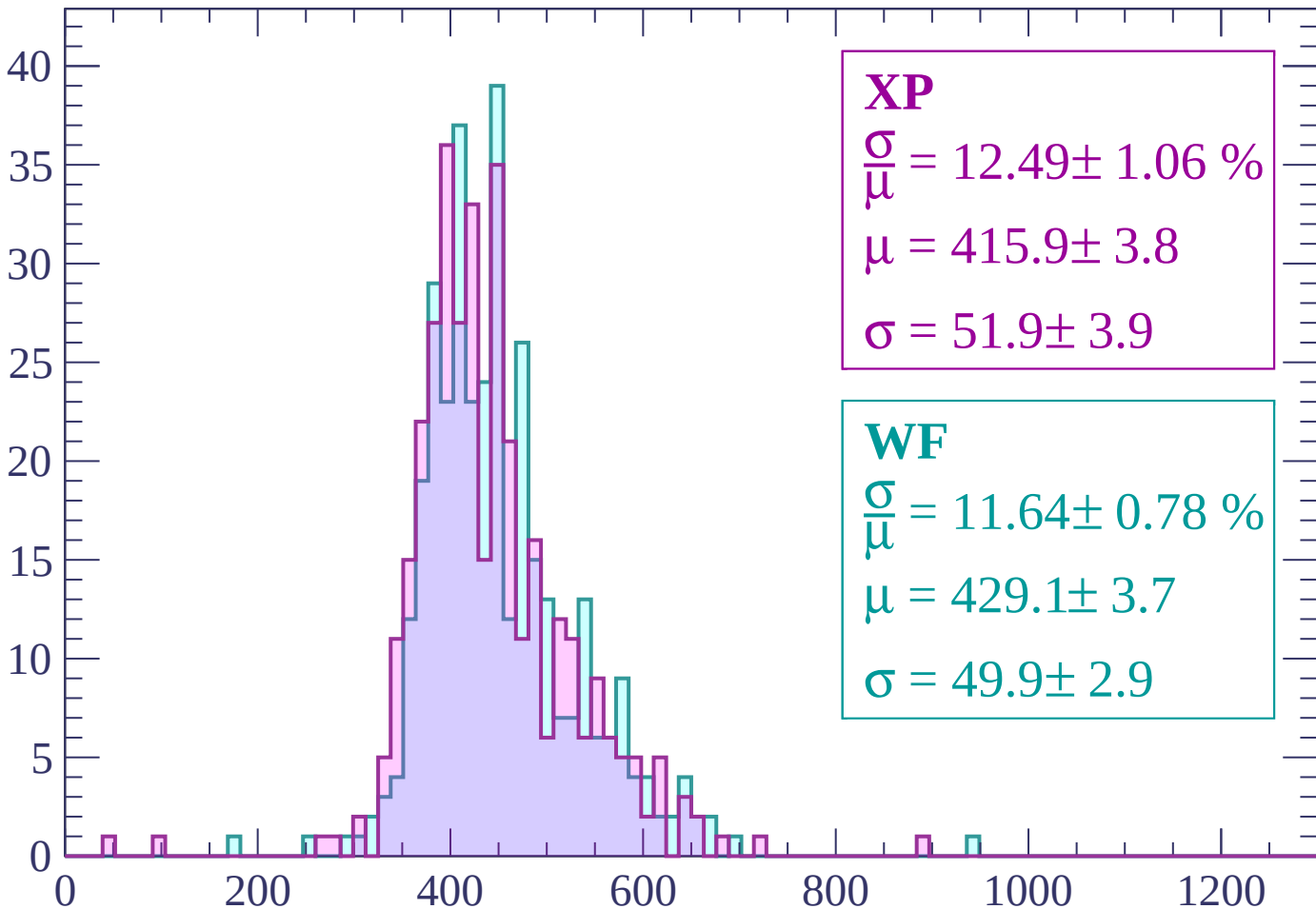
Energy loss | $-840 < p < -780$

Count



Energy loss | $-780 < p < -720$

Count



XP

$$\frac{\sigma}{\mu} = 12.49 \pm 1.06 \%$$

$$\mu = 415.9 \pm 3.8$$

$$\sigma = 51.9 \pm 3.9$$

WF

$$\frac{\sigma}{\mu} = 11.64 \pm 0.78 \%$$

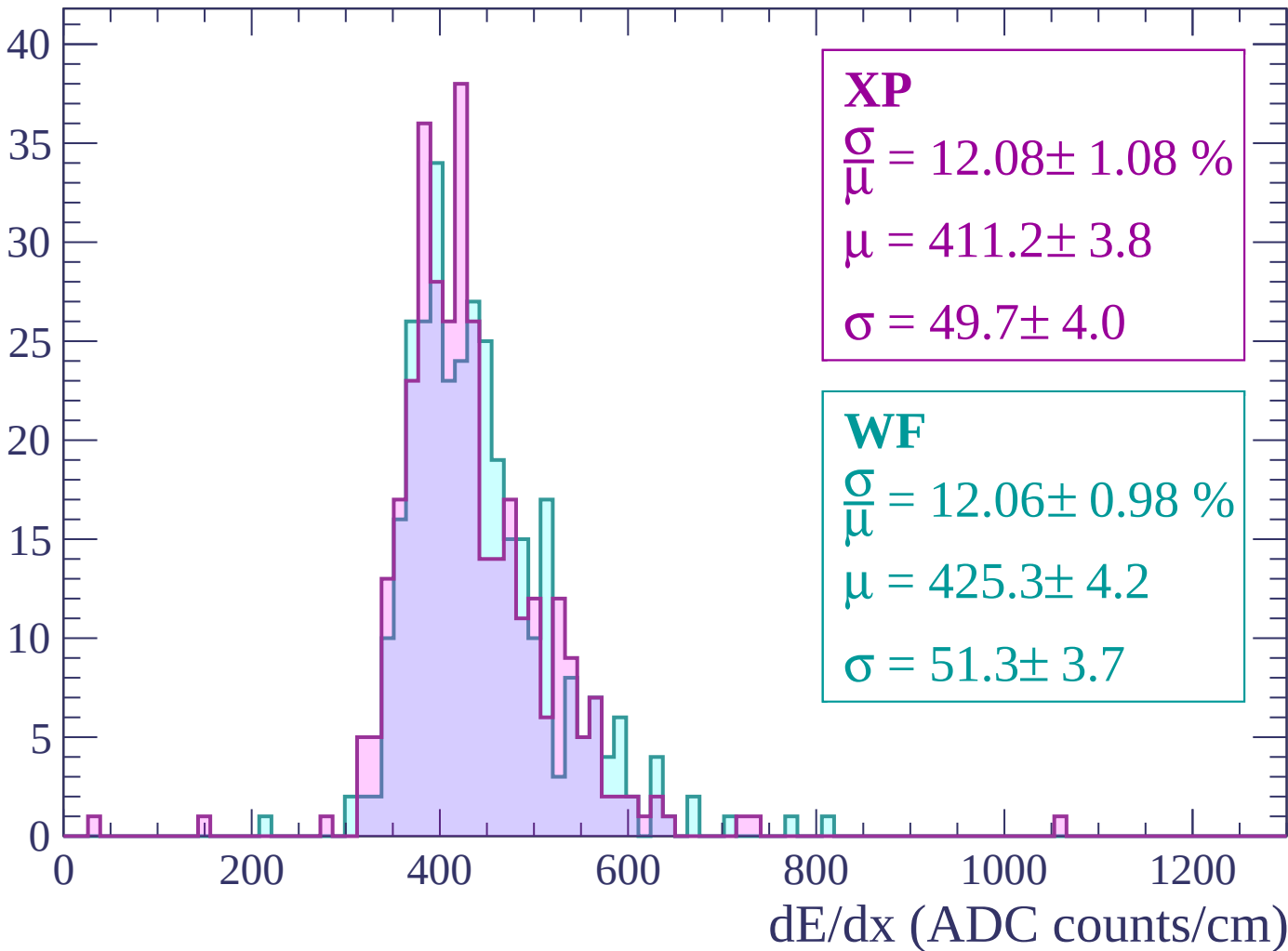
$$\mu = 429.1 \pm 3.7$$

$$\sigma = 49.9 \pm 2.9$$

dE/dx (ADC counts/cm)

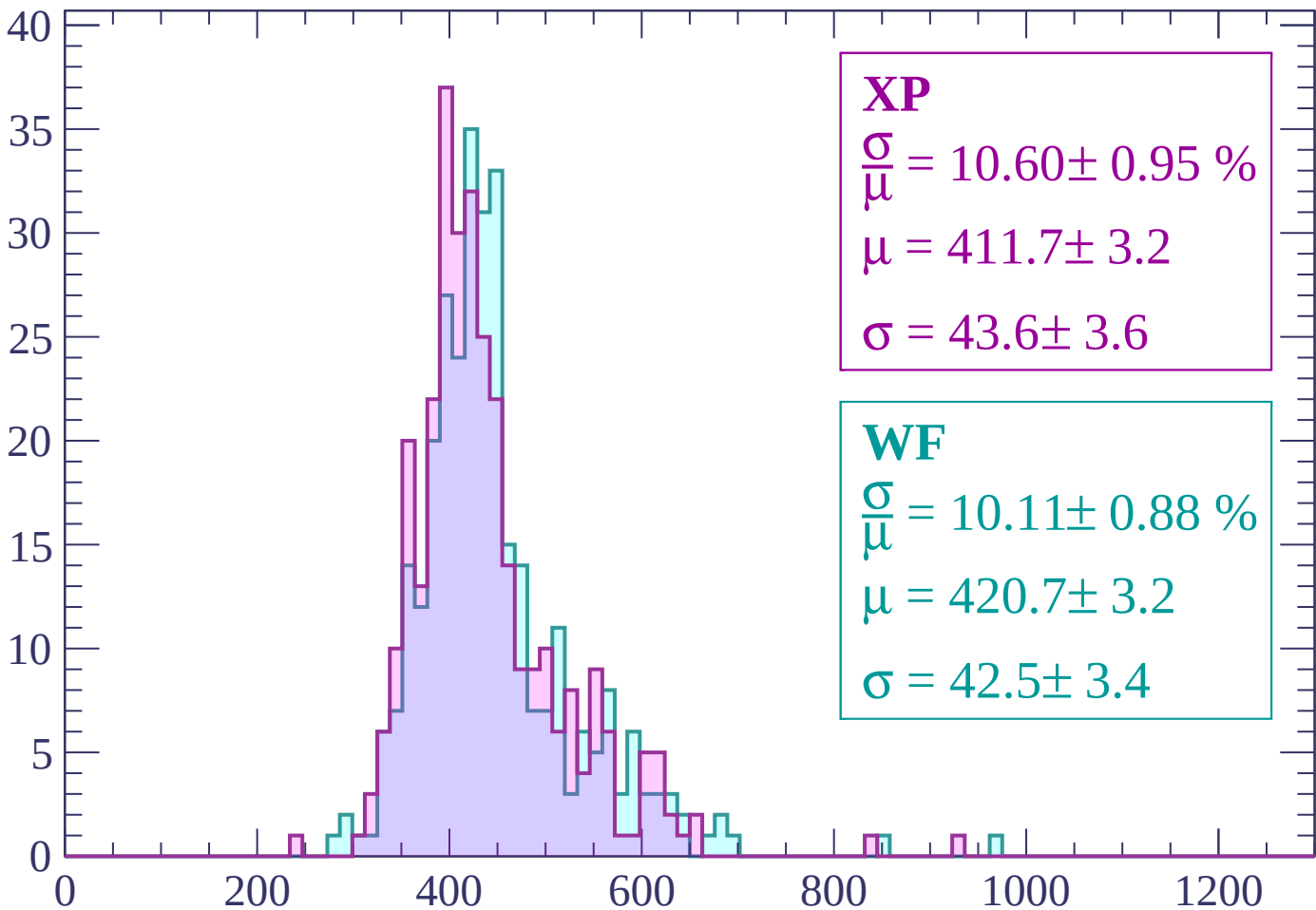
Energy loss | -720 < p < -660

Count



Energy loss | $-660 < p < -600$

Count



XP

$$\frac{\sigma}{\mu} = 10.60 \pm 0.95 \%$$

$$\mu = 411.7 \pm 3.2$$

$$\sigma = 43.6 \pm 3.6$$

WF

$$\frac{\sigma}{\mu} = 10.11 \pm 0.88 \%$$

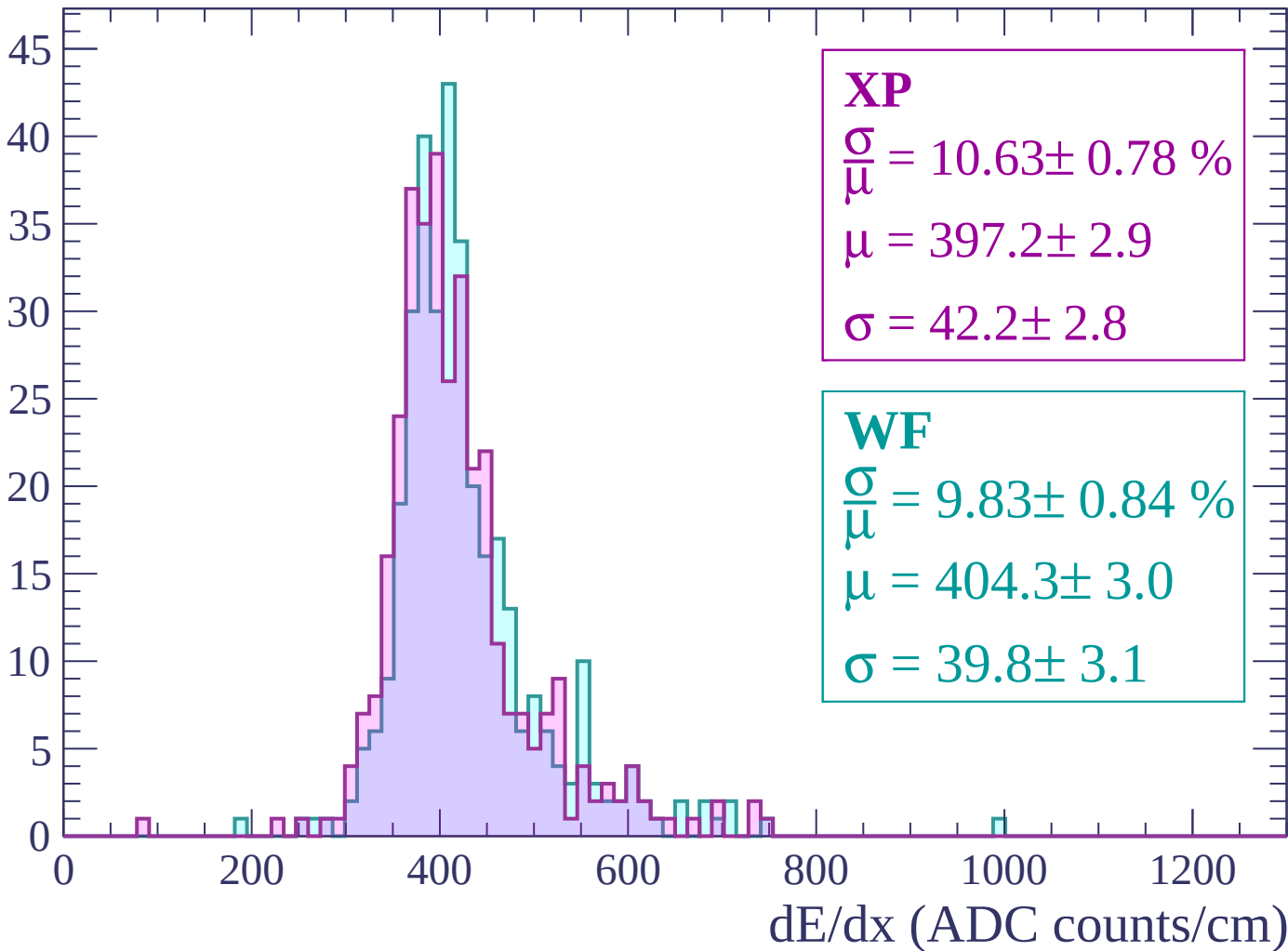
$$\mu = 420.7 \pm 3.2$$

$$\sigma = 42.5 \pm 3.4$$

dE/dx (ADC counts/cm)

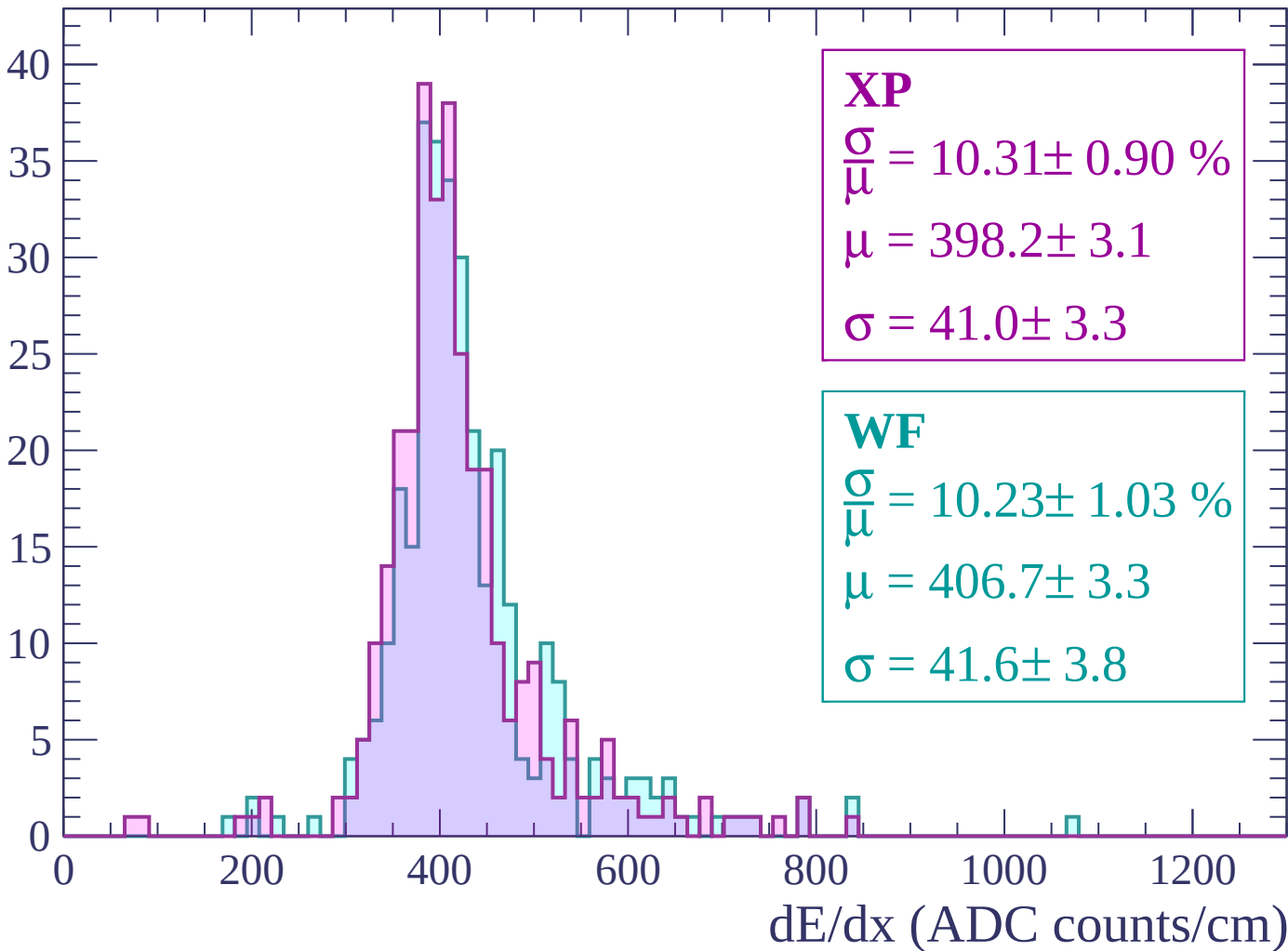
Energy loss | $-600 < p < -540$

Count



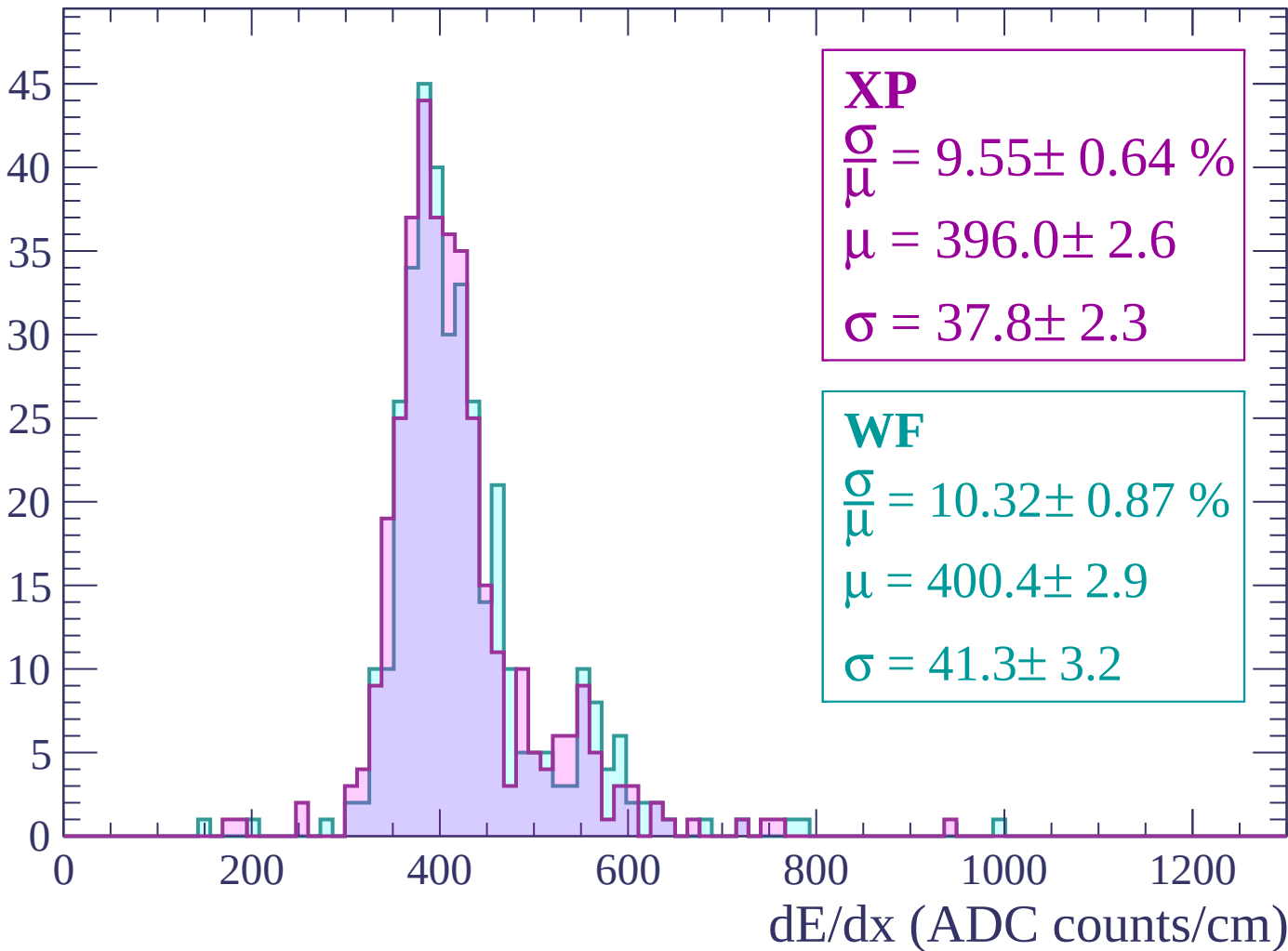
Energy loss | $-540 < p < -480$

Count



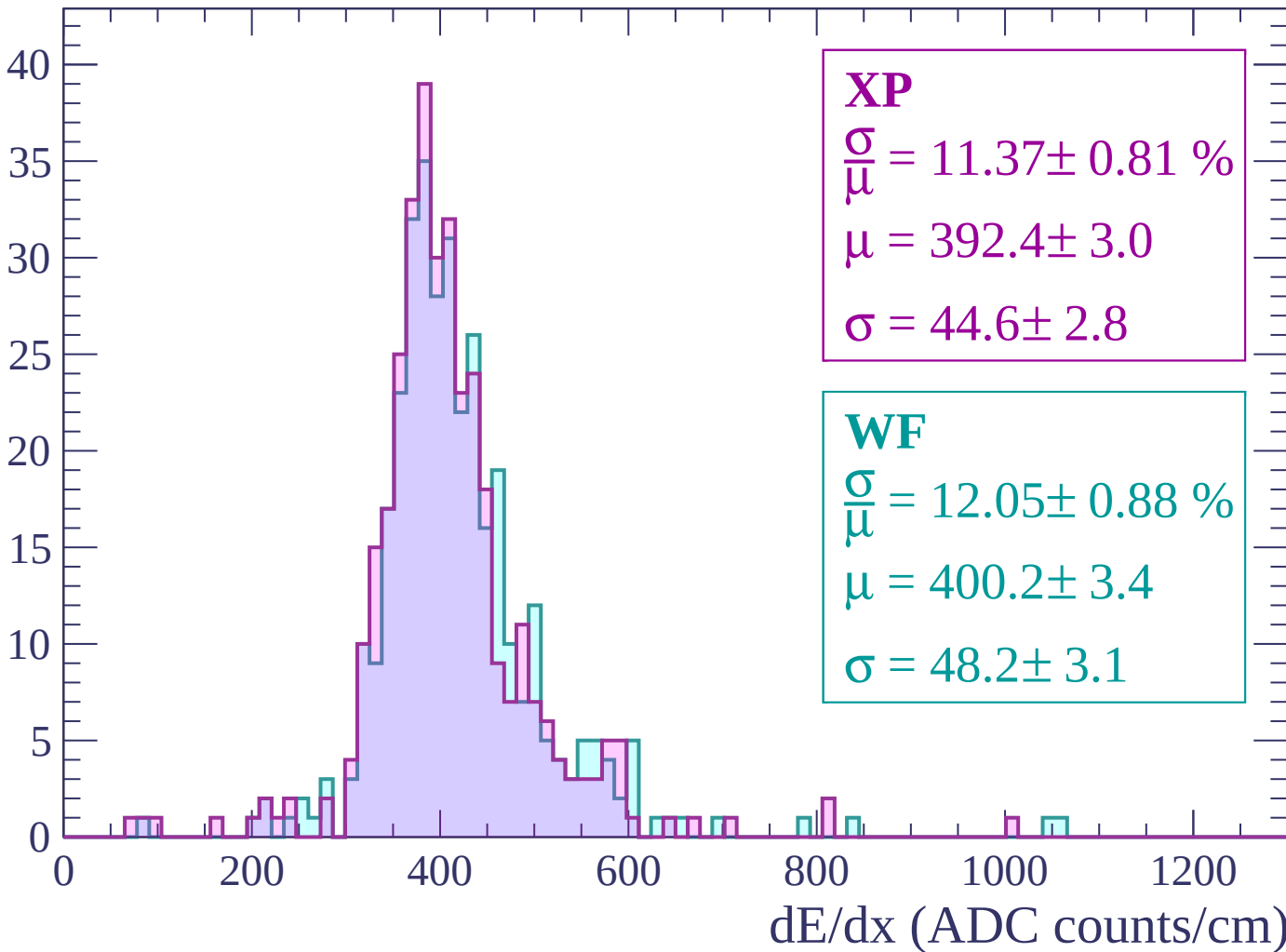
Energy loss | $-480 < p < -420$

Count



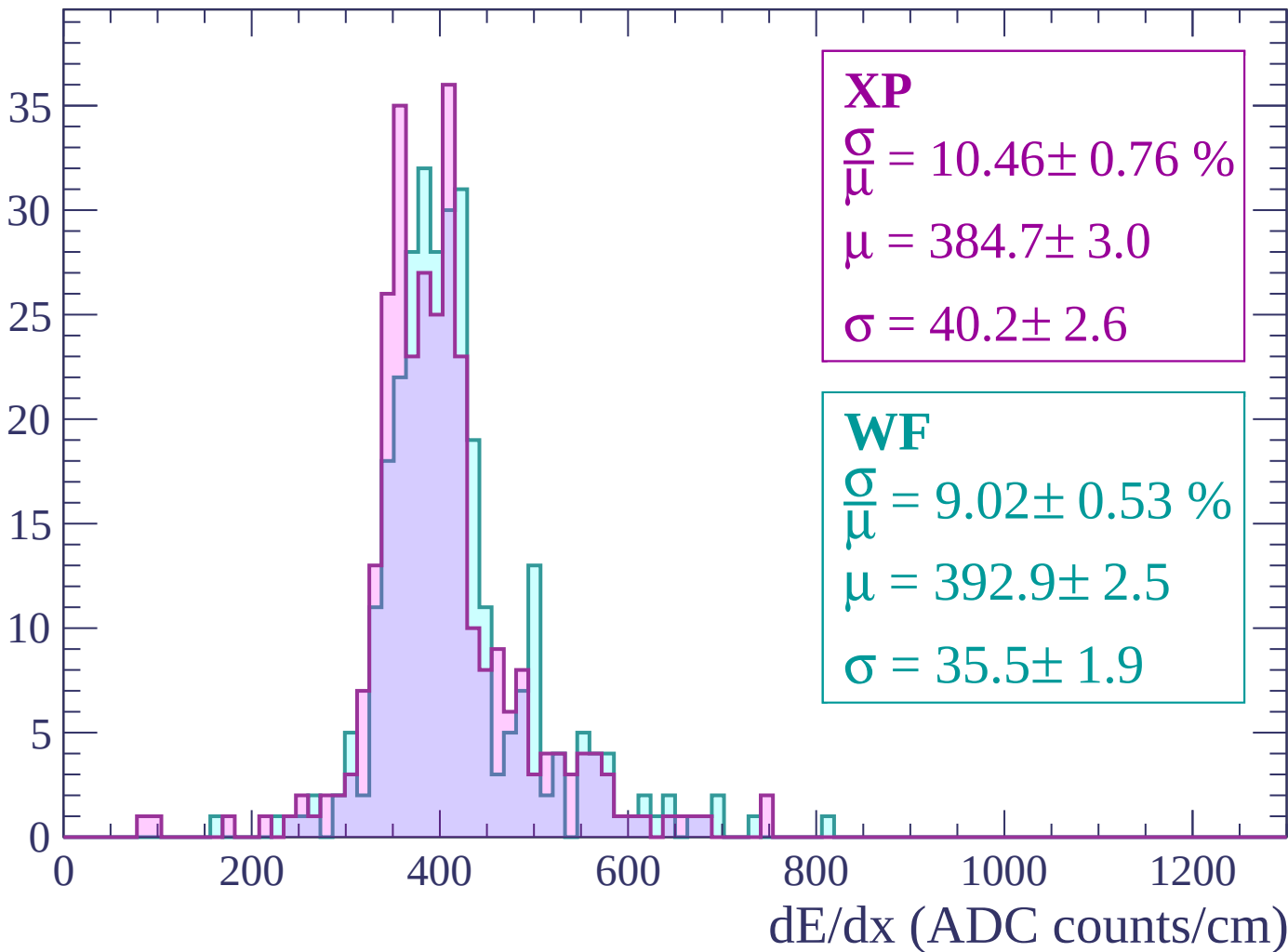
Energy loss | $-420 < p < -360$

Count



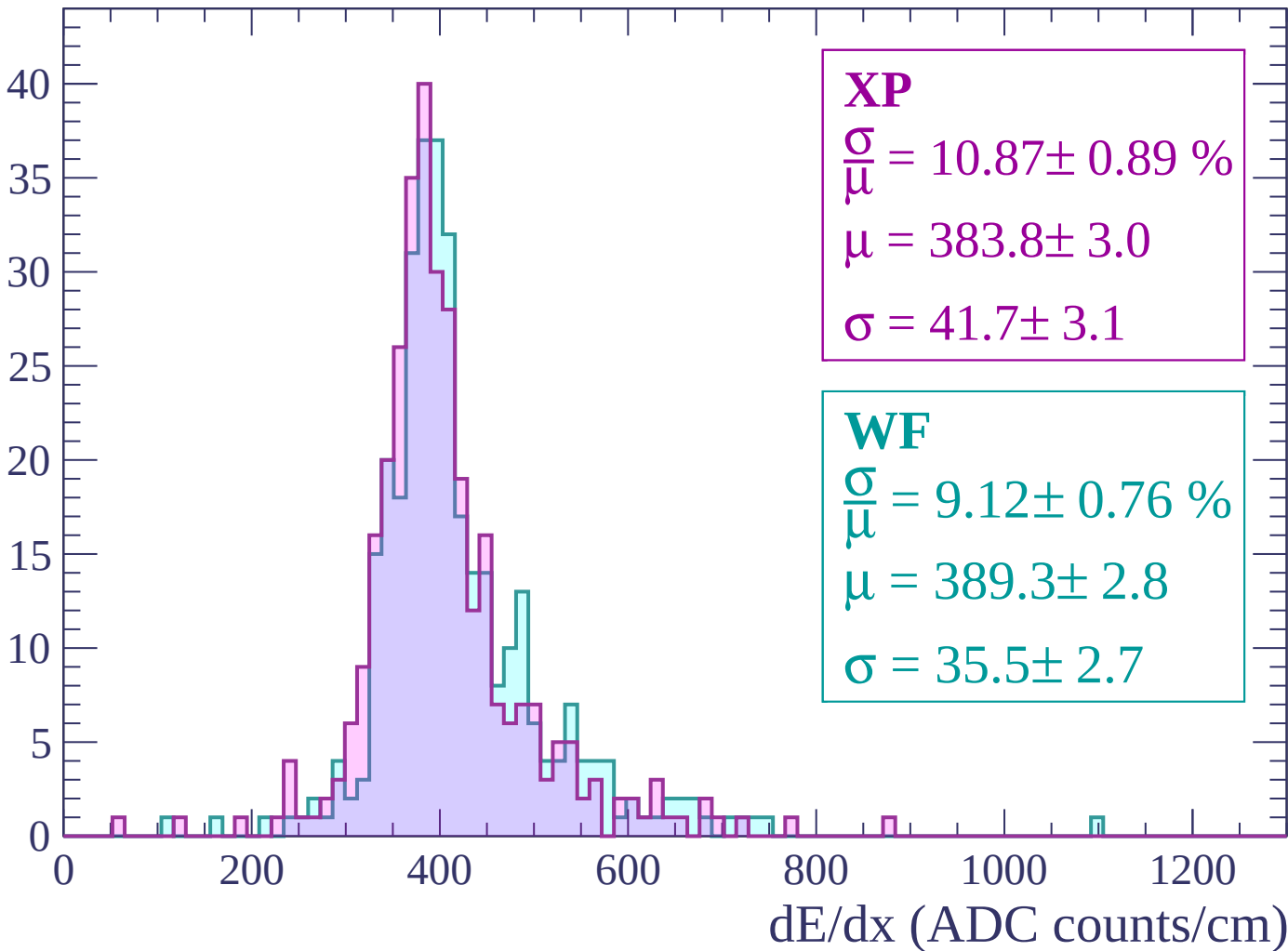
Energy loss | $-360 < p < -300$

Count



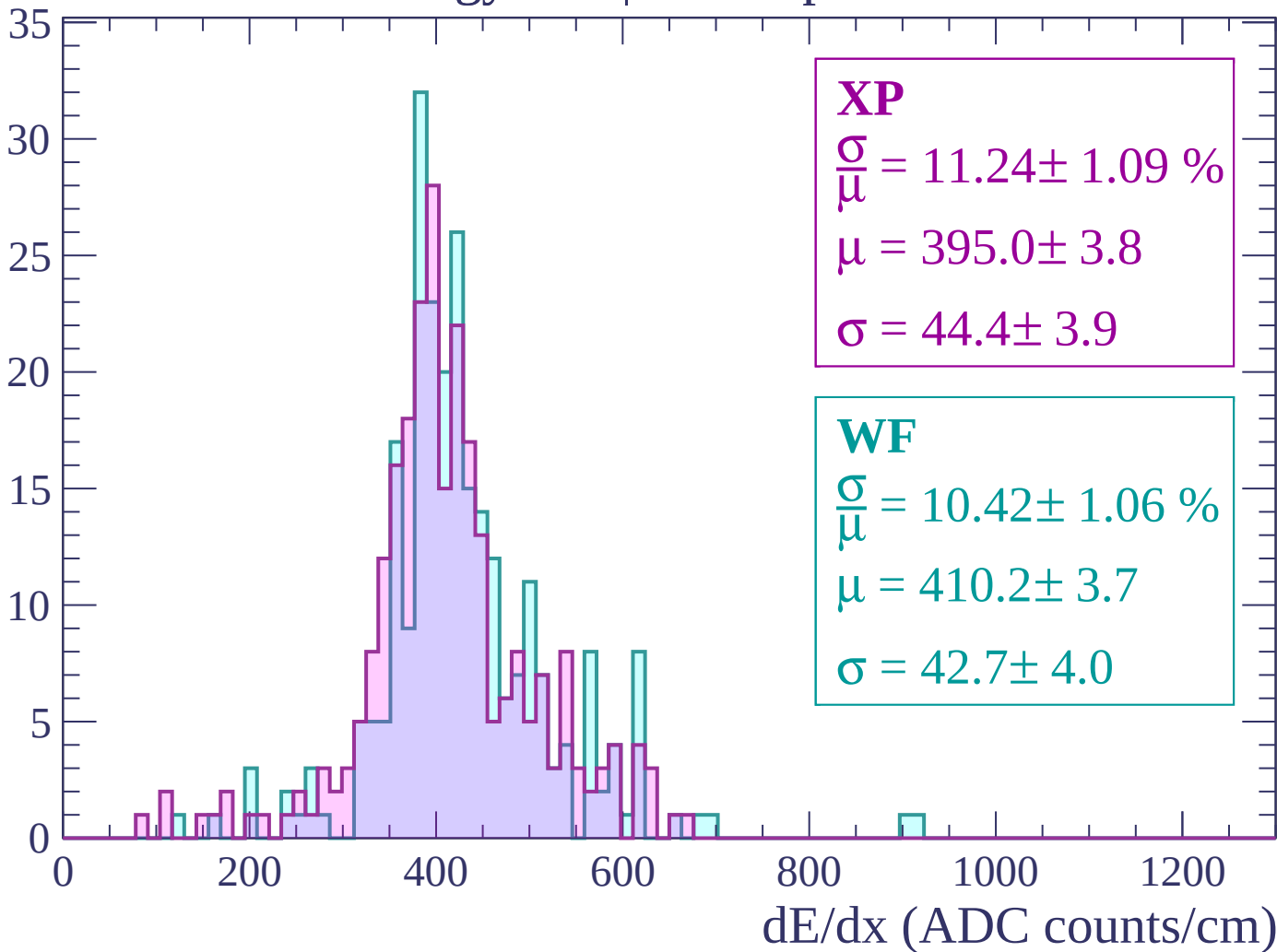
Energy loss | $-300 < p < -240$

Count



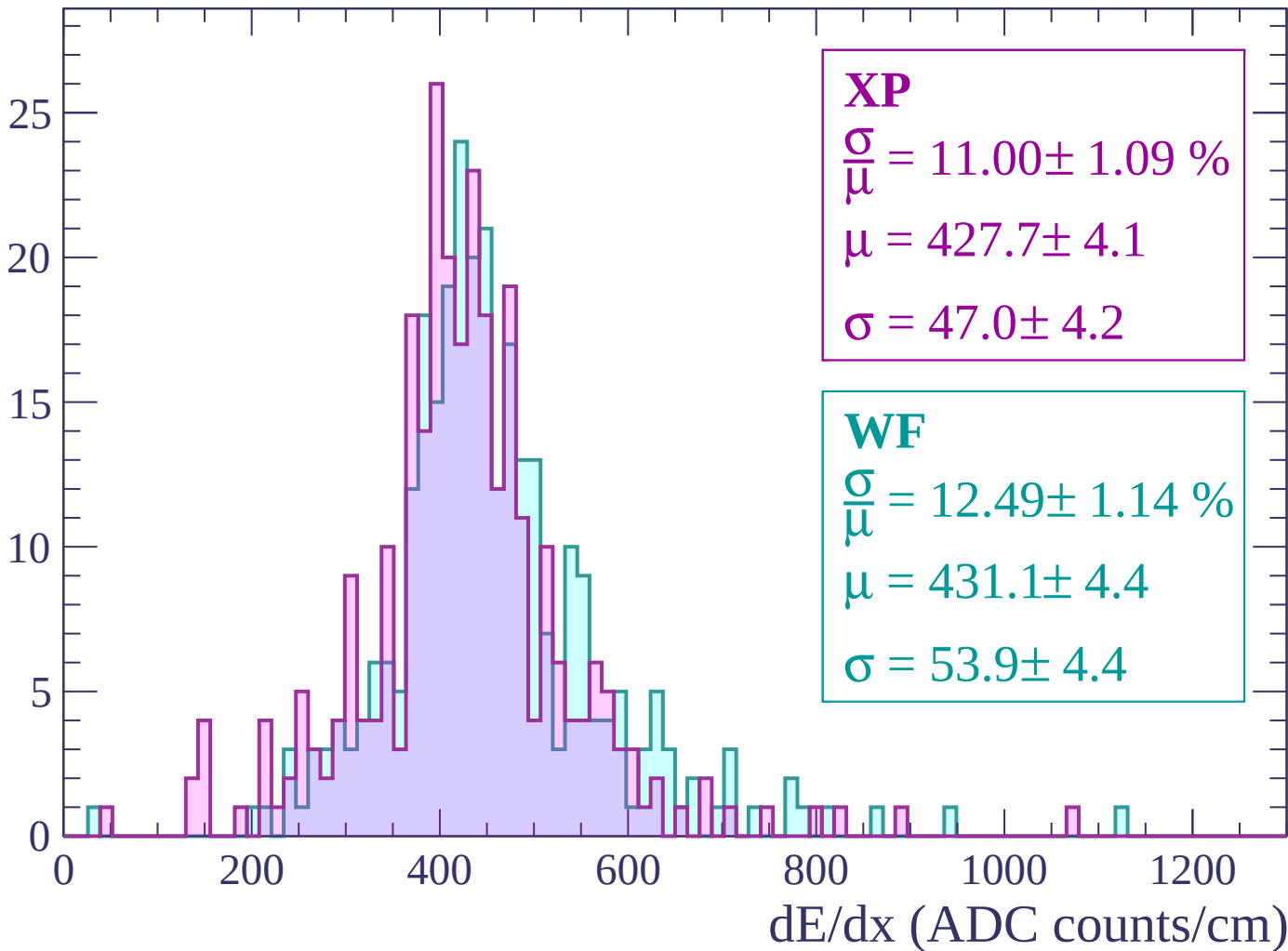
Energy loss | -240 < p < -180

Count



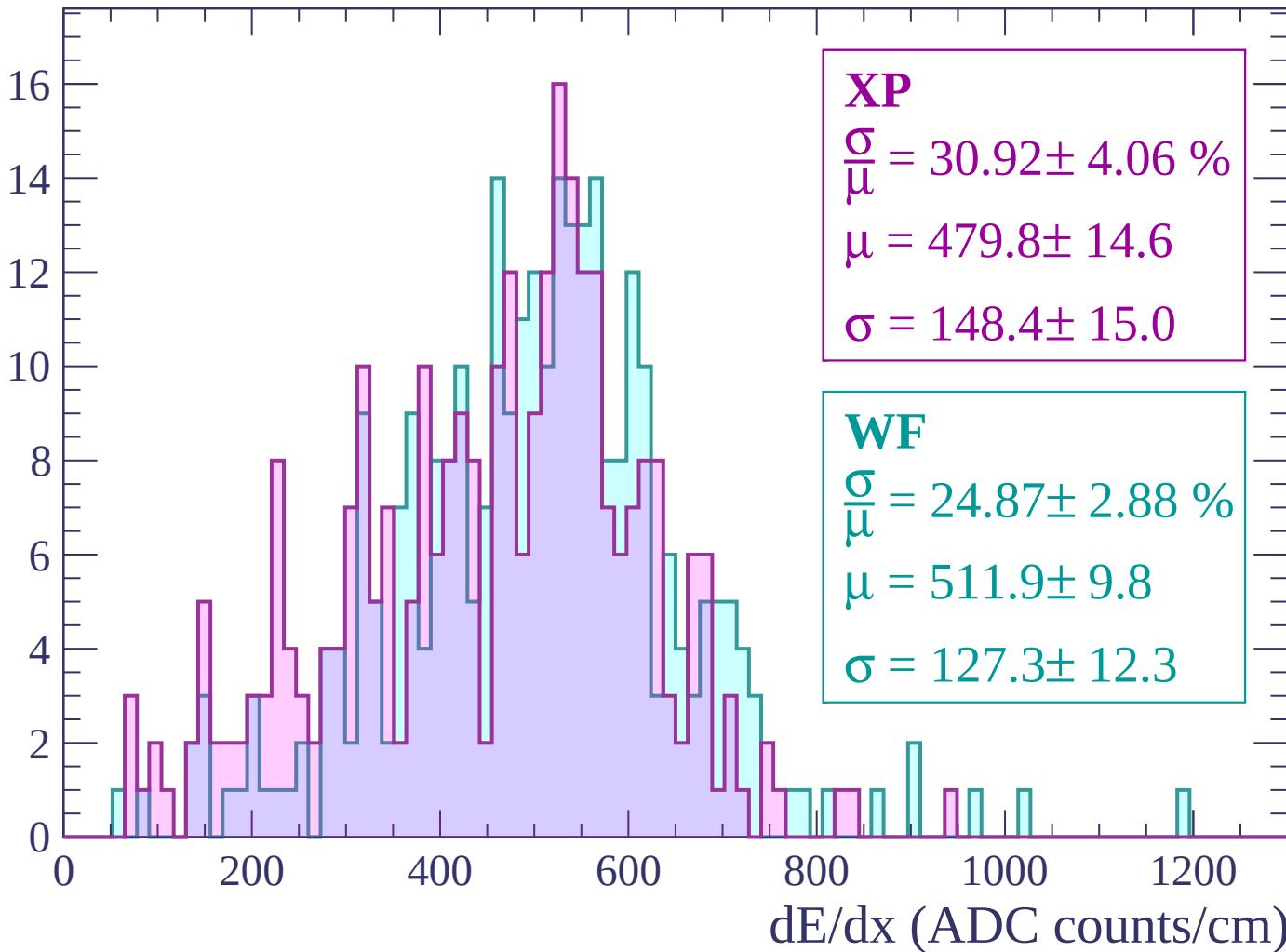
Energy loss | $-180 < p < -120$

Count



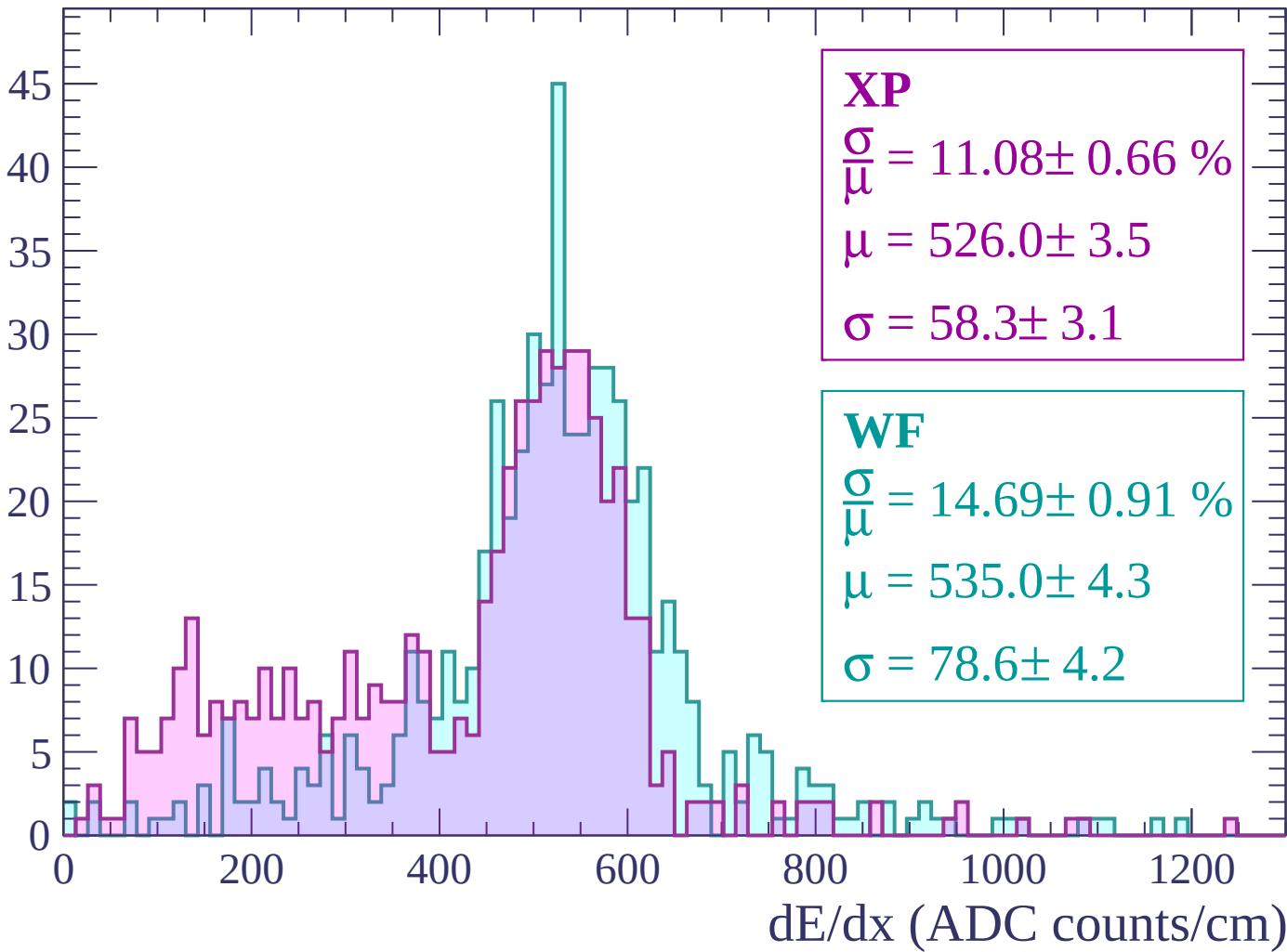
Energy loss | $-120 < p < -60$

Count



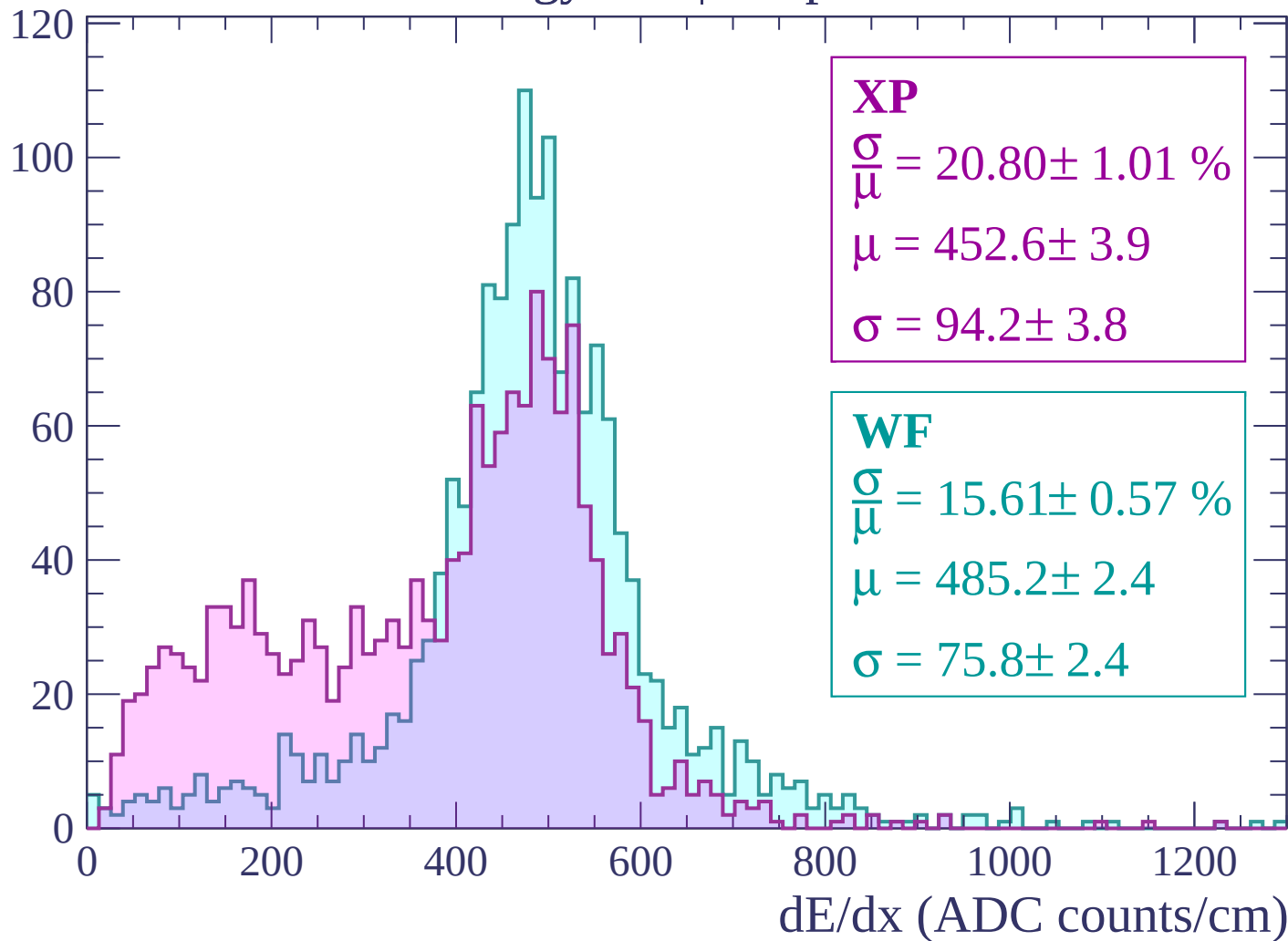
Energy loss | $-60 < p < 0$

Count



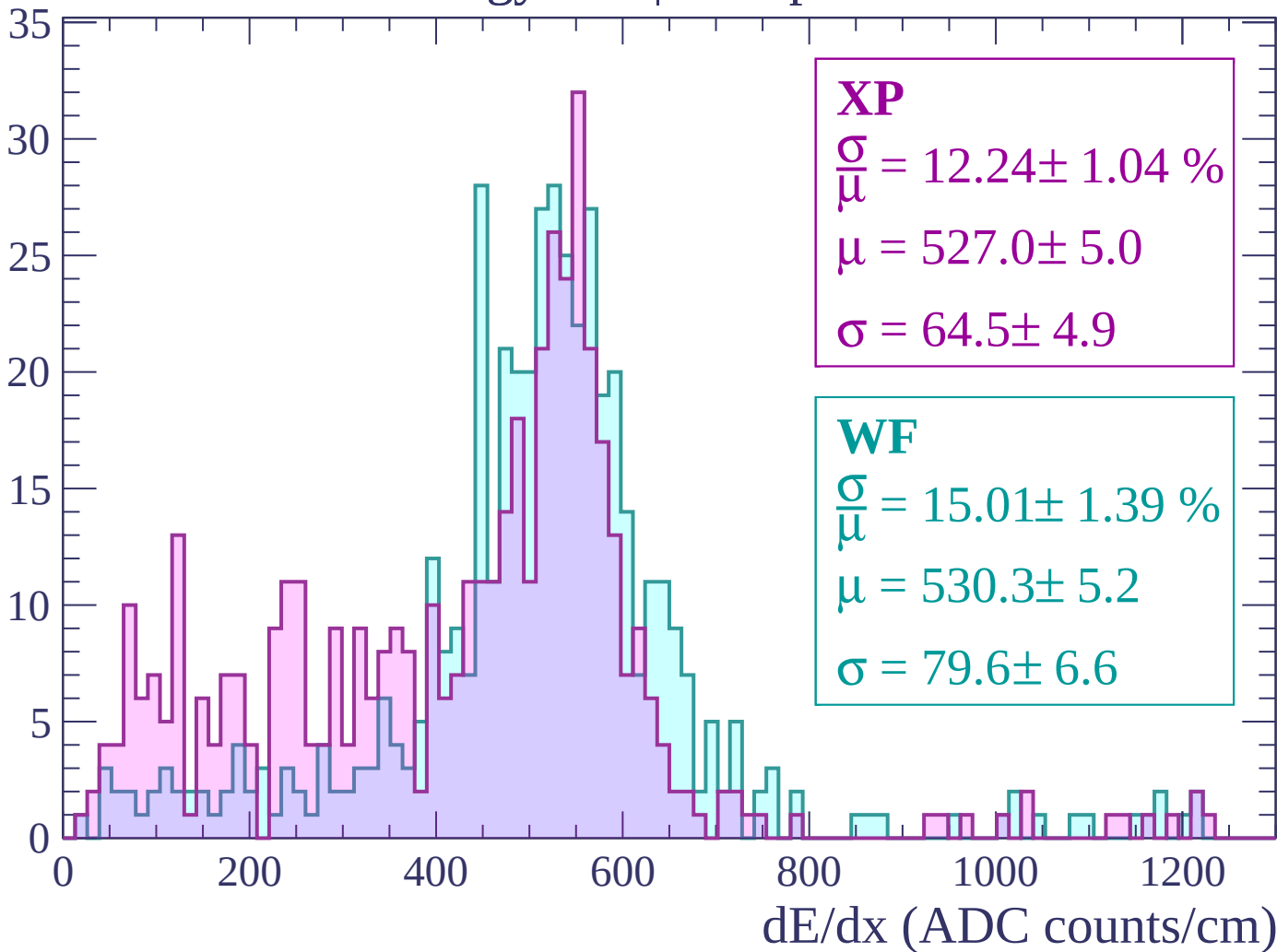
Energy loss | $0 < p < 60$

Count



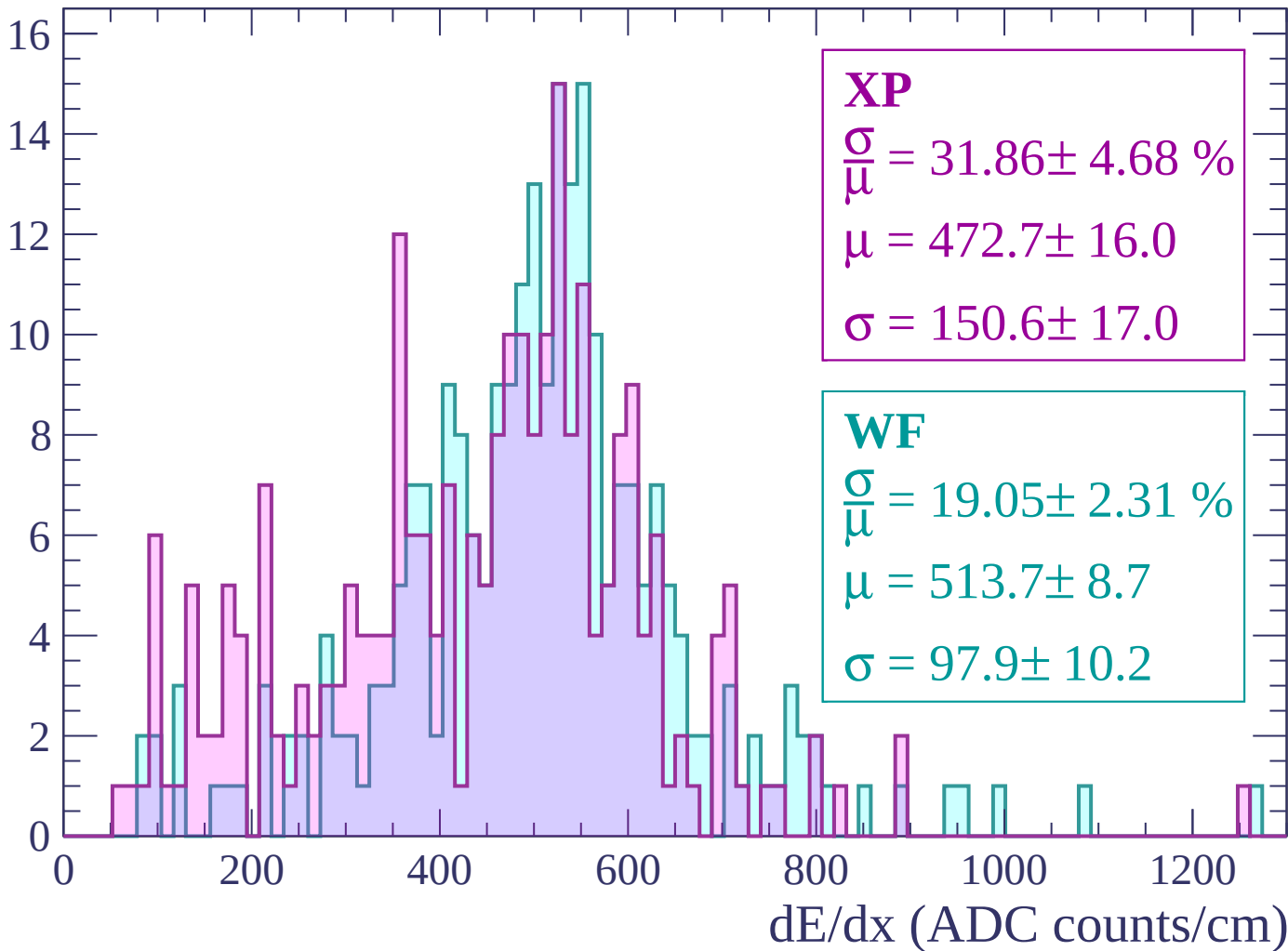
Energy loss | $60 < p < 120$

Count



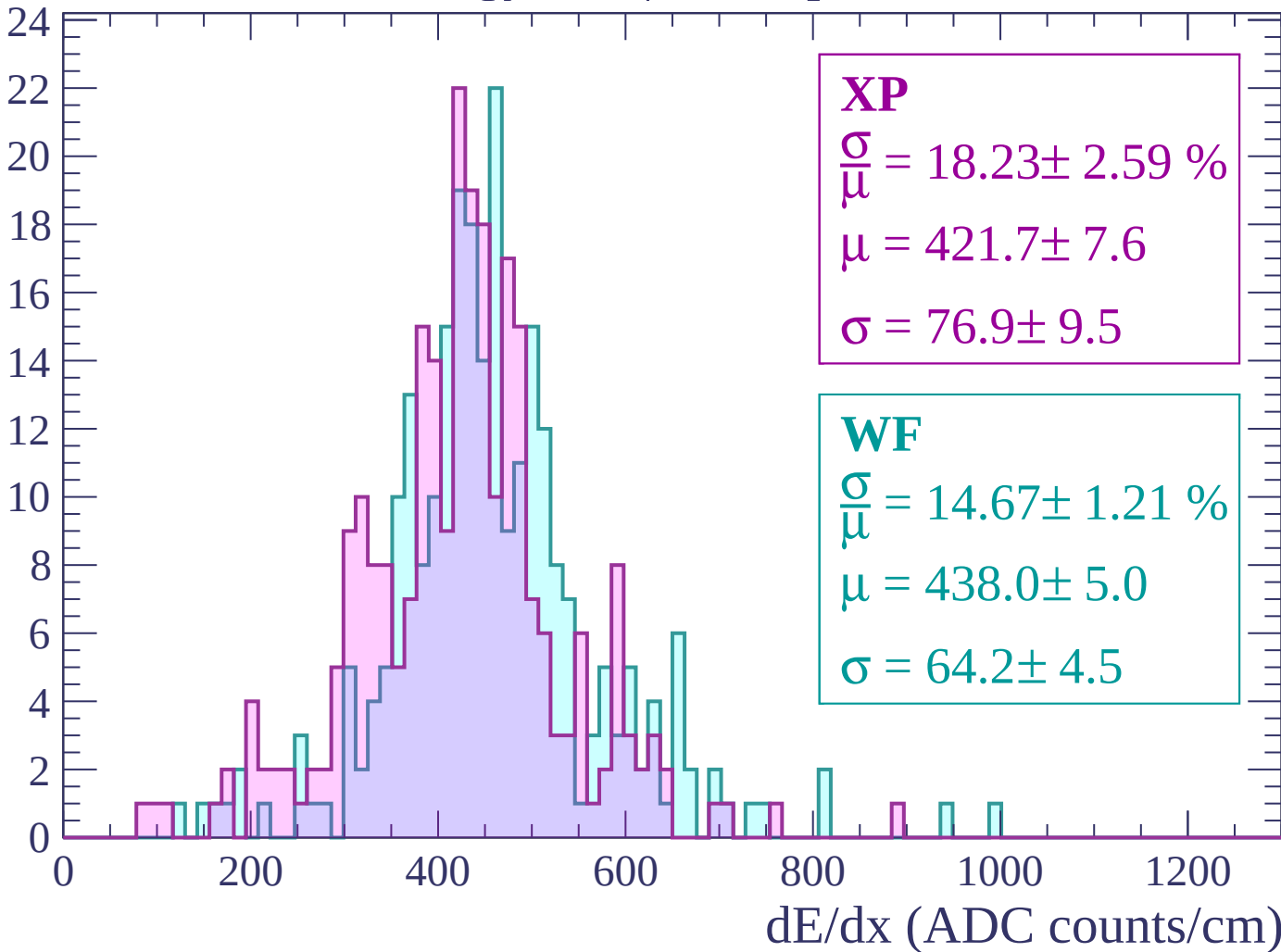
Energy loss | $120 < p < 180$

Count



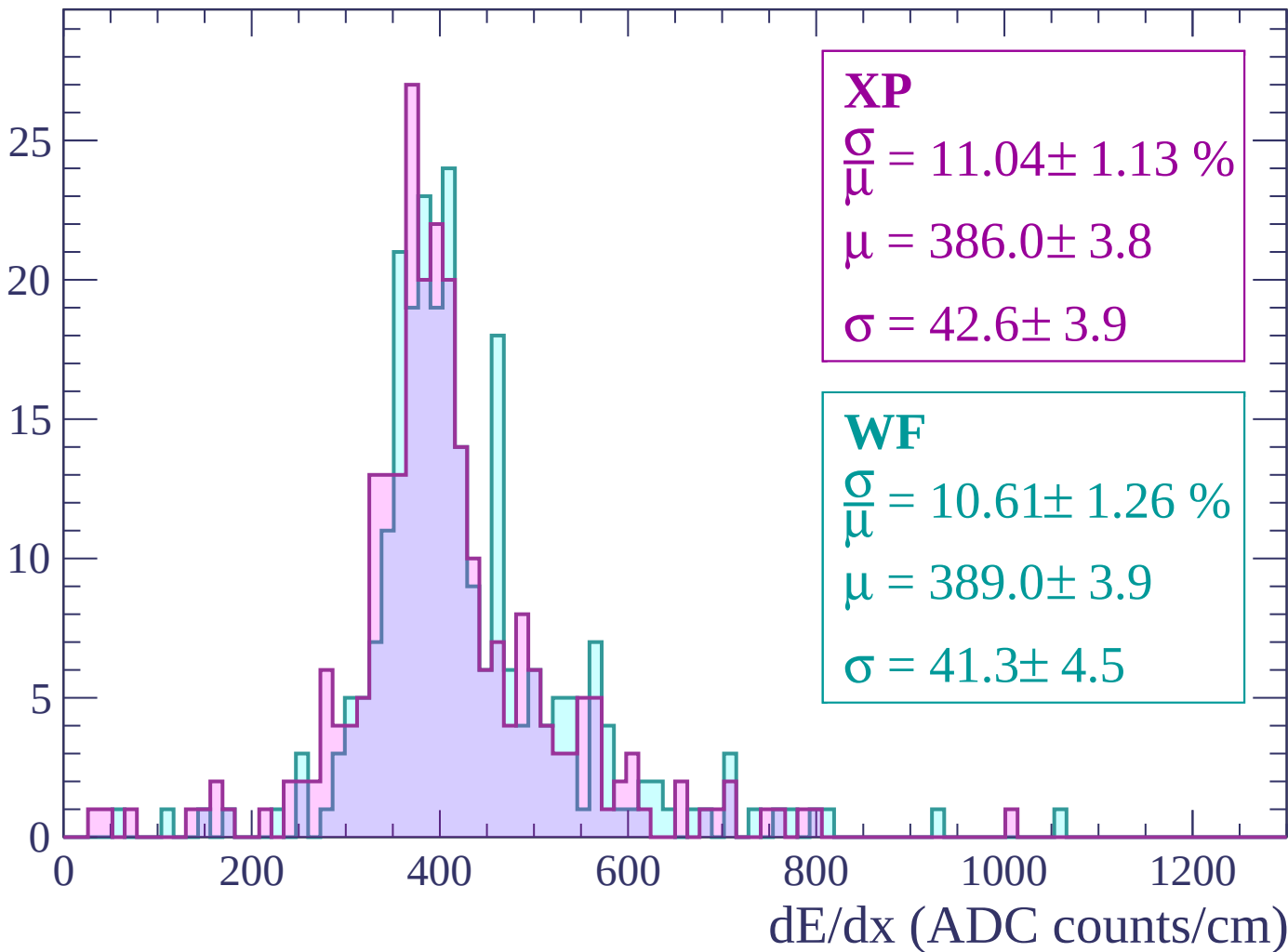
Energy loss | $180 < p < 240$

Count



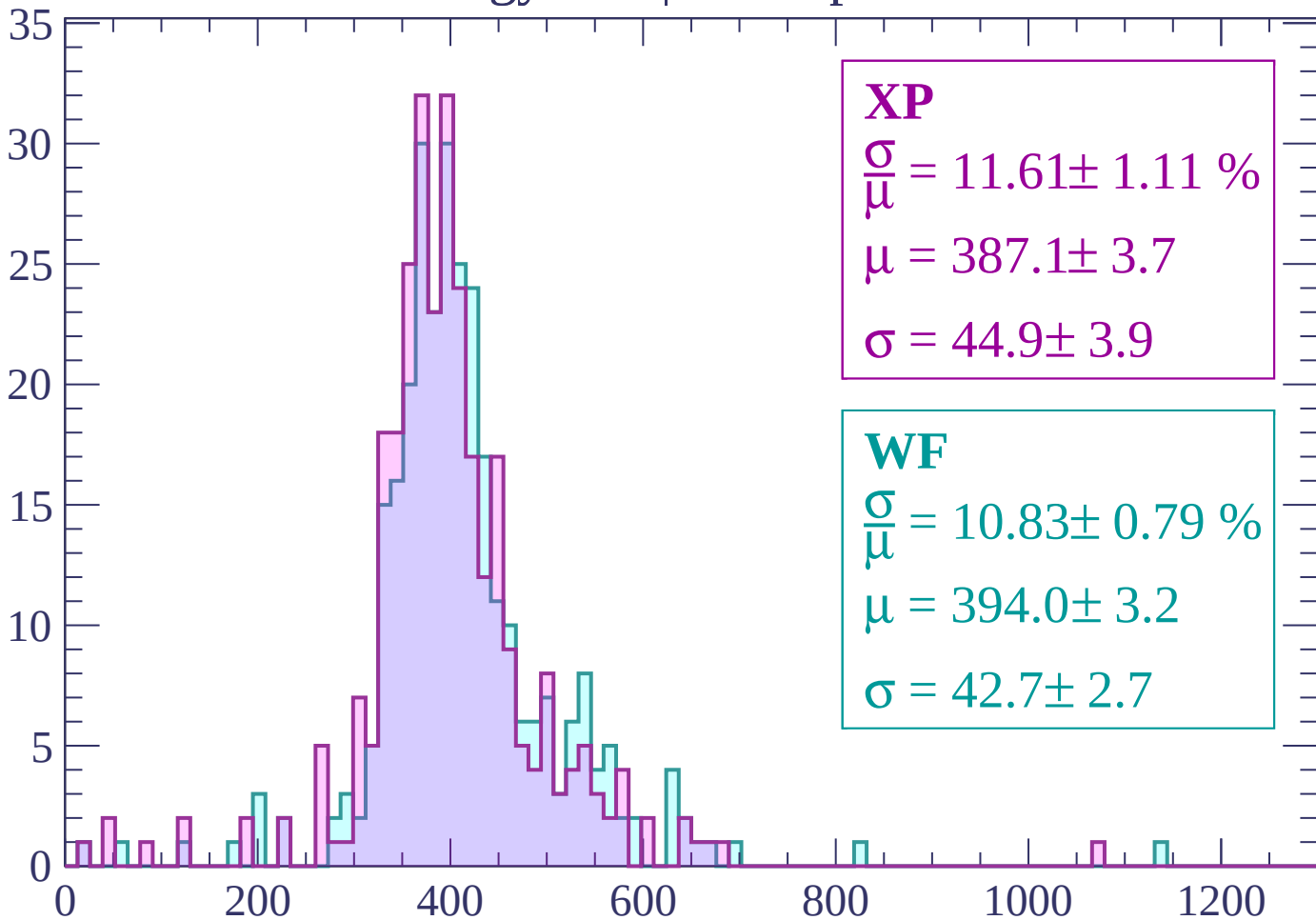
Energy loss | $240 < p < 300$

Count



Energy loss | $300 < p < 360$

Count



XP

$$\frac{\sigma}{\mu} = 11.61 \pm 1.11 \%$$

$$\mu = 387.1 \pm 3.7$$

$$\sigma = 44.9 \pm 3.9$$

WF

$$\frac{\sigma}{\mu} = 10.83 \pm 0.79 \%$$

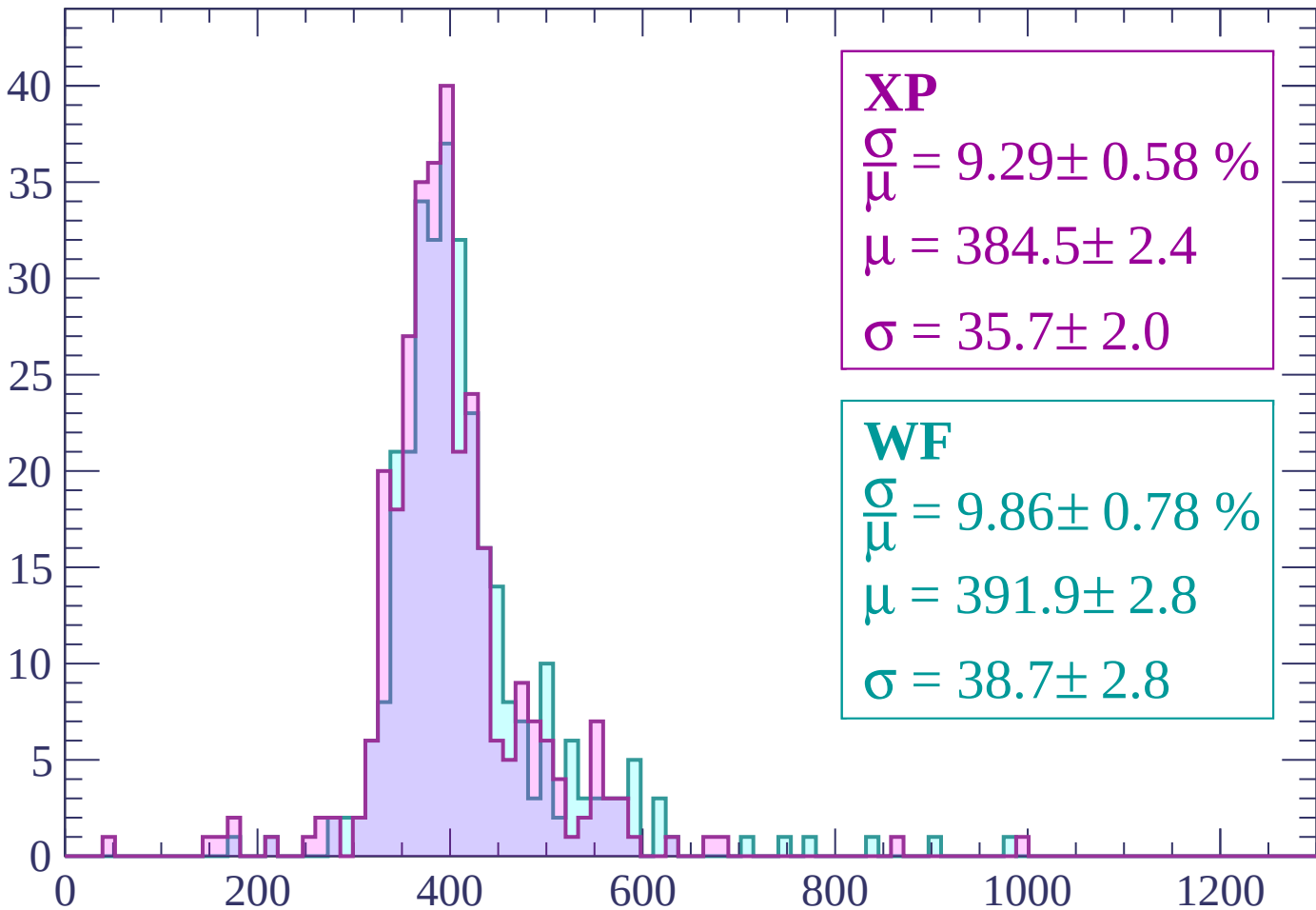
$$\mu = 394.0 \pm 3.2$$

$$\sigma = 42.7 \pm 2.7$$

dE/dx (ADC counts/cm)

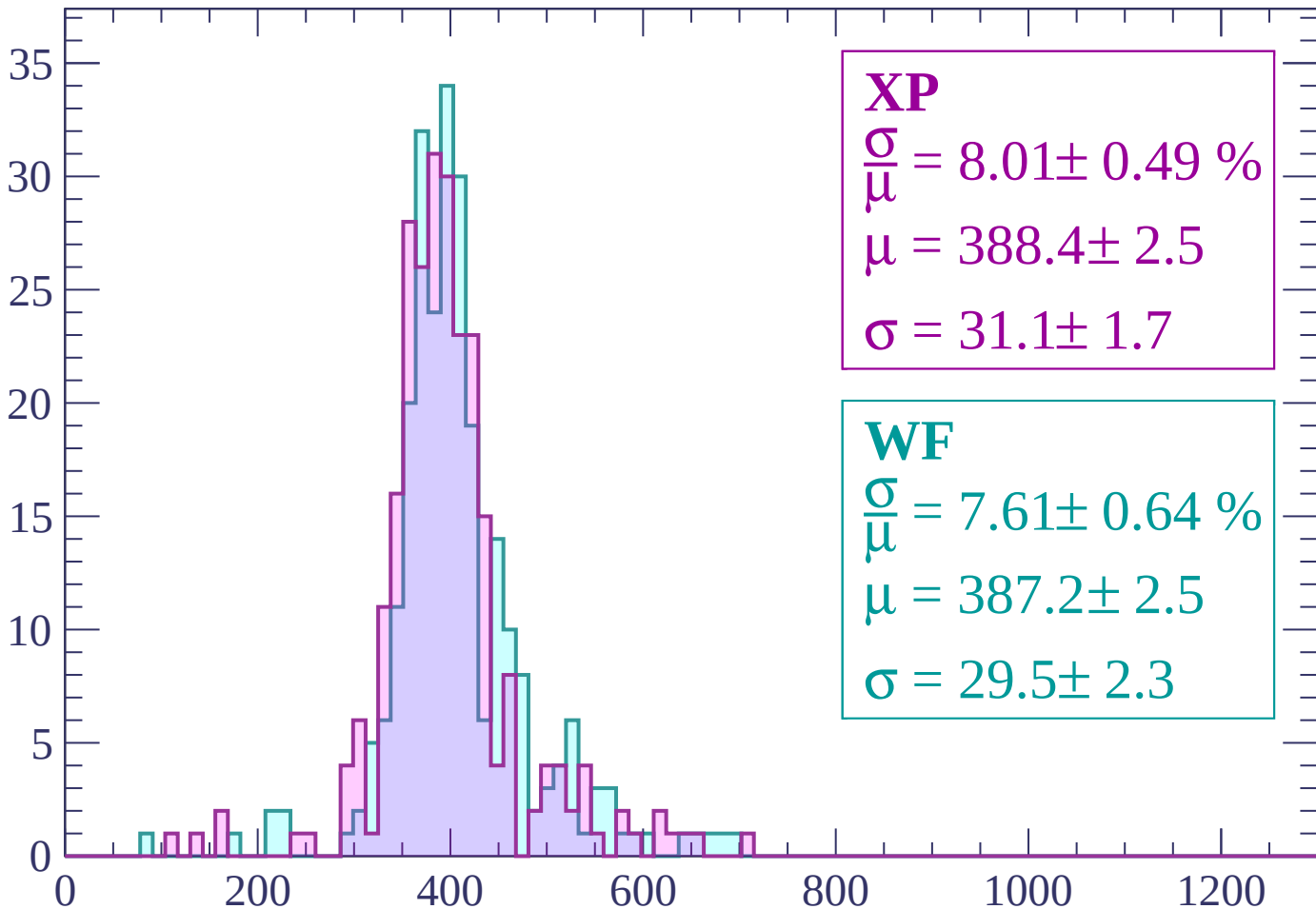
Energy loss | $360 < p < 420$

Count



Energy loss | $420 < p < 480$

Count



XP

$$\frac{\sigma}{\mu} = 8.01 \pm 0.49 \%$$

$$\mu = 388.4 \pm 2.5$$

$$\sigma = 31.1 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 7.61 \pm 0.64 \%$$

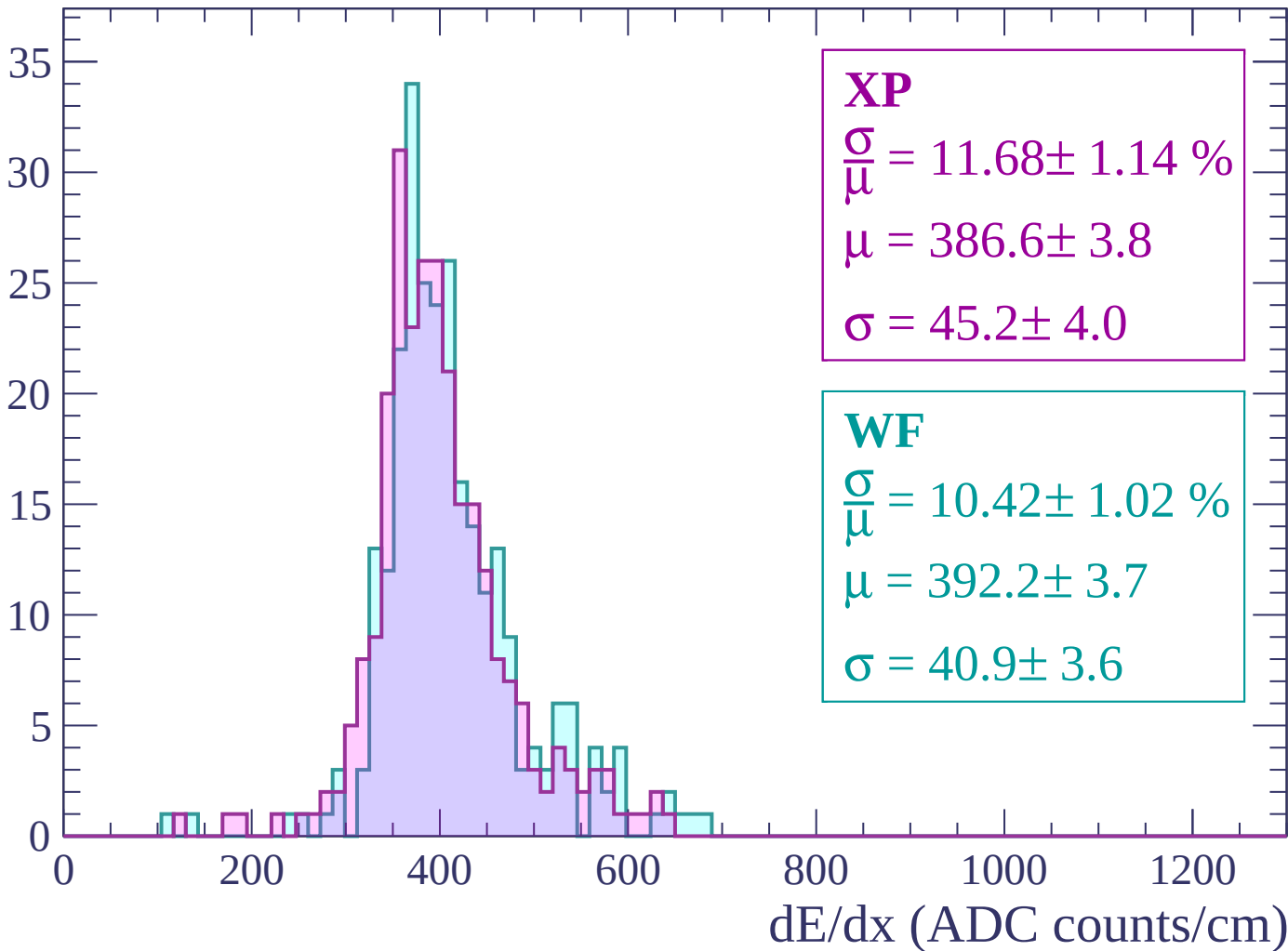
$$\mu = 387.2 \pm 2.5$$

$$\sigma = 29.5 \pm 2.3$$

dE/dx (ADC counts/cm)

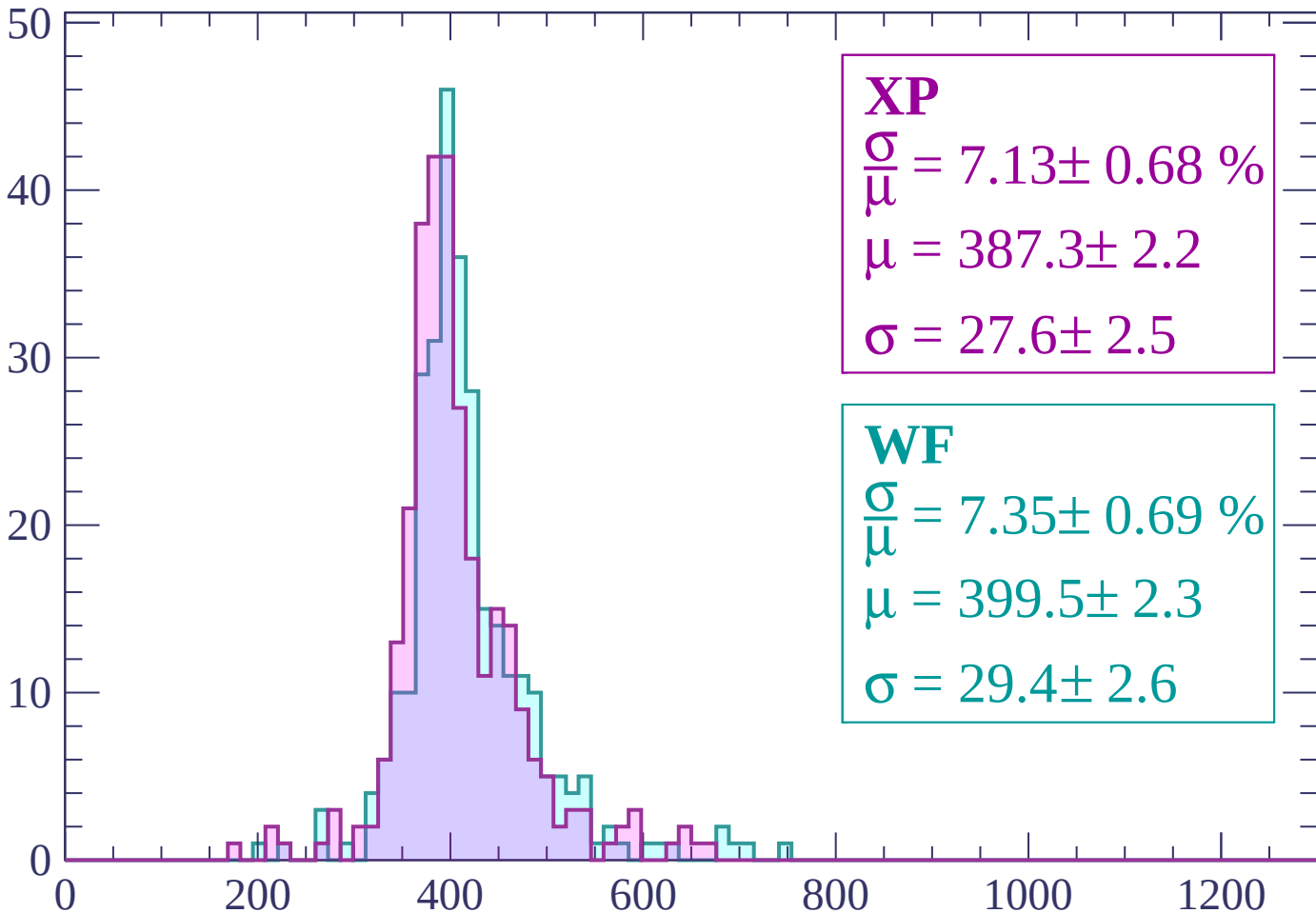
Energy loss | $480 < p < 540$

Count



Energy loss | $540 < p < 600$

Count



XP

$$\frac{\sigma}{\mu} = 7.13 \pm 0.68 \%$$

$$\mu = 387.3 \pm 2.2$$

$$\sigma = 27.6 \pm 2.5$$

WF

$$\frac{\sigma}{\mu} = 7.35 \pm 0.69 \%$$

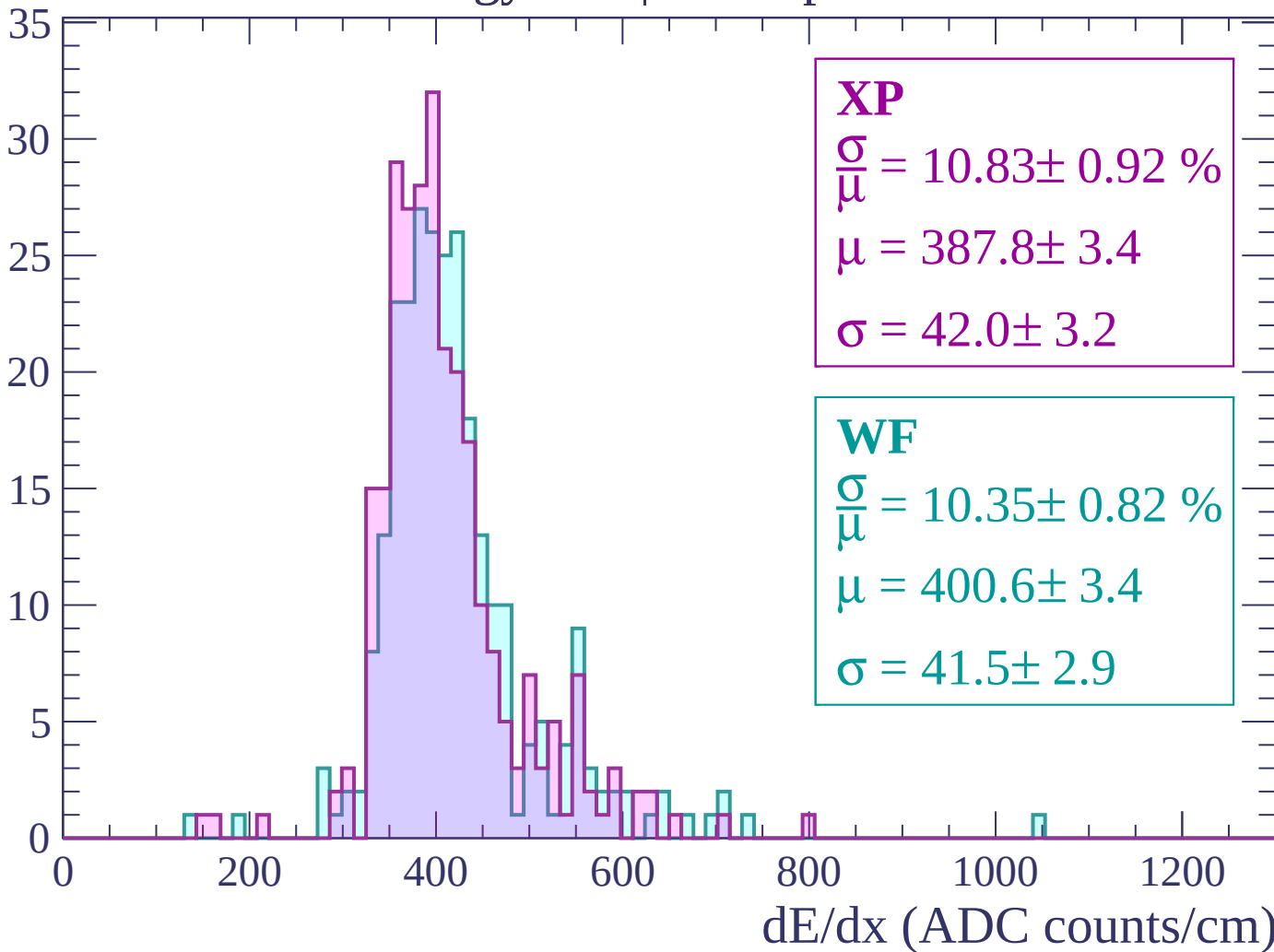
$$\mu = 399.5 \pm 2.3$$

$$\sigma = 29.4 \pm 2.6$$

dE/dx (ADC counts/cm)

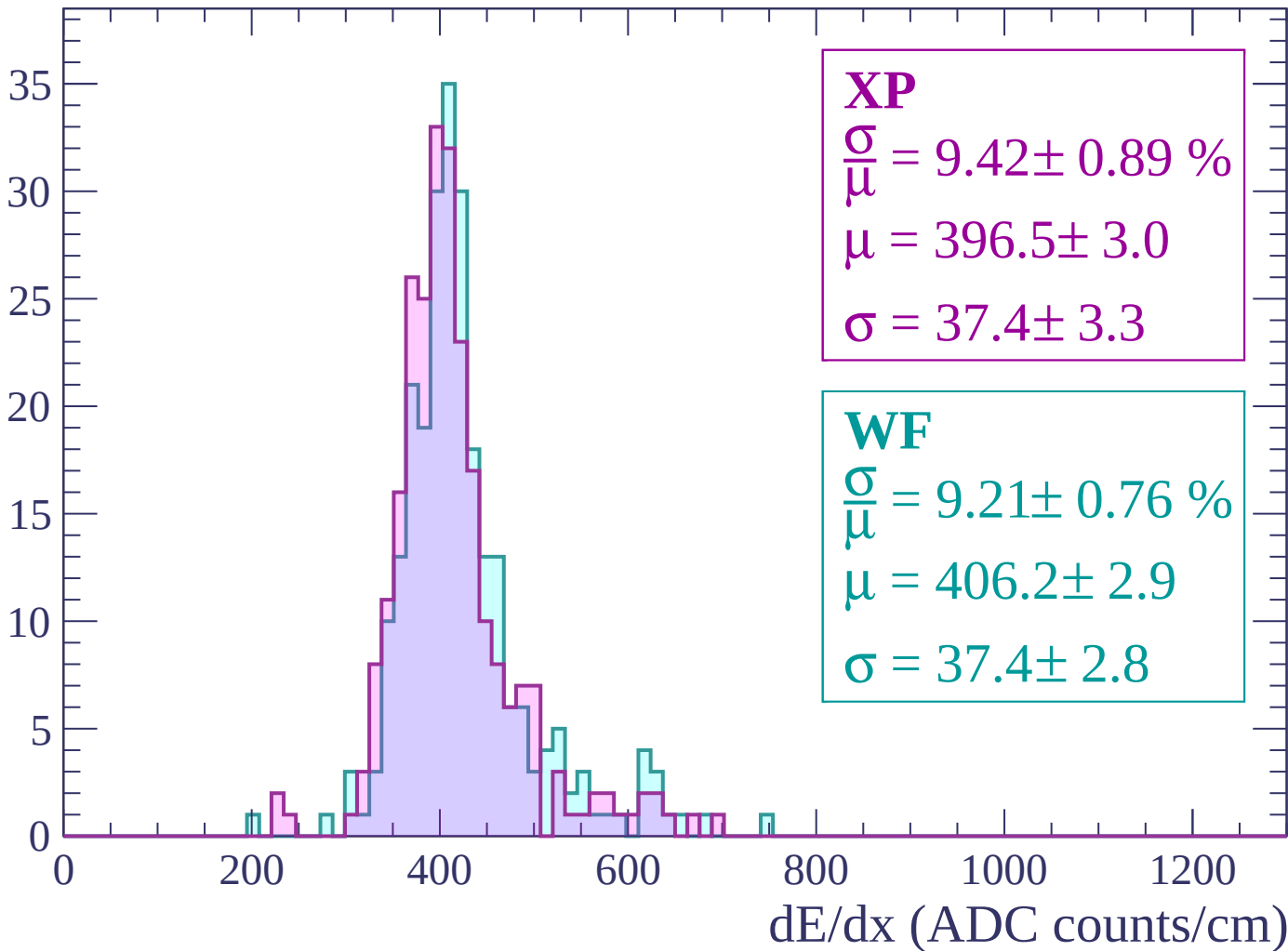
Energy loss | $600 < p < 660$

Count



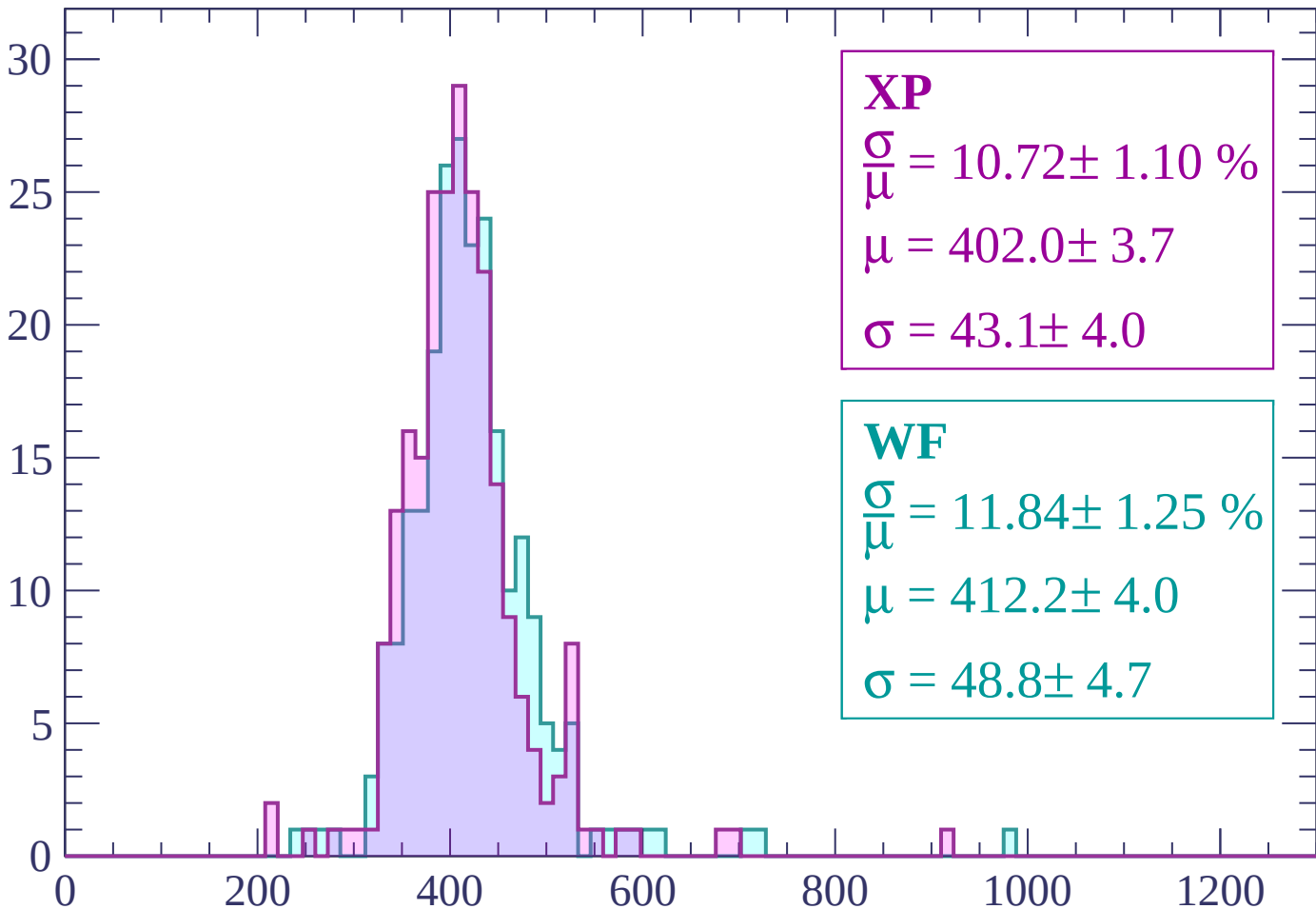
Energy loss | $660 < p < 720$

Count



Energy loss | $720 < p < 780$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 10.72 \pm 1.10 \%$$

$$\mu = 402.0 \pm 3.7$$

$$\sigma = 43.1 \pm 4.0$$

WF

$$\frac{\sigma}{\bar{\mu}} = 11.84 \pm 1.25 \%$$

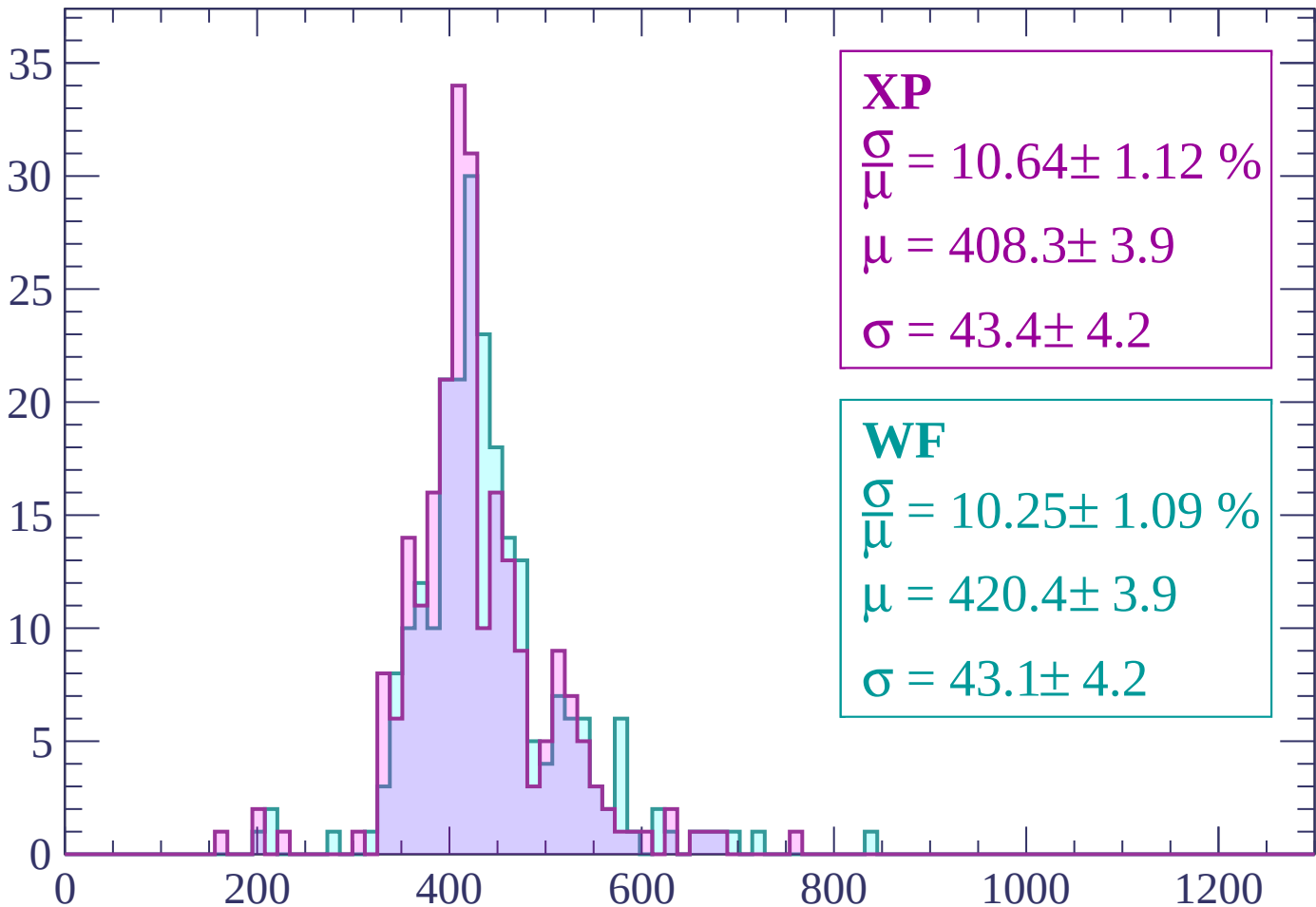
$$\mu = 412.2 \pm 4.0$$

$$\sigma = 48.8 \pm 4.7$$

dE/dx (ADC counts/cm)

Energy loss | $780 < p < 840$

Count



XP

$$\frac{\sigma}{\mu} = 10.64 \pm 1.12 \%$$

$$\mu = 408.3 \pm 3.9$$

$$\sigma = 43.4 \pm 4.2$$

WF

$$\frac{\sigma}{\mu} = 10.25 \pm 1.09 \%$$

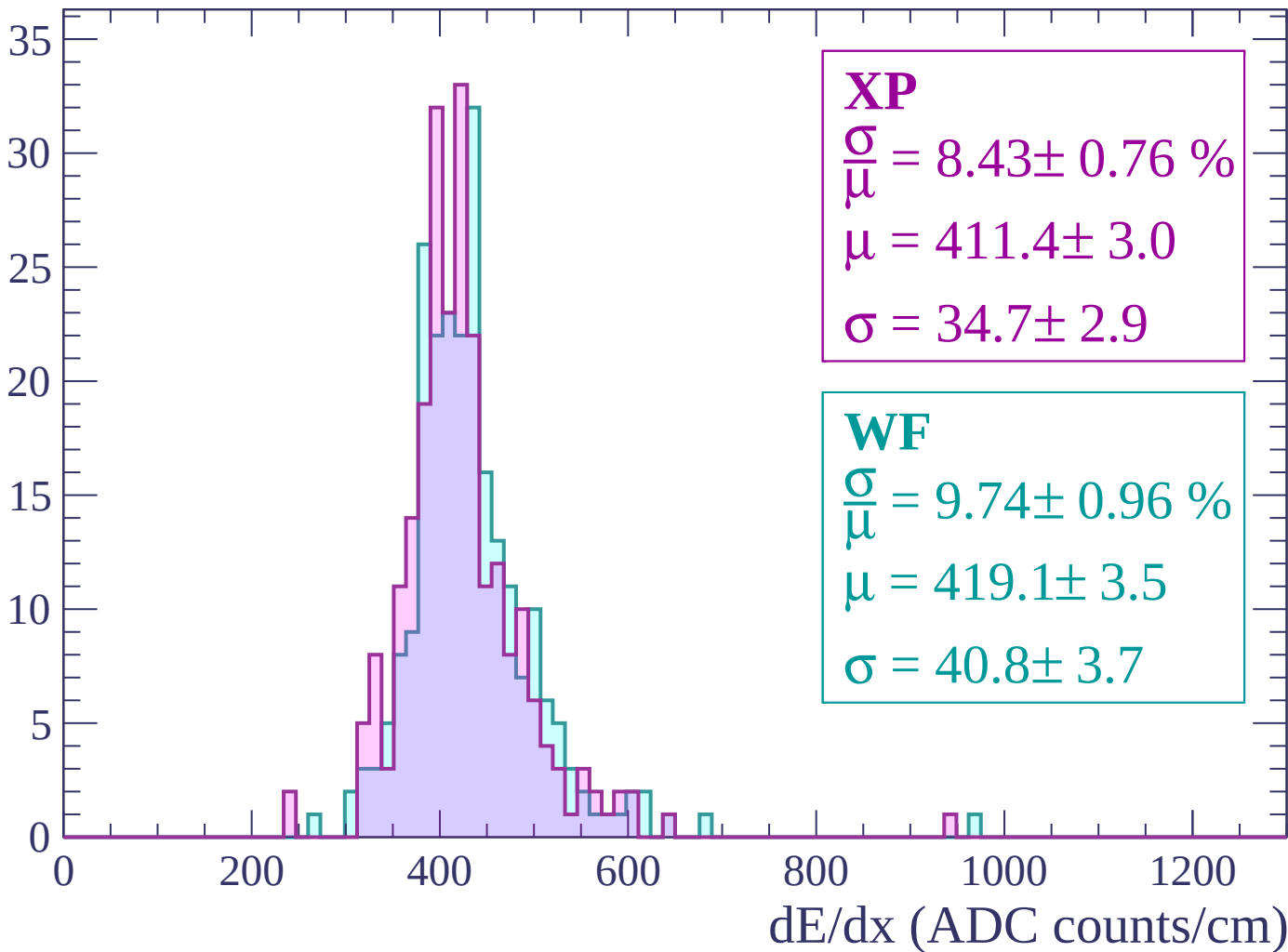
$$\mu = 420.4 \pm 3.9$$

$$\sigma = 43.1 \pm 4.2$$

dE/dx (ADC counts/cm)

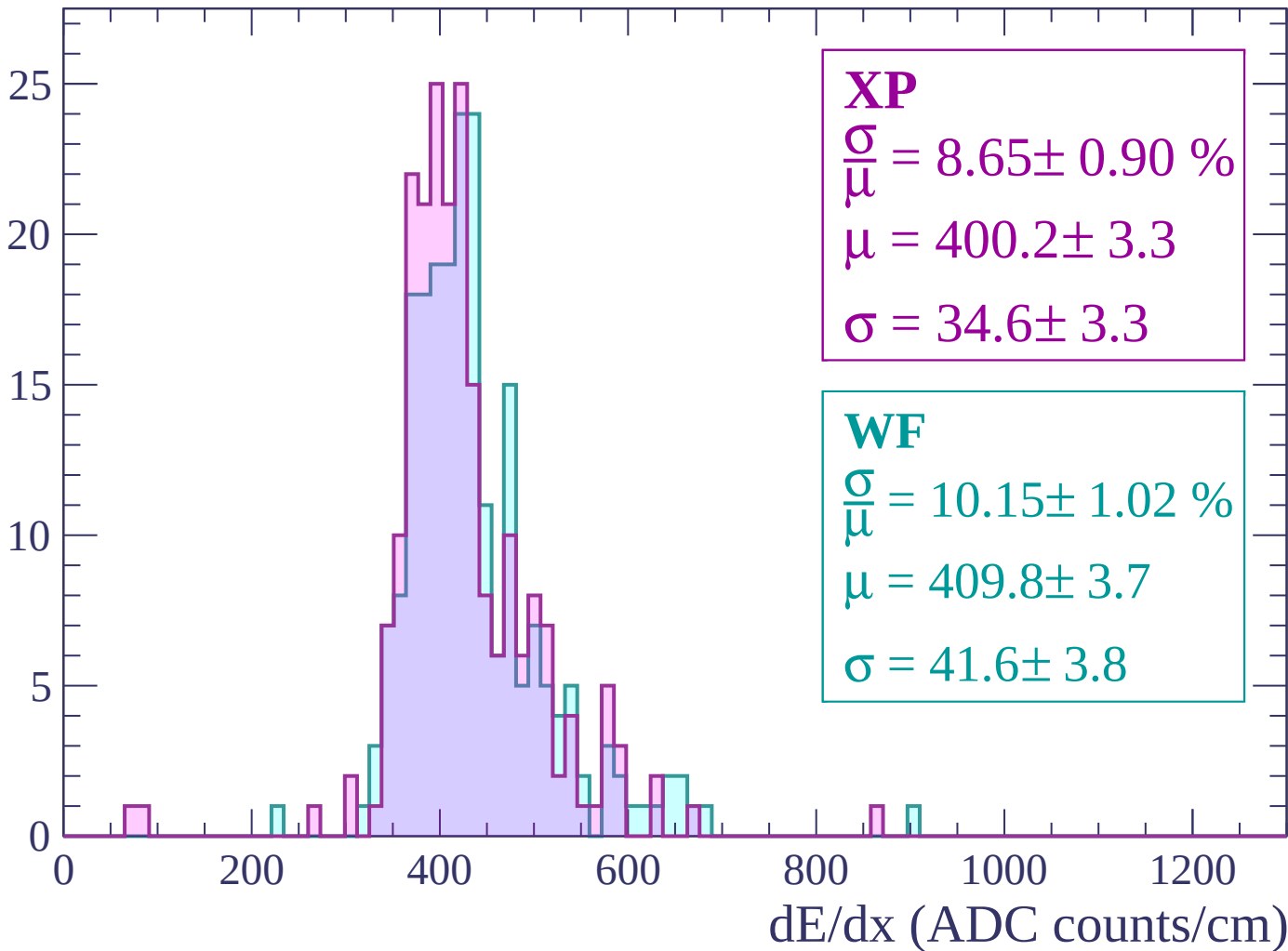
Energy loss | $840 < p < 900$

Count



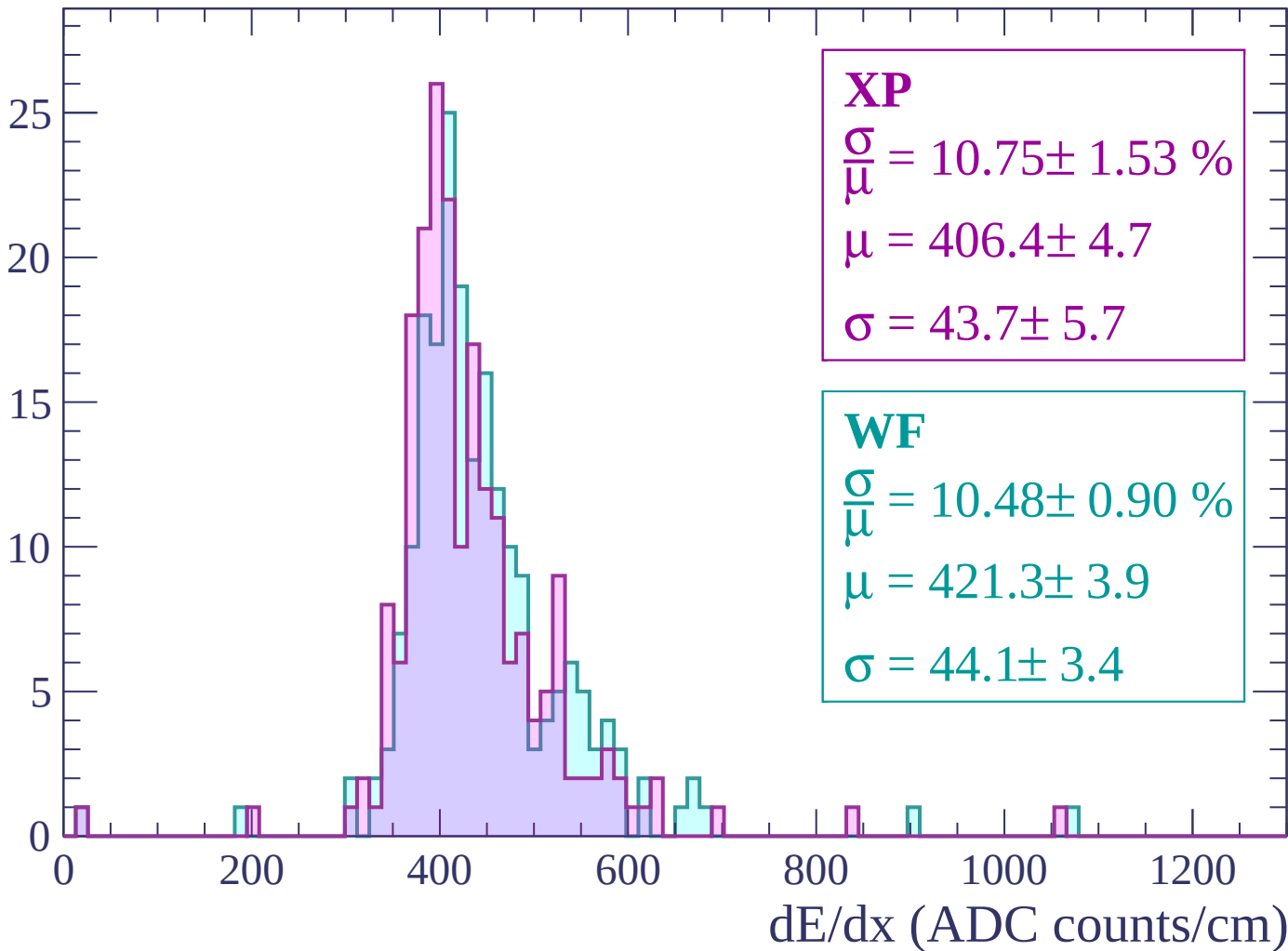
Energy loss | $900 < p < 960$

Count



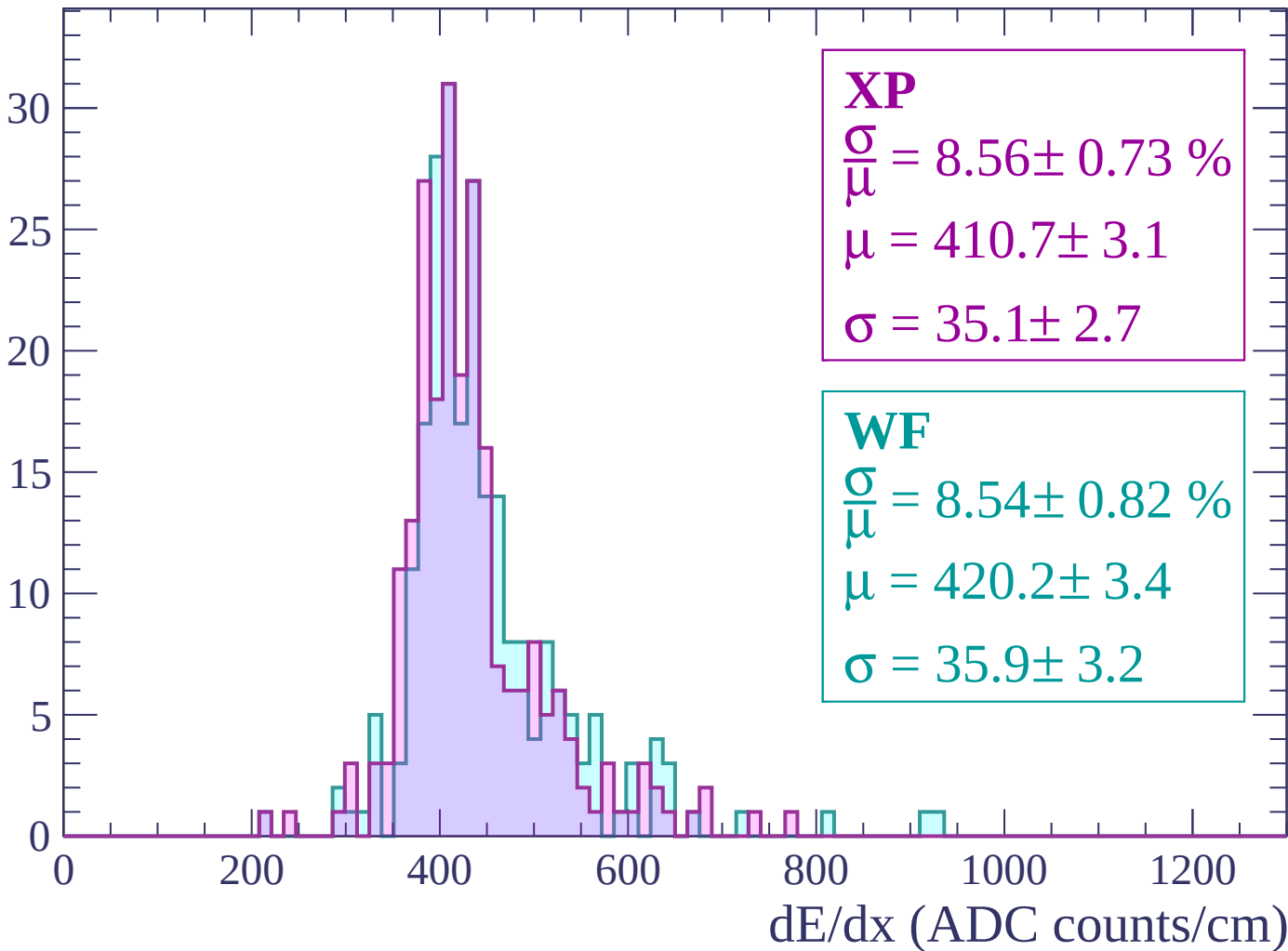
Energy loss | $960 < p < 1020$

Count



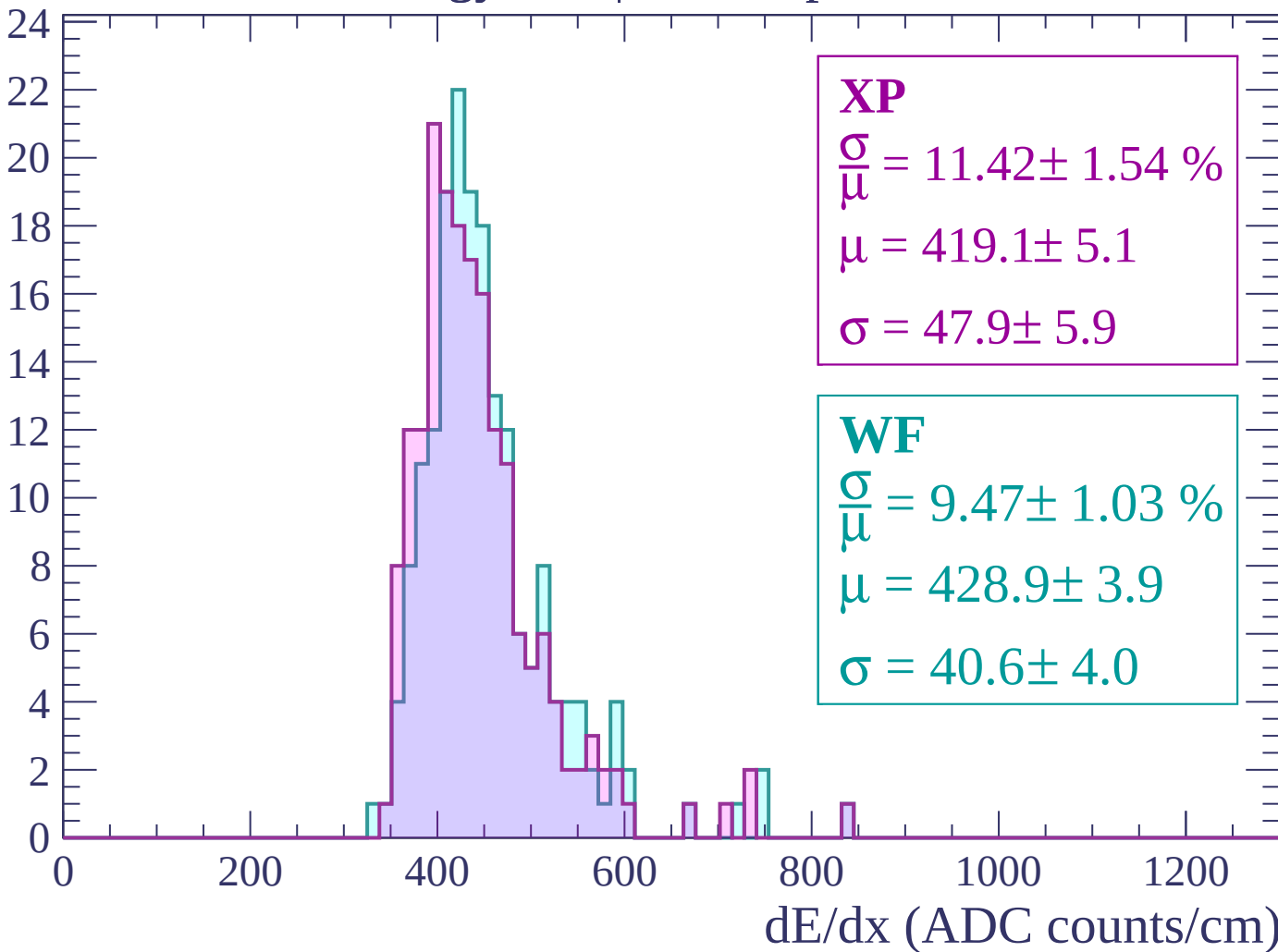
Energy loss | $1020 < p < 1080$

Count



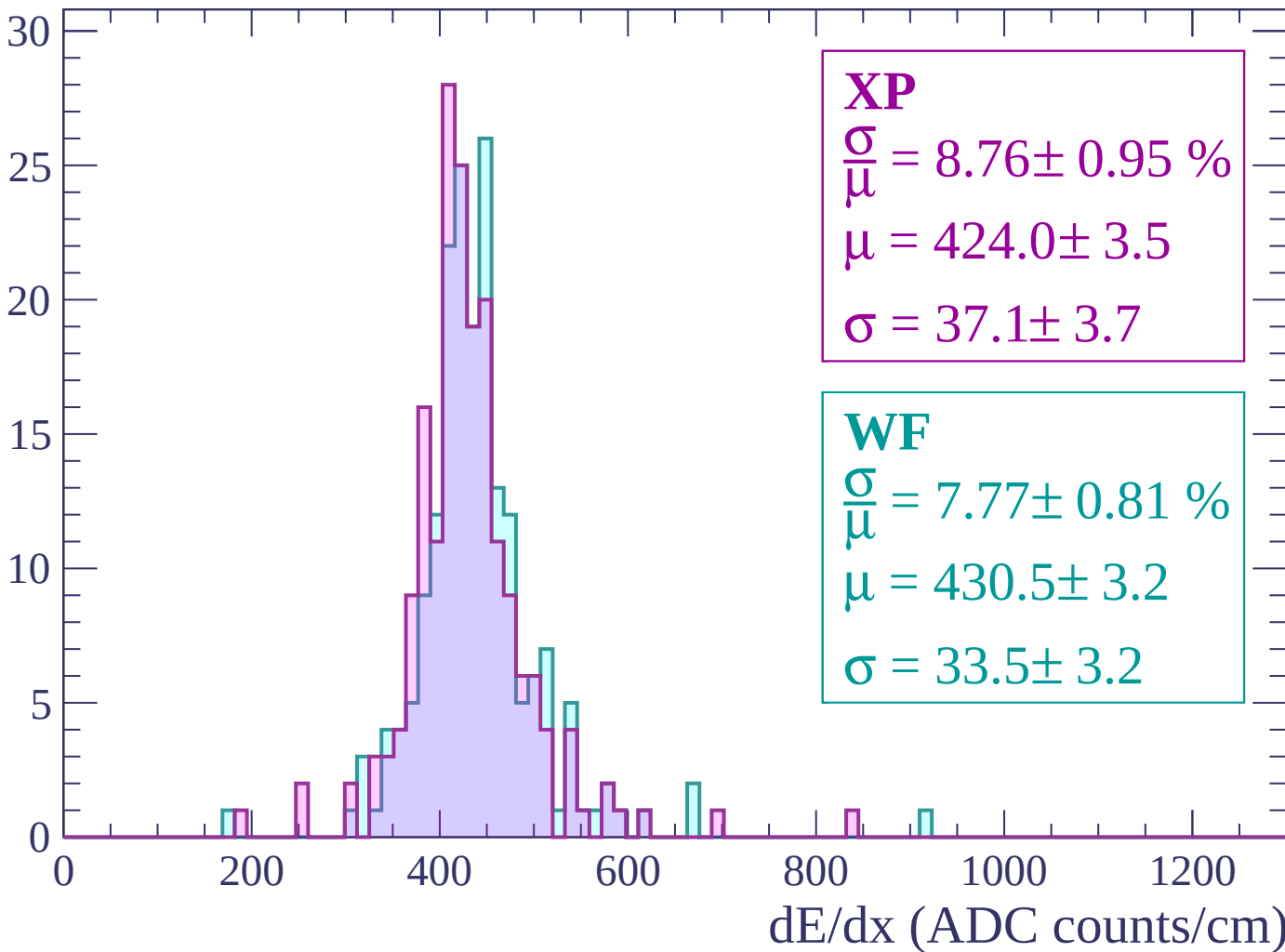
Energy loss | $1080 < p < 1140$

Count



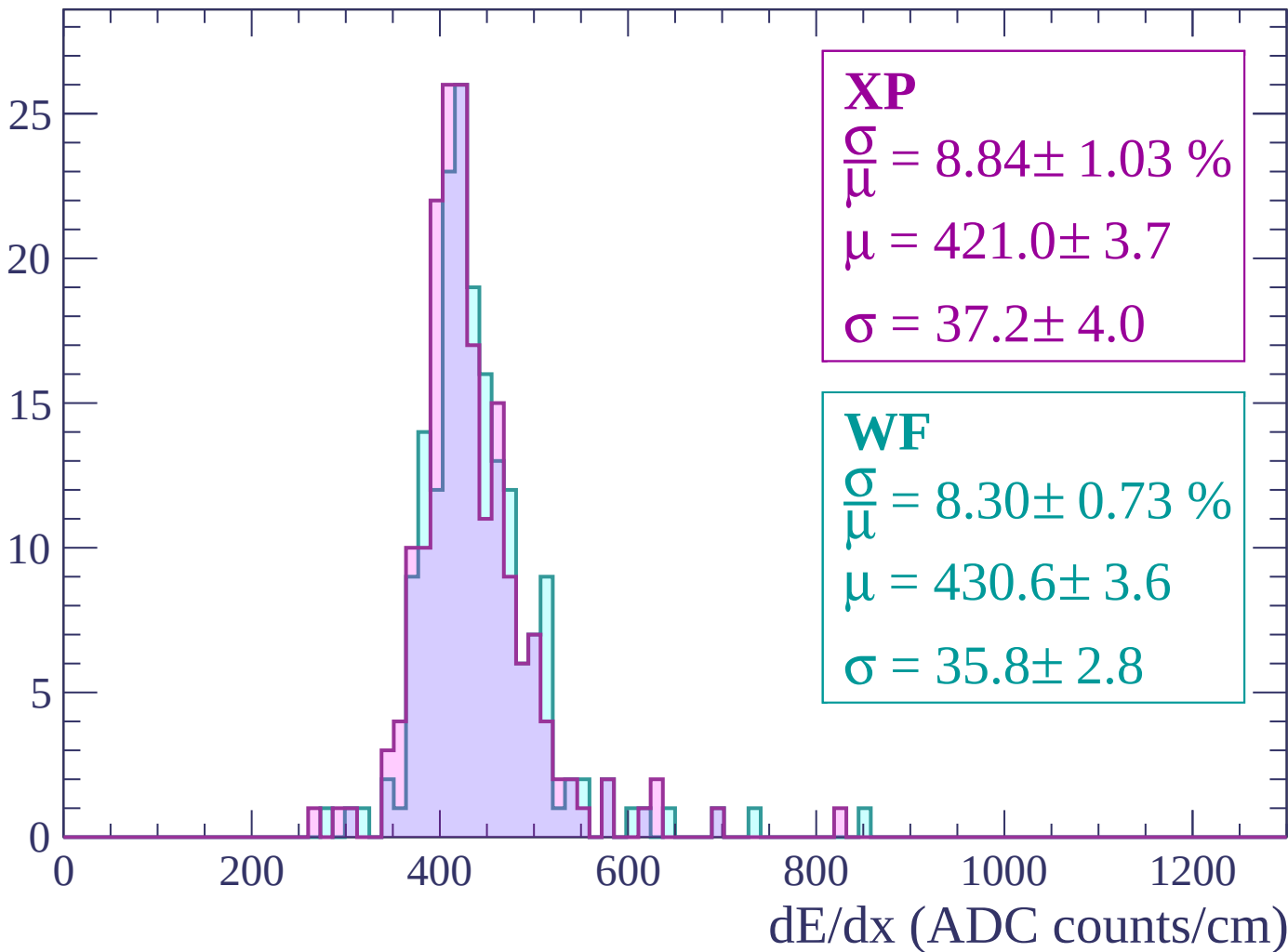
Energy loss | $1140 < p < 1200$

Count



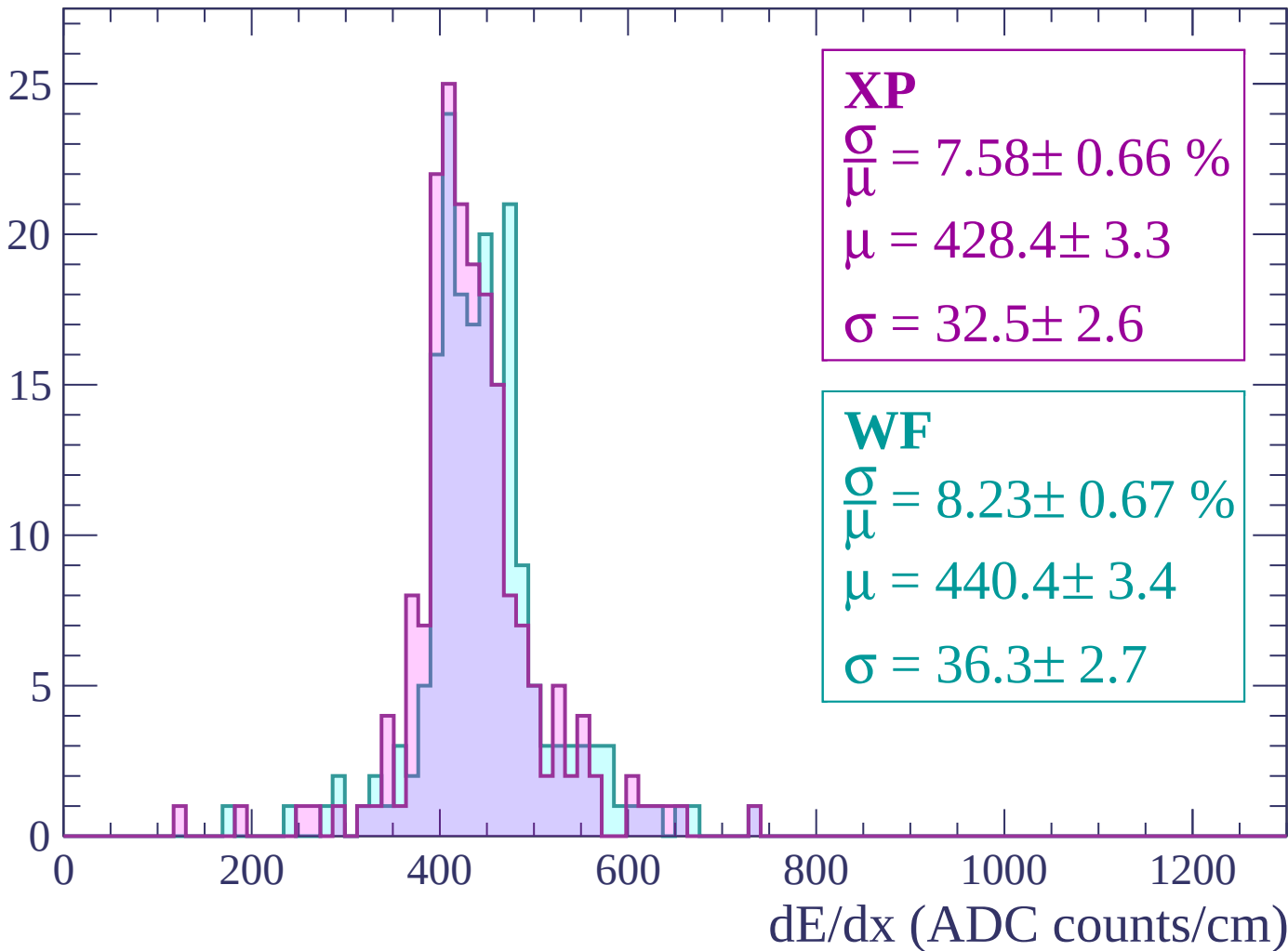
Energy loss | $1200 < p < 1260$

Count



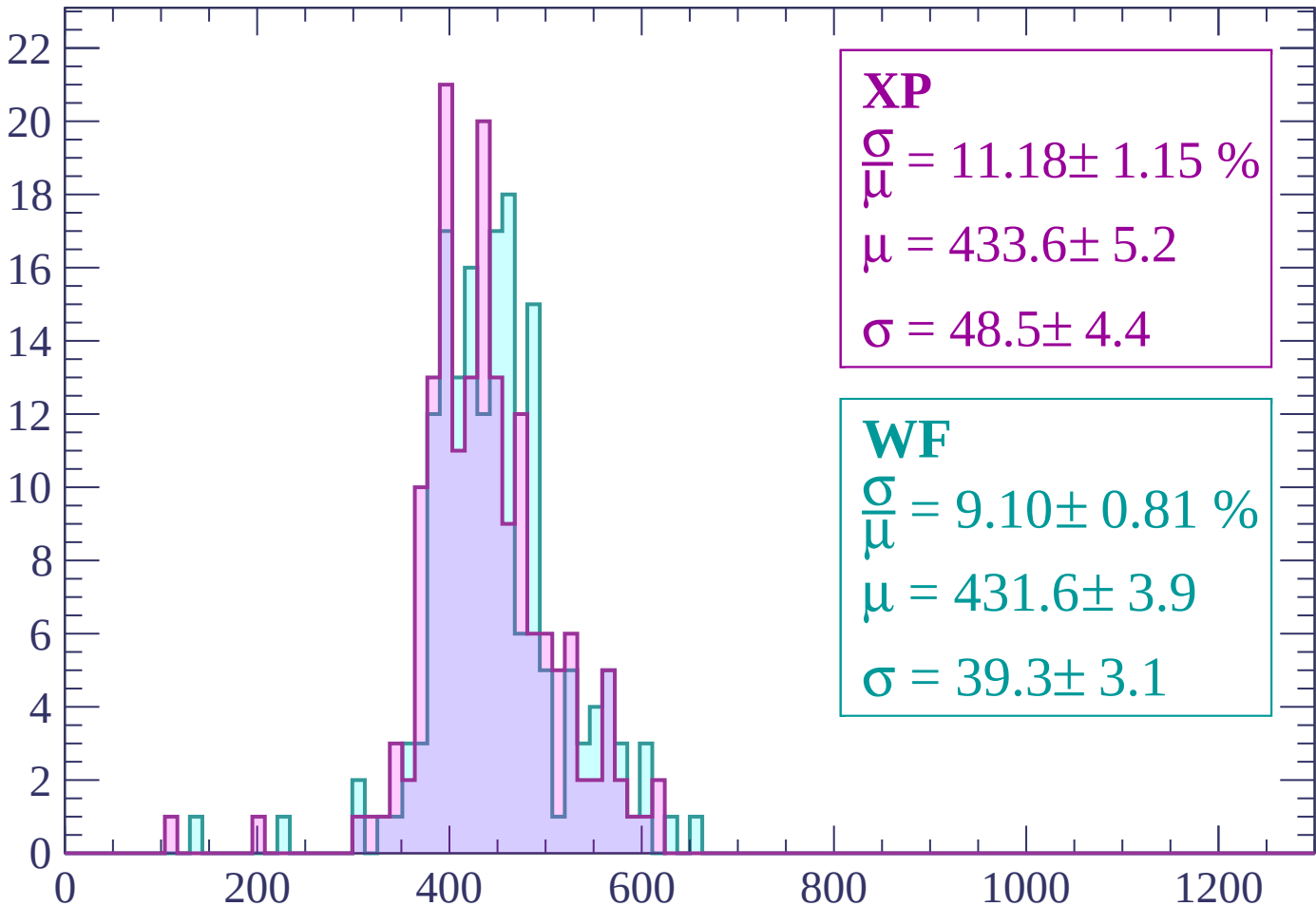
Energy loss | $1260 < p < 1320$

Count



Energy loss | $1320 < p < 1380$

Count



XP

$$\frac{\sigma}{\mu} = 11.18 \pm 1.15 \%$$

$$\mu = 433.6 \pm 5.2$$

$$\sigma = 48.5 \pm 4.4$$

WF

$$\frac{\sigma}{\mu} = 9.10 \pm 0.81 \%$$

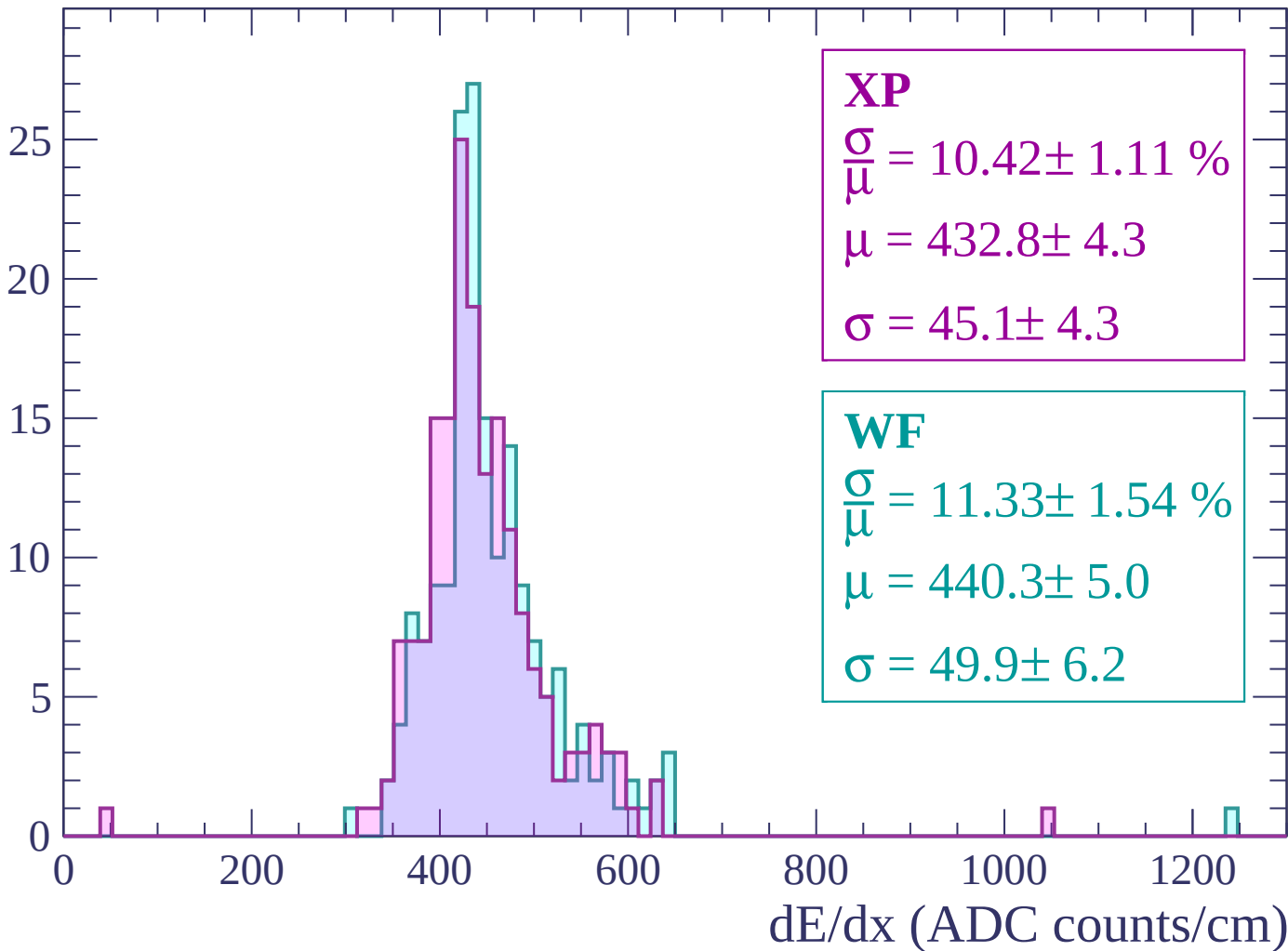
$$\mu = 431.6 \pm 3.9$$

$$\sigma = 39.3 \pm 3.1$$

dE/dx (ADC counts/cm)

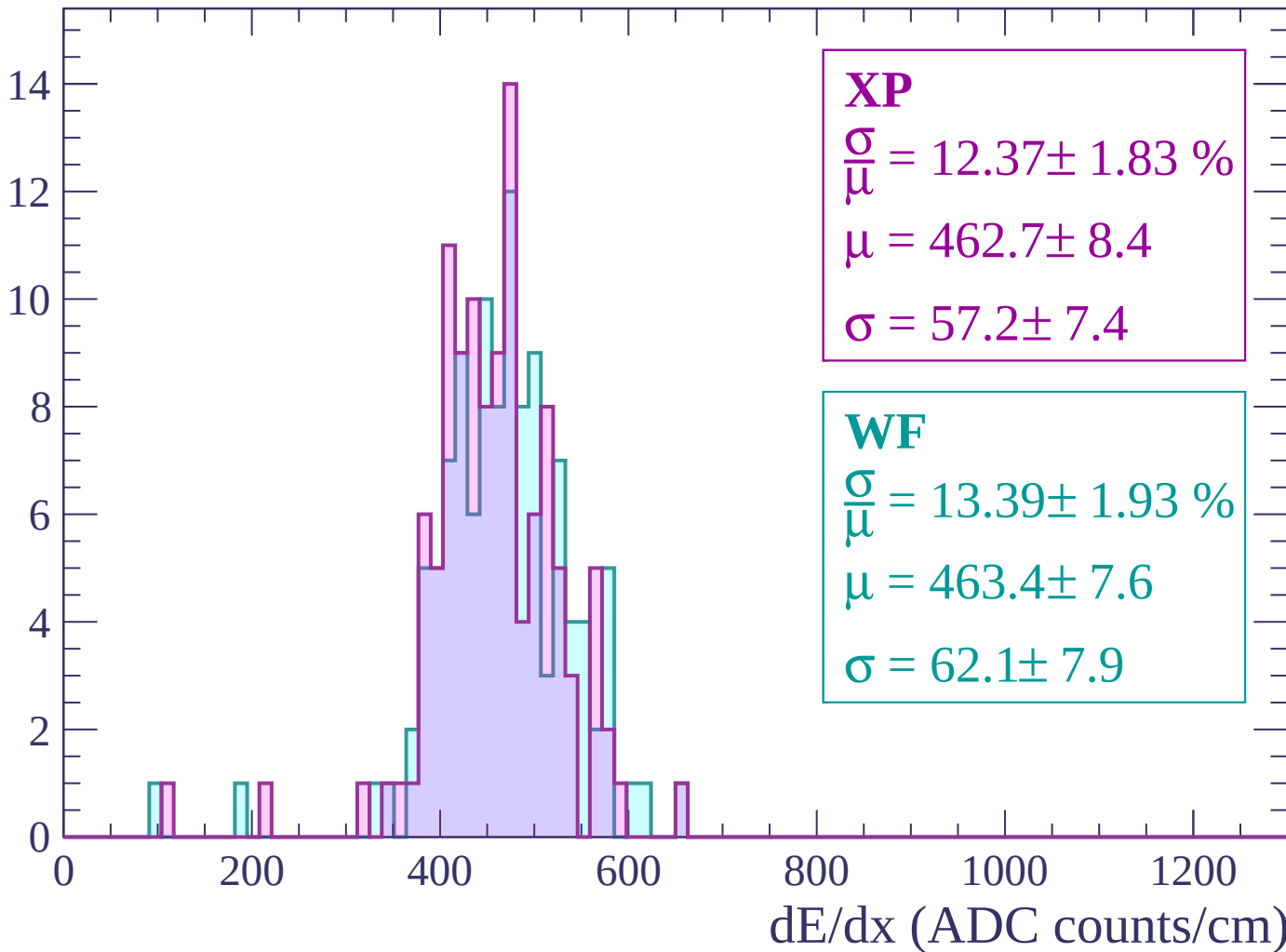
Energy loss | $1380 < p < 1440$

Count



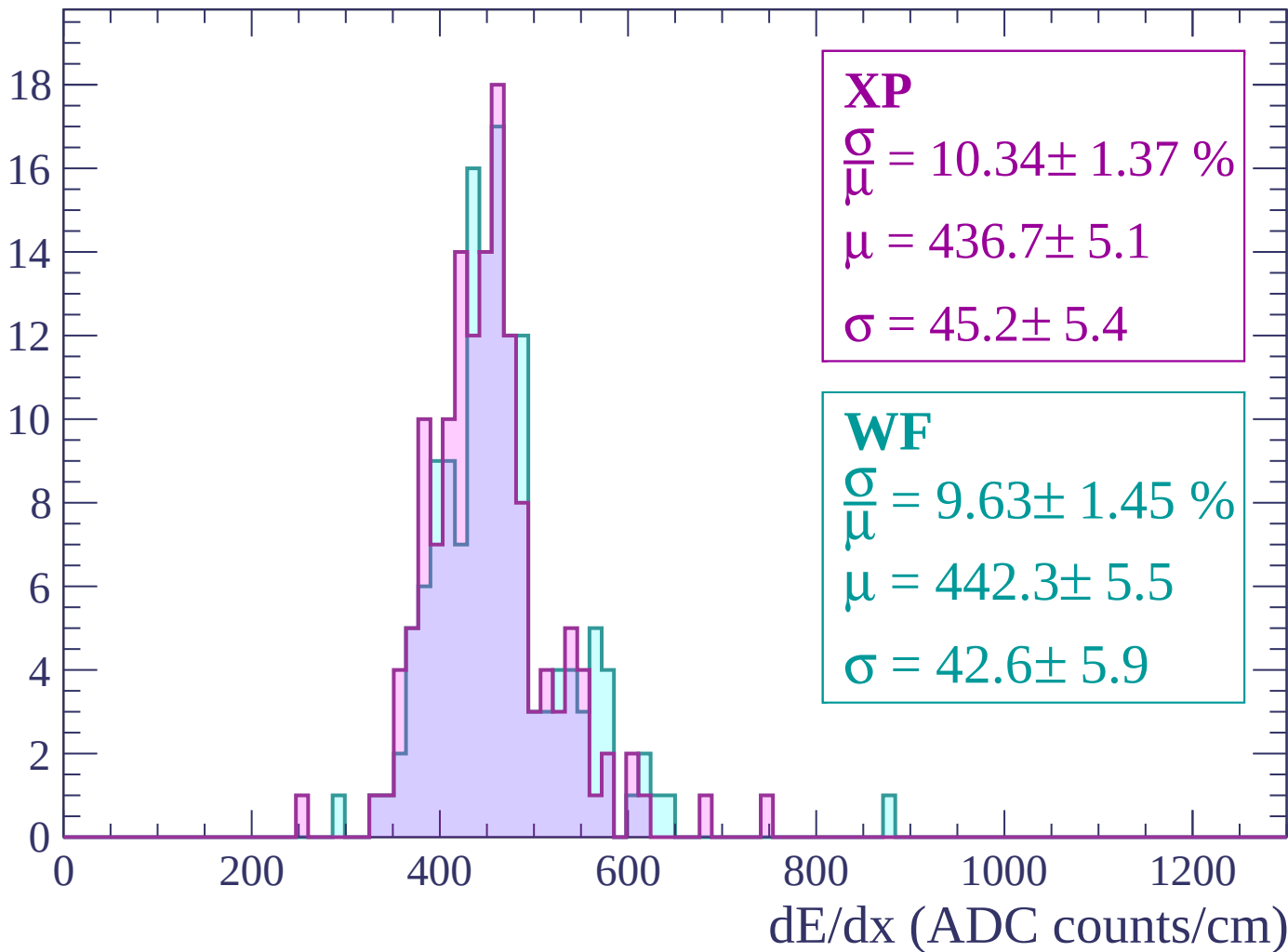
Energy loss | $1440 < p < 1500$

Count



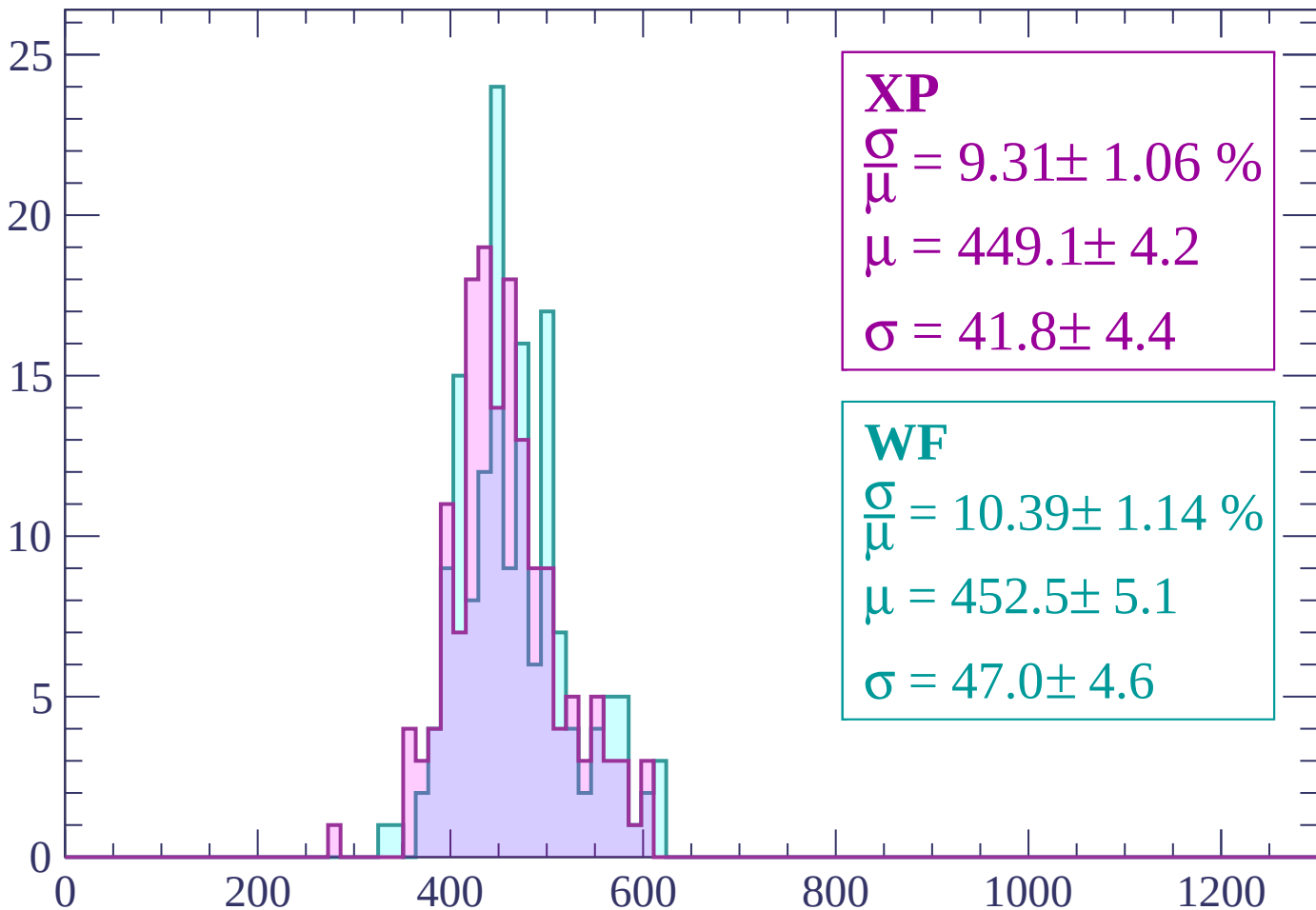
Energy loss | $1500 < p < 1560$

Count



Energy loss | $1560 < p < 1620$

Count



XP

$$\frac{\sigma}{\mu} = 9.31 \pm 1.06 \%$$

$$\mu = 449.1 \pm 4.2$$

$$\sigma = 41.8 \pm 4.4$$

WF

$$\frac{\sigma}{\mu} = 10.39 \pm 1.14 \%$$

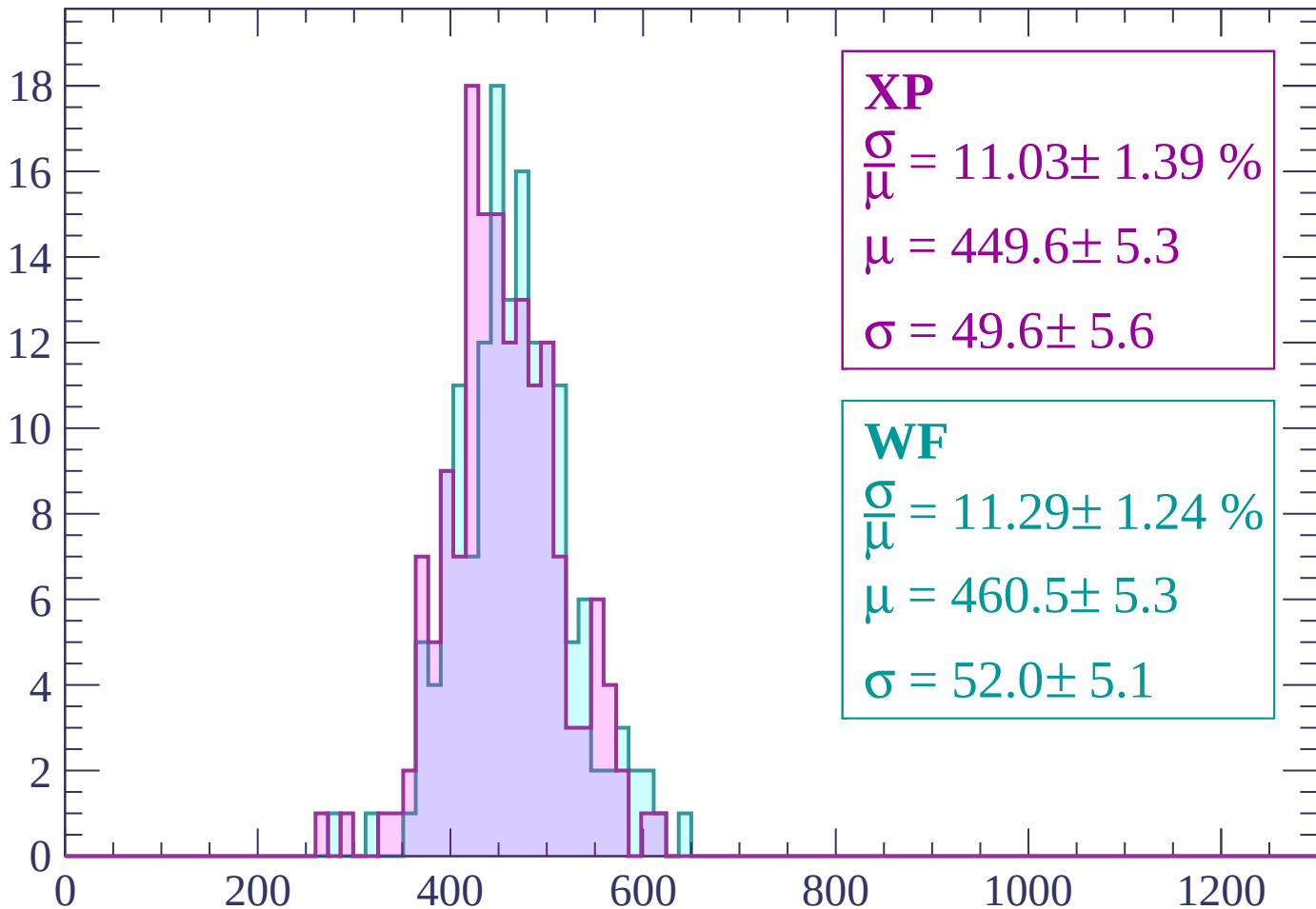
$$\mu = 452.5 \pm 5.1$$

$$\sigma = 47.0 \pm 4.6$$

dE/dx (ADC counts/cm)

Energy loss | $1620 < p < 1680$

Count



XP

$$\frac{\sigma}{\mu} = 11.03 \pm 1.39 \%$$

$$\mu = 449.6 \pm 5.3$$

$$\sigma = 49.6 \pm 5.6$$

WF

$$\frac{\sigma}{\mu} = 11.29 \pm 1.24 \%$$

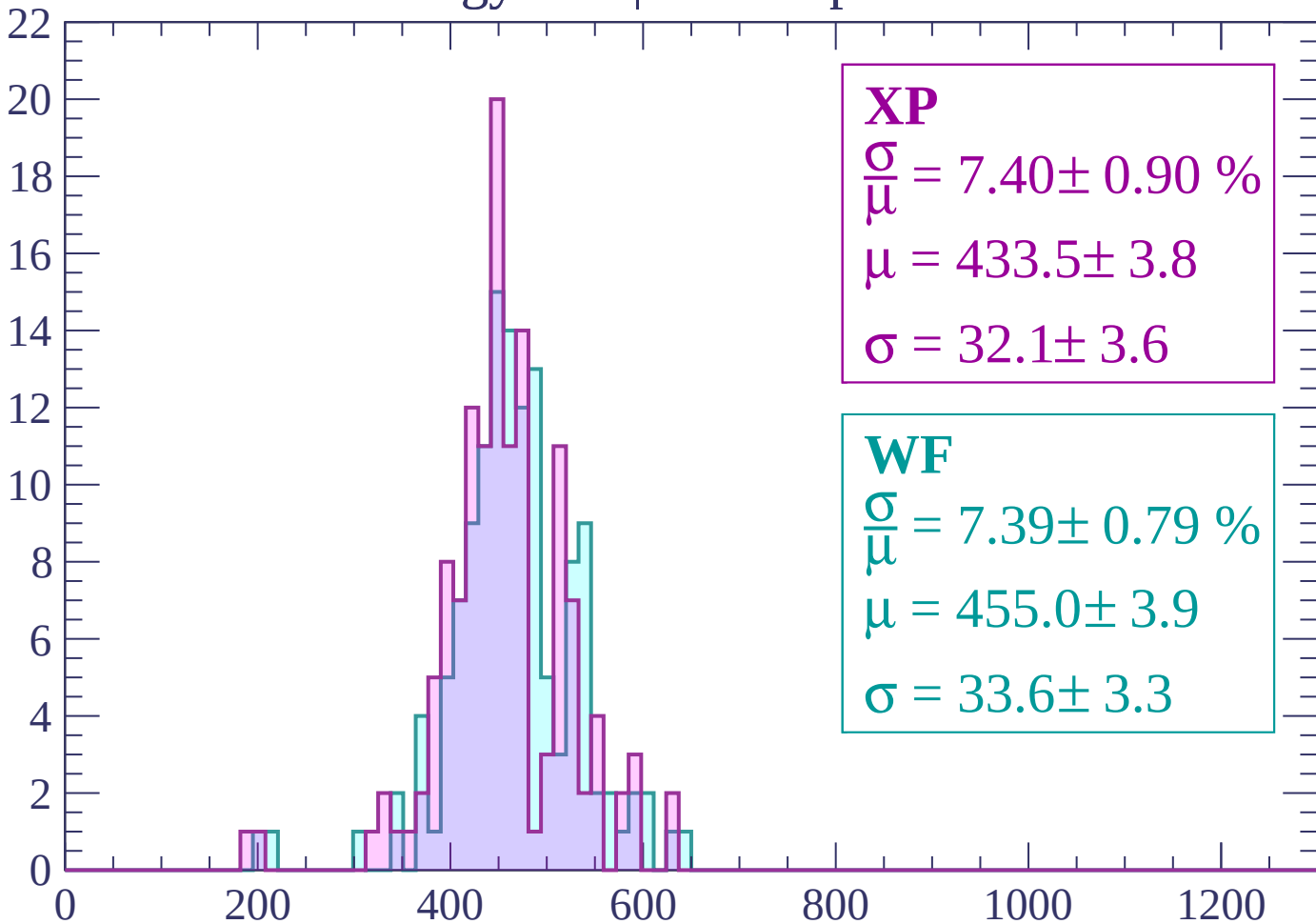
$$\mu = 460.5 \pm 5.3$$

$$\sigma = 52.0 \pm 5.1$$

dE/dx (ADC counts/cm)

Energy loss | $1680 < p < 1740$

Count



XP

$$\frac{\sigma}{\mu} = 7.40 \pm 0.90 \%$$

$$\mu = 433.5 \pm 3.8$$

$$\sigma = 32.1 \pm 3.6$$

WF

$$\frac{\sigma}{\mu} = 7.39 \pm 0.79 \%$$

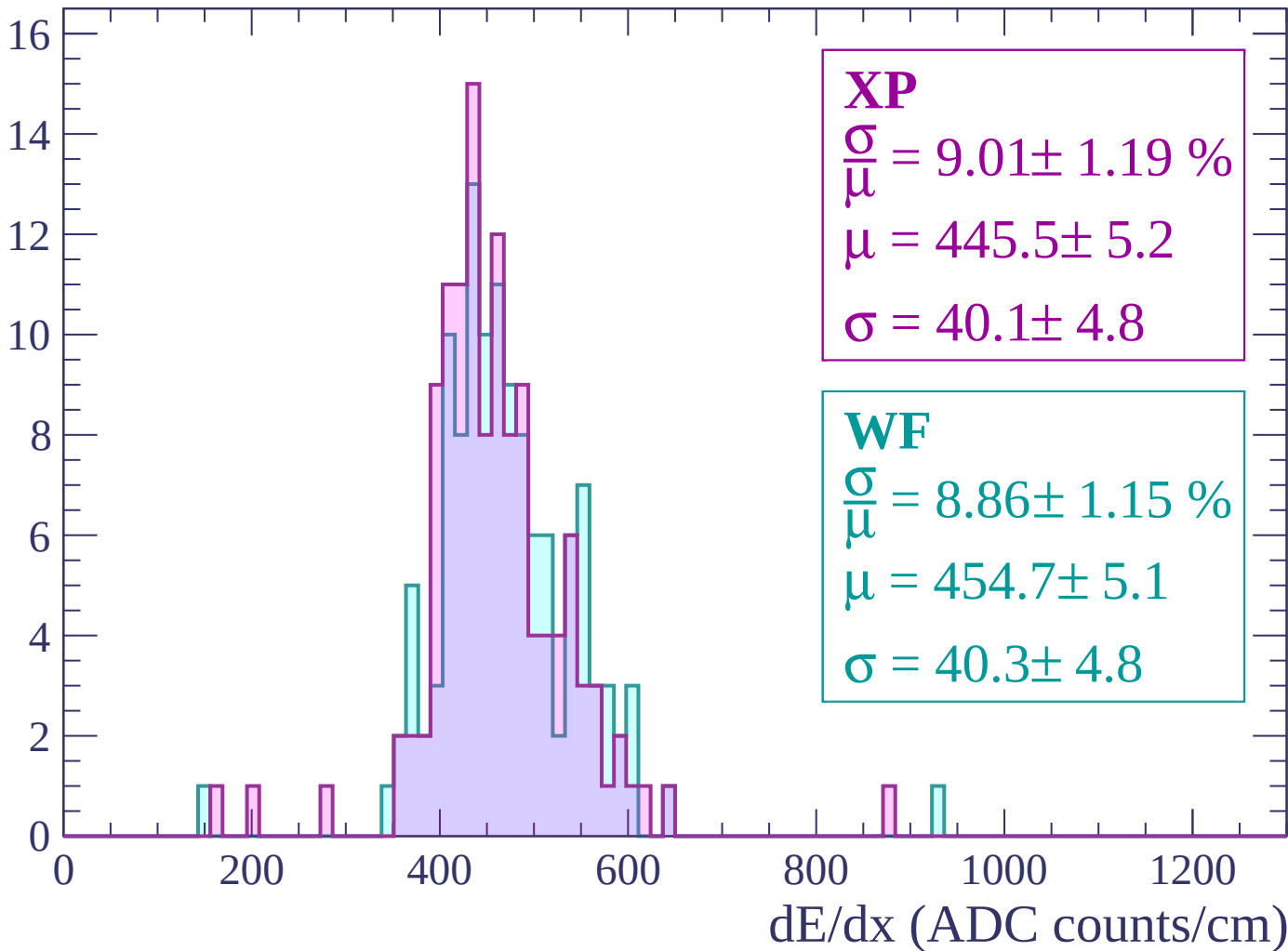
$$\mu = 455.0 \pm 3.9$$

$$\sigma = 33.6 \pm 3.3$$

dE/dx (ADC counts/cm)

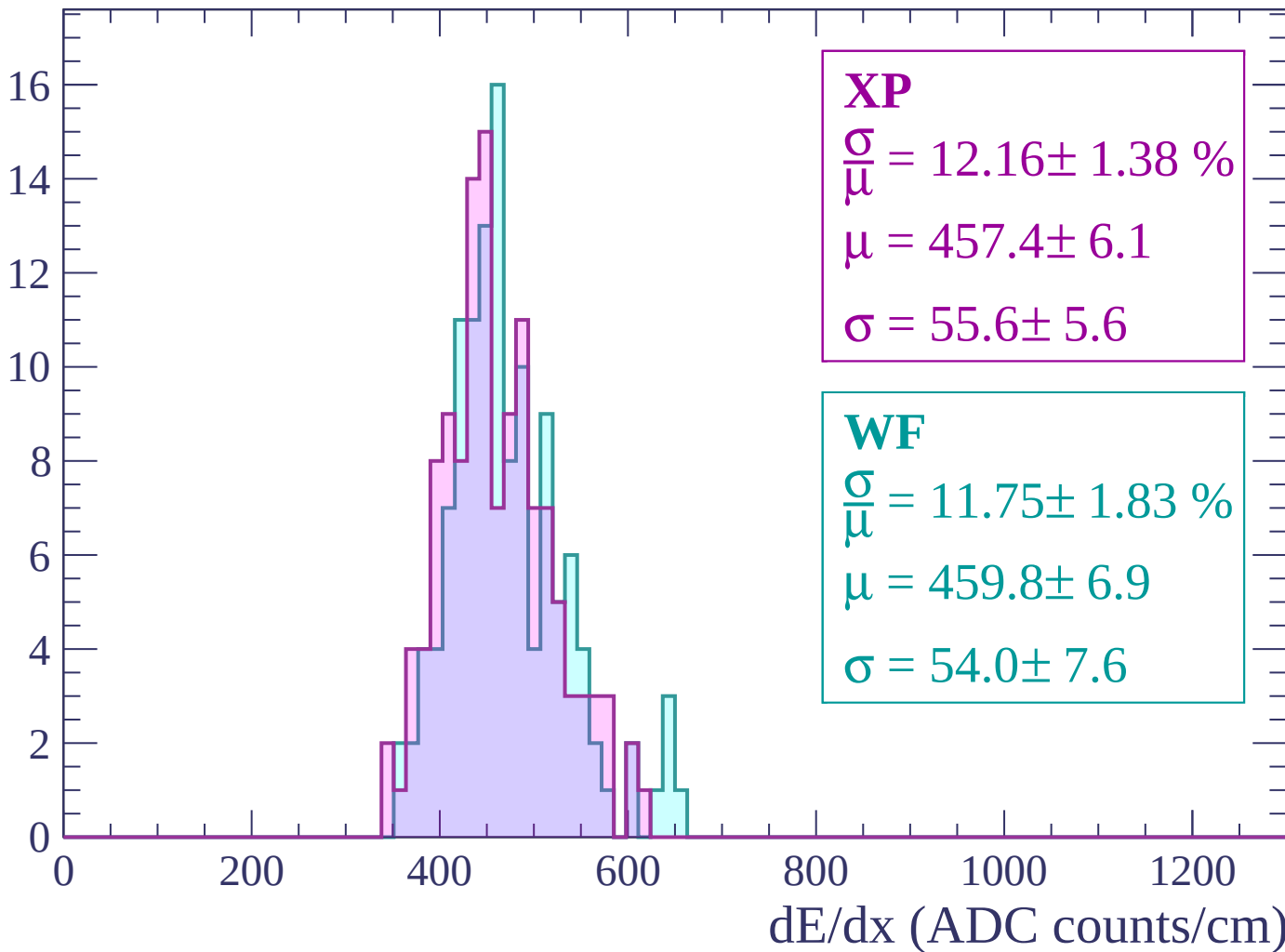
Energy loss | $1740 < p < 1800$

Count



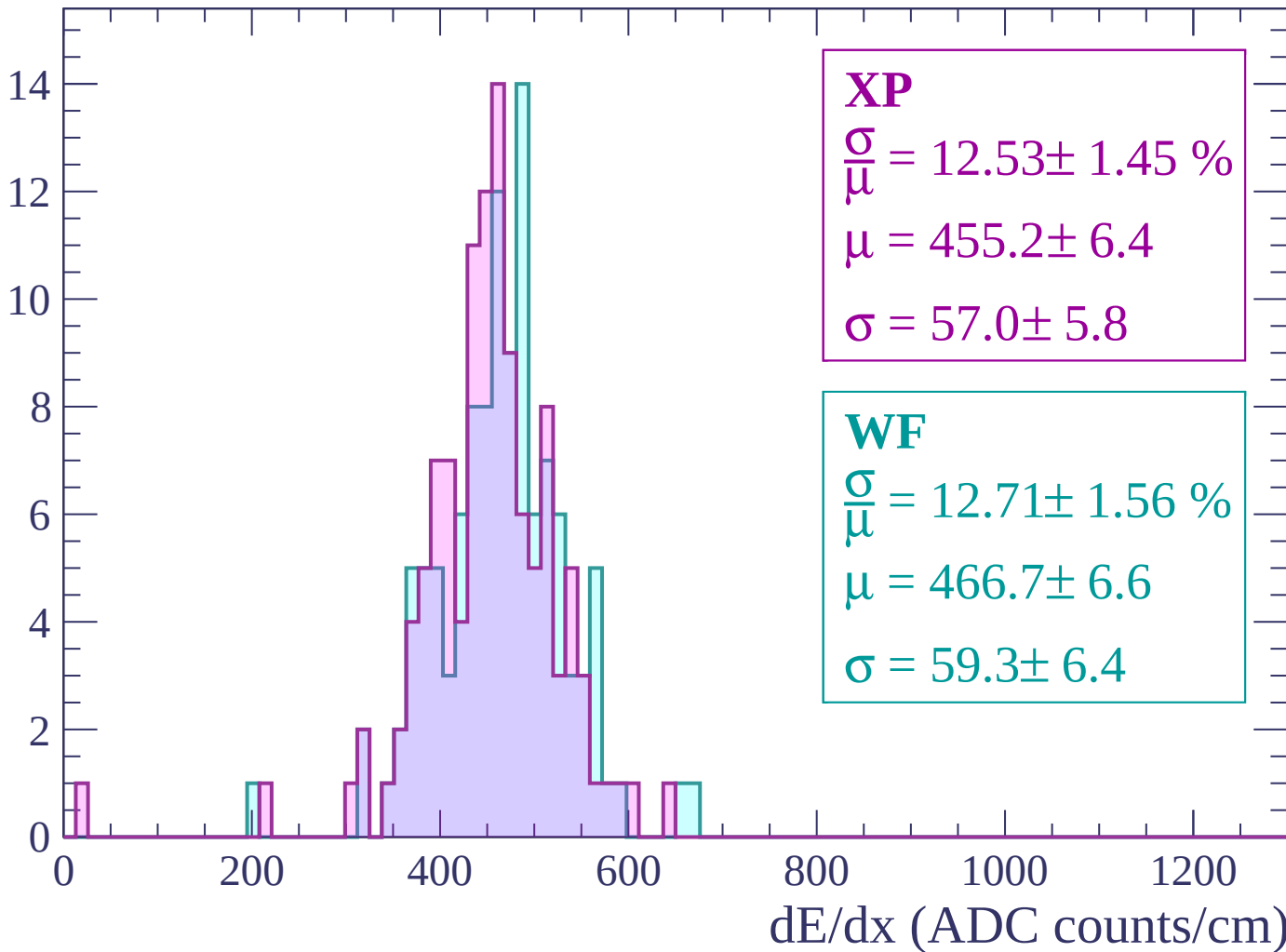
Energy loss | $1800 < p < 1860$

Count



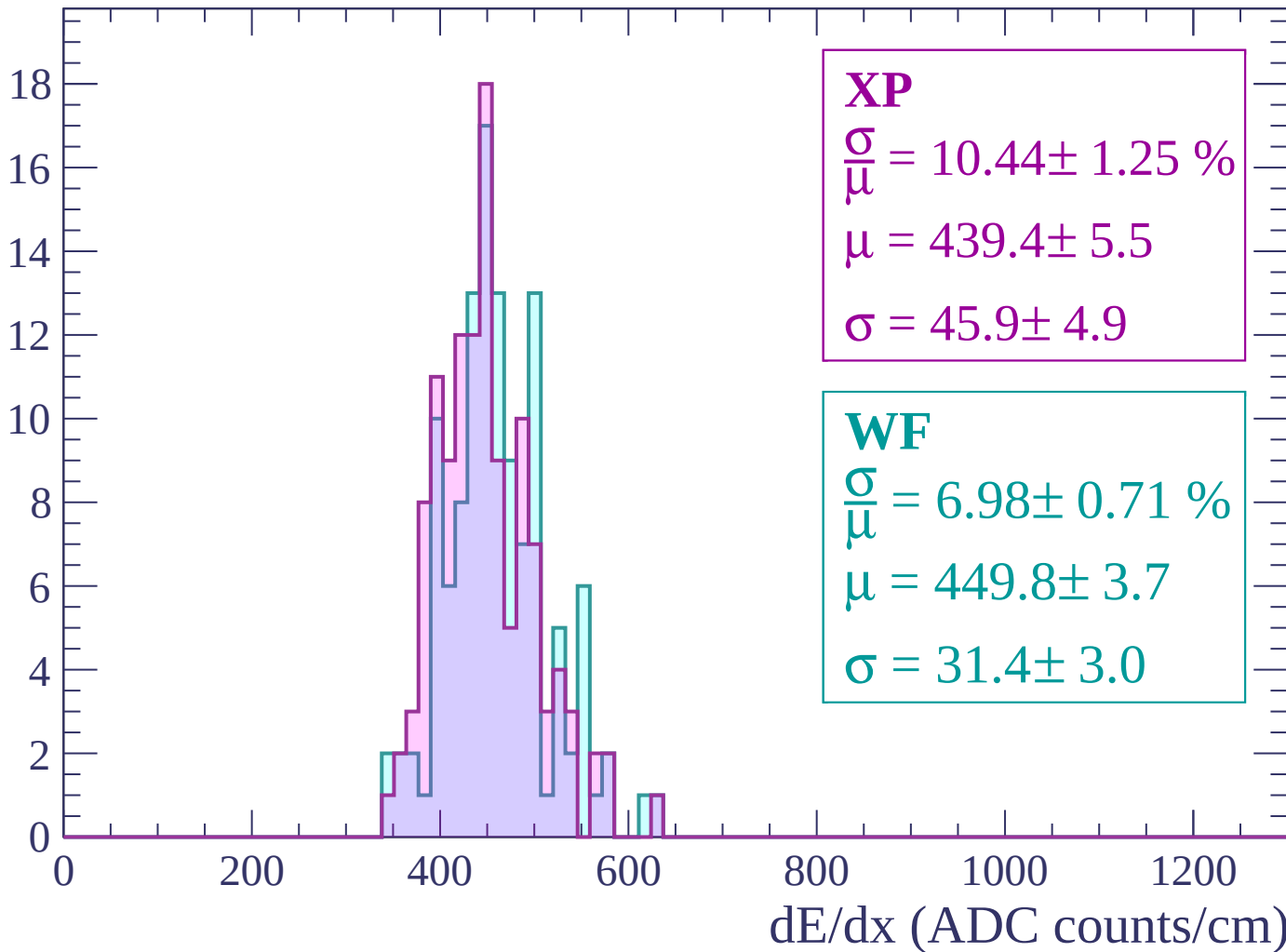
Energy loss | $1860 < p < 1920$

Count



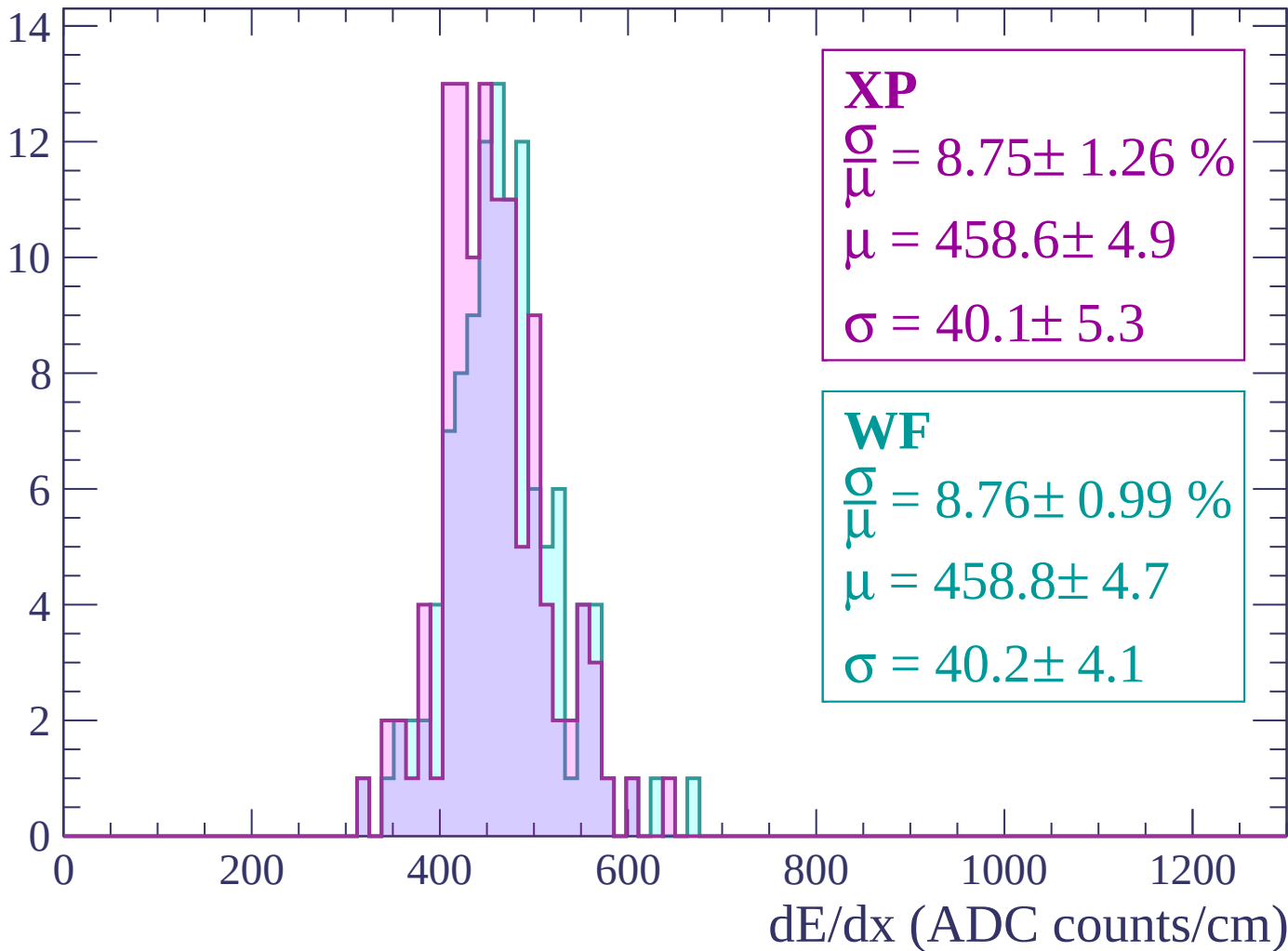
Energy loss | $1920 < p < 1980$

Count



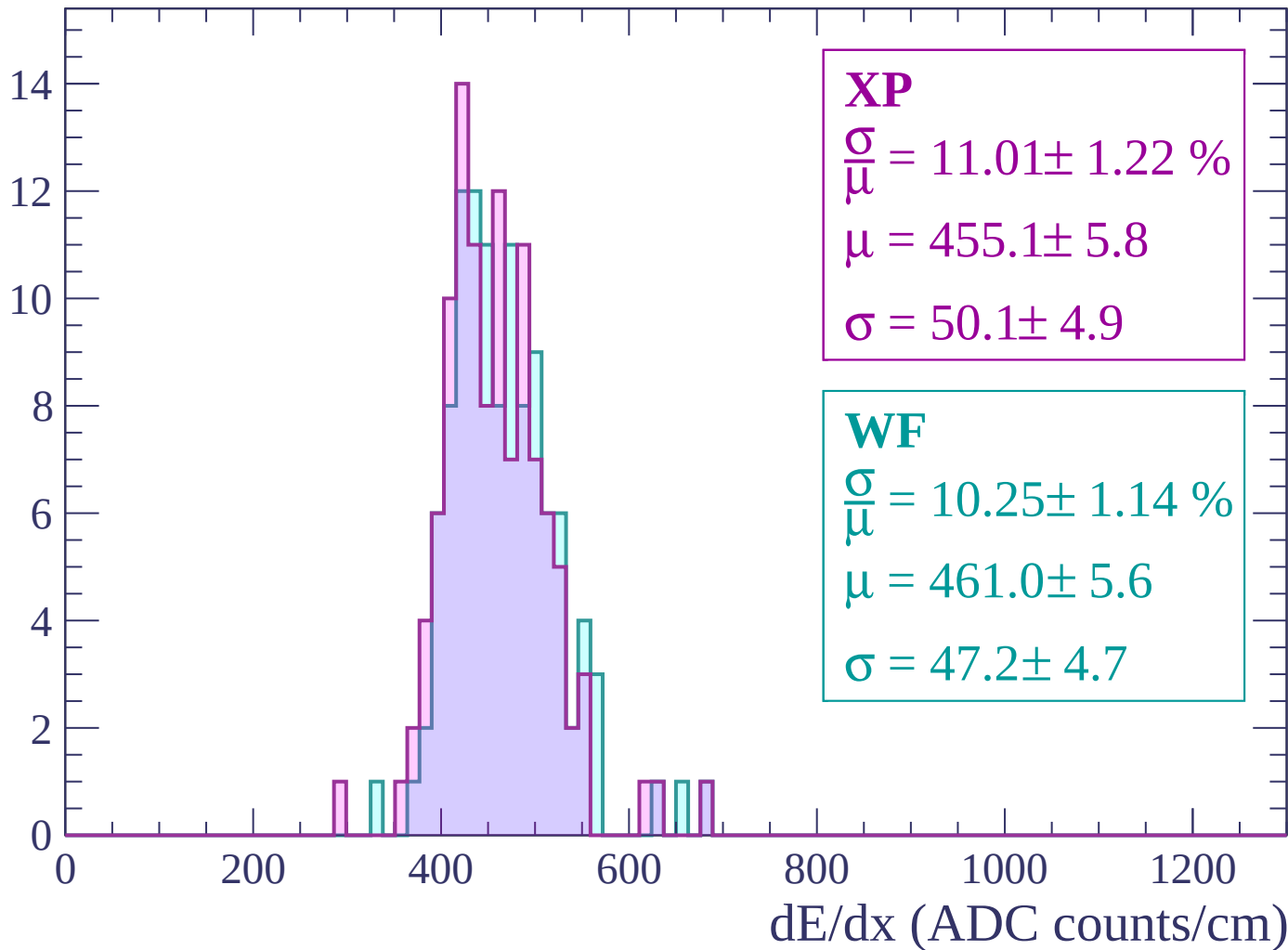
Energy loss | $1980 < p < 2040$

Count



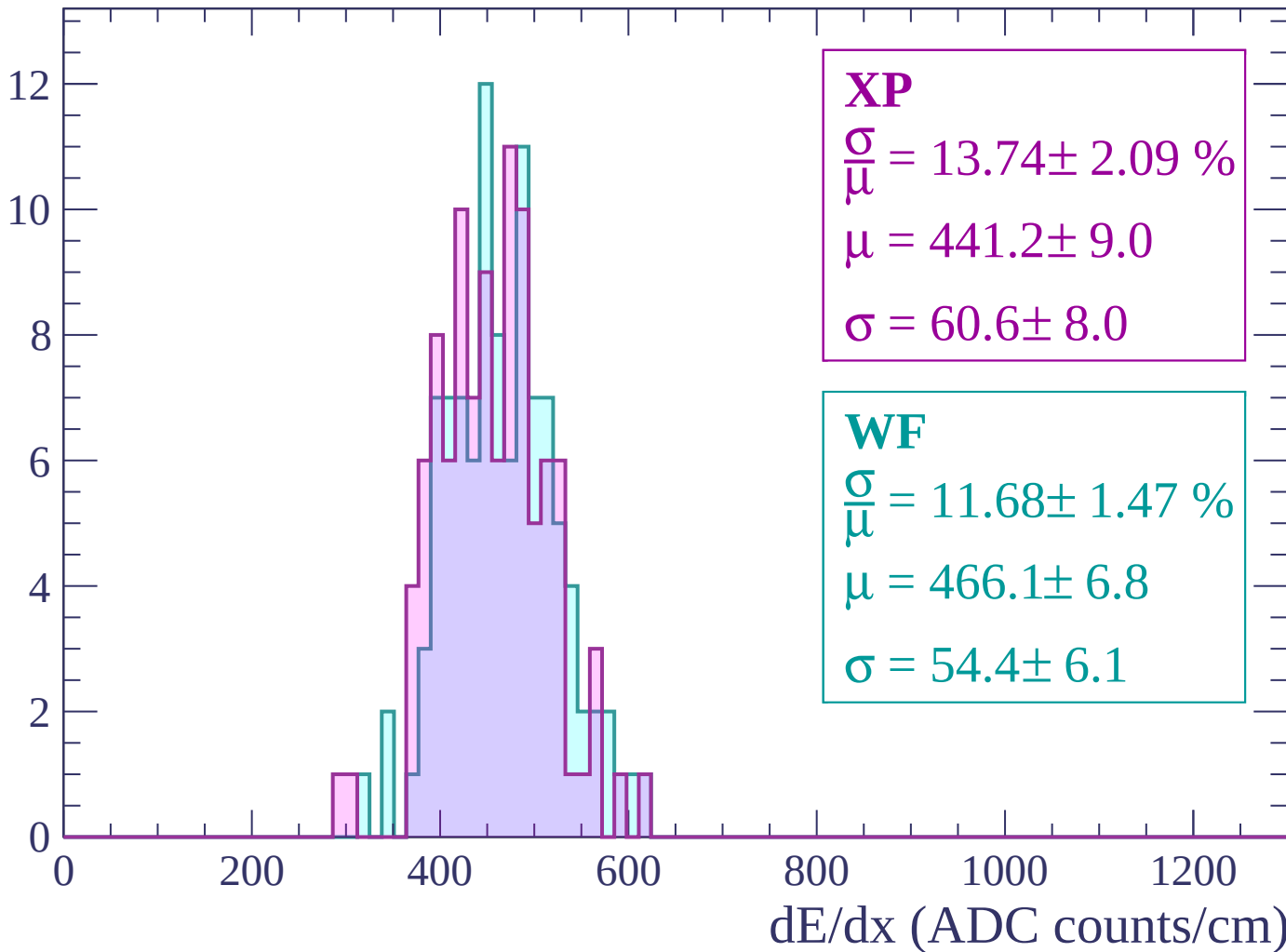
Energy loss | $2040 < p < 2100$

Count



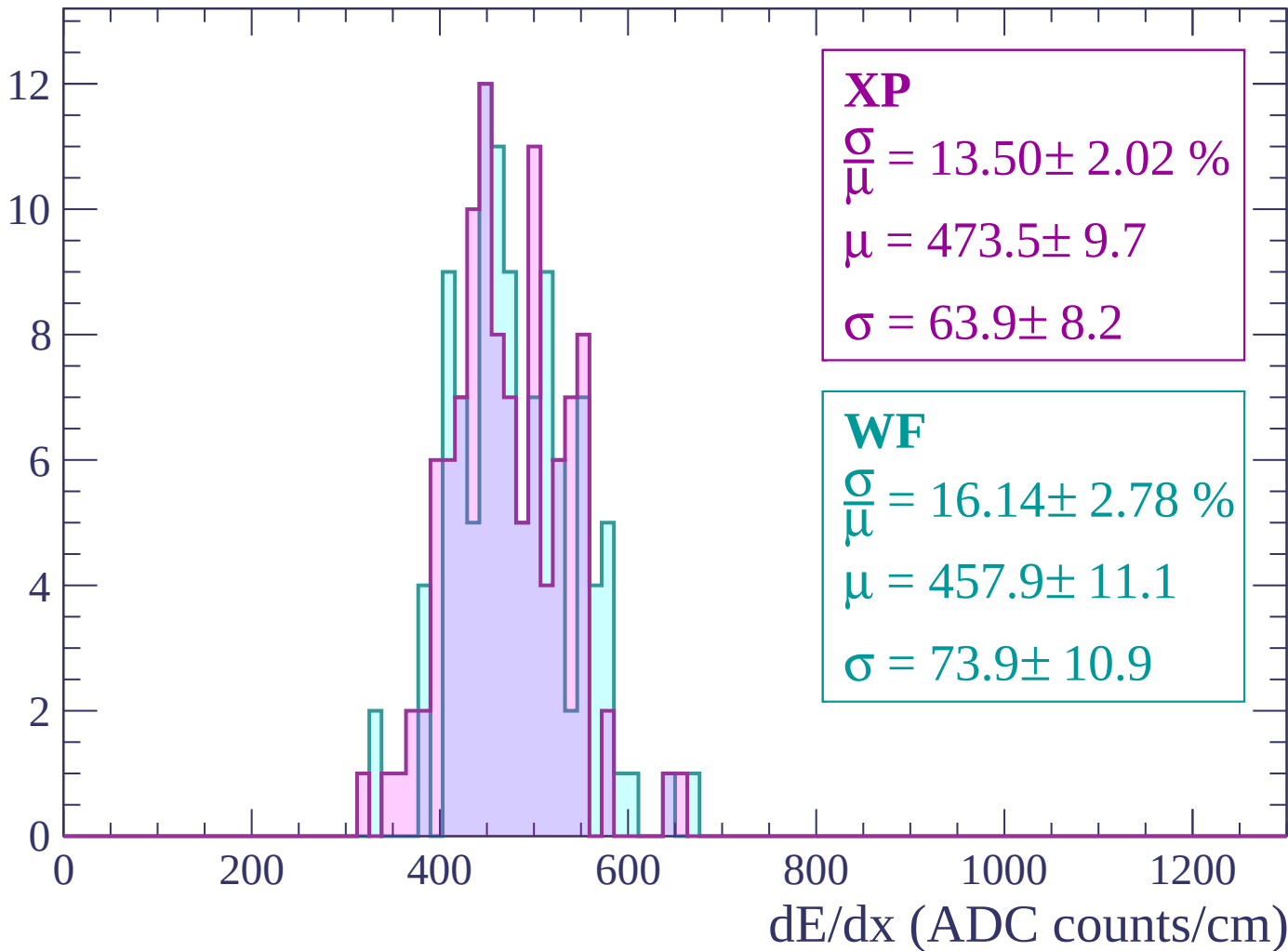
Energy loss | $2100 < p < 2160$

Count



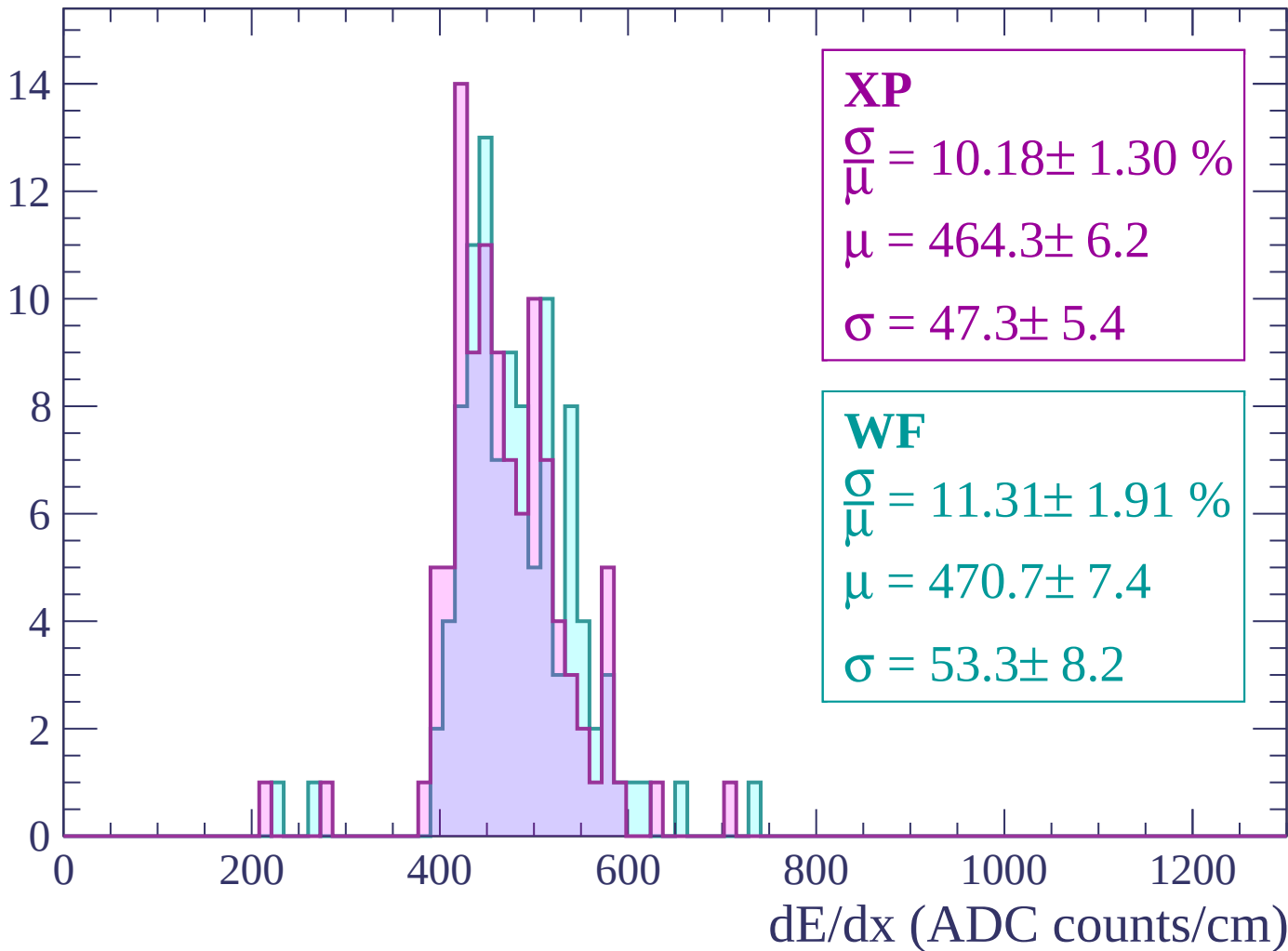
Energy loss | $2160 < p < 2220$

Count



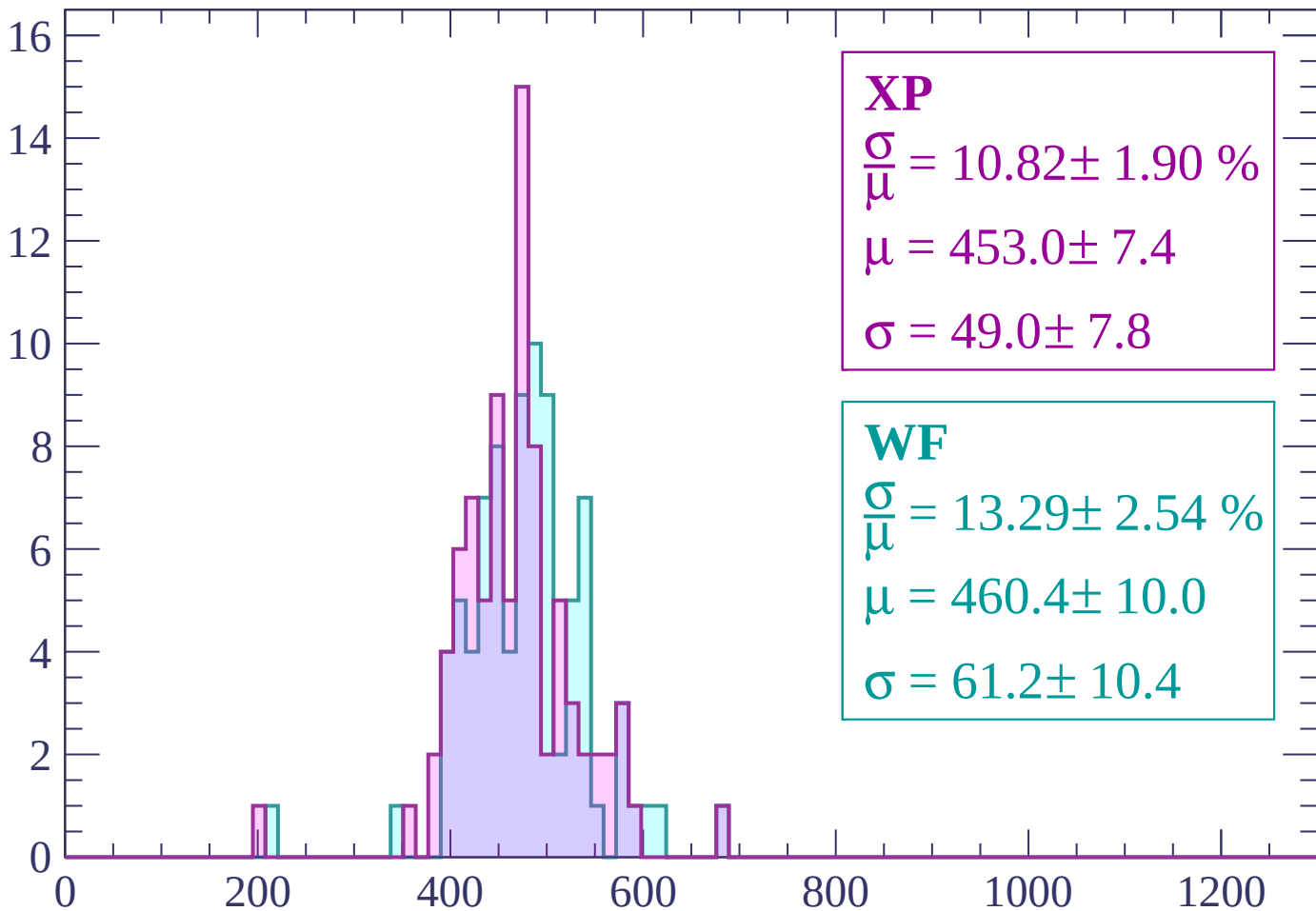
Energy loss | $2220 < p < 2280$

Count



Energy loss | $2280 < p < 2340$

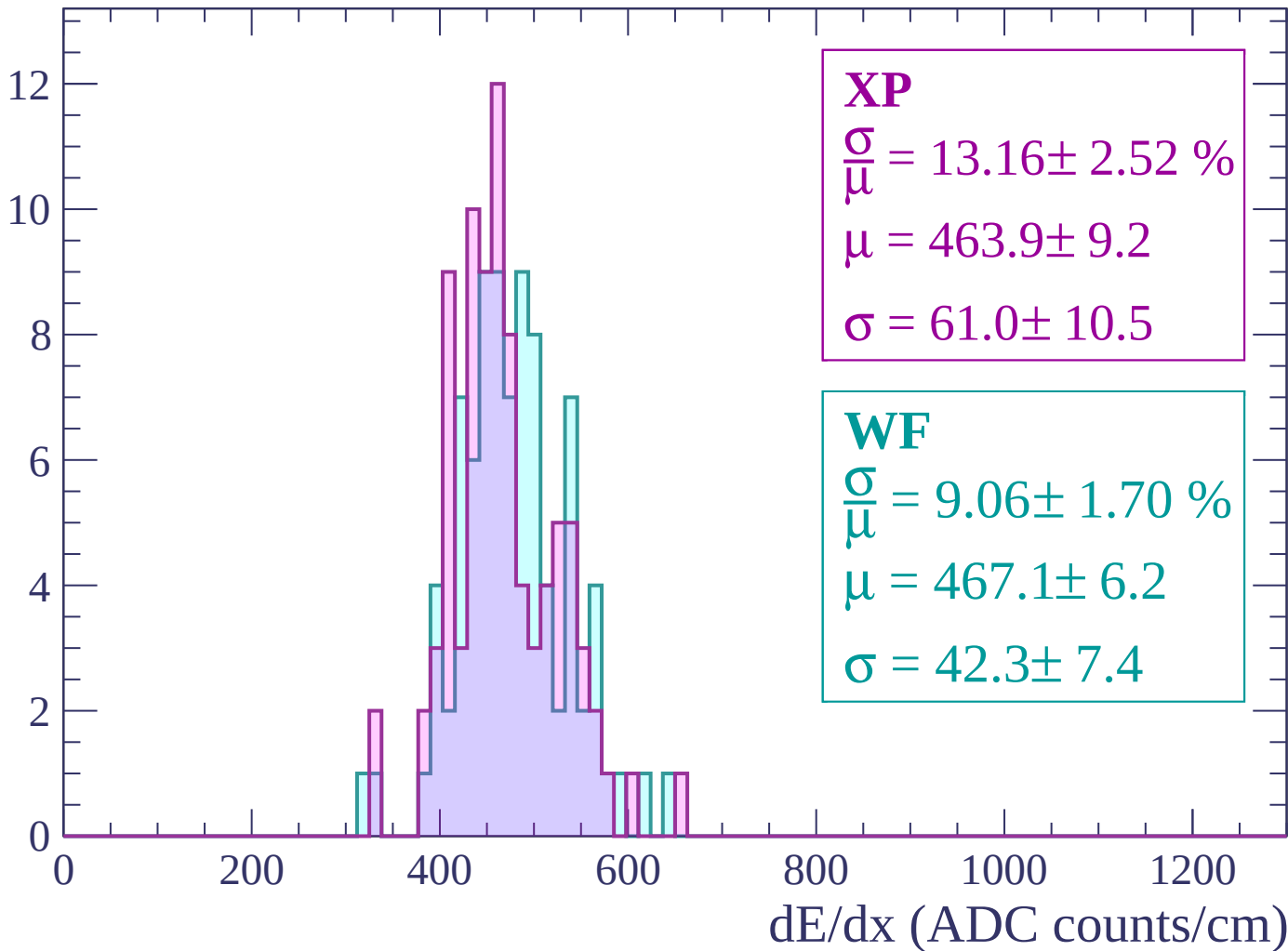
Count



dE/dx (ADC counts/cm)

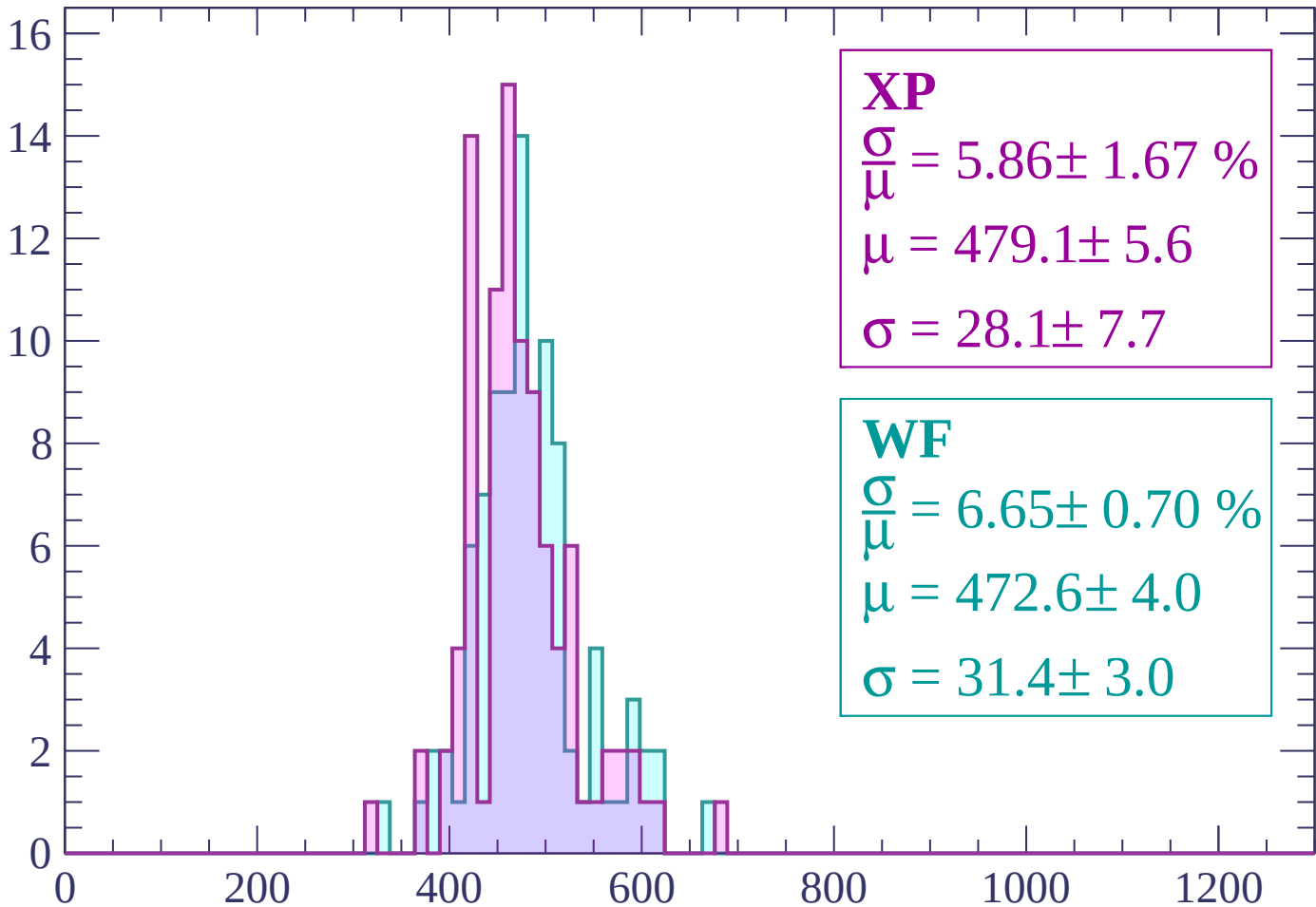
Energy loss | $2340 < p < 2400$

Count



Energy loss | $2400 < p < 2460$

Count



XP

$$\frac{\sigma}{\mu} = 5.86 \pm 1.67 \%$$

$$\mu = 479.1 \pm 5.6$$

$$\sigma = 28.1 \pm 7.7$$

WF

$$\frac{\sigma}{\mu} = 6.65 \pm 0.70 \%$$

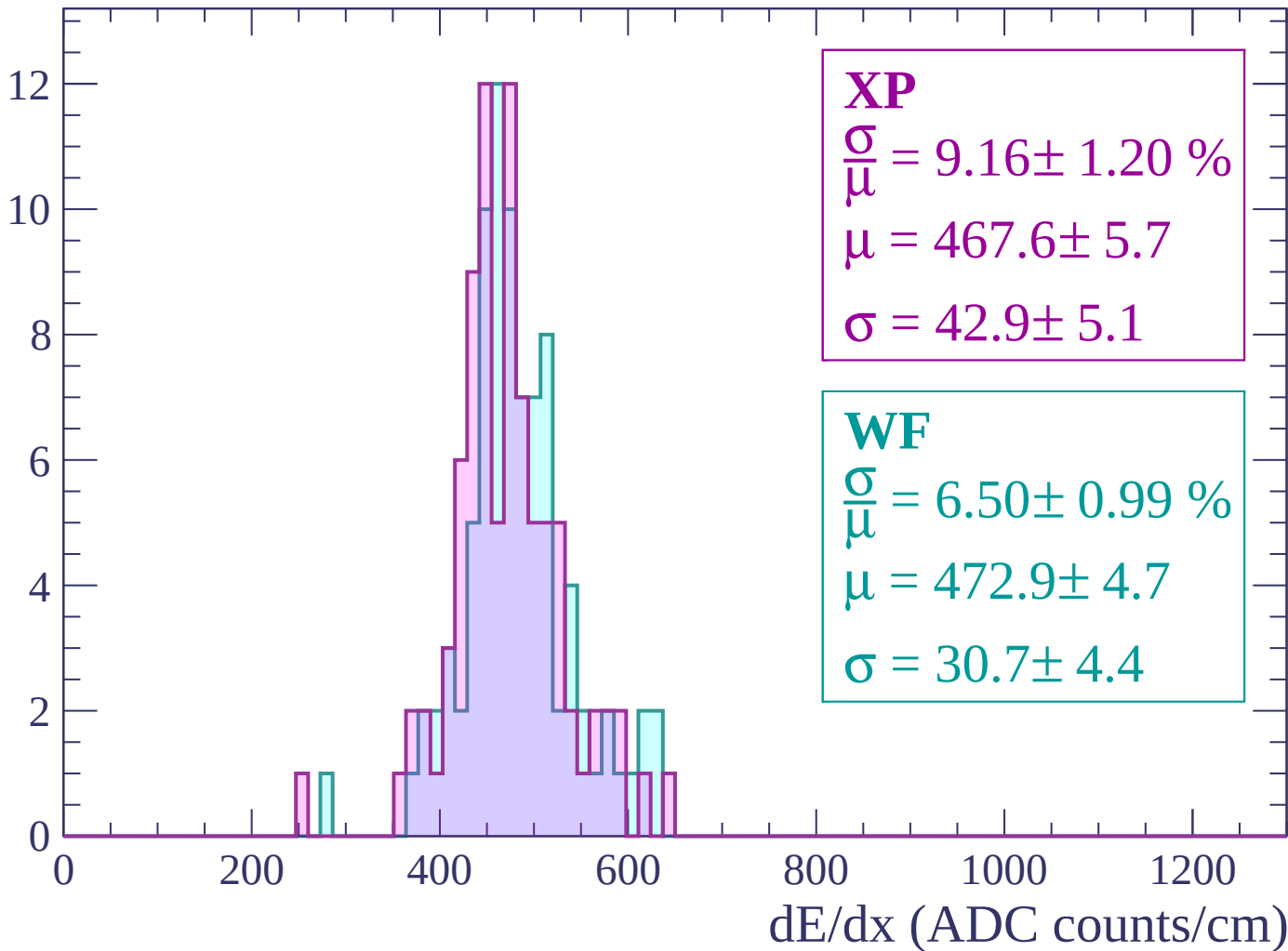
$$\mu = 472.6 \pm 4.0$$

$$\sigma = 31.4 \pm 3.0$$

dE/dx (ADC counts/cm)

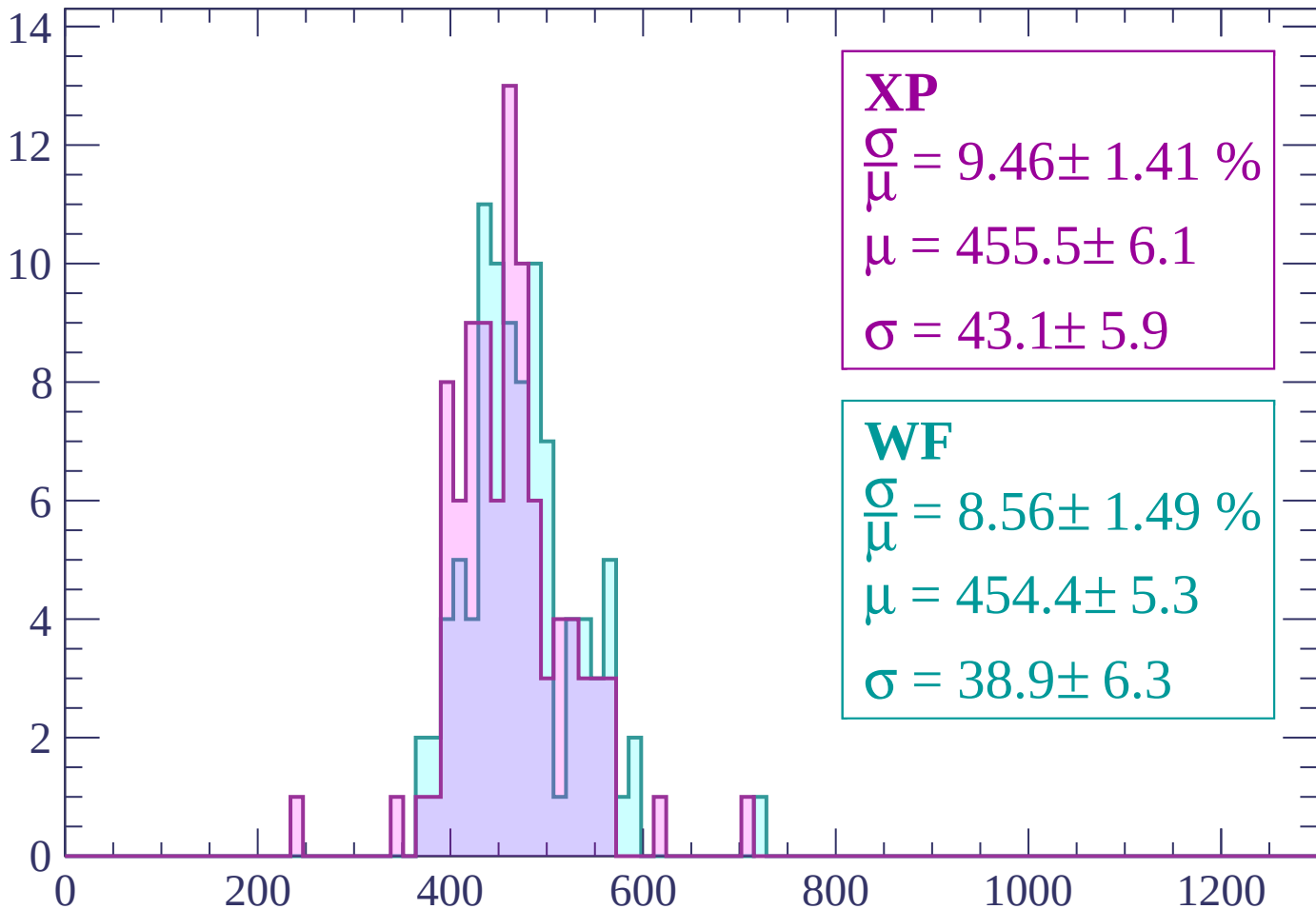
Energy loss | $2460 < p < 2520$

Count



Energy loss | $2520 < p < 2580$

Count



XP

$$\frac{\sigma}{\mu} = 9.46 \pm 1.41 \%$$

$$\mu = 455.5 \pm 6.1$$

$$\sigma = 43.1 \pm 5.9$$

WF

$$\frac{\sigma}{\mu} = 8.56 \pm 1.49 \%$$

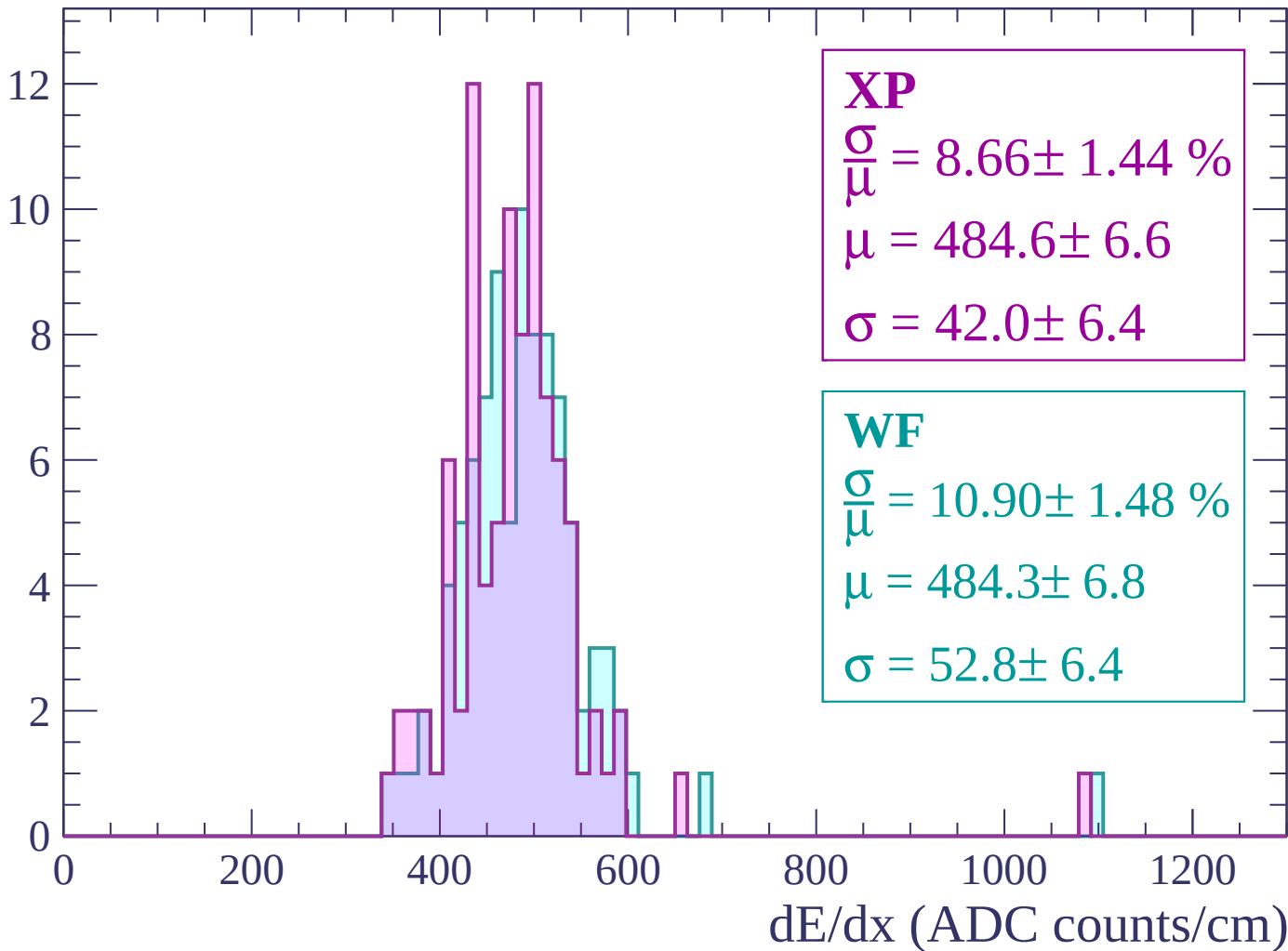
$$\mu = 454.4 \pm 5.3$$

$$\sigma = 38.9 \pm 6.3$$

dE/dx (ADC counts/cm)

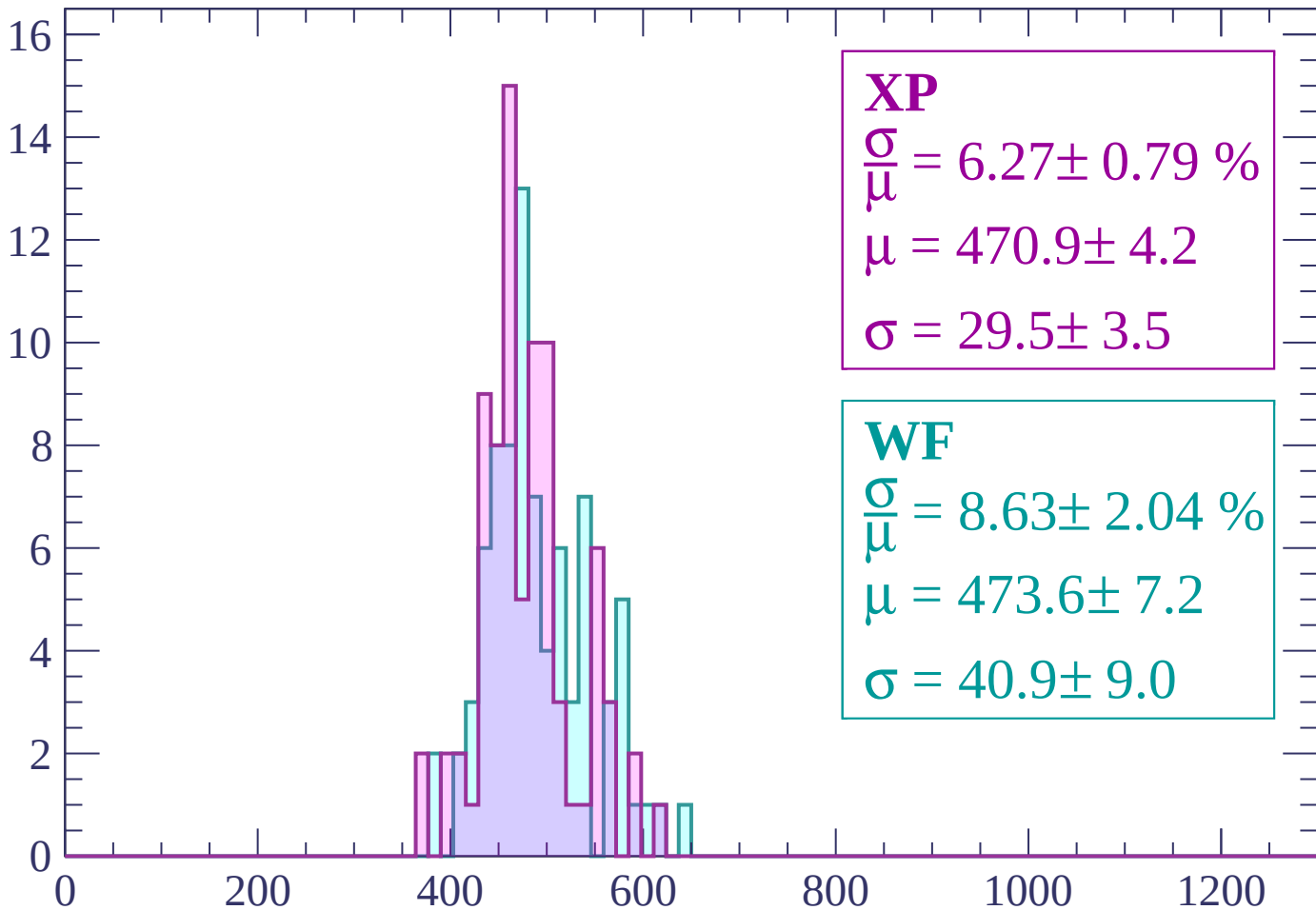
Energy loss | $2580 < p < 2640$

Count



Energy loss | $2640 < p < 2700$

Count



XP

$$\frac{\sigma}{\mu} = 6.27 \pm 0.79 \%$$

$$\mu = 470.9 \pm 4.2$$

$$\sigma = 29.5 \pm 3.5$$

WF

$$\frac{\sigma}{\mu} = 8.63 \pm 2.04 \%$$

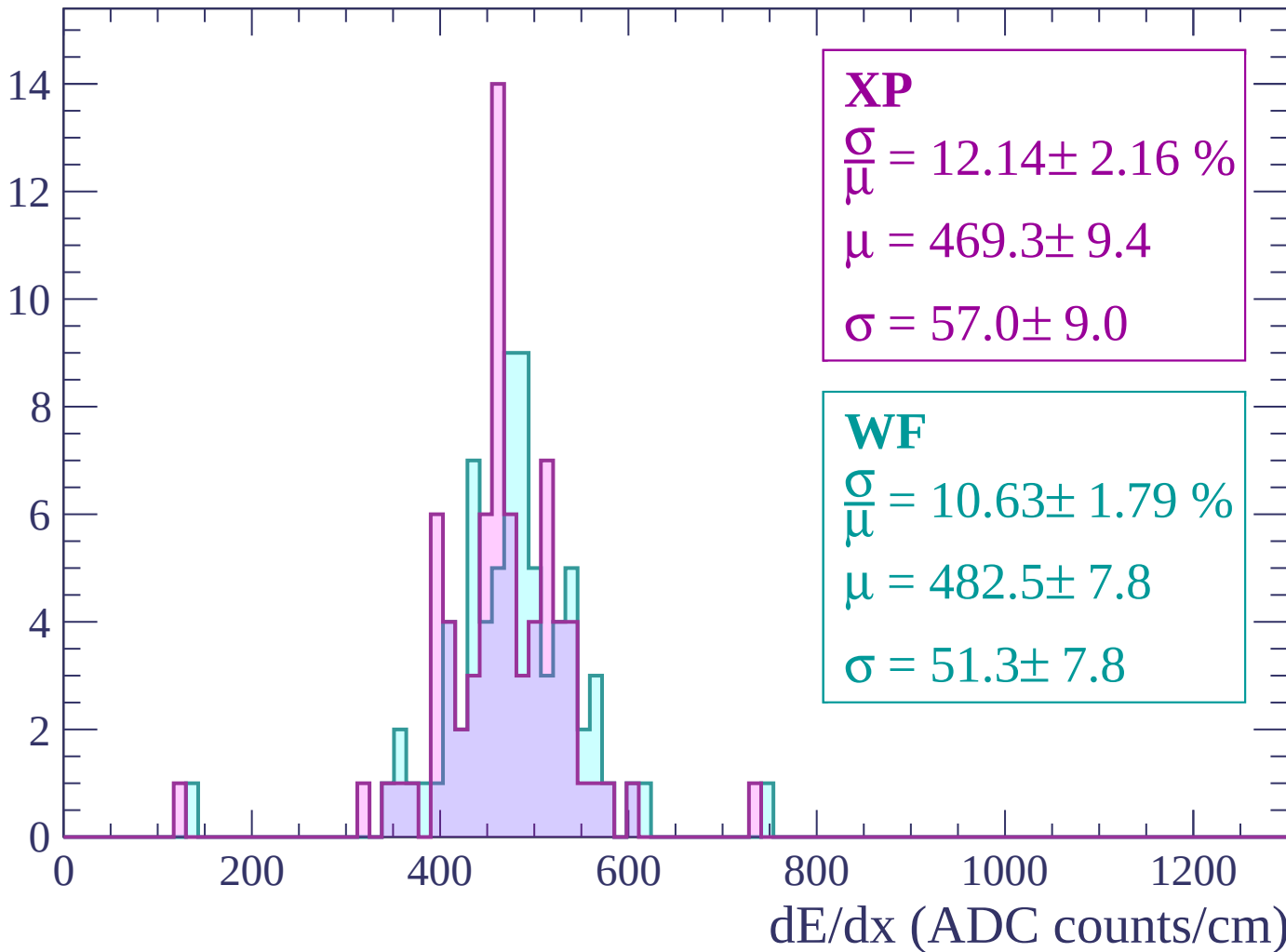
$$\mu = 473.6 \pm 7.2$$

$$\sigma = 40.9 \pm 9.0$$

dE/dx (ADC counts/cm)

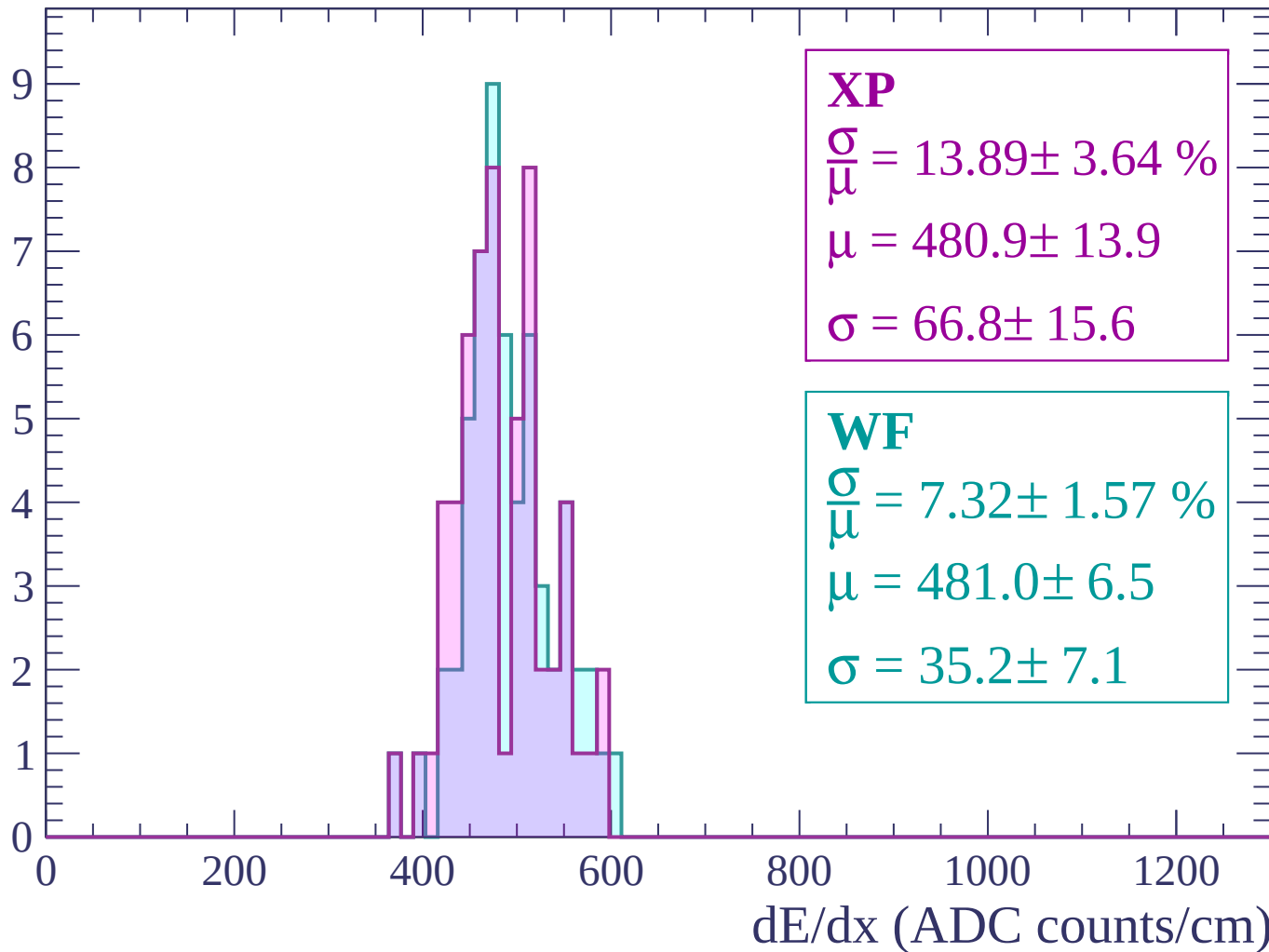
Energy loss | $2700 < p < 2760$

Count



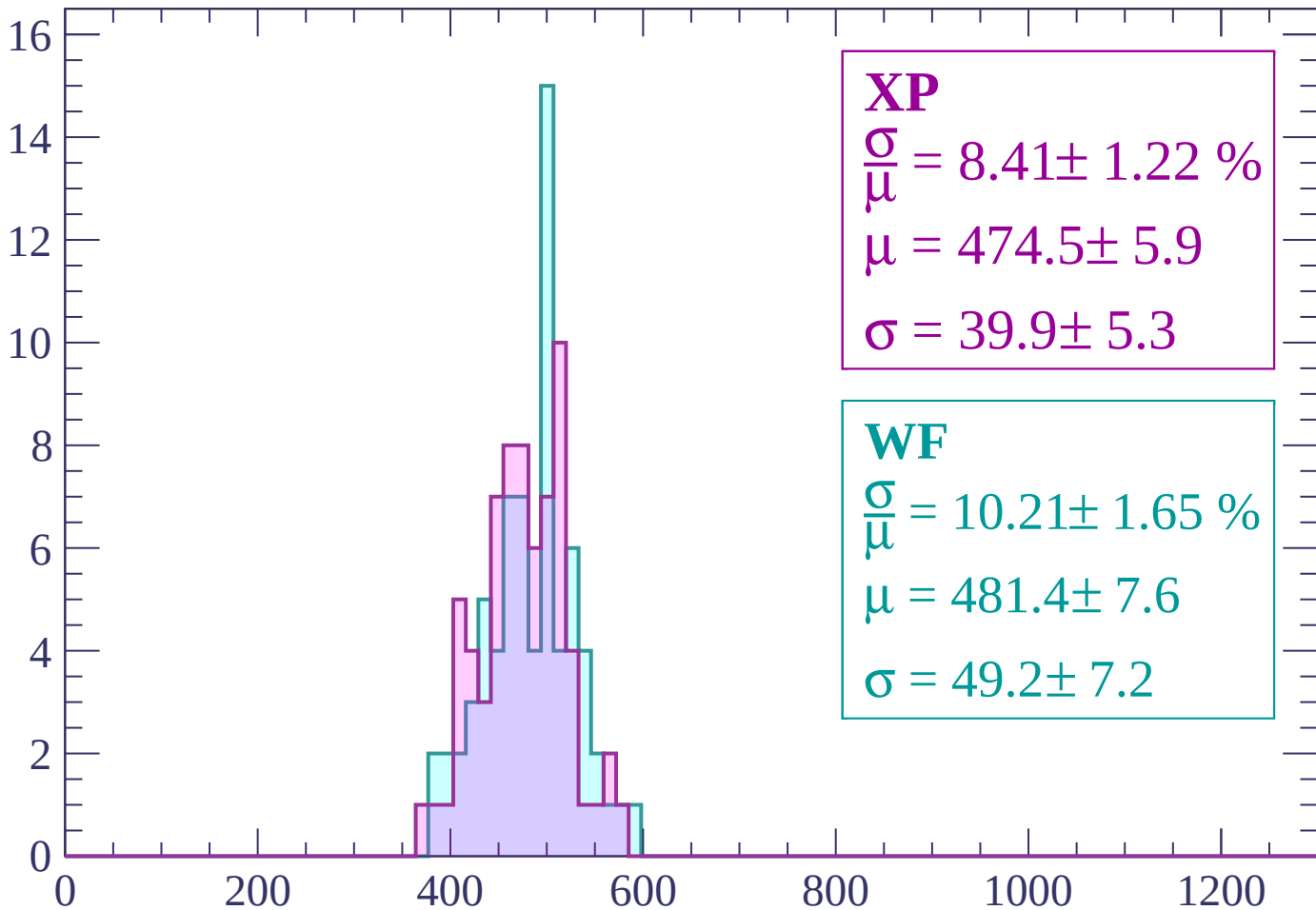
Energy loss | $2760 < p < 2820$

Count



Energy loss | $2820 < p < 2880$

Count



XP

$$\frac{\sigma}{\mu} = 8.41 \pm 1.22 \%$$

$$\mu = 474.5 \pm 5.9$$

$$\sigma = 39.9 \pm 5.3$$

WF

$$\frac{\sigma}{\mu} = 10.21 \pm 1.65 \%$$

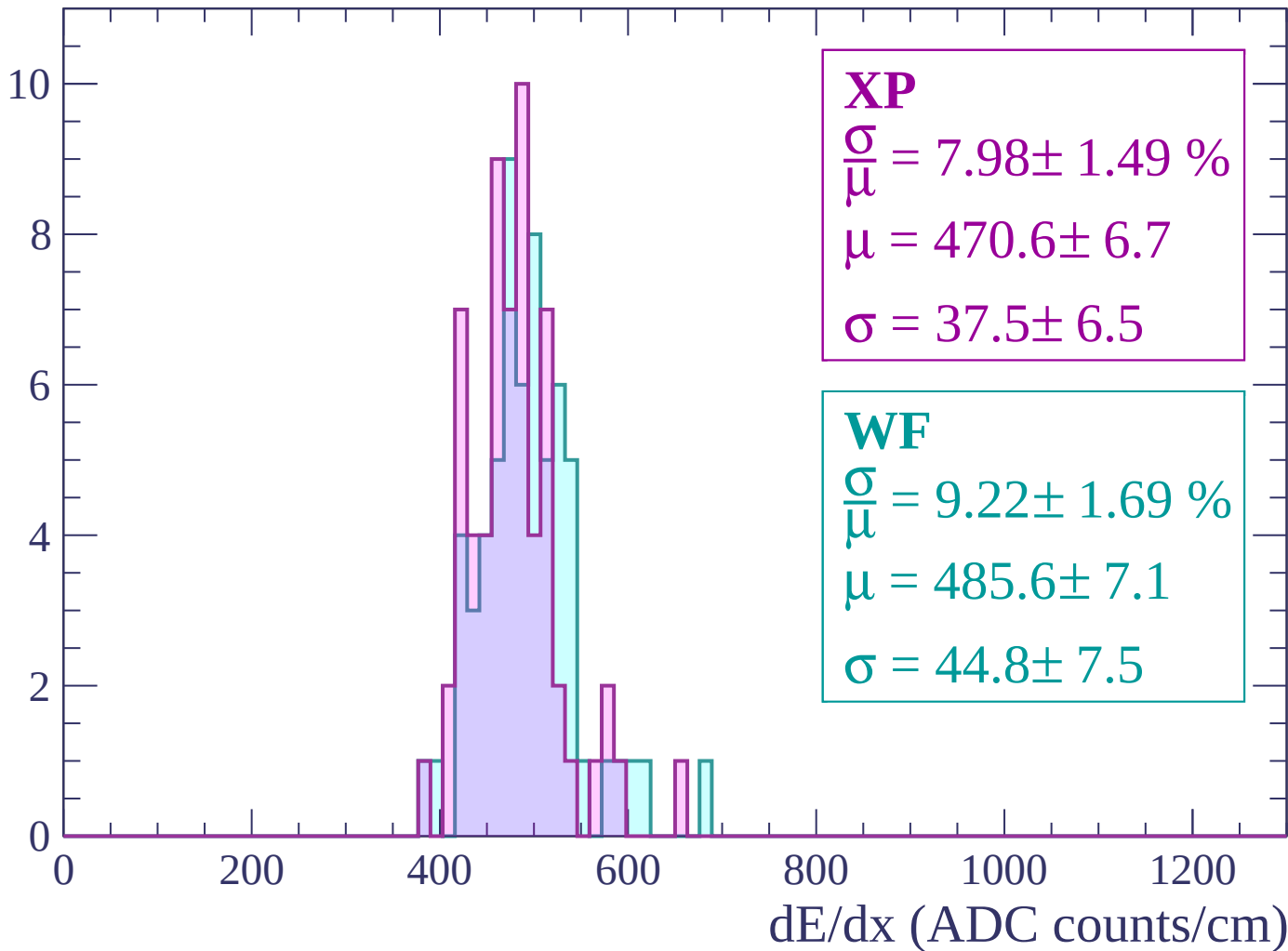
$$\mu = 481.4 \pm 7.6$$

$$\sigma = 49.2 \pm 7.2$$

dE/dx (ADC counts/cm)

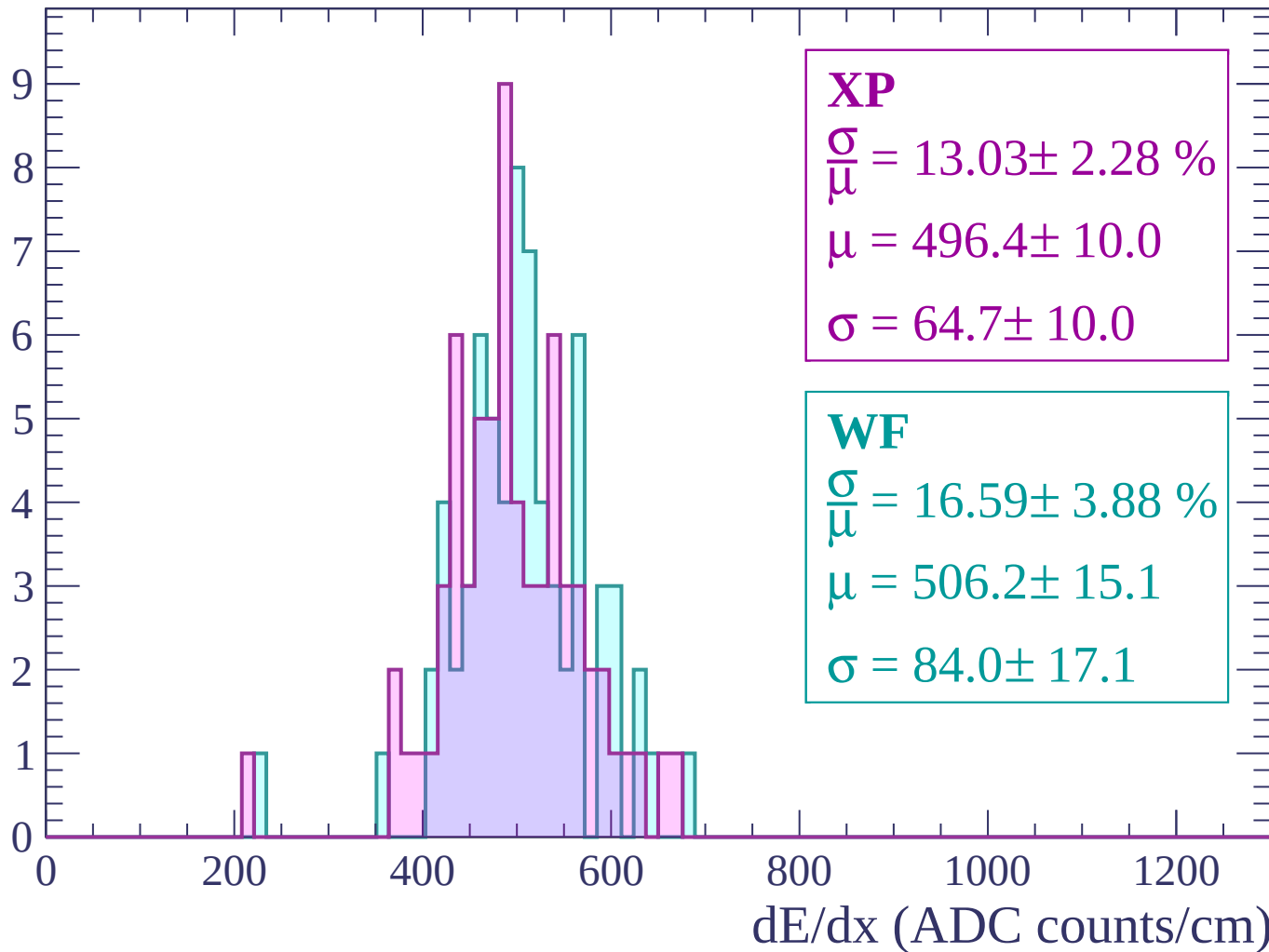
Energy loss | $2880 < p < 2940$

Count



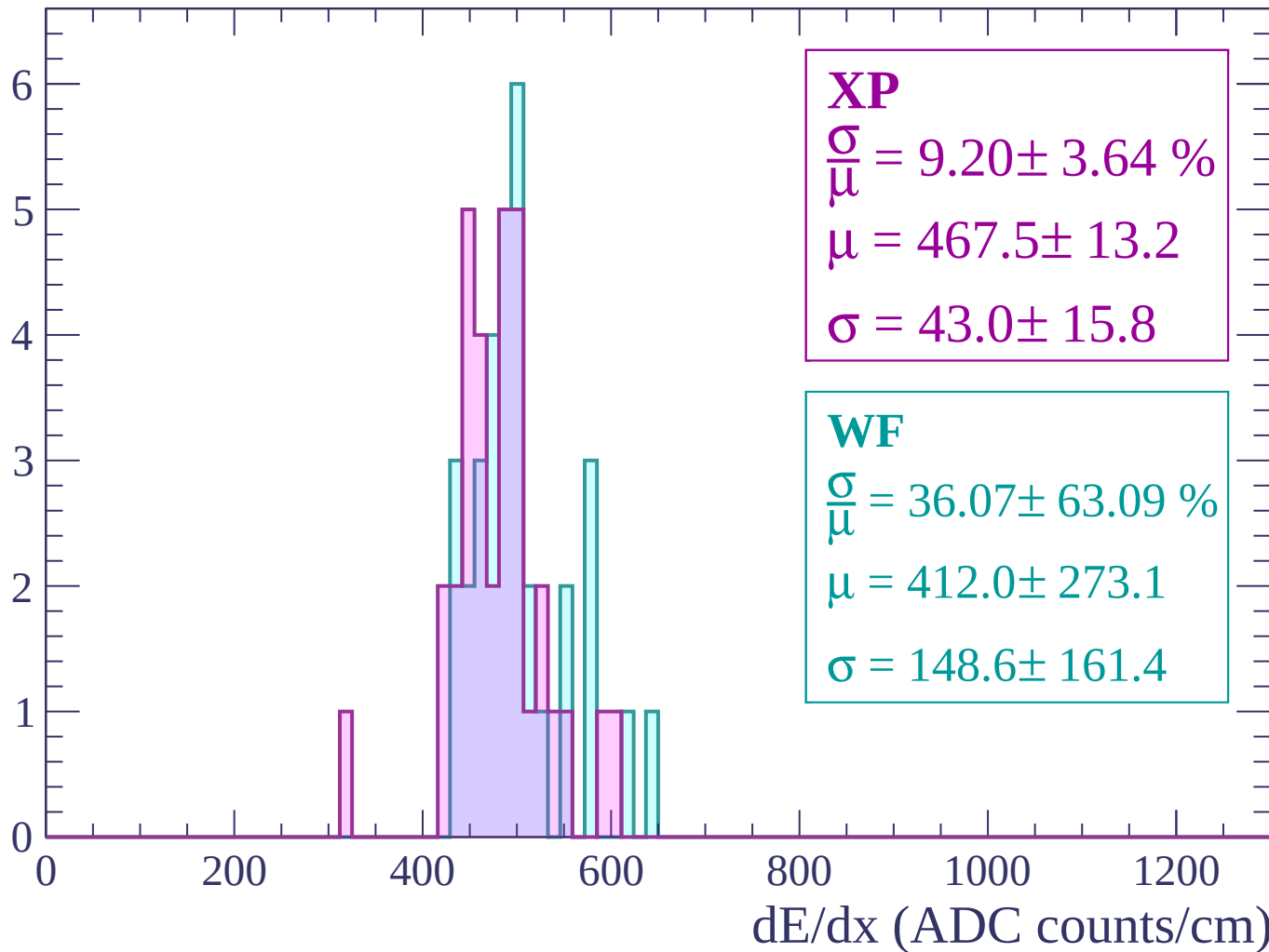
Energy loss | $2940 < p < 3000$

Count



Energy loss | $3000 < p < 3060$

Count



Energy loss | $-90 < \theta < -88$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-88 < \theta < -86$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-86 < \theta < -84$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-84 < \theta < -82$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-82 < \theta < -80$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-80 < \theta < -78$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-78 < \theta < -76$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

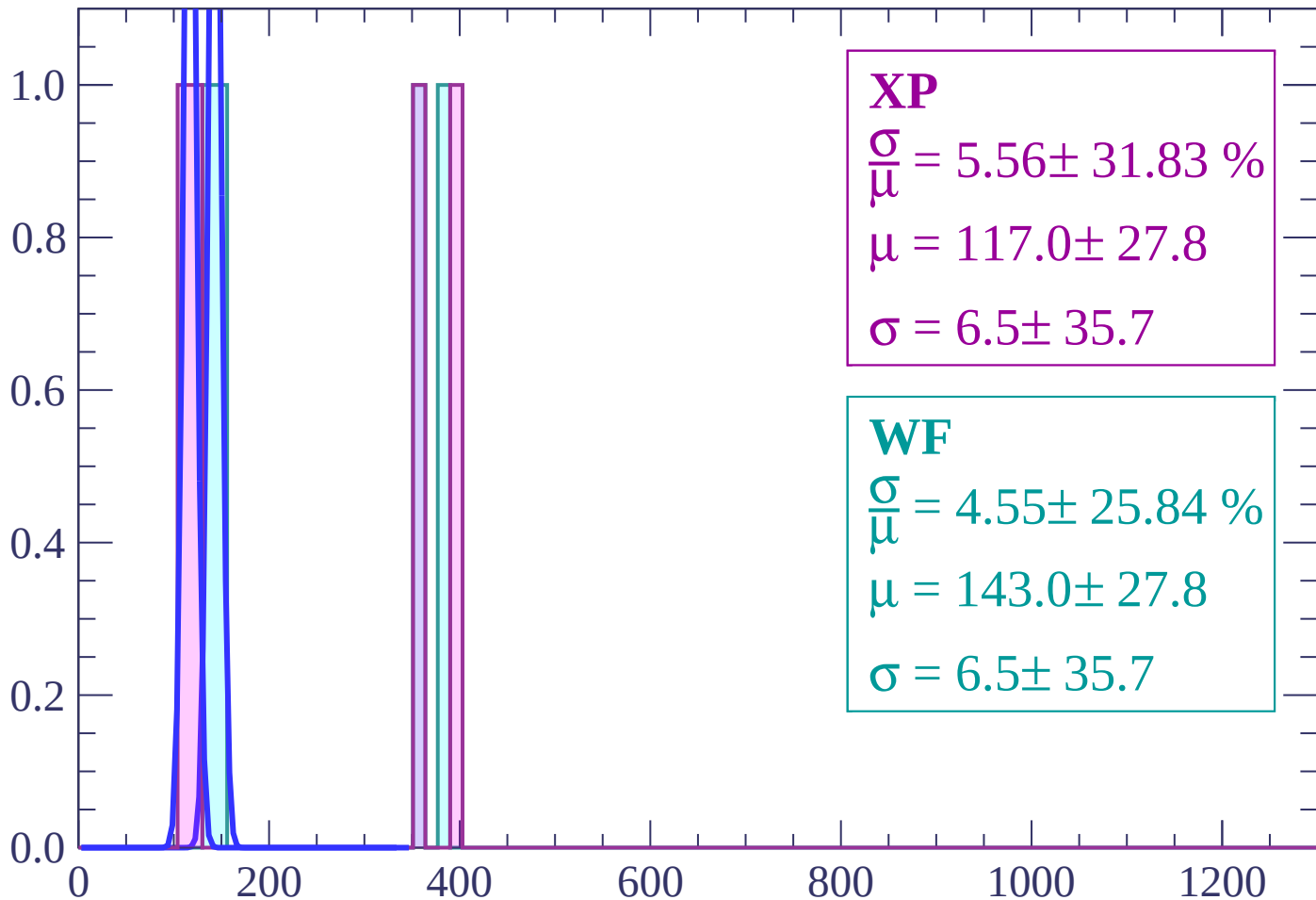
$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-76 < \theta < -74$

Count



XP

$$\frac{\sigma}{\mu} = 5.56 \pm 31.83 \%$$

$$\mu = 117.0 \pm 27.8$$

$$\sigma = 6.5 \pm 35.7$$

WF

$$\frac{\sigma}{\mu} = 4.55 \pm 25.84 \%$$

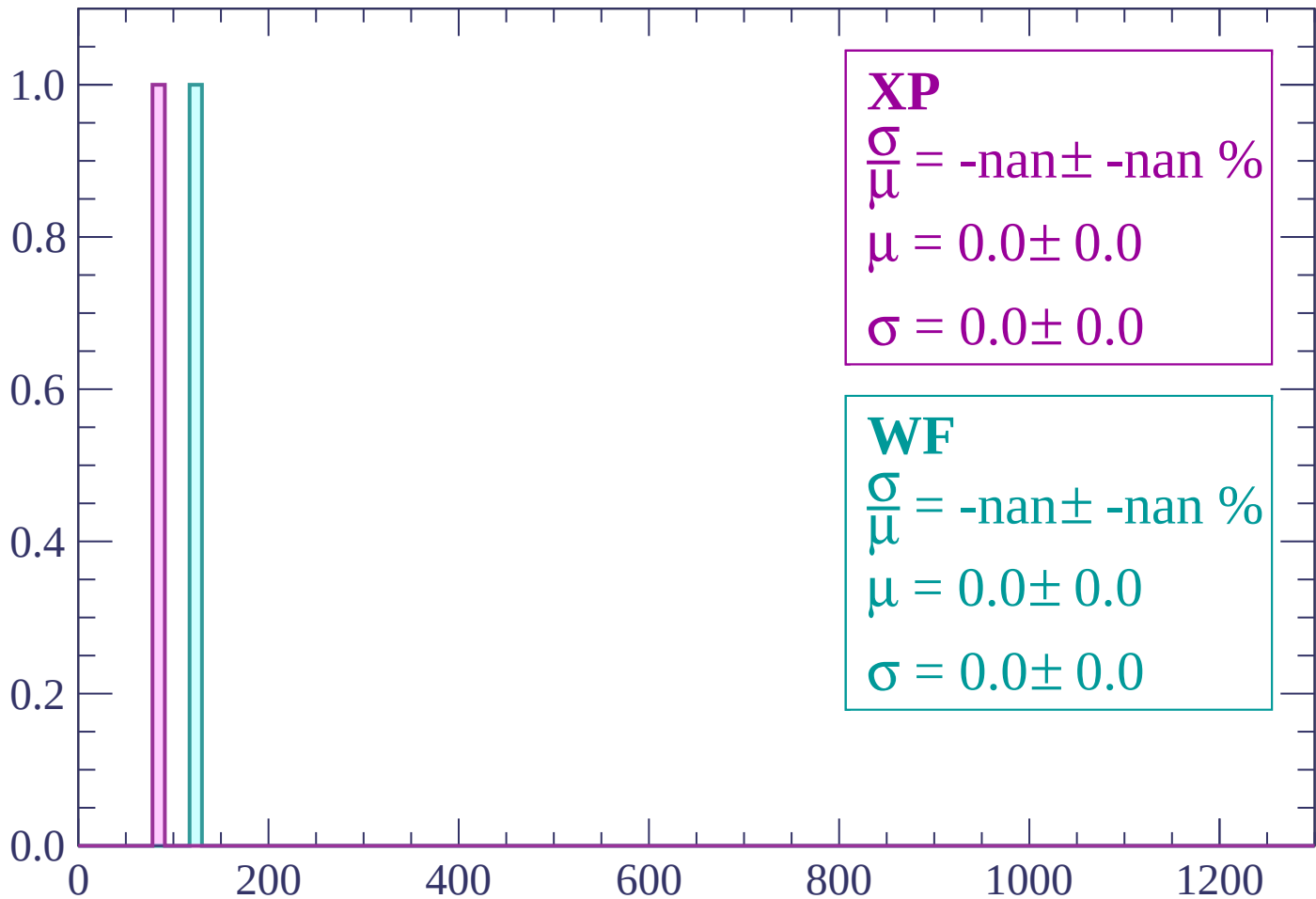
$$\mu = 143.0 \pm 27.8$$

$$\sigma = 6.5 \pm 35.7$$

dE/dx (ADC counts/cm)

Energy loss | $-74 < \theta < -72$

Count



XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

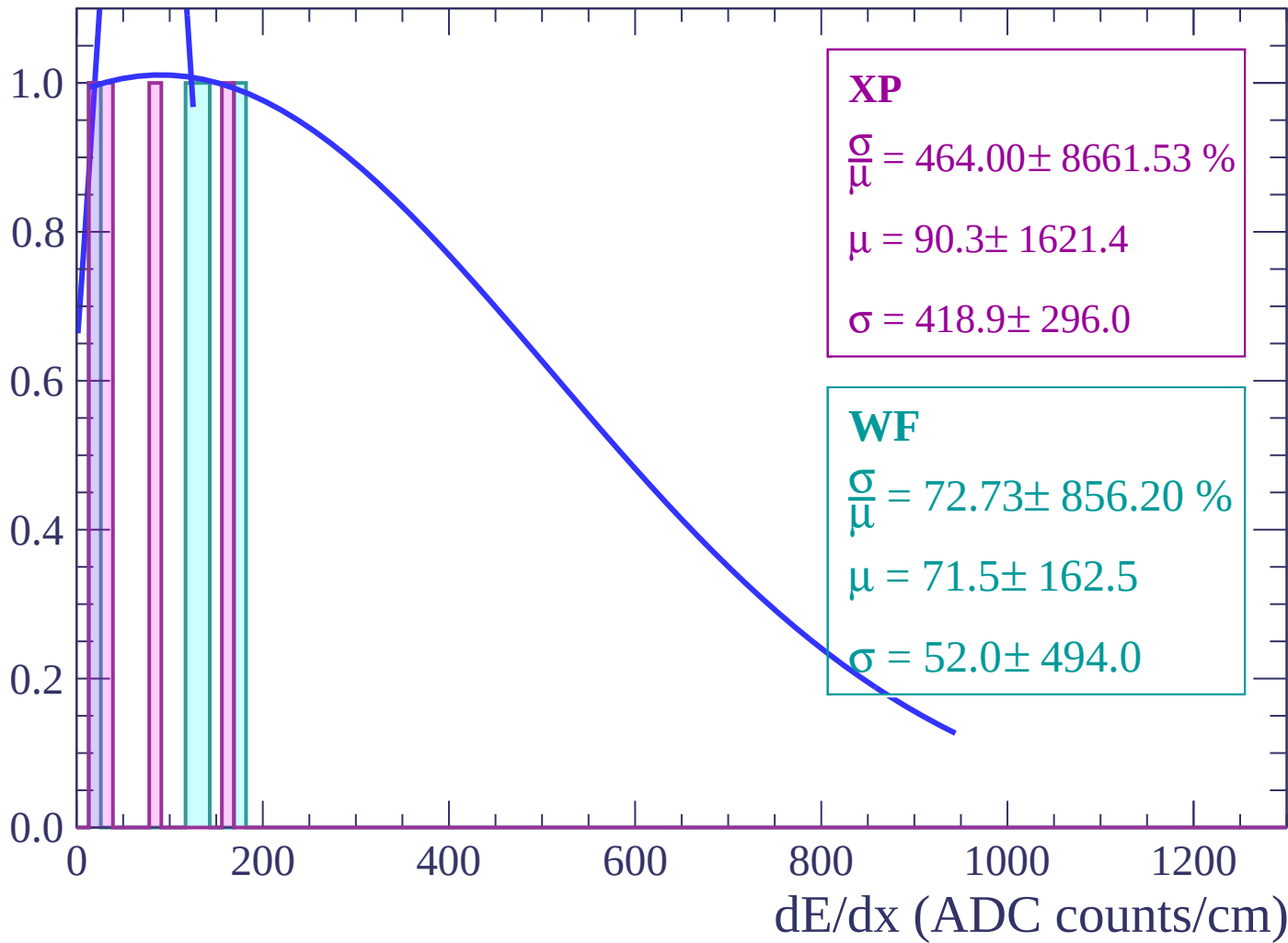
$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

dE/dx (ADC counts/cm)

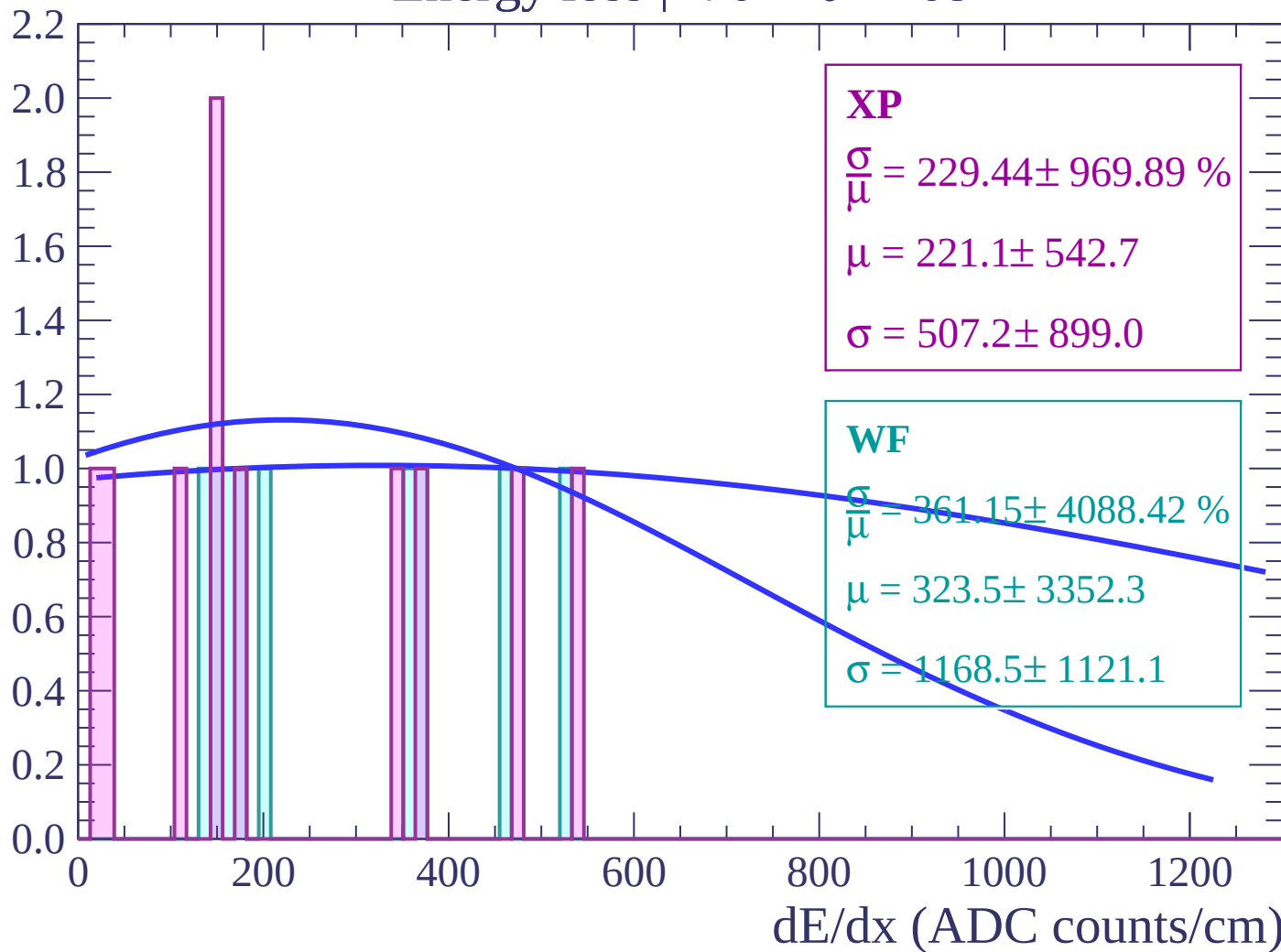
Energy loss | $-72 < \theta < -70$

Count



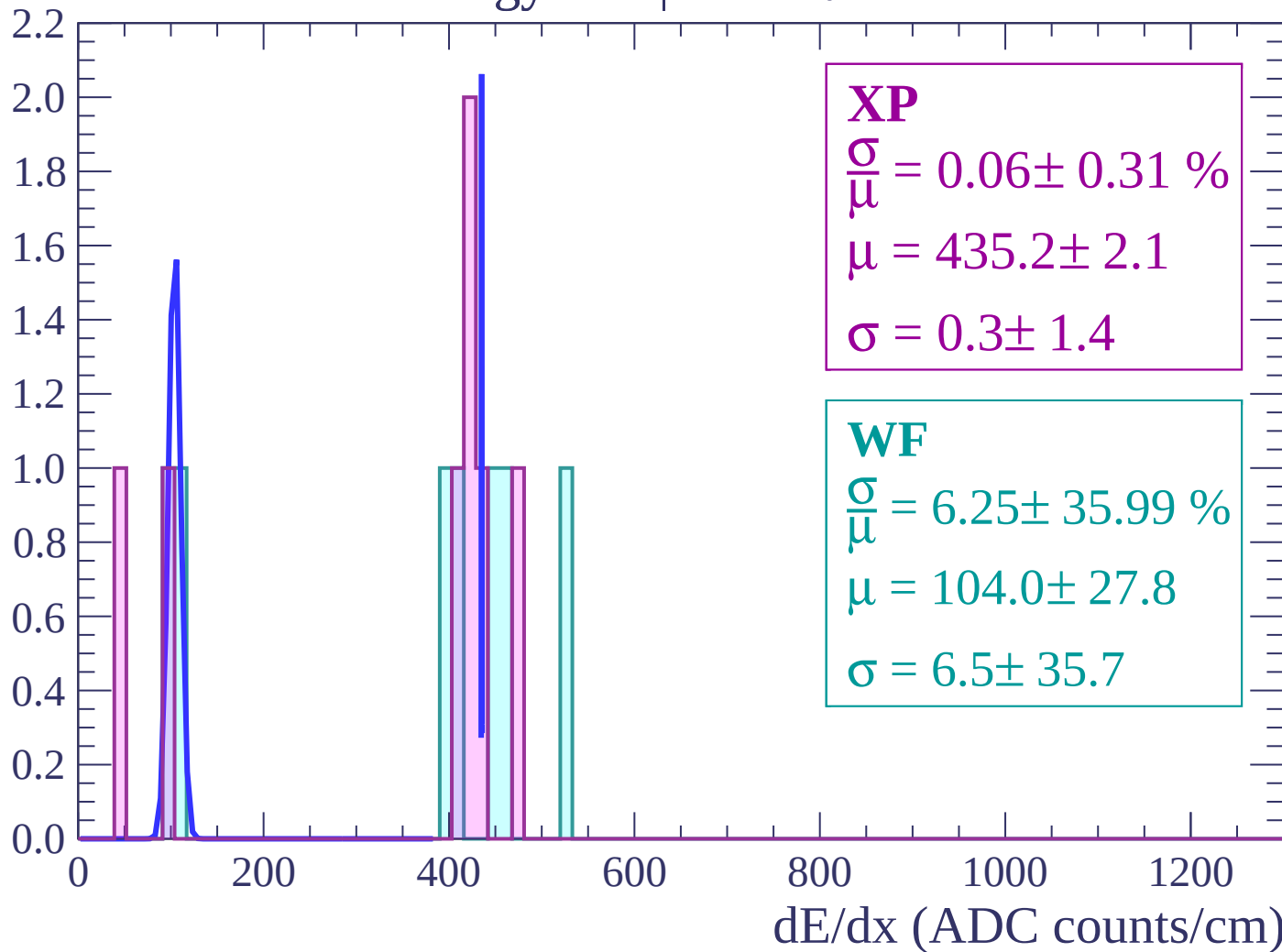
Energy loss | $-70 < \theta < -68$

Count



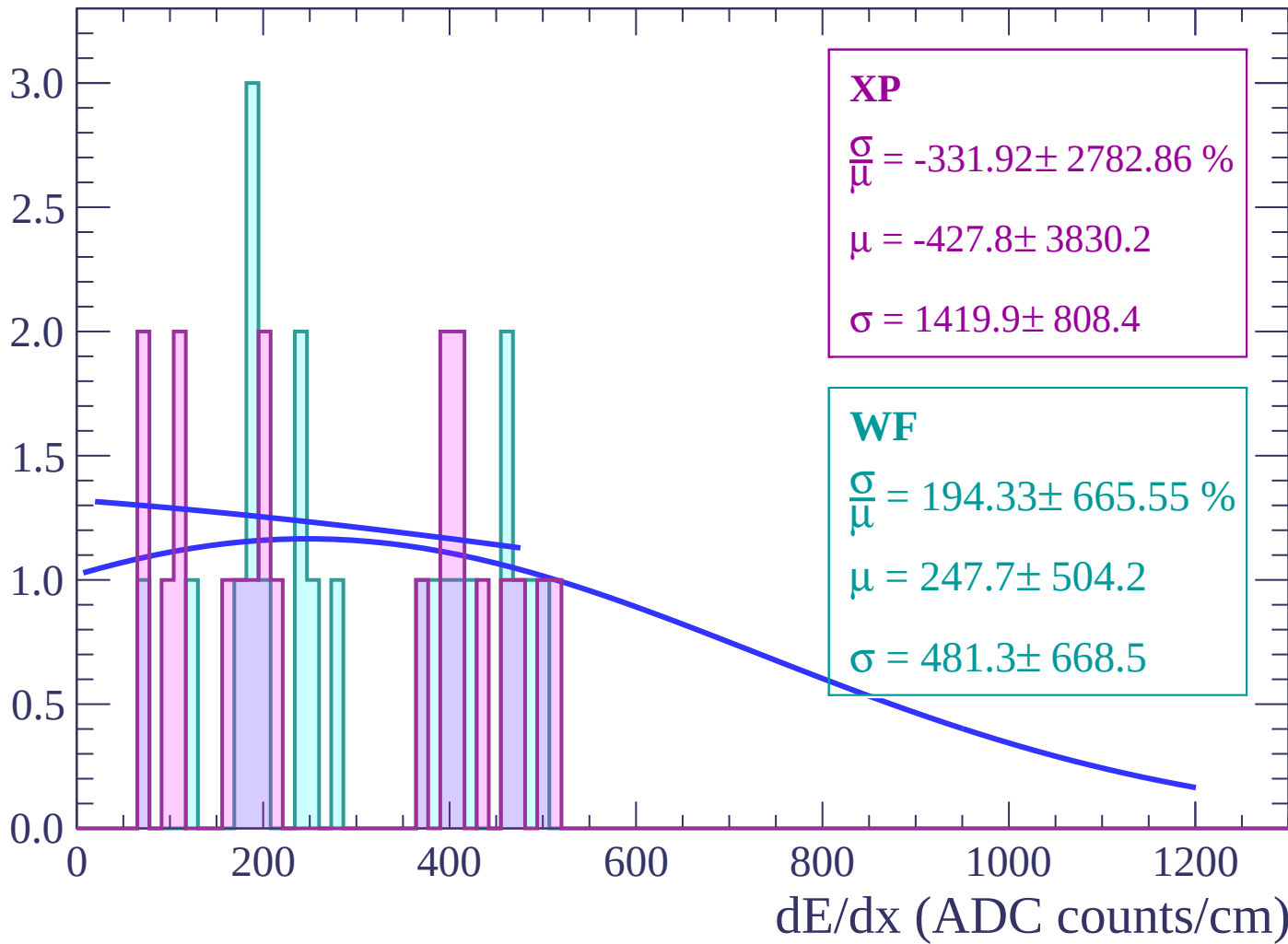
Energy loss | $-68 < \theta < -66$

Count



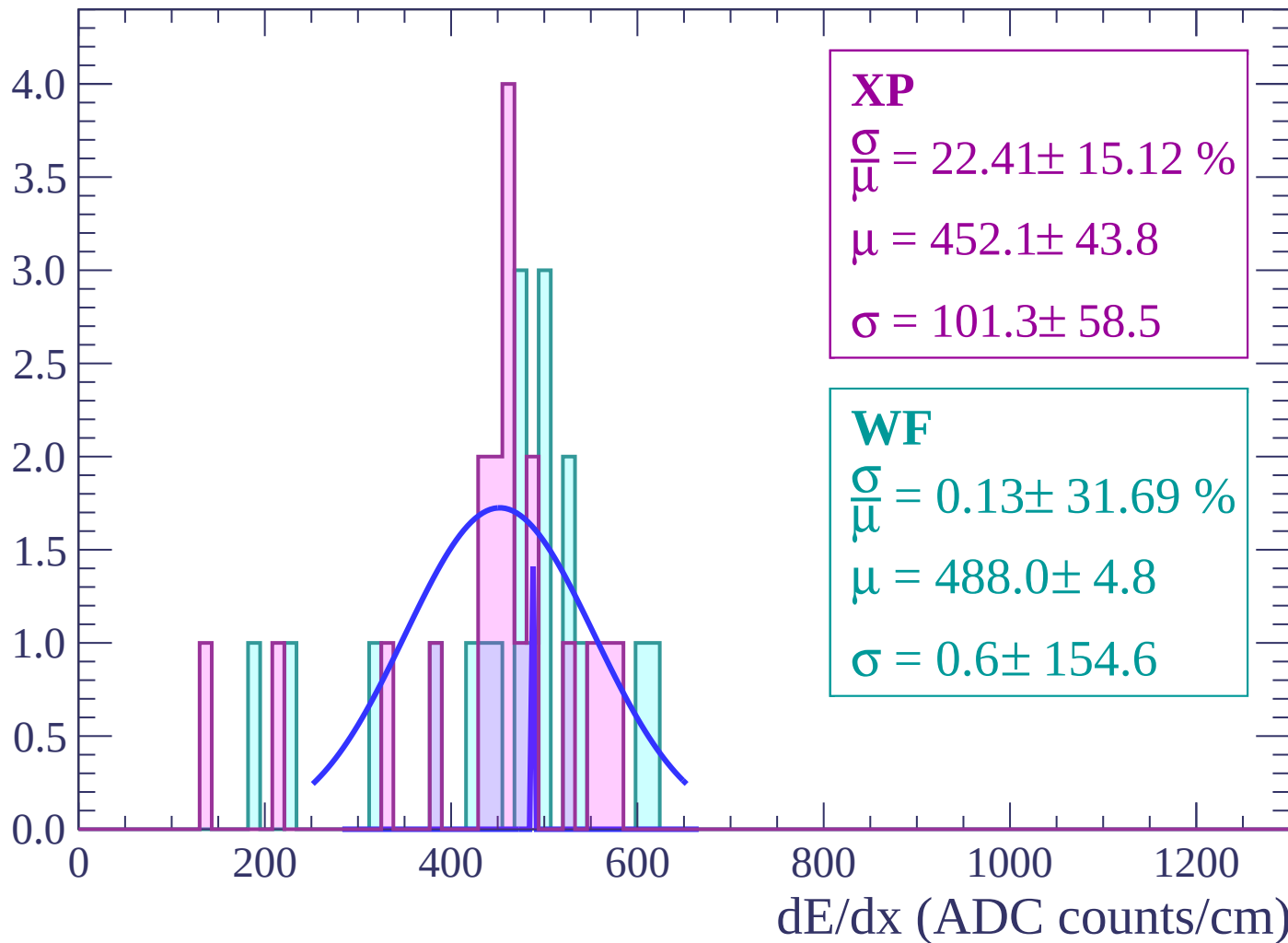
Energy loss | $-66 < \theta < -64$

Count



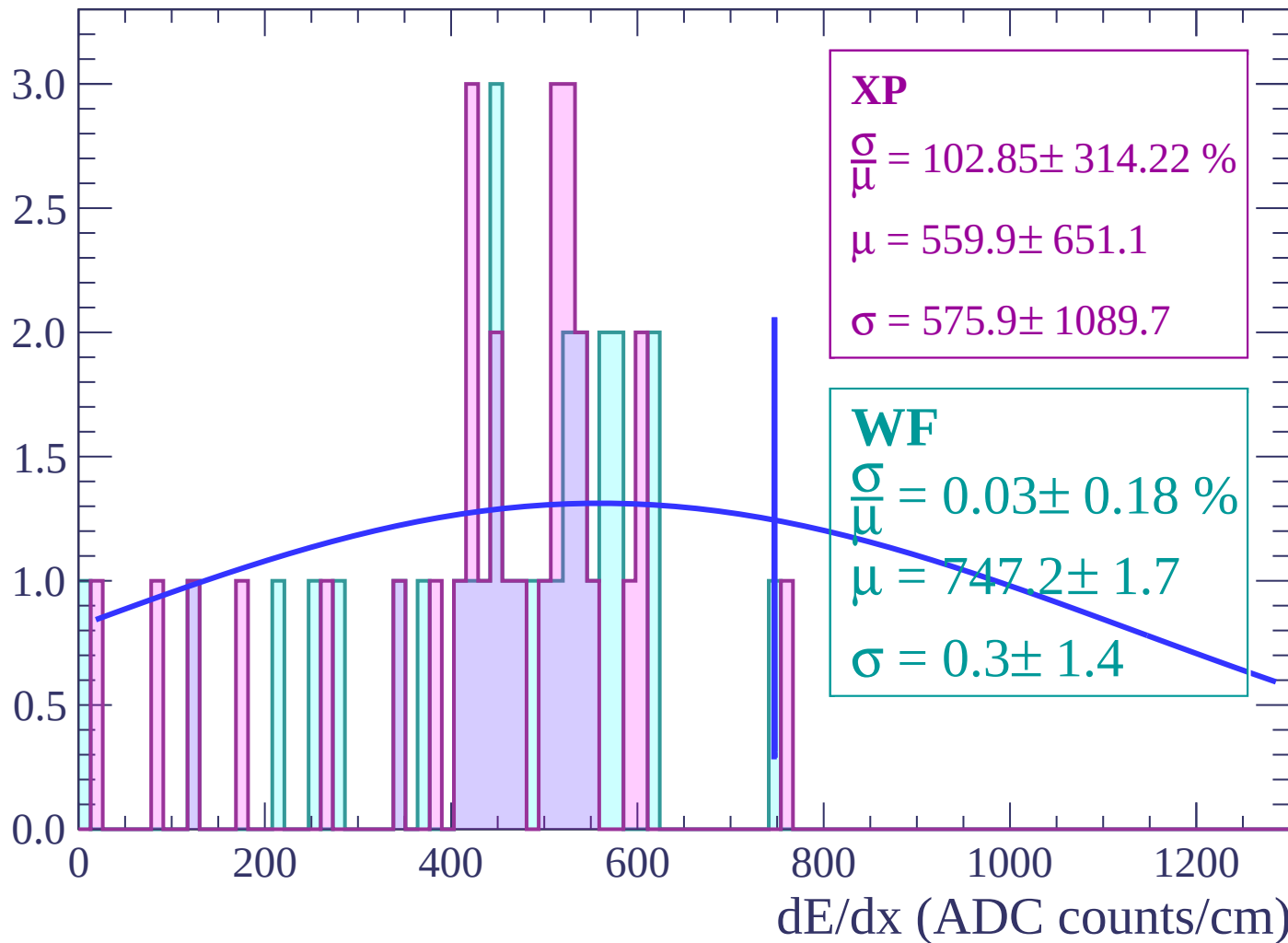
Energy loss | $-64 < \theta < -62$

Count



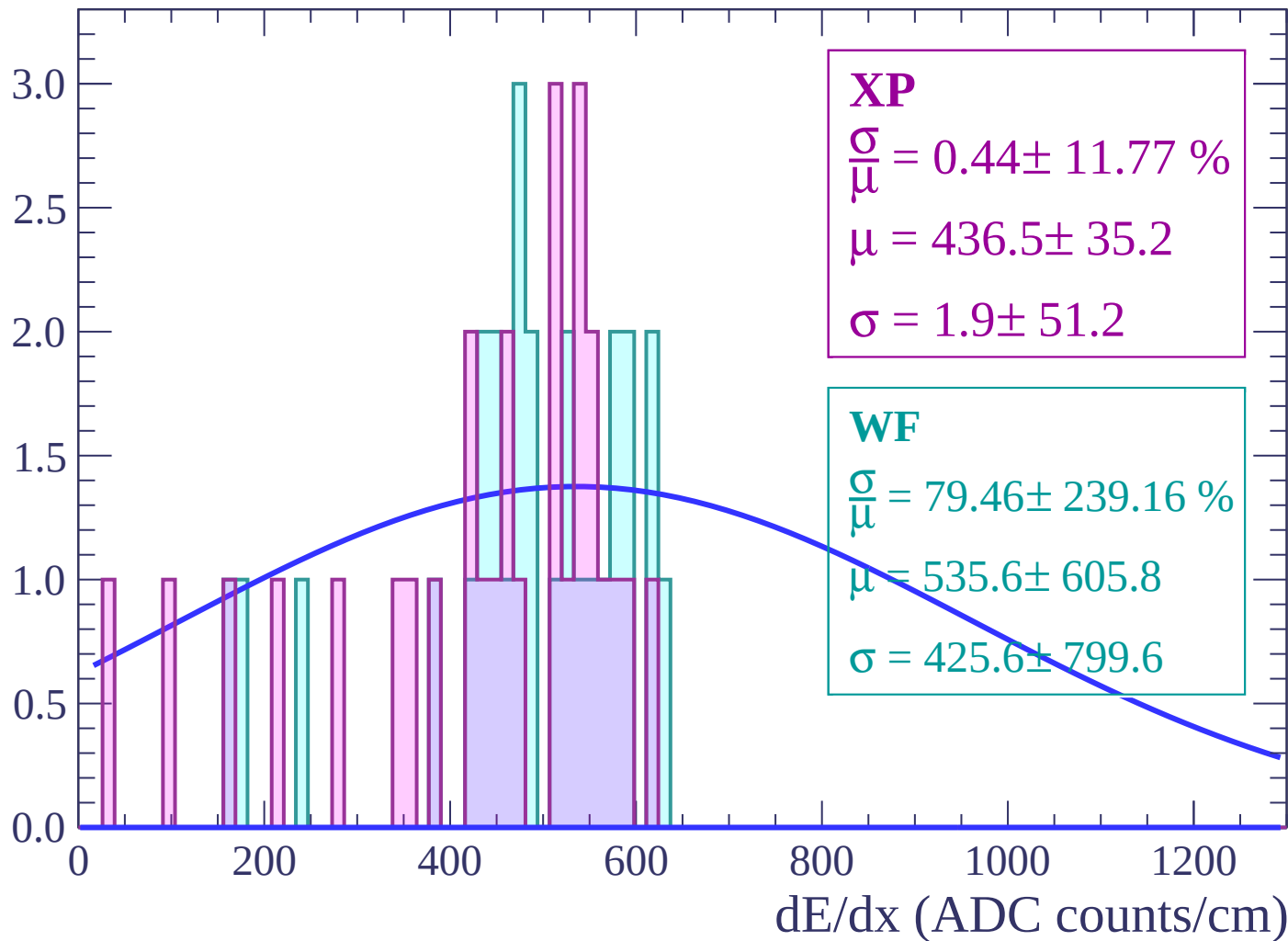
Energy loss | $-62 < \theta < -60$

Count



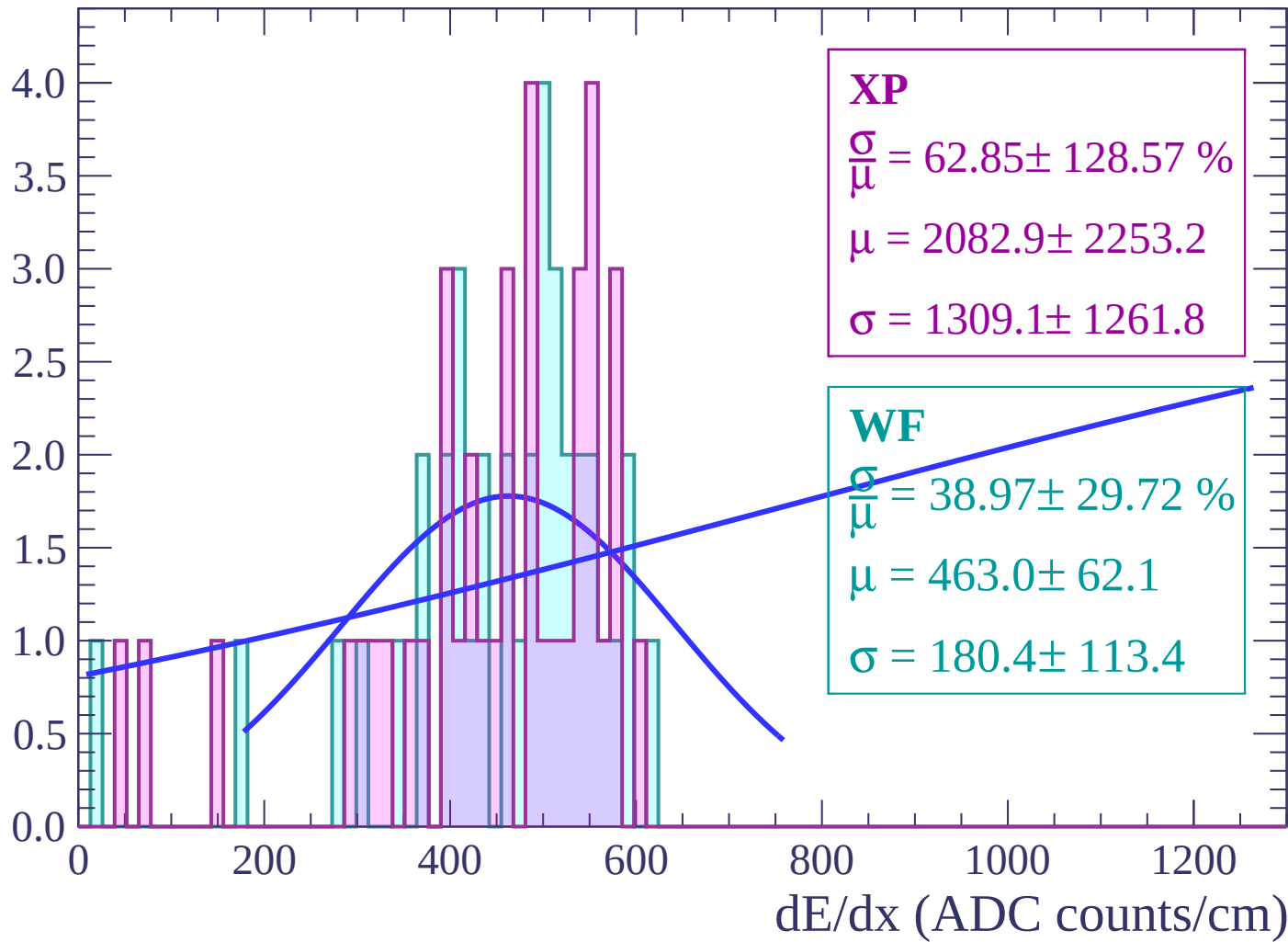
Energy loss | $-60 < \theta < -58$

Count



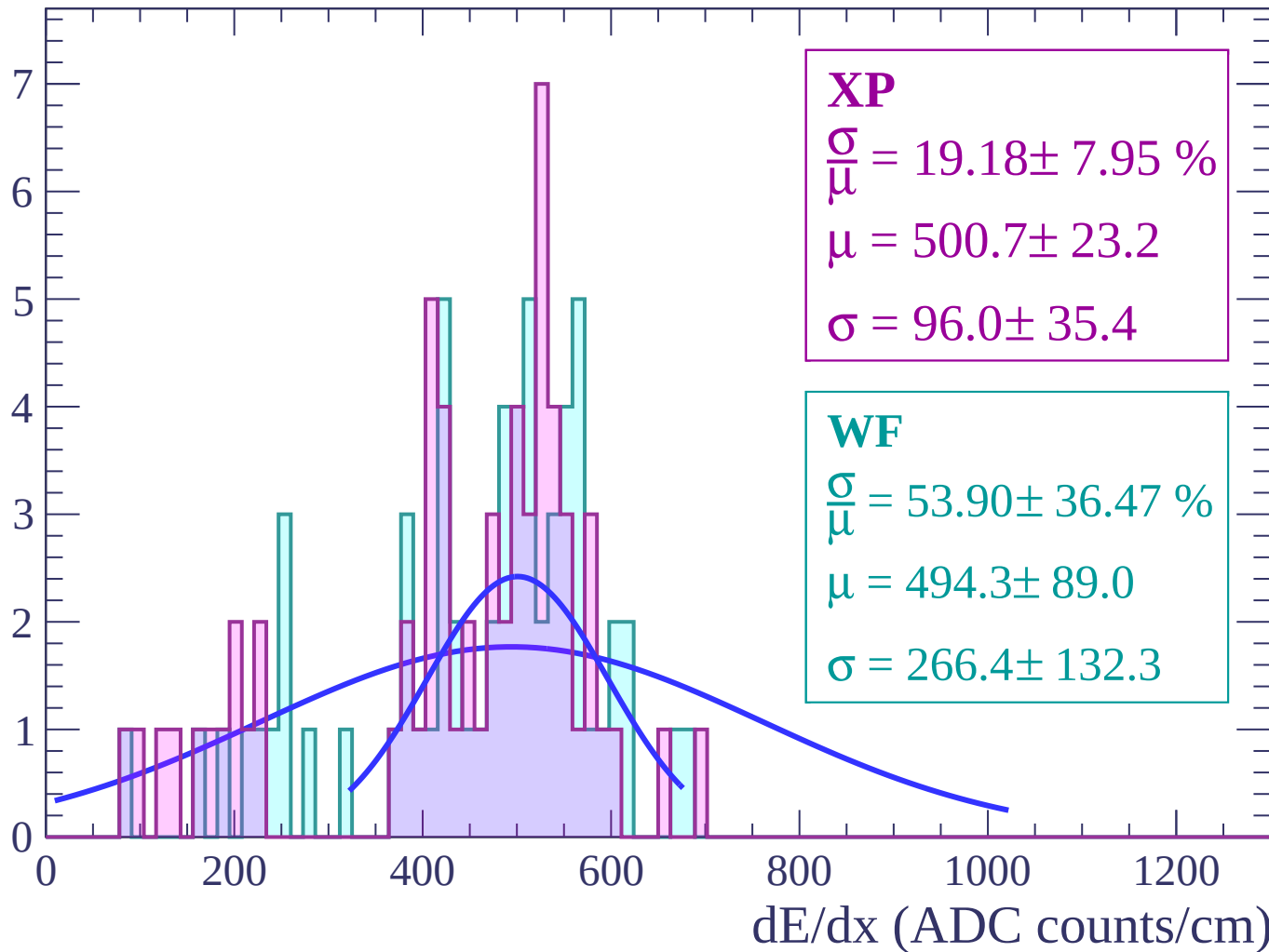
Energy loss | $-58 < \theta < -56$

Count



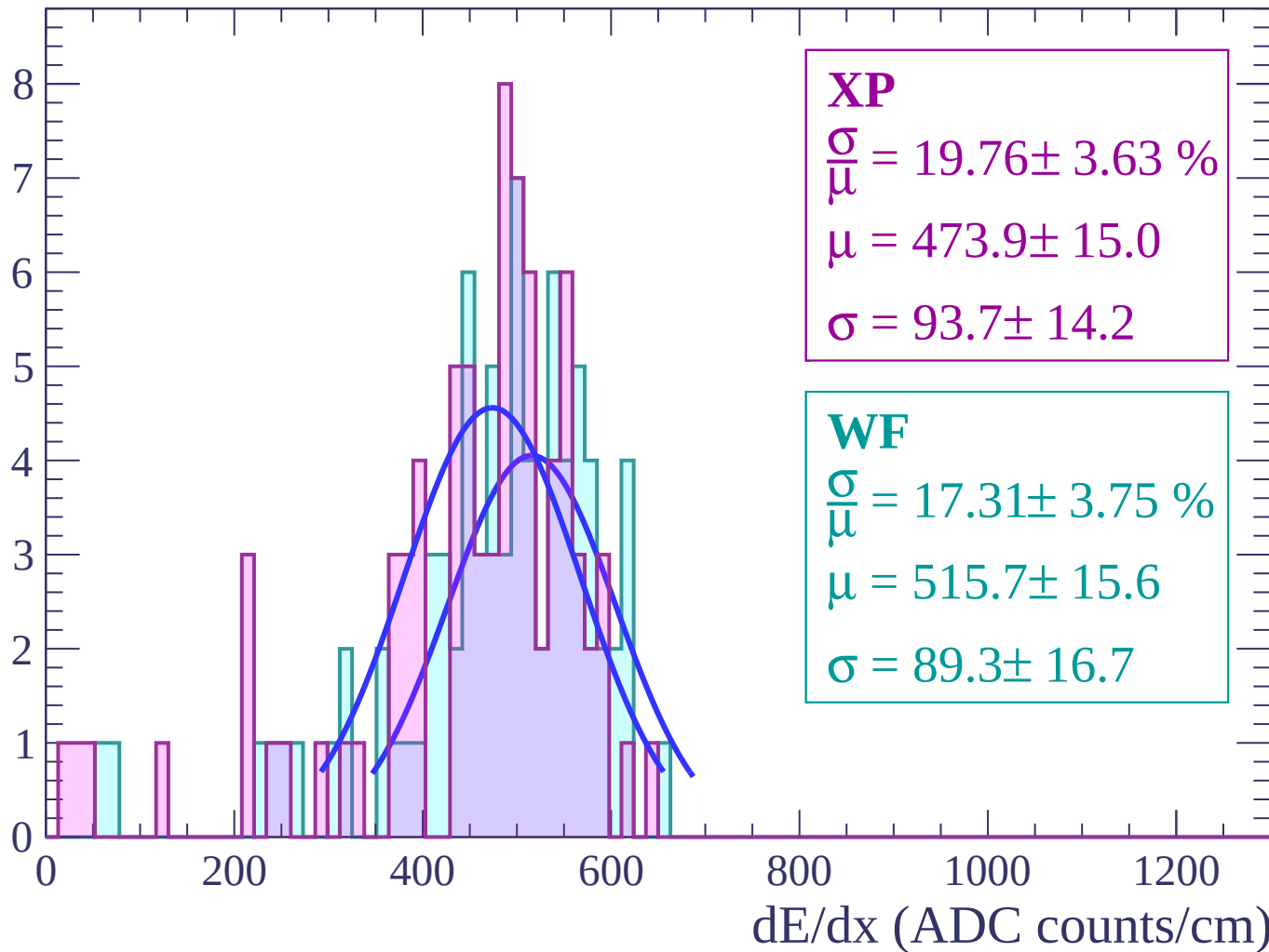
Energy loss | $-56 < \theta < -54$

Count



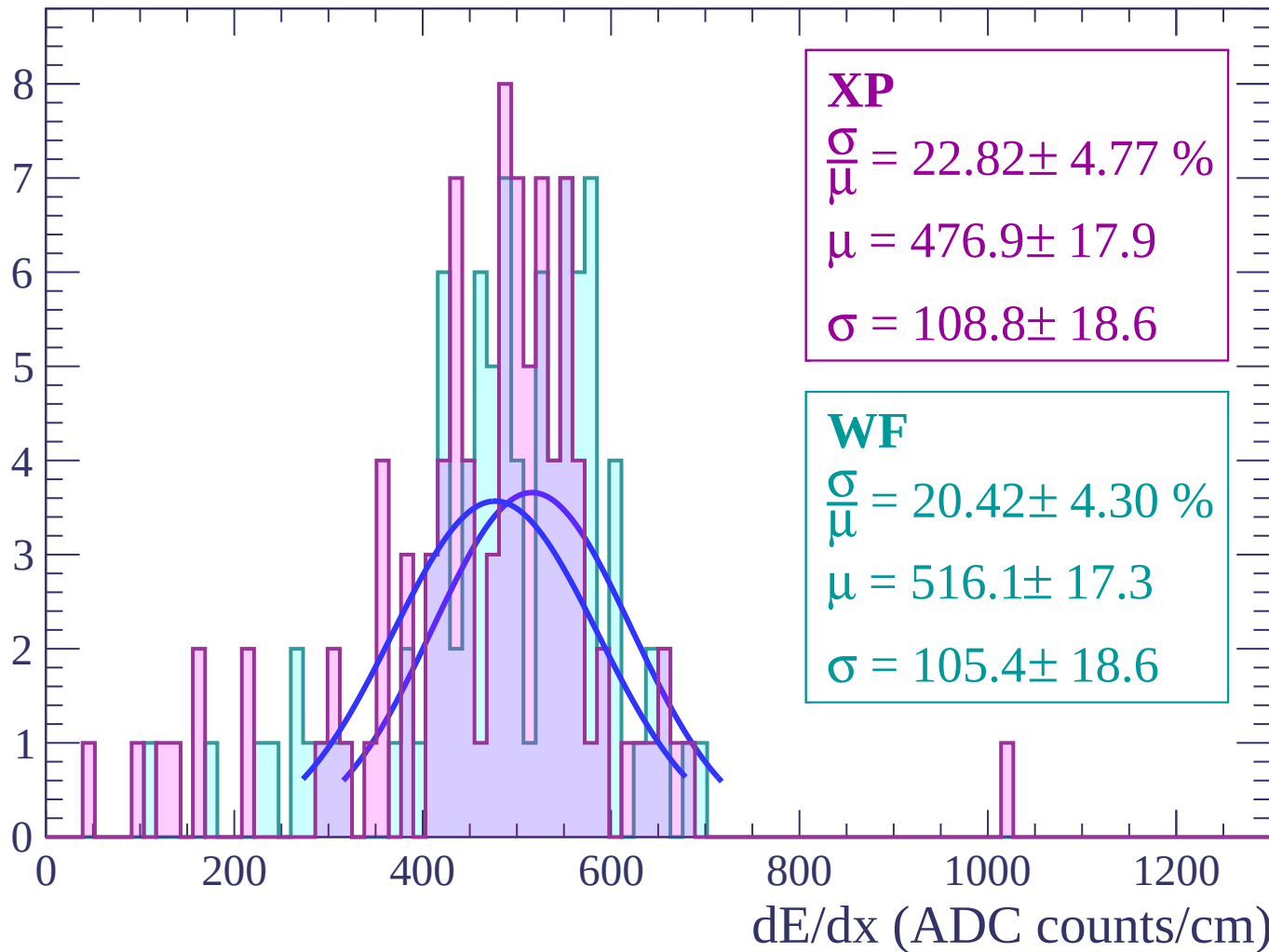
Energy loss | $-54 < \theta < -52$

Count



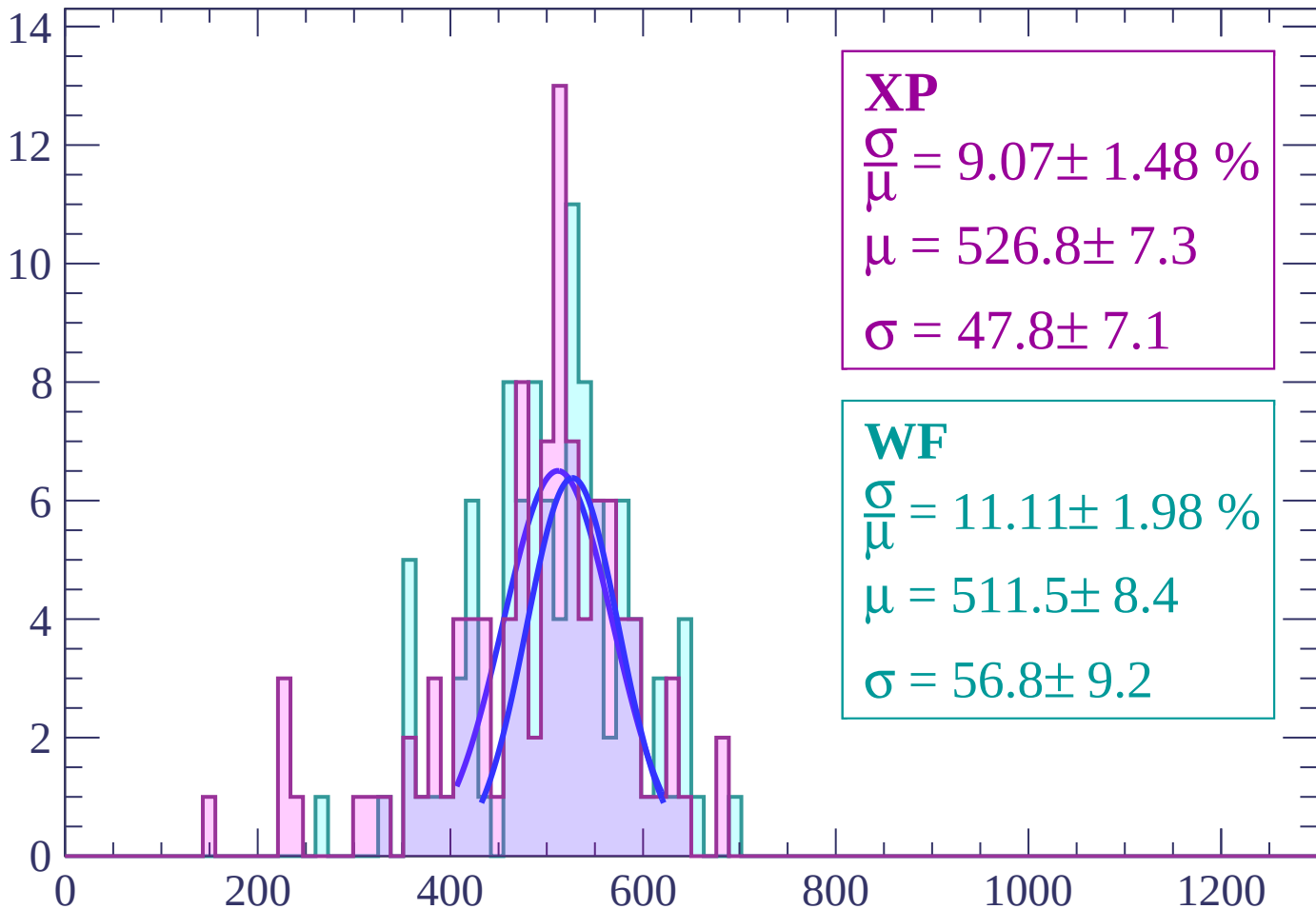
Energy loss | $-52 < \theta < -50$

Count



Energy loss | $-50 < \theta < -48$

Count



XP

$$\frac{\sigma}{\mu} = 9.07 \pm 1.48 \%$$

$$\mu = 526.8 \pm 7.3$$

$$\sigma = 47.8 \pm 7.1$$

WF

$$\frac{\sigma}{\mu} = 11.11 \pm 1.98 \%$$

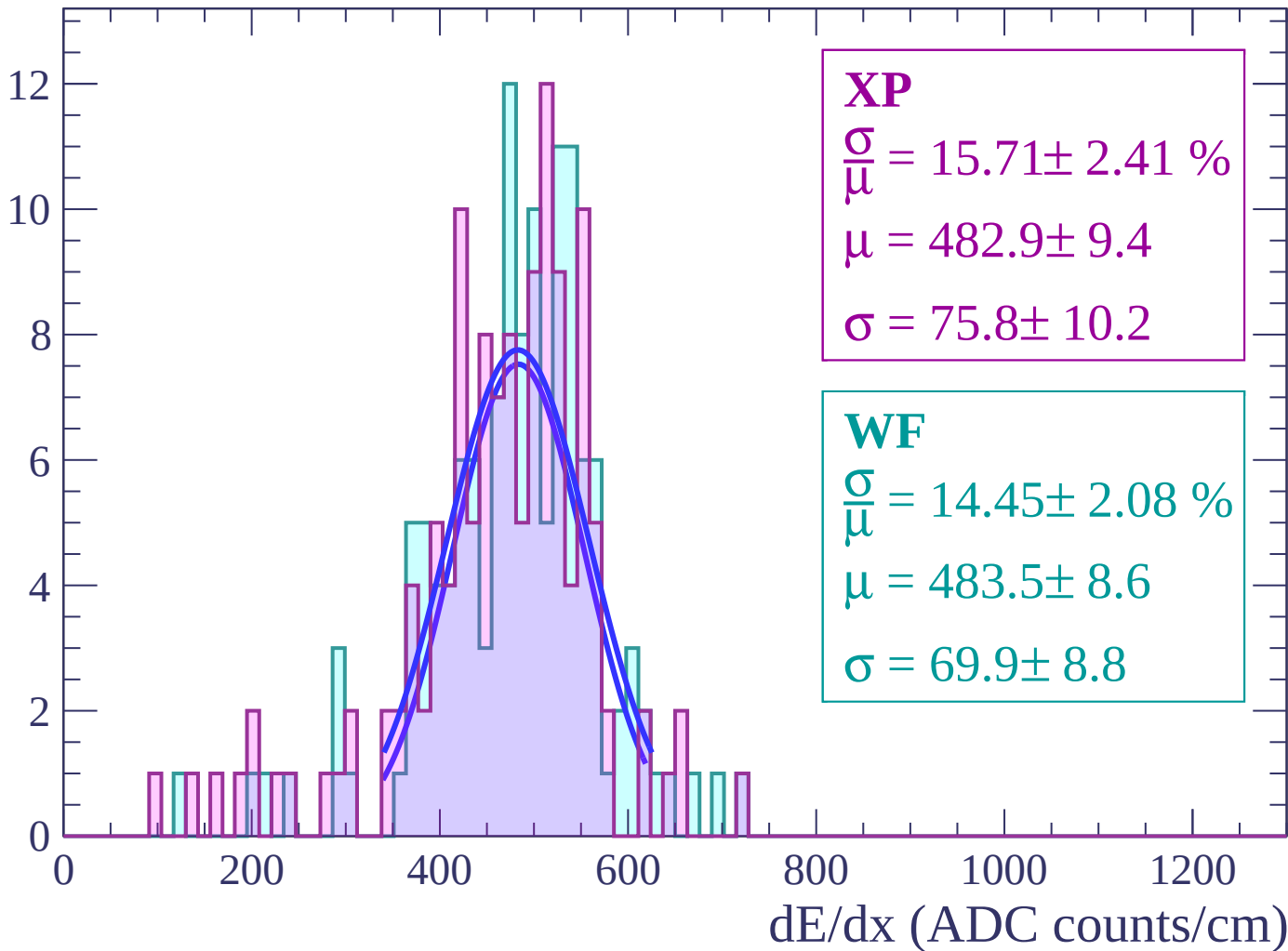
$$\mu = 511.5 \pm 8.4$$

$$\sigma = 56.8 \pm 9.2$$

dE/dx (ADC counts/cm)

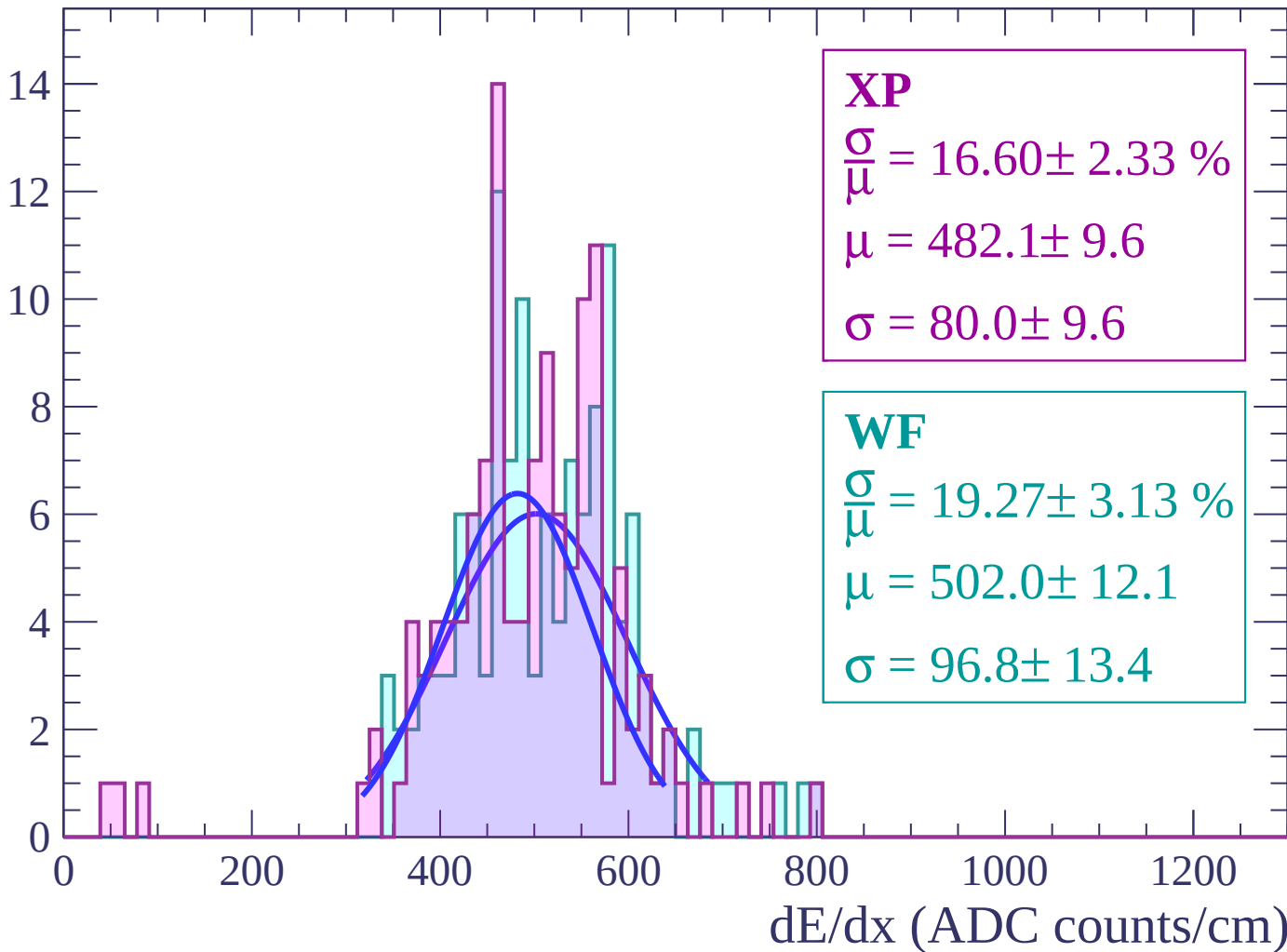
Energy loss | $-48 < \theta < -46$

Count



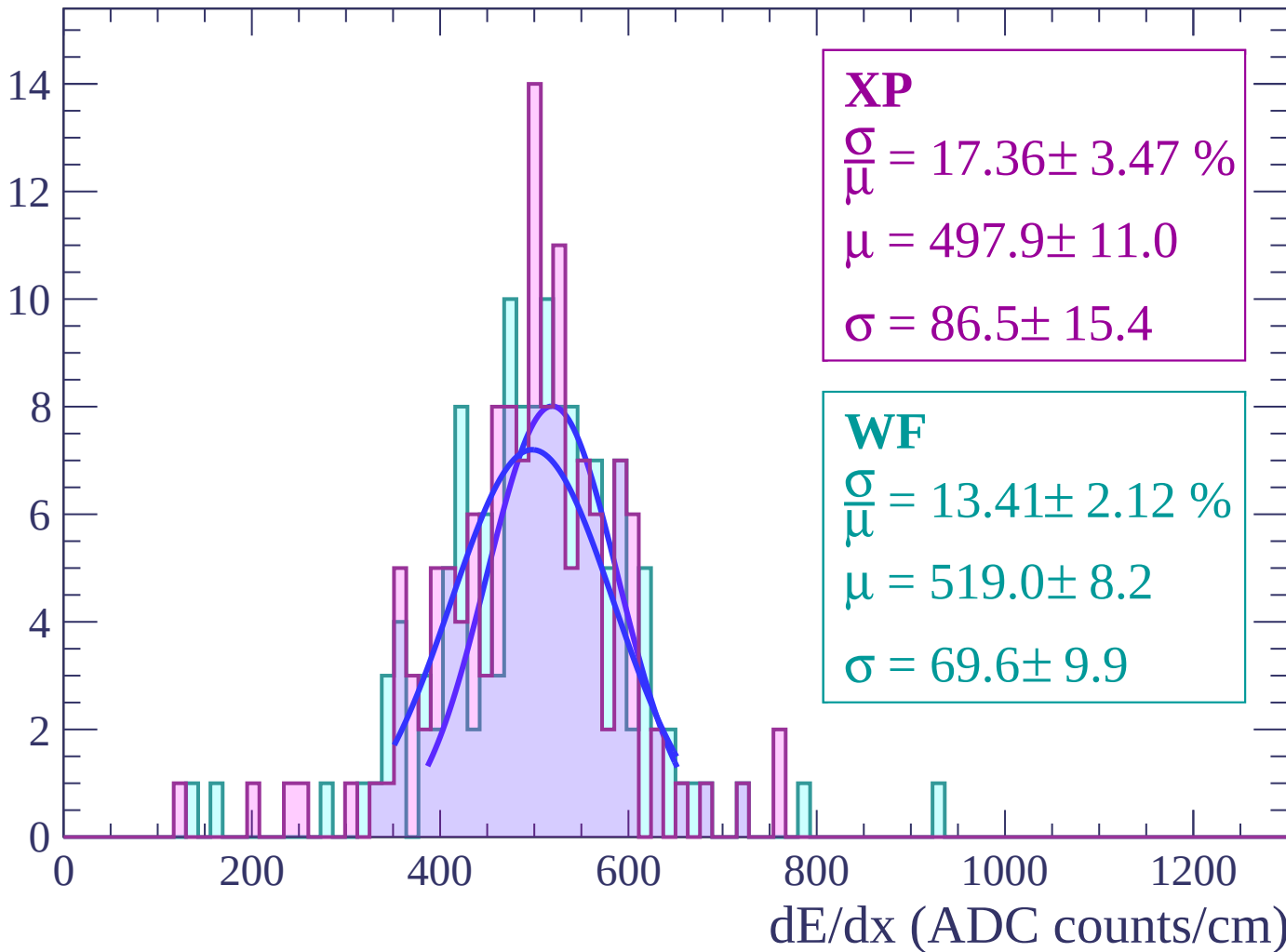
Energy loss | $-46 < \theta < -44$

Count



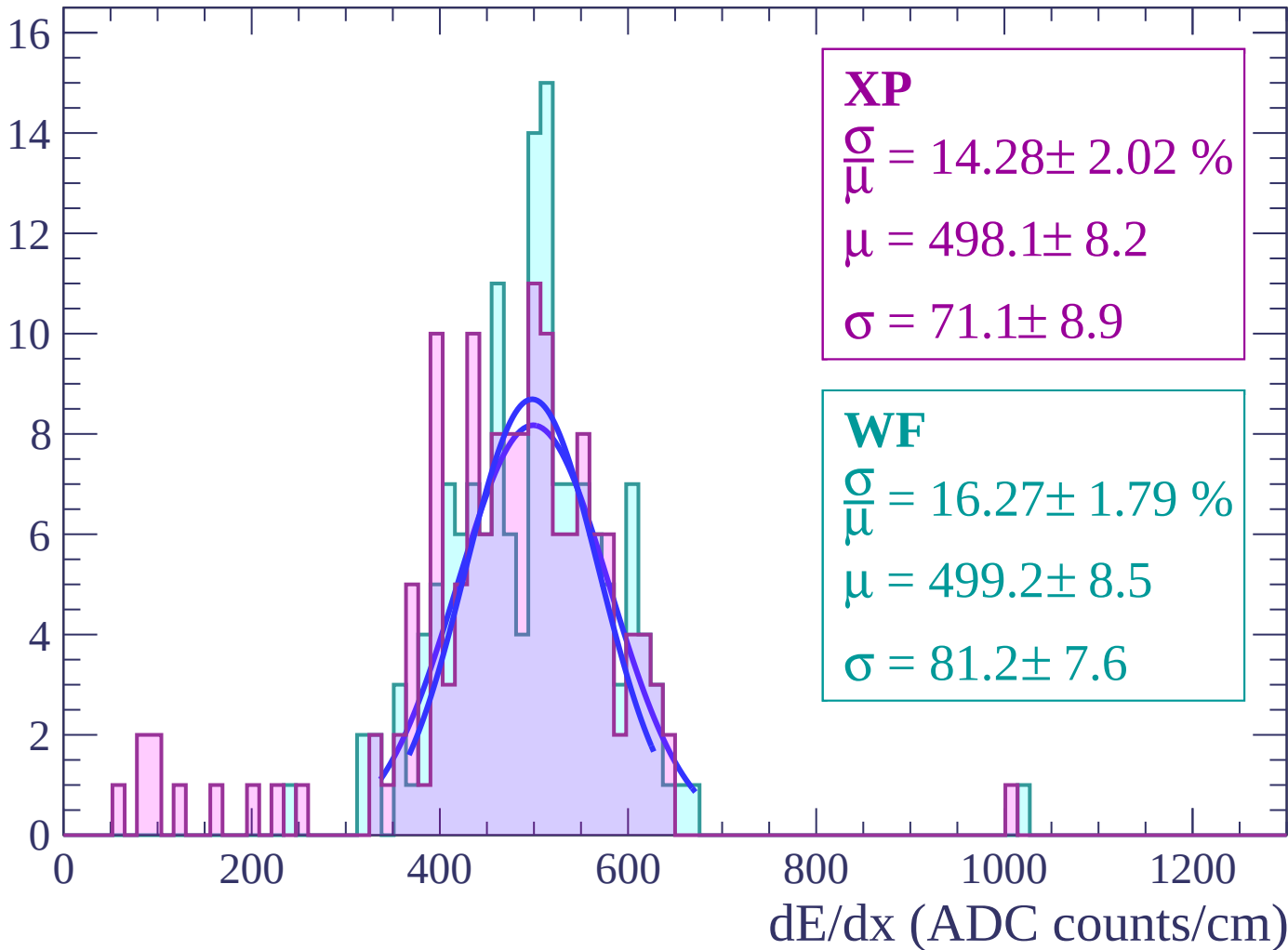
Energy loss | $-44 < \theta < -42$

Count



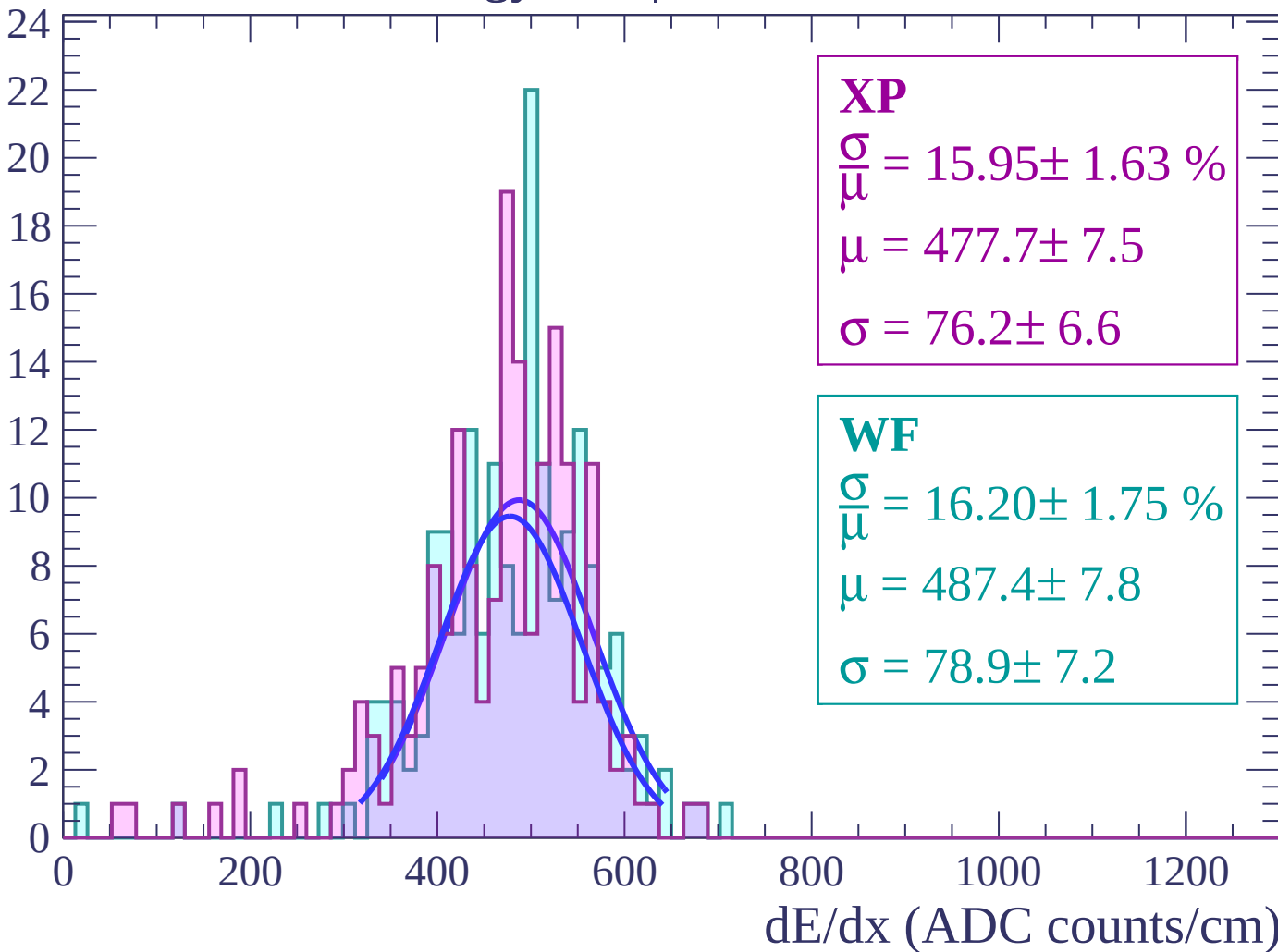
Energy loss | $-42 < \theta < -40$

Count



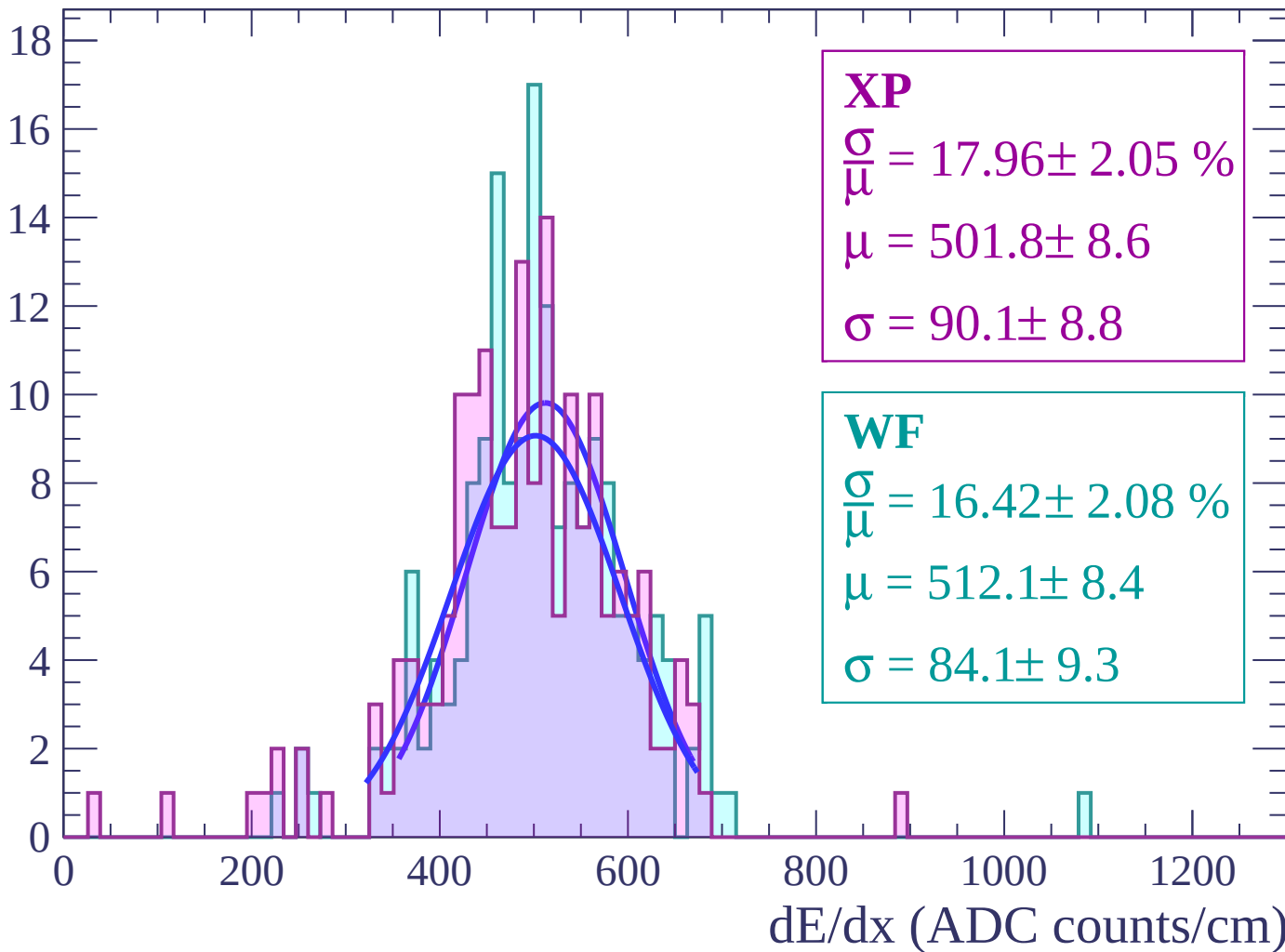
Energy loss | $-40 < \theta < -38$

Count



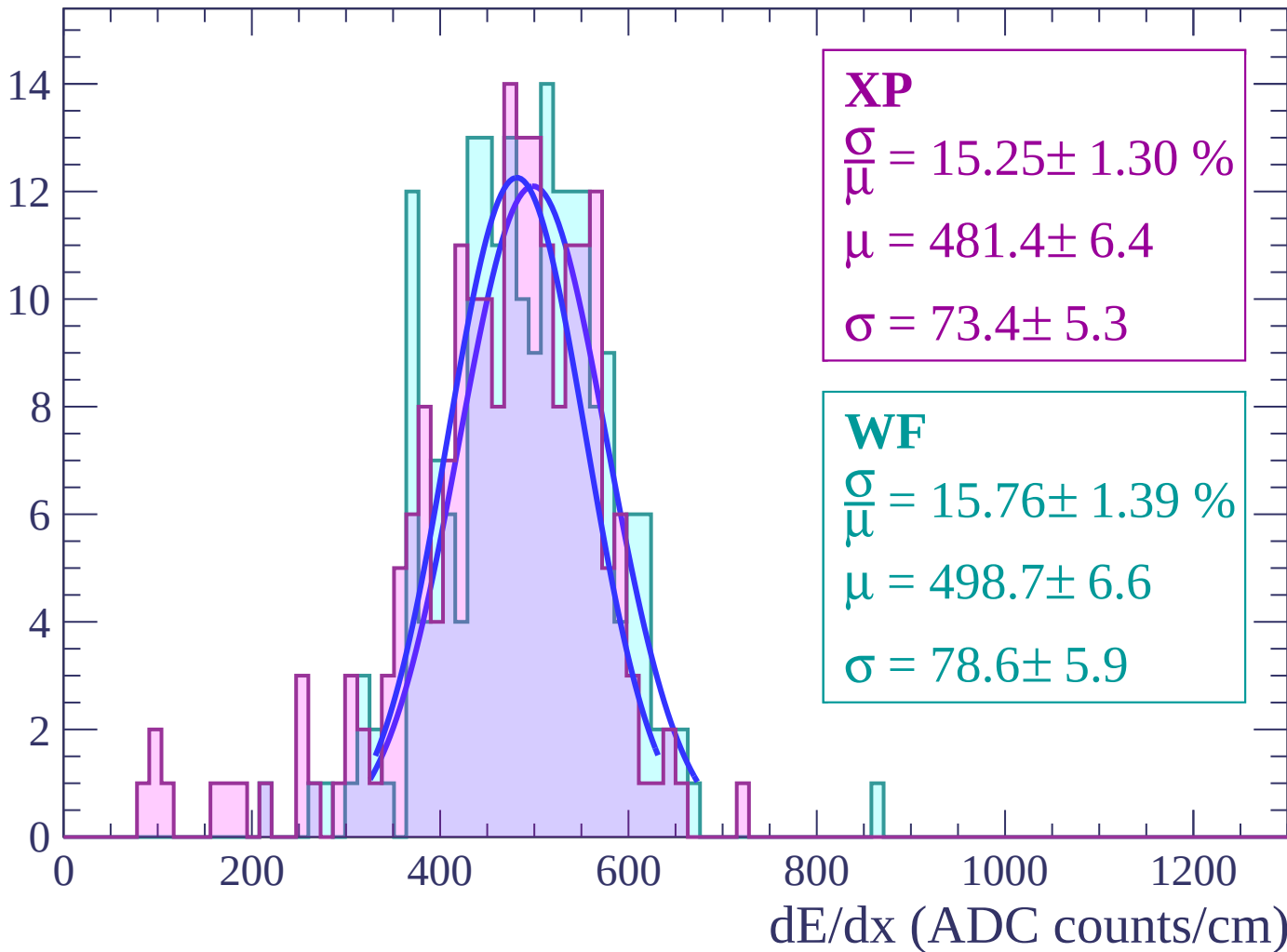
Energy loss | $-38 < \theta < -36$

Count



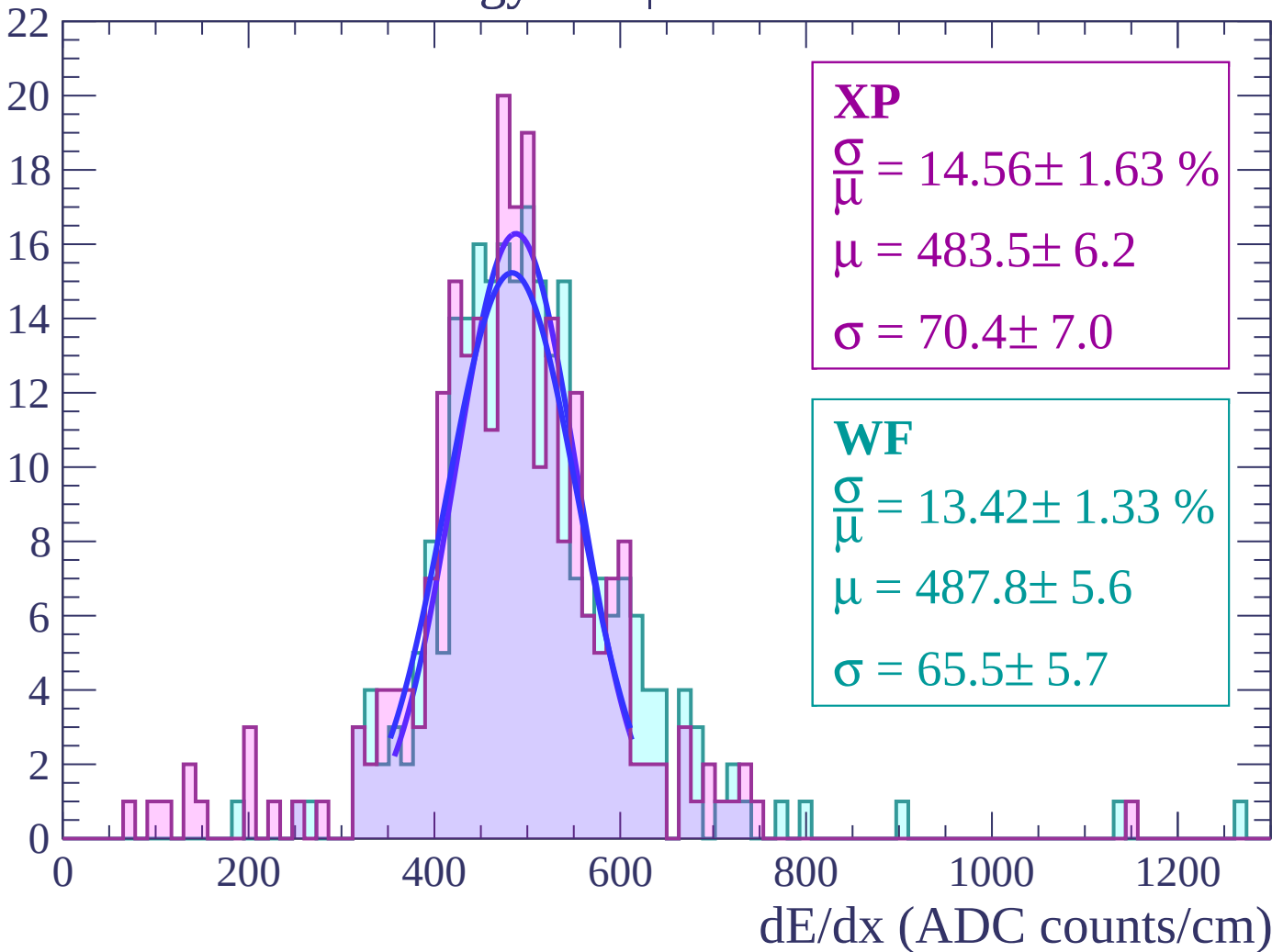
Energy loss | $-36 < \theta < -34$

Count



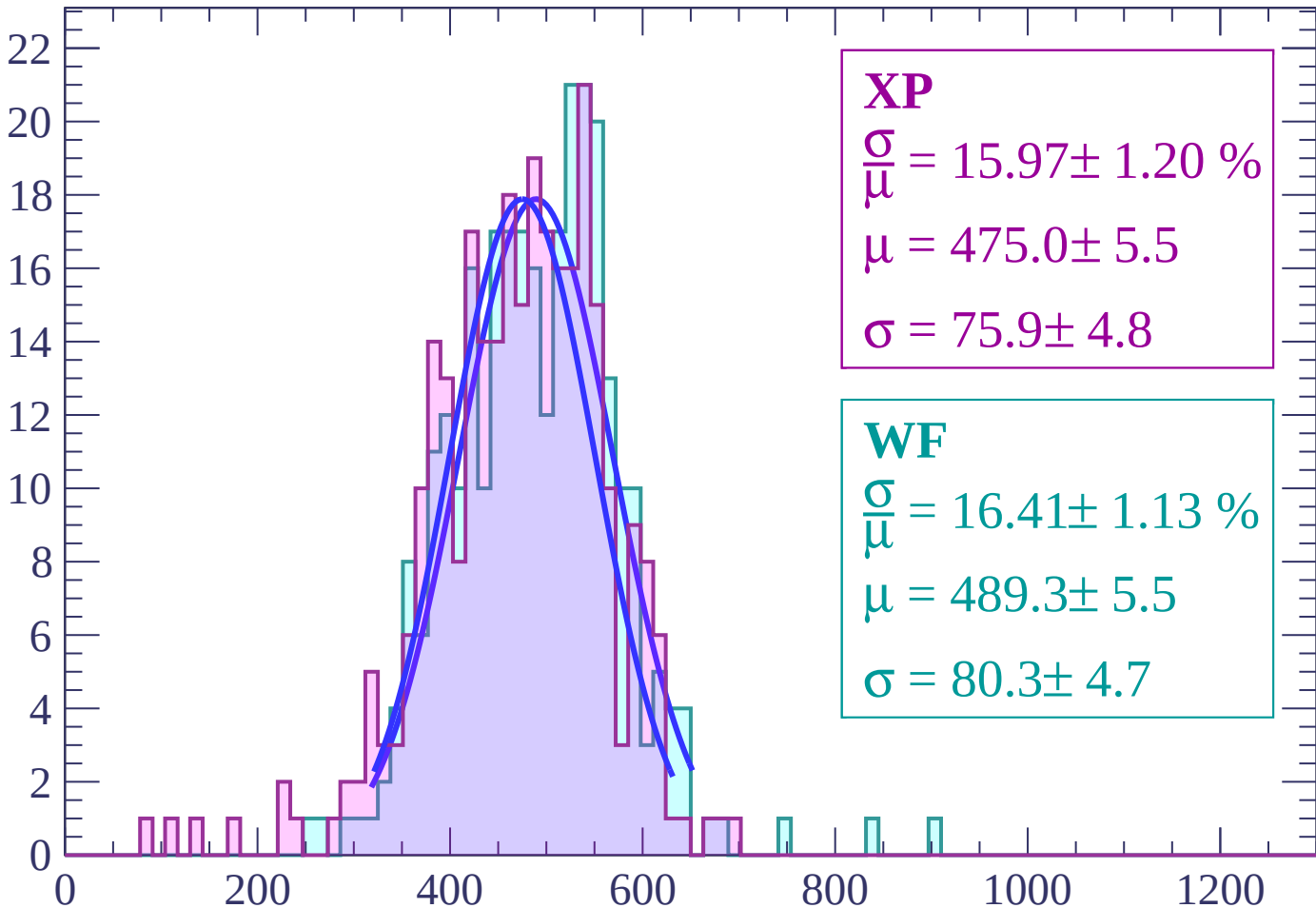
Energy loss | $-34 < \theta < -32$

Count



Energy loss | $-32 < \theta < -30$

Count



XP

$$\frac{\sigma}{\mu} = 15.97 \pm 1.20 \%$$

$$\mu = 475.0 \pm 5.5$$

$$\sigma = 75.9 \pm 4.8$$

WF

$$\frac{\sigma}{\mu} = 16.41 \pm 1.13 \%$$

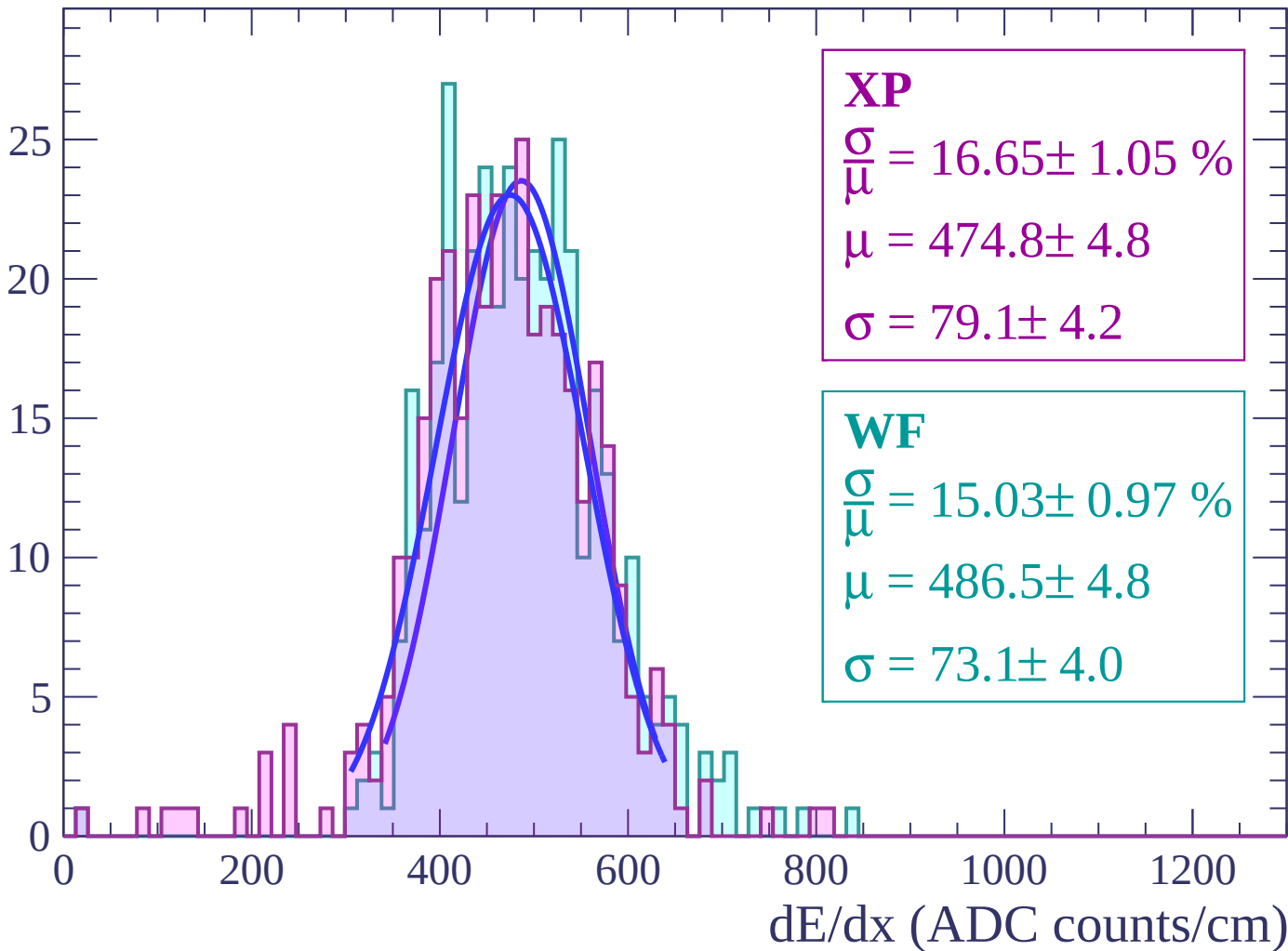
$$\mu = 489.3 \pm 5.5$$

$$\sigma = 80.3 \pm 4.7$$

dE/dx (ADC counts/cm)

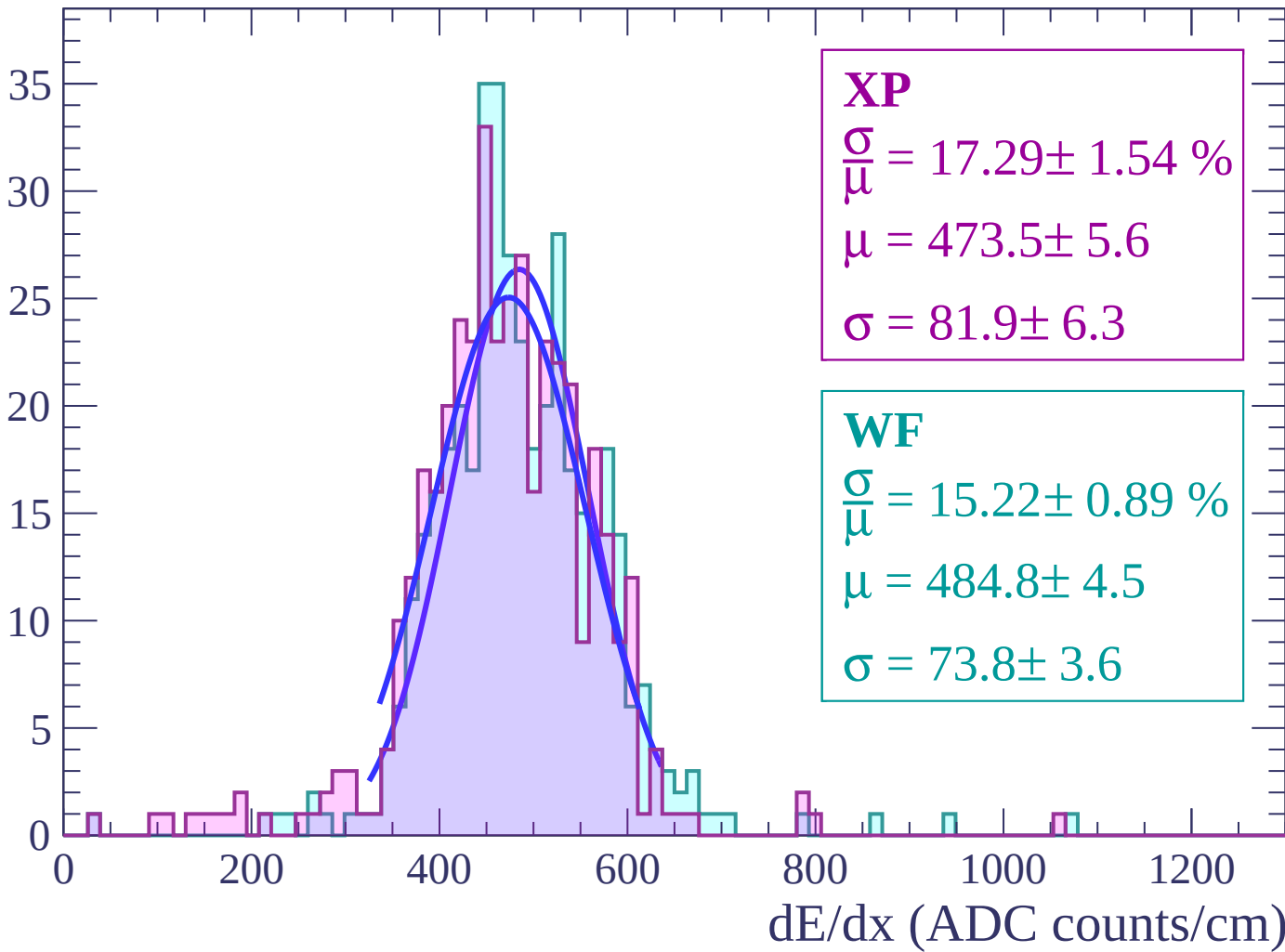
Energy loss | $-30 < \theta < -28$

Count



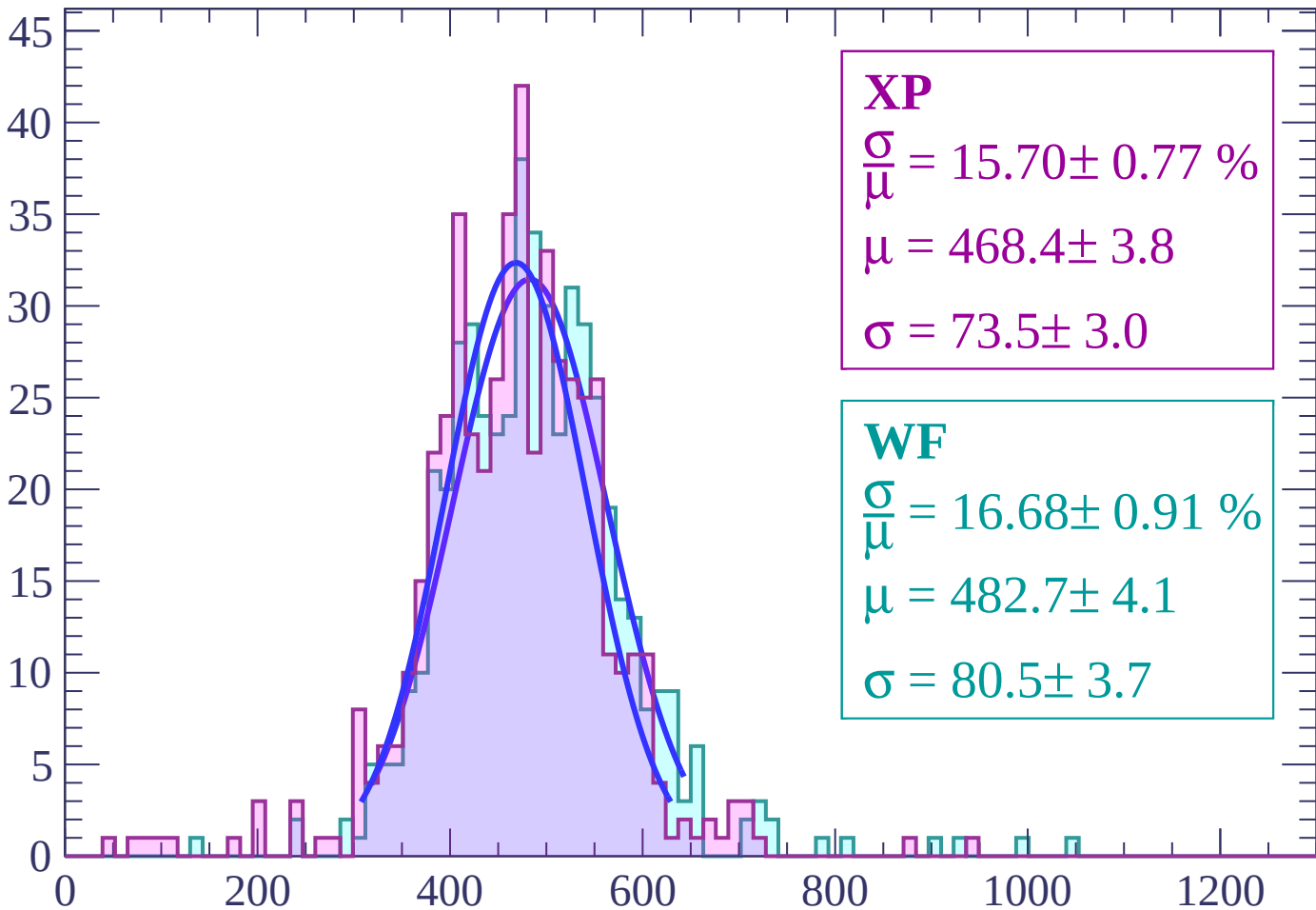
Energy loss | $-28 < \theta < -26$

Count



Energy loss | $-26 < \theta < -24$

Count



XP

$$\frac{\sigma}{\mu} = 15.70 \pm 0.77 \%$$

$$\mu = 468.4 \pm 3.8$$

$$\sigma = 73.5 \pm 3.0$$

WF

$$\frac{\sigma}{\mu} = 16.68 \pm 0.91 \%$$

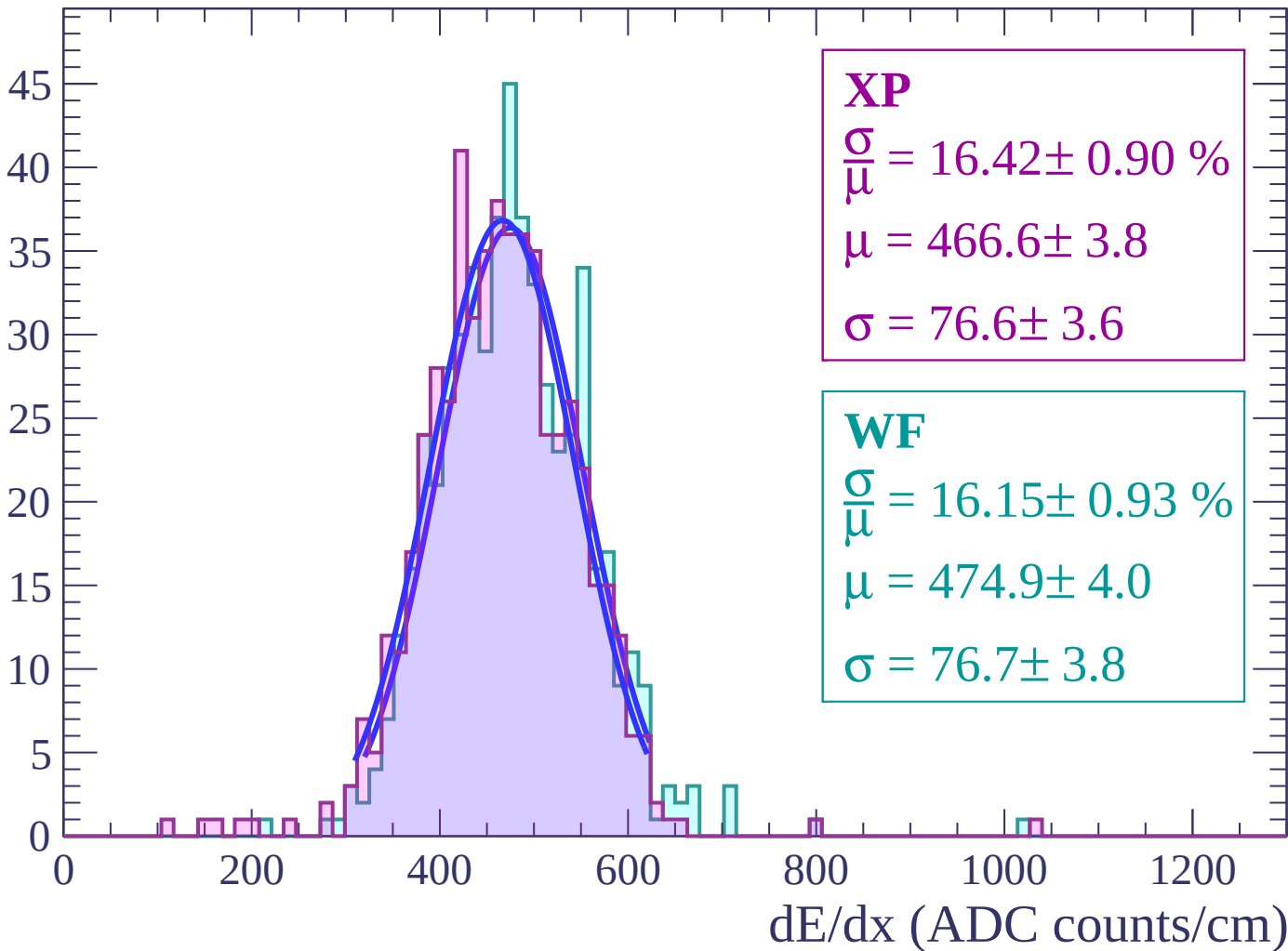
$$\mu = 482.7 \pm 4.1$$

$$\sigma = 80.5 \pm 3.7$$

dE/dx (ADC counts/cm)

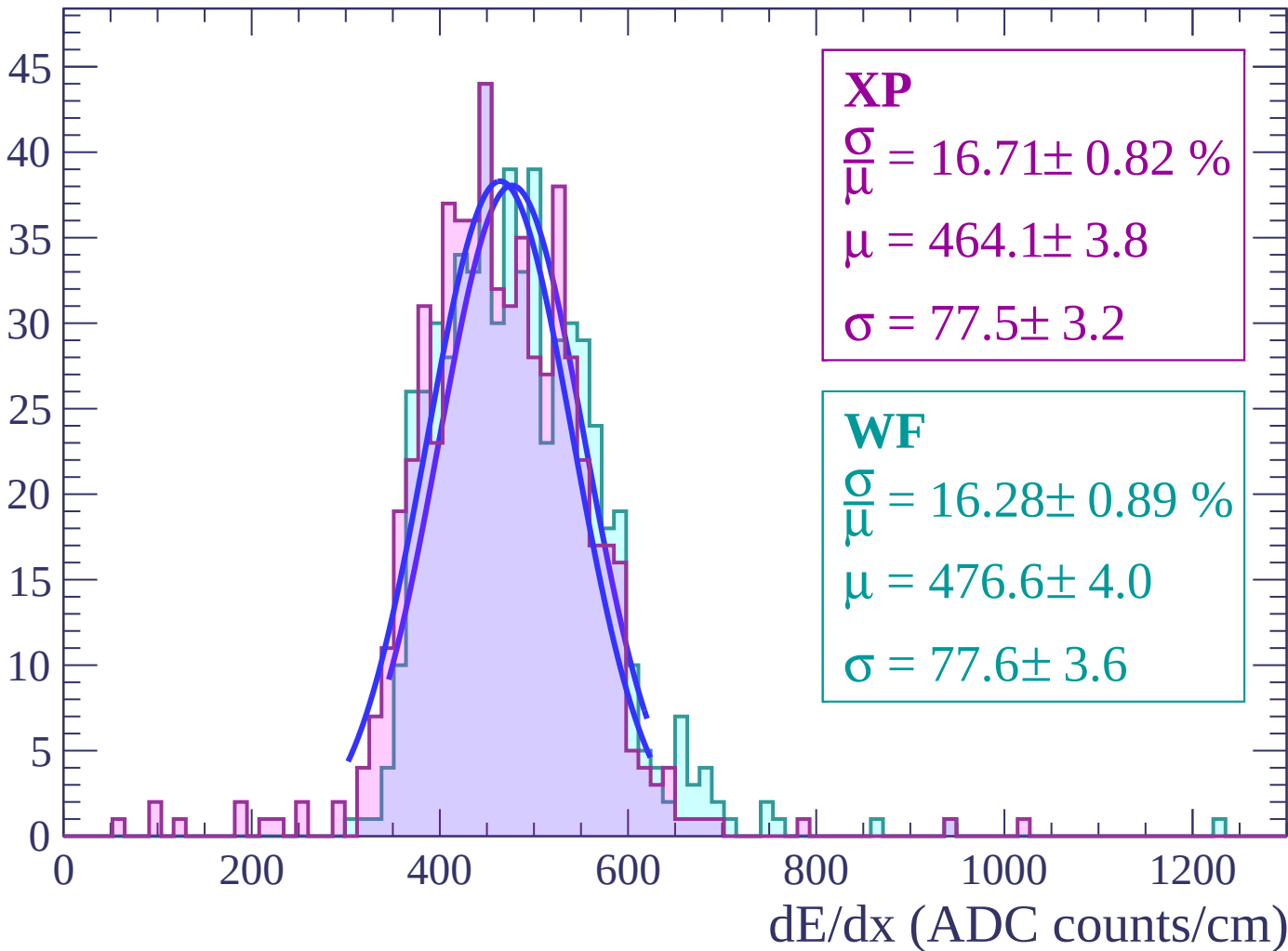
Energy loss | $-24 < \theta < -22$

Count



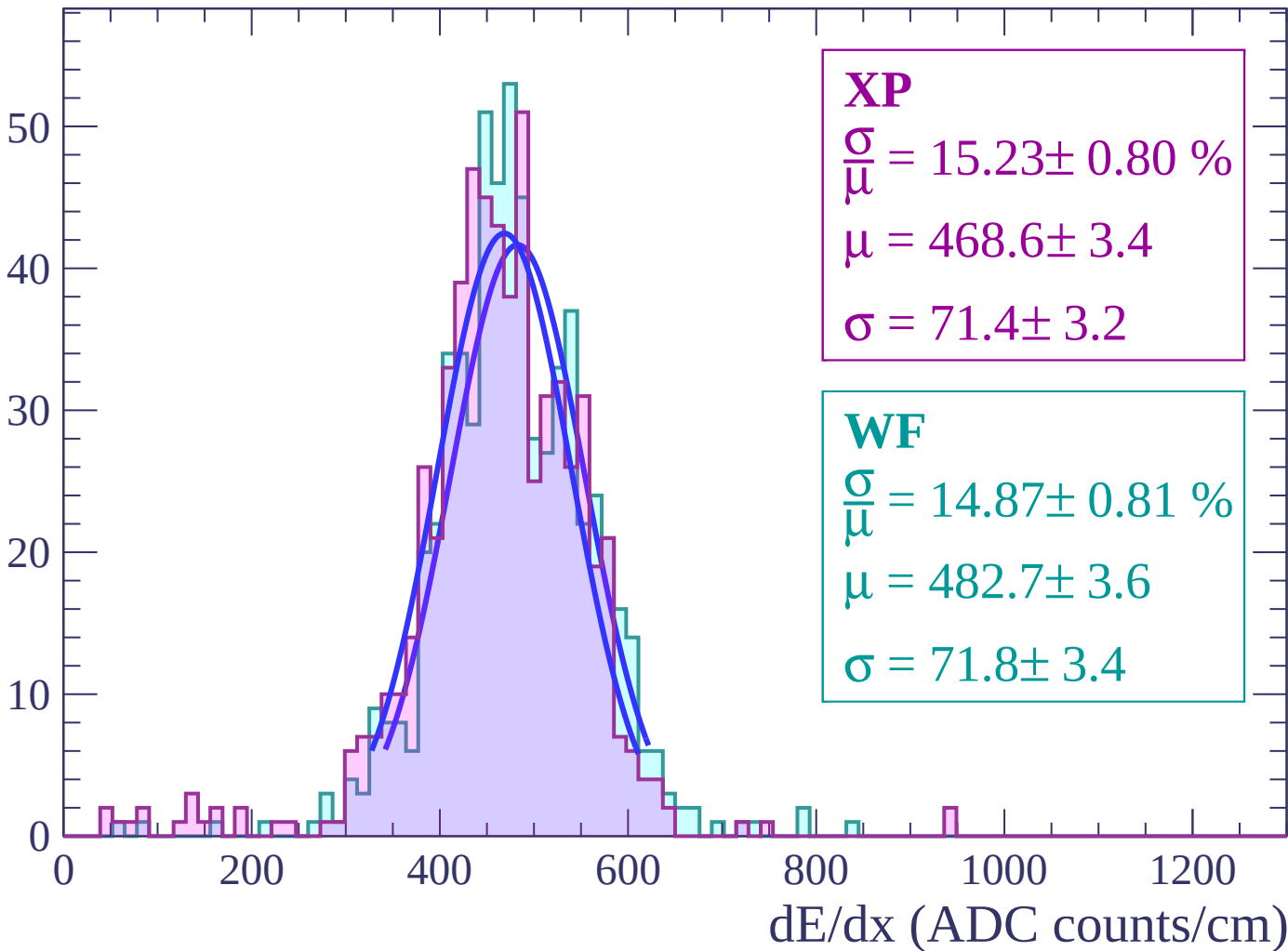
Energy loss | $-22 < \theta < -20$

Count



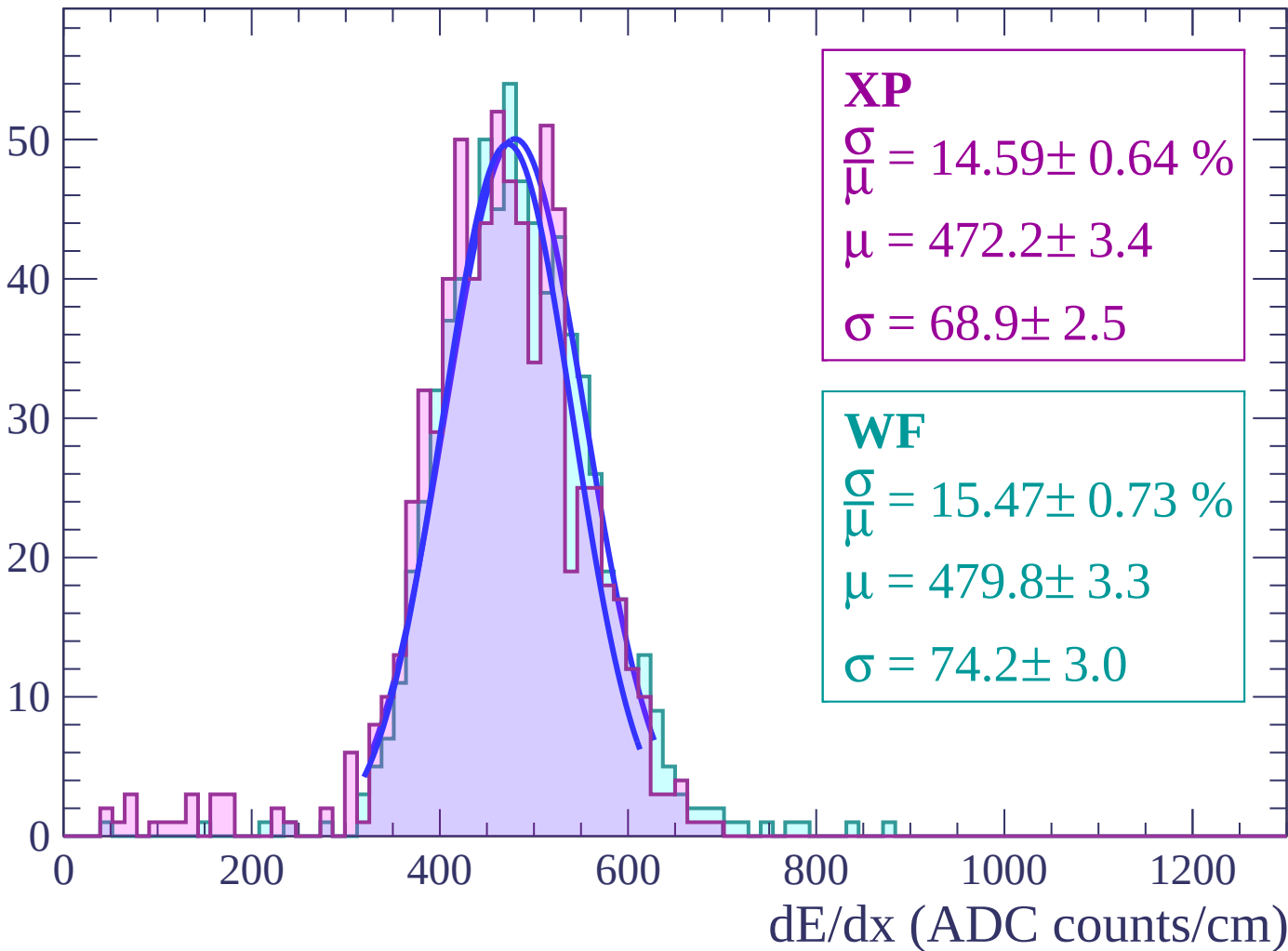
Energy loss | $-20 < \theta < -18$

Count



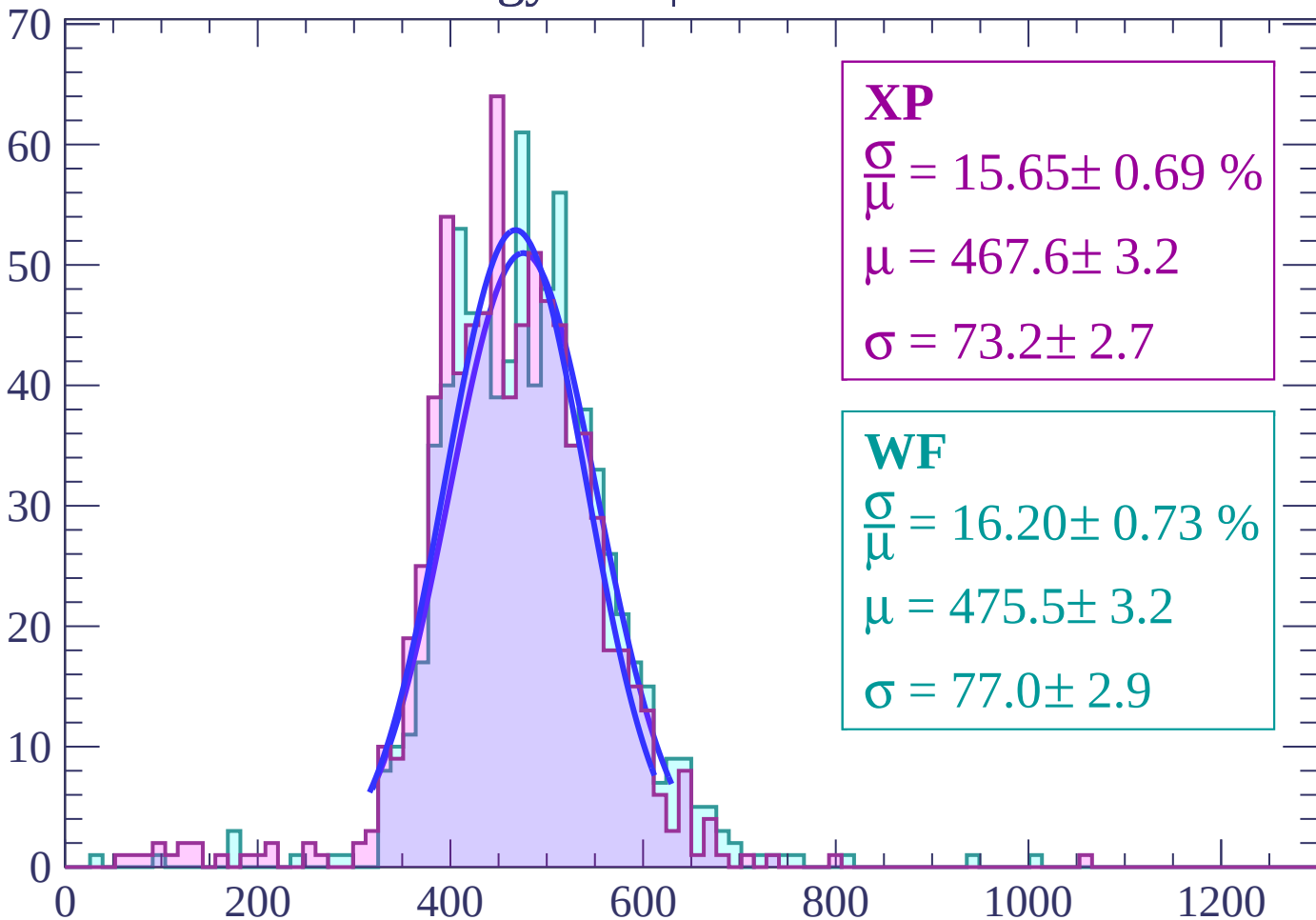
Energy loss | $-18 < \theta < -16$

Count



Energy loss | $-16 < \theta < -14$

Count



XP

$$\frac{\sigma}{\mu} = 15.65 \pm 0.69 \%$$

$$\mu = 467.6 \pm 3.2$$

$$\sigma = 73.2 \pm 2.7$$

WF

$$\frac{\sigma}{\mu} = 16.20 \pm 0.73 \%$$

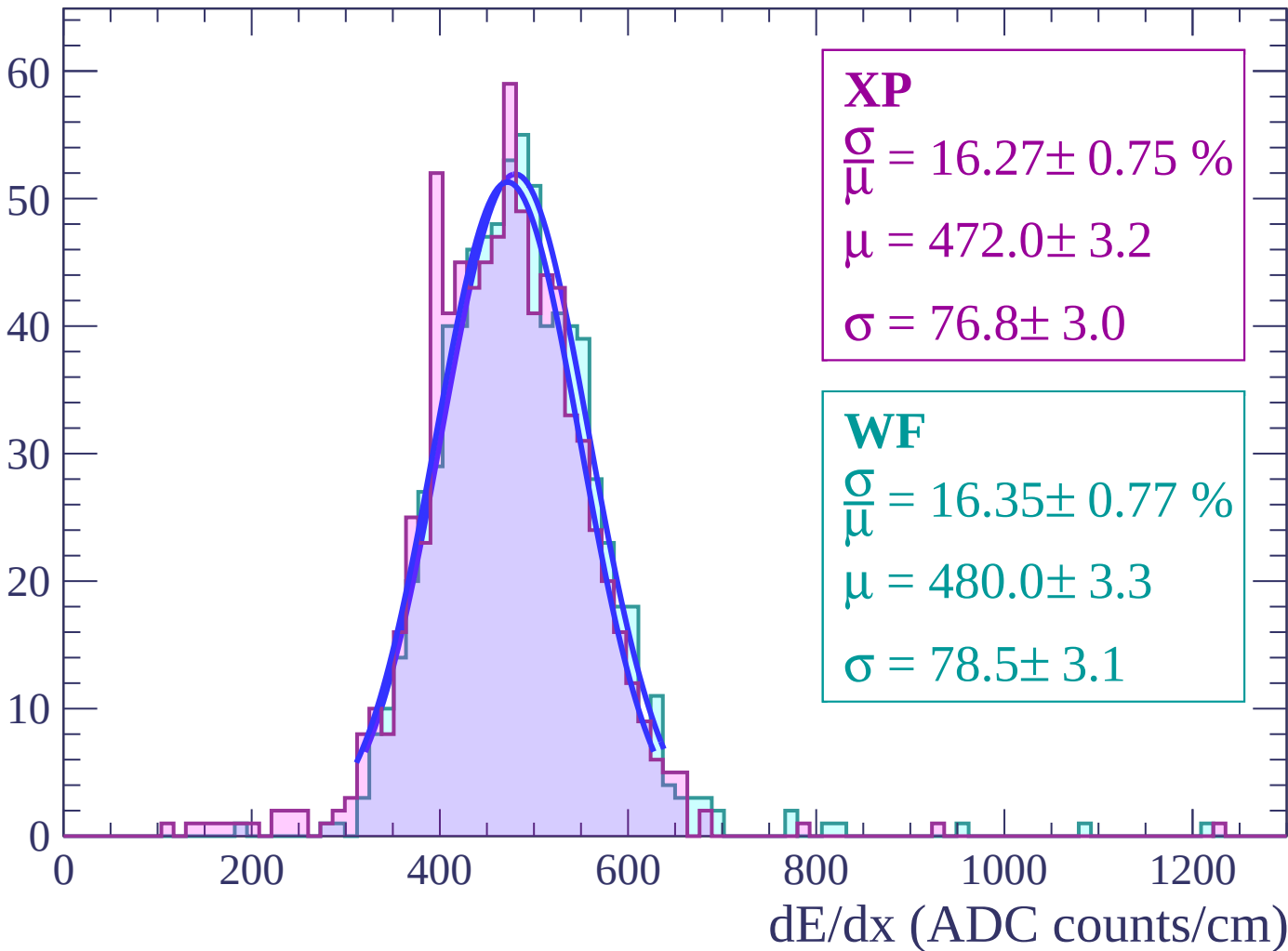
$$\mu = 475.5 \pm 3.2$$

$$\sigma = 77.0 \pm 2.9$$

dE/dx (ADC counts/cm)

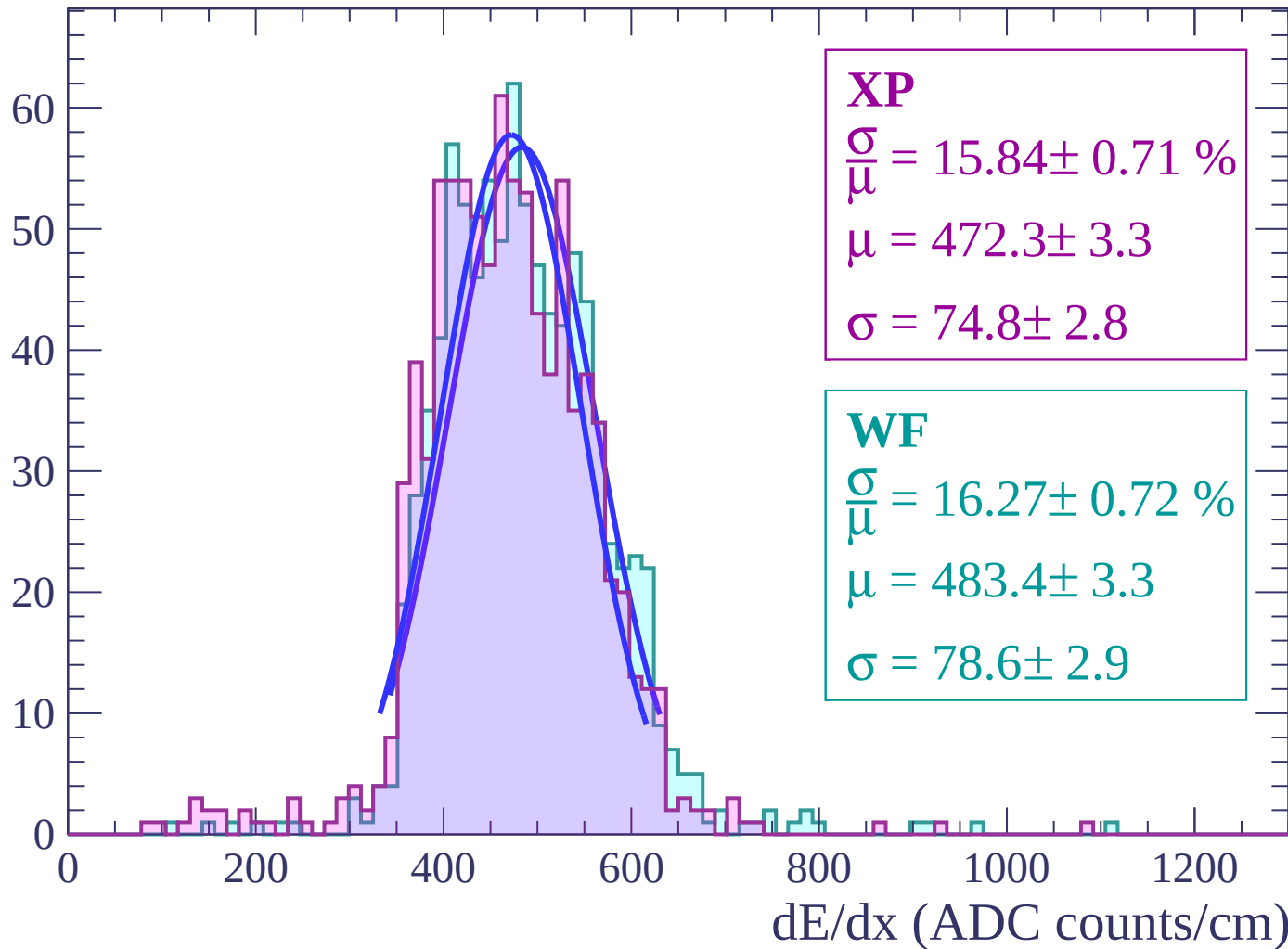
Energy loss | $-14 < \theta < -12$

Count



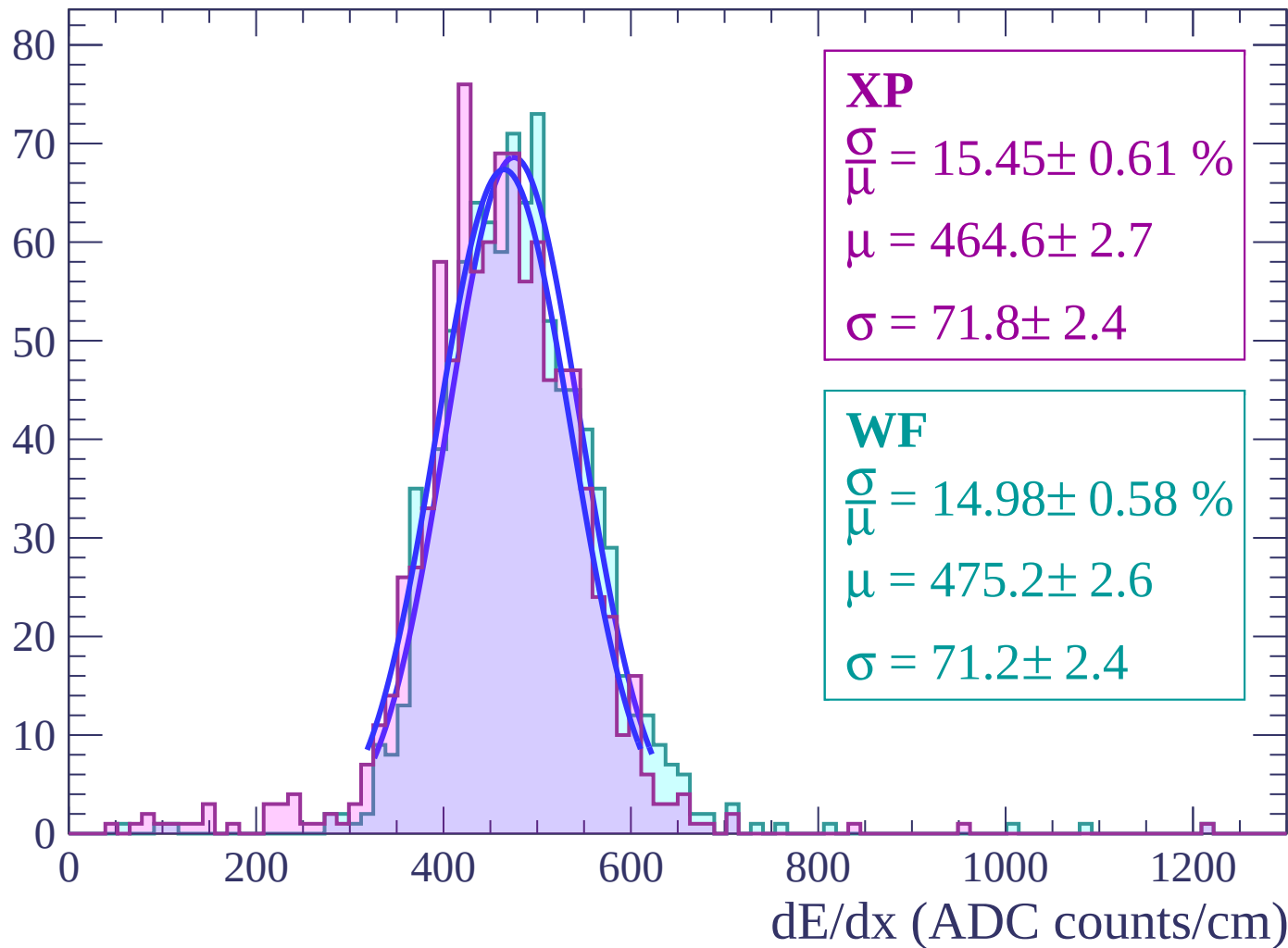
Energy loss | $-12 < \theta < -10$

Count



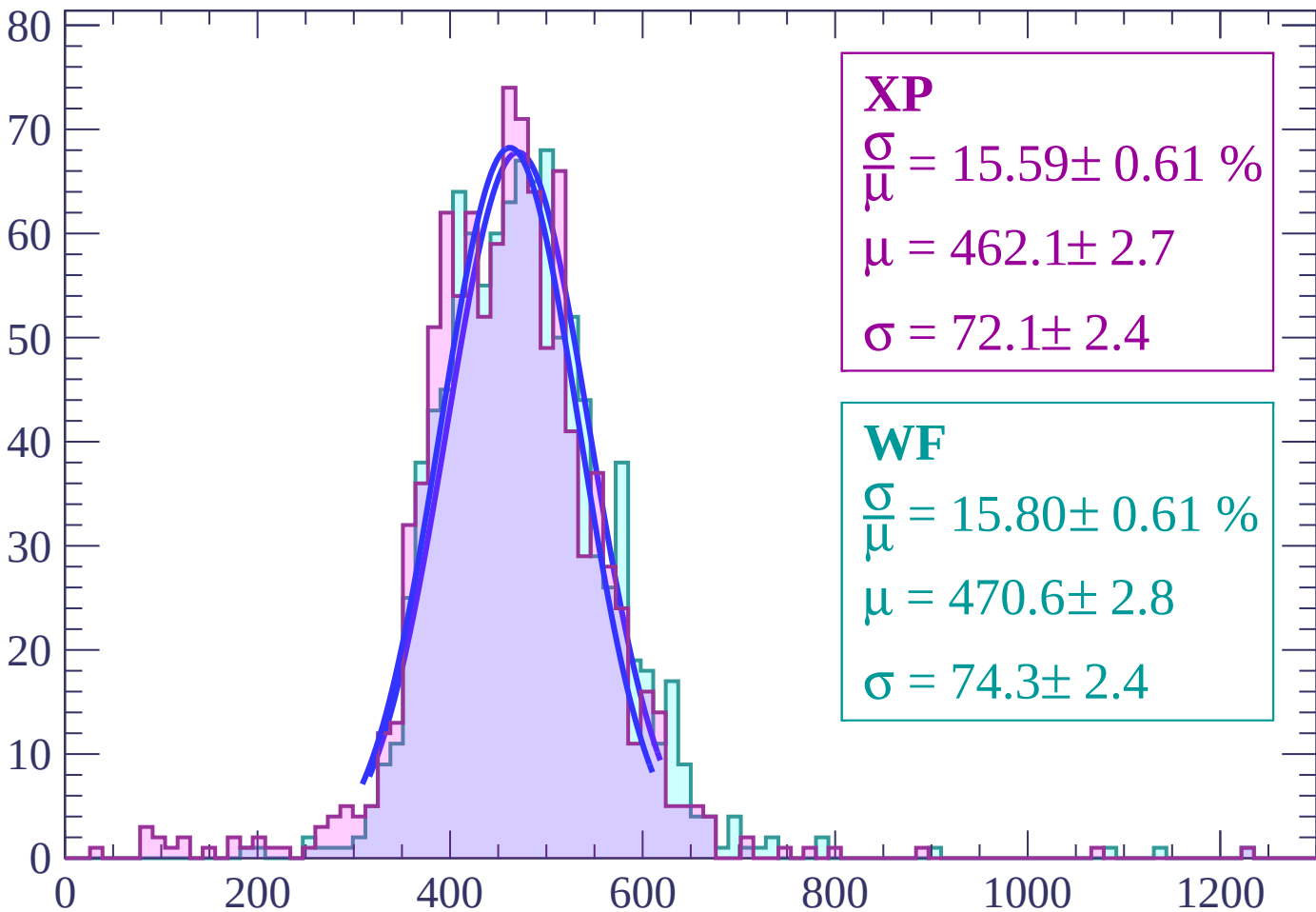
Energy loss | $-10 < \theta < -8$

Count



Energy loss | $-8 < \theta < -6$

Count



XP

$$\frac{\sigma}{\mu} = 15.59 \pm 0.61 \%$$

$$\mu = 462.1 \pm 2.7$$

$$\sigma = 72.1 \pm 2.4$$

WF

$$\frac{\sigma}{\mu} = 15.80 \pm 0.61 \%$$

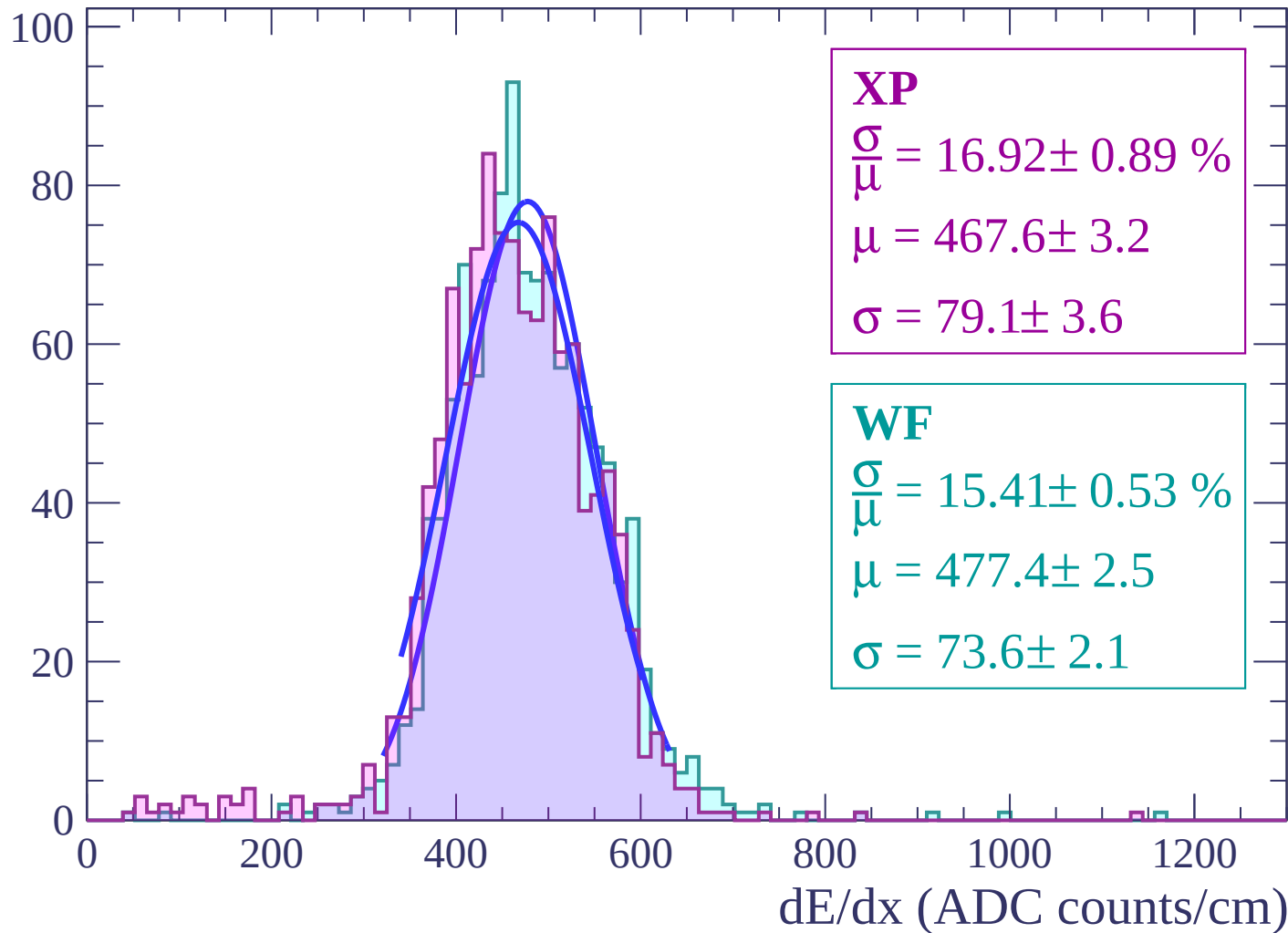
$$\mu = 470.6 \pm 2.8$$

$$\sigma = 74.3 \pm 2.4$$

dE/dx (ADC counts/cm)

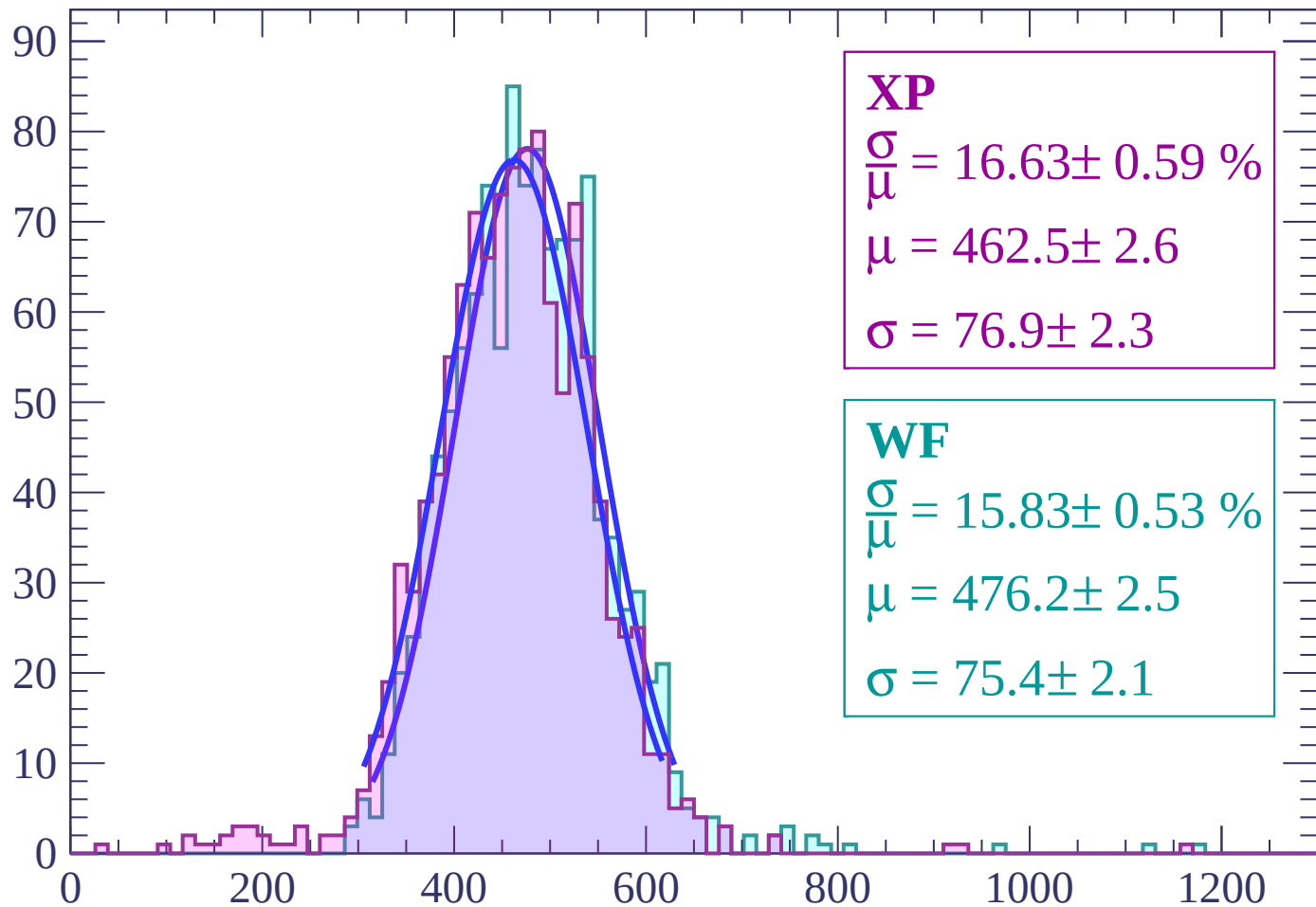
Energy loss | $-6 < \theta < -4$

Count



Energy loss | $-4 < \theta < -2$

Count



XP

$$\frac{\sigma}{\mu} = 16.63 \pm 0.59 \%$$

$$\mu = 462.5 \pm 2.6$$

$$\sigma = 76.9 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 15.83 \pm 0.53 \%$$

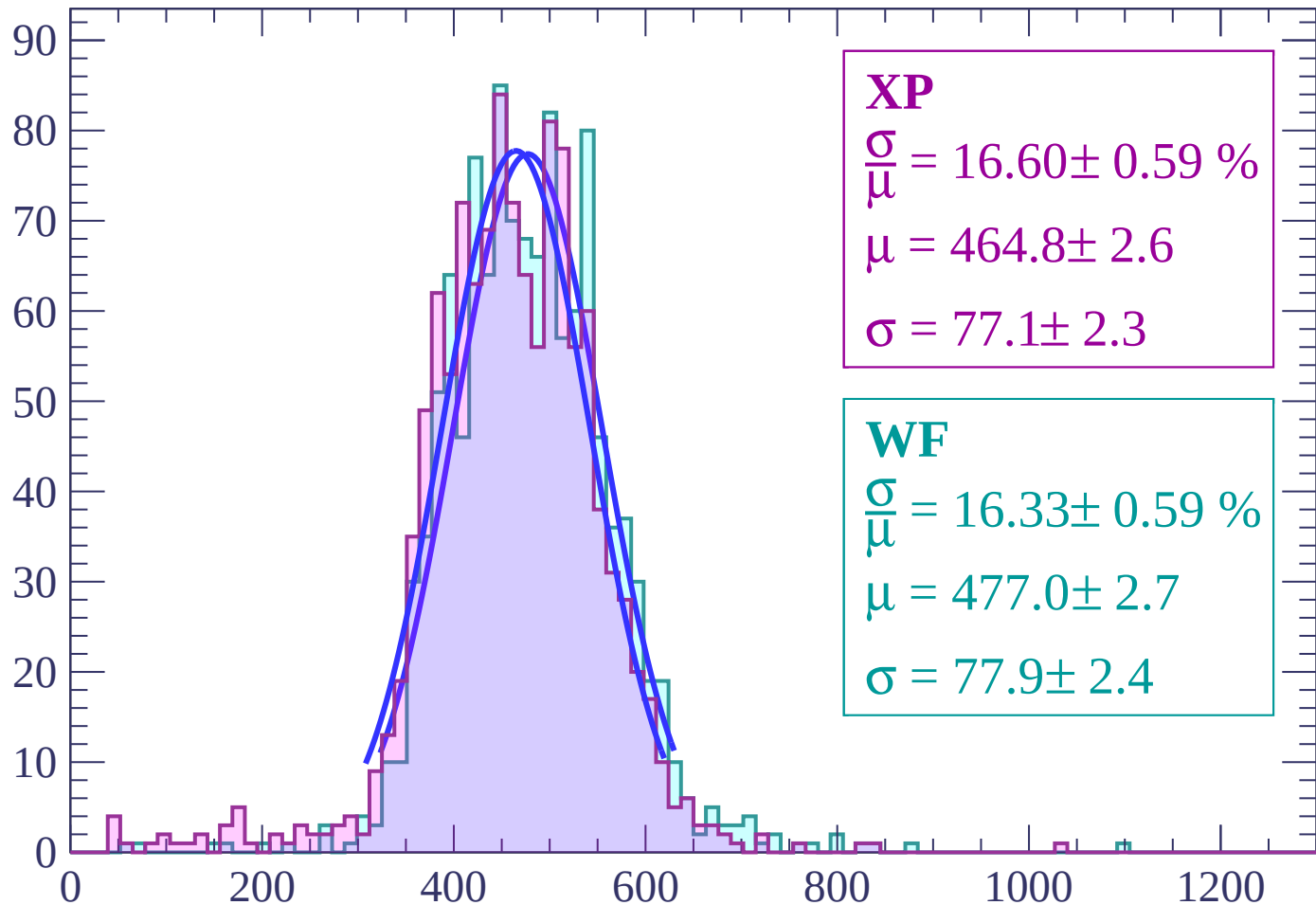
$$\mu = 476.2 \pm 2.5$$

$$\sigma = 75.4 \pm 2.1$$

dE/dx (ADC counts/cm)

Energy loss | $-2 < \theta < 0$

Count



XP

$$\frac{\sigma}{\mu} = 16.60 \pm 0.59 \%$$

$$\mu = 464.8 \pm 2.6$$

$$\sigma = 77.1 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 16.33 \pm 0.59 \%$$

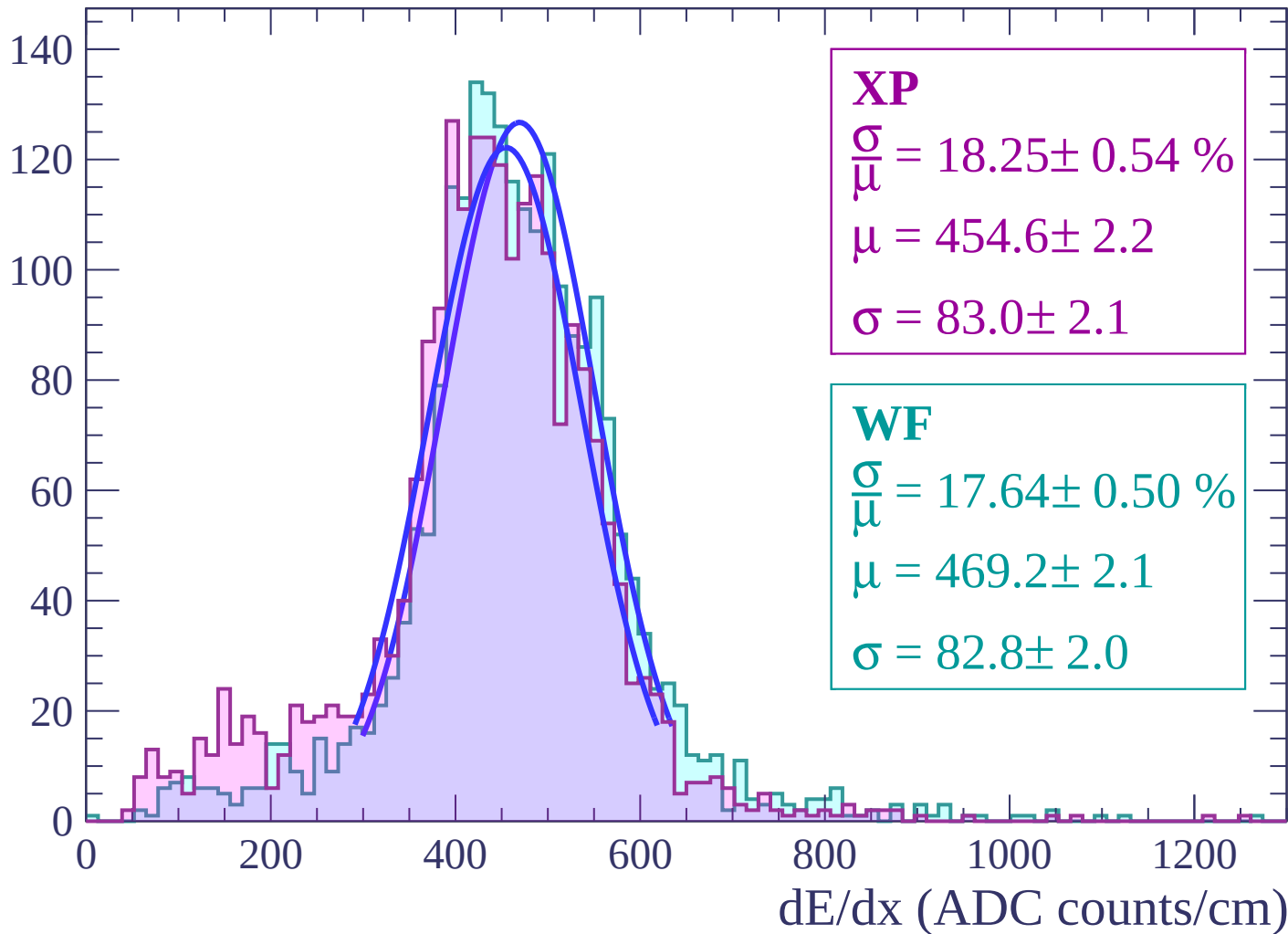
$$\mu = 477.0 \pm 2.7$$

$$\sigma = 77.9 \pm 2.4$$

dE/dx (ADC counts/cm)

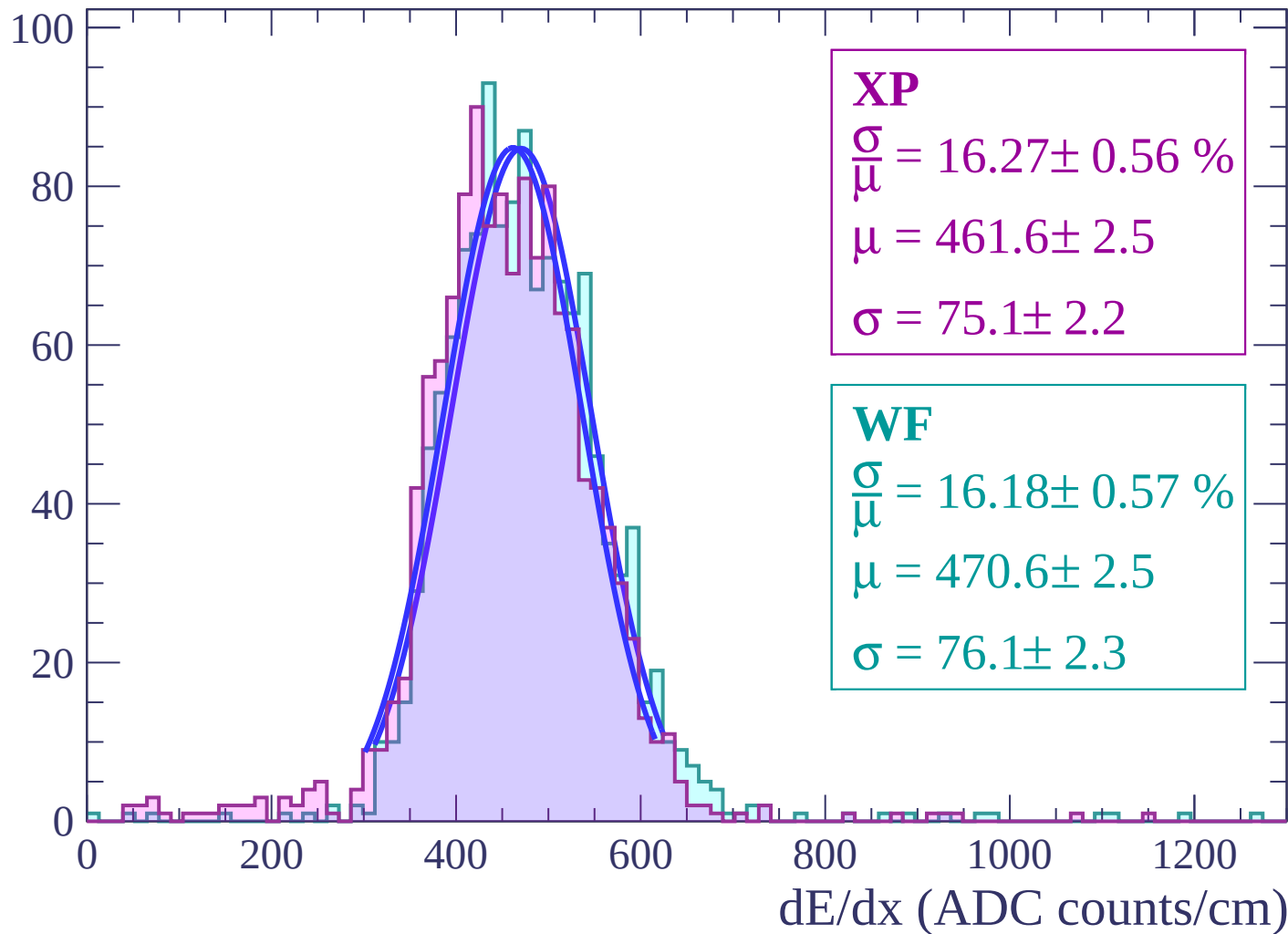
Energy loss | $0 < \theta < 2$

Count



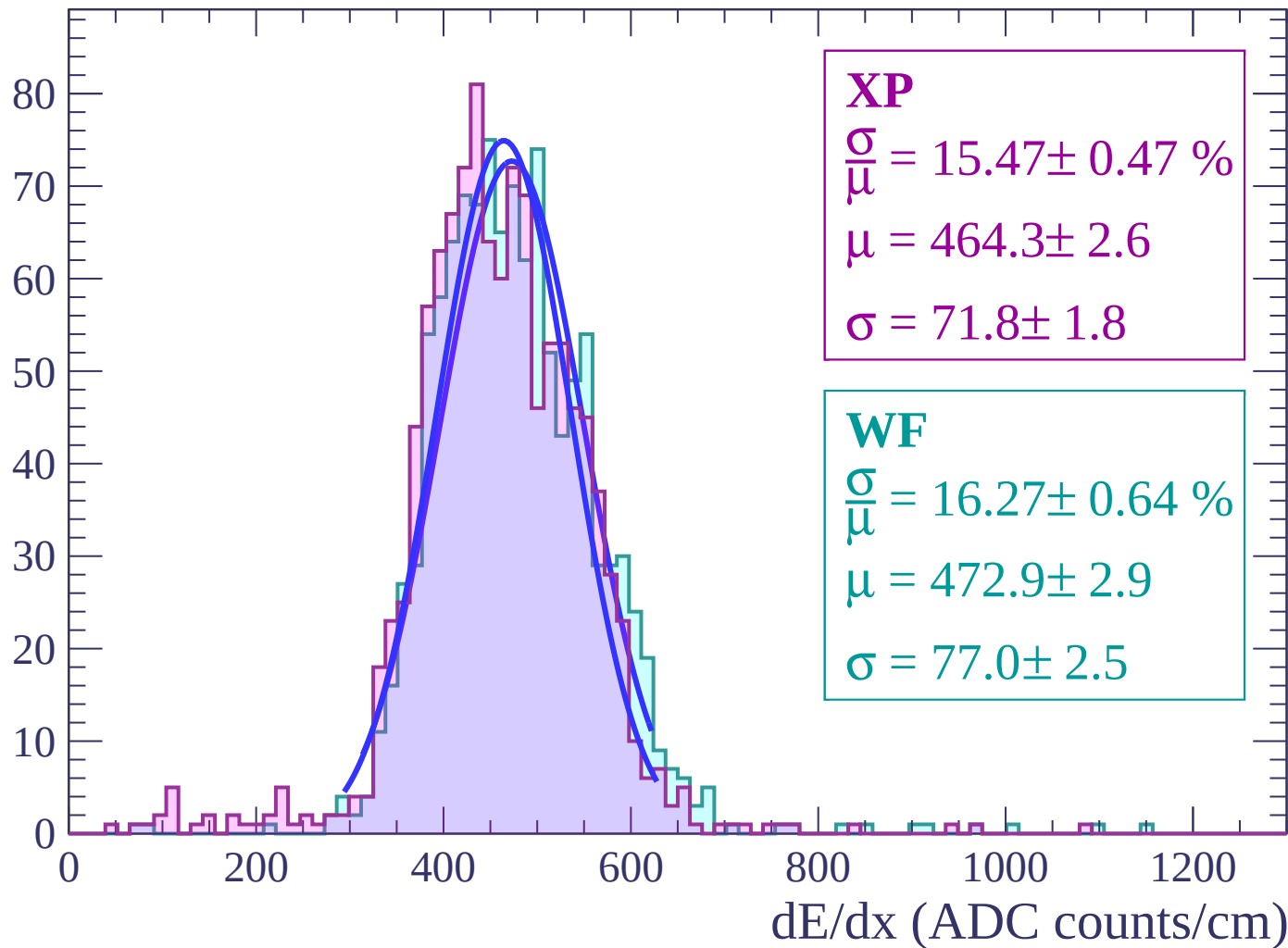
Energy loss | $2 < \theta < 4$

Count



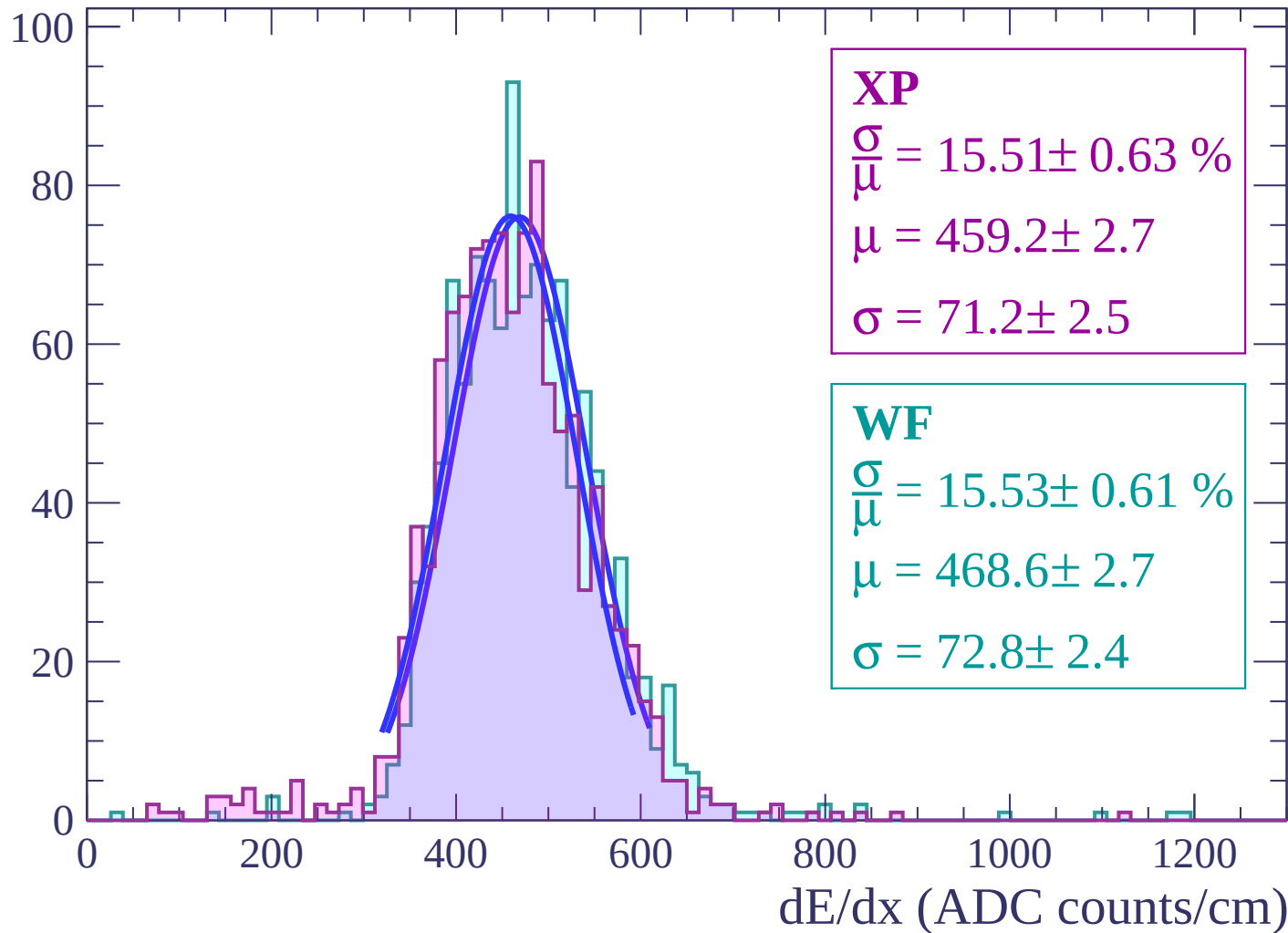
Energy loss | $4 < \theta < 6$

Count



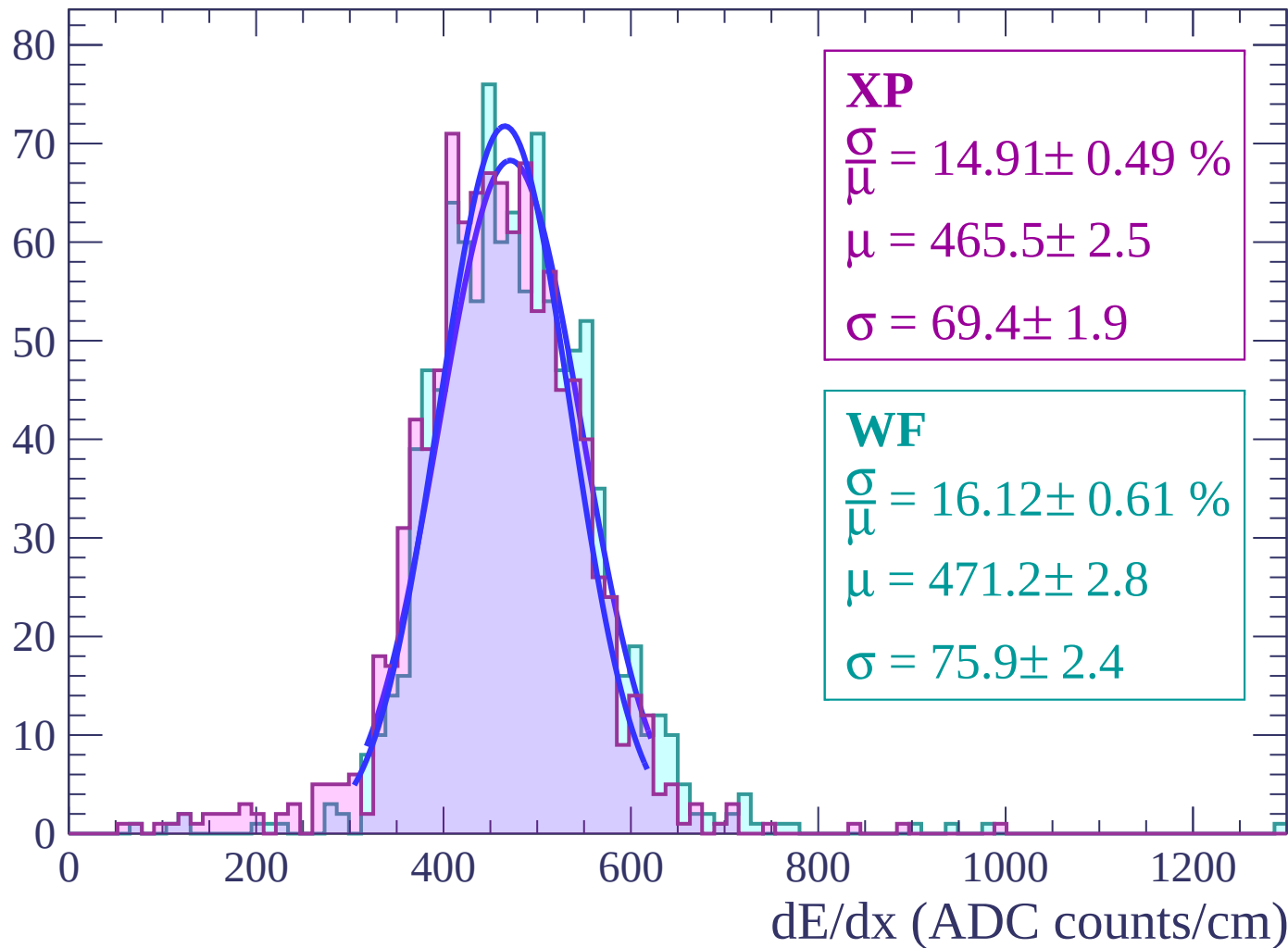
Energy loss | $6 < \theta < 8$

Count



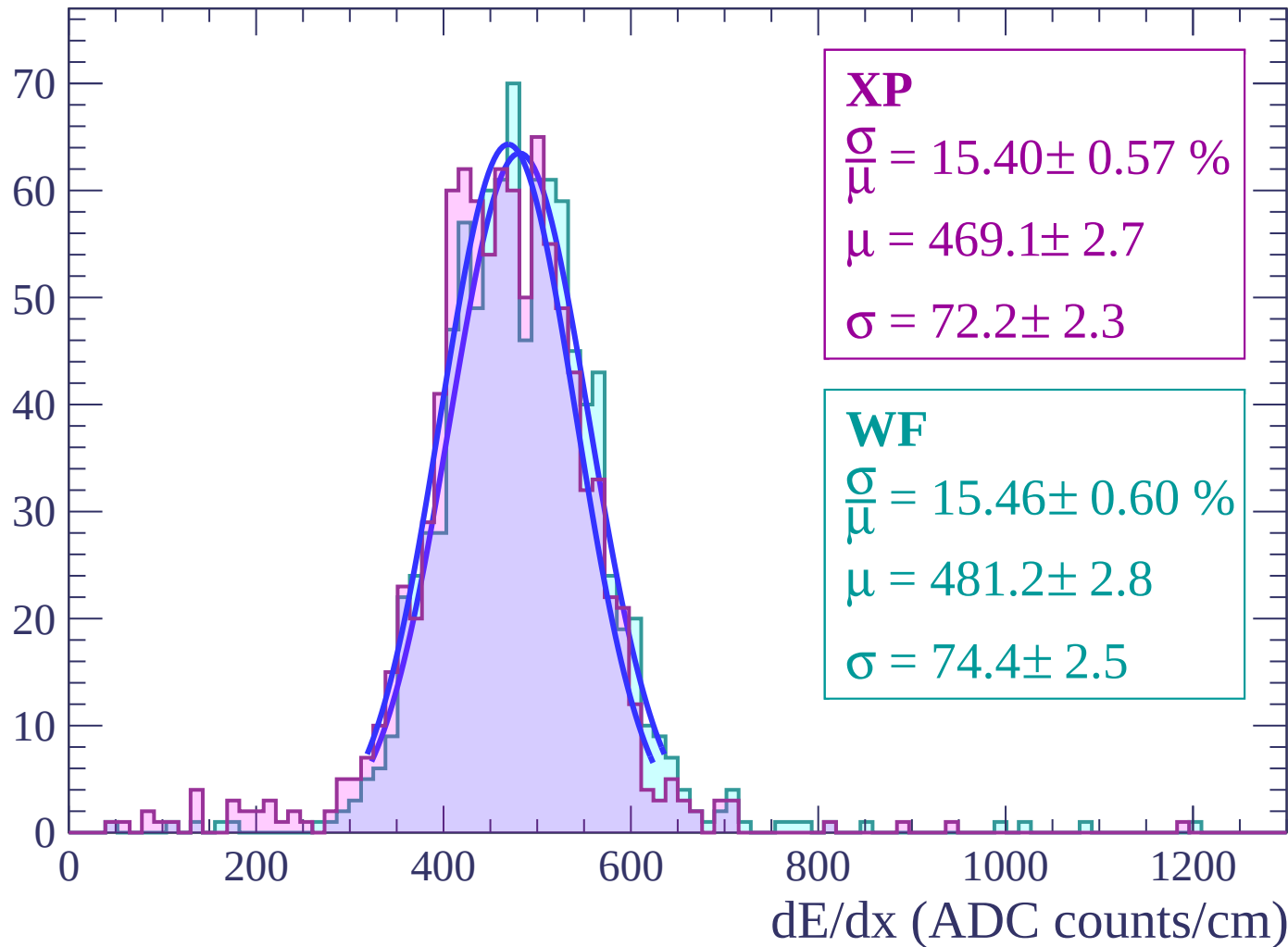
Energy loss | $8 < \theta < 10$

Count



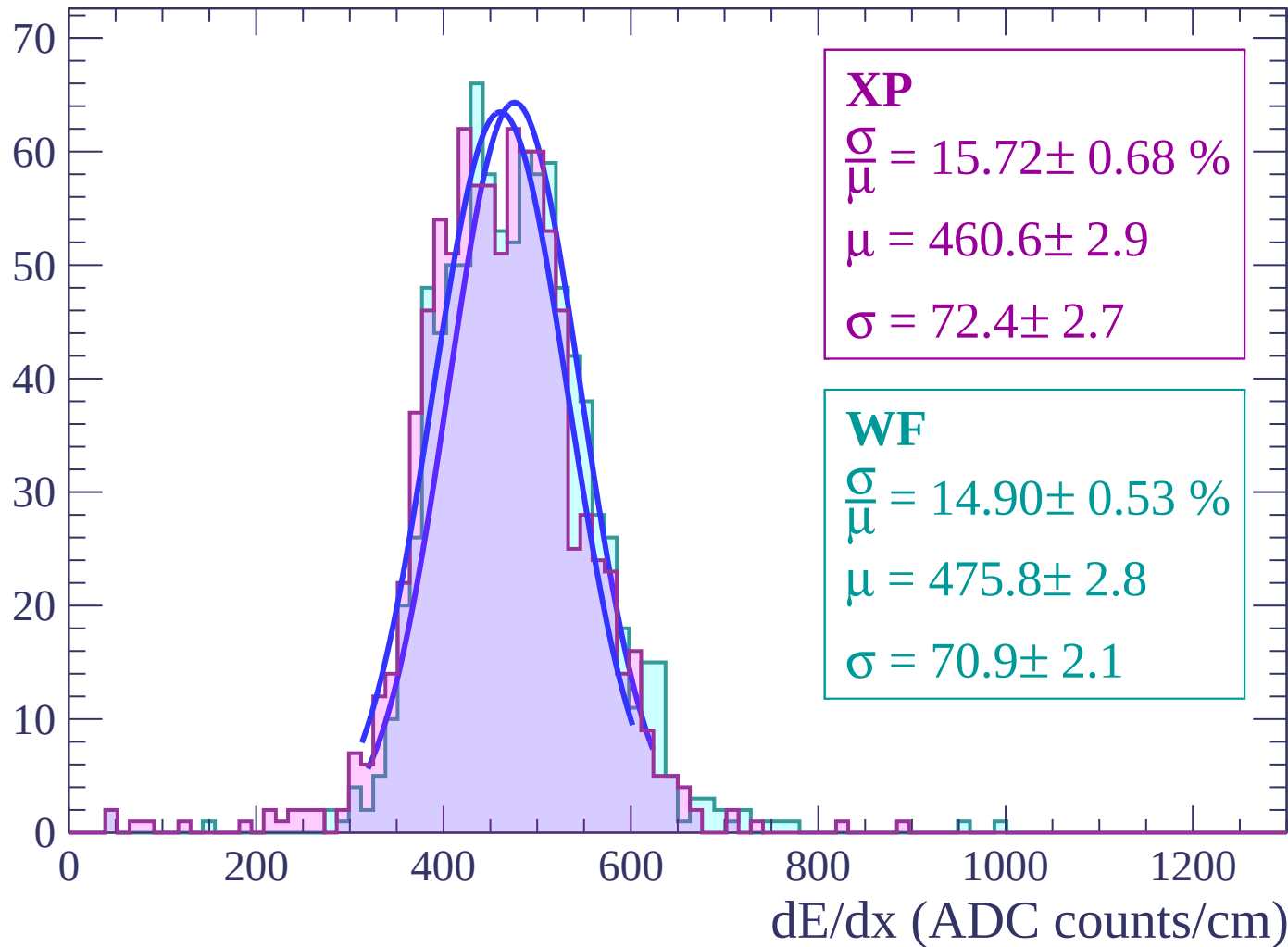
Energy loss | $10 < \theta < 12$

Count



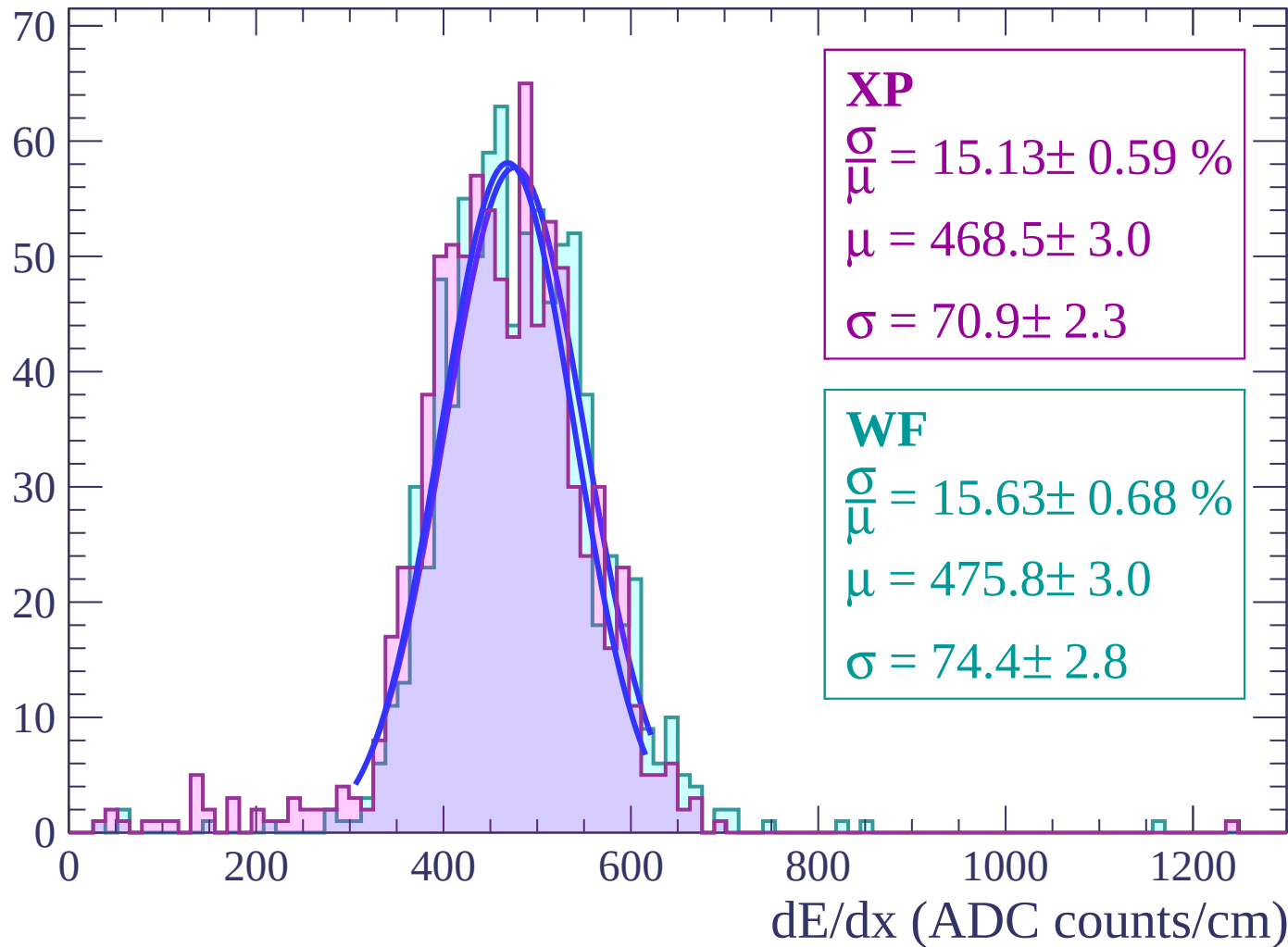
Energy loss | $12 < \theta < 14$

Count



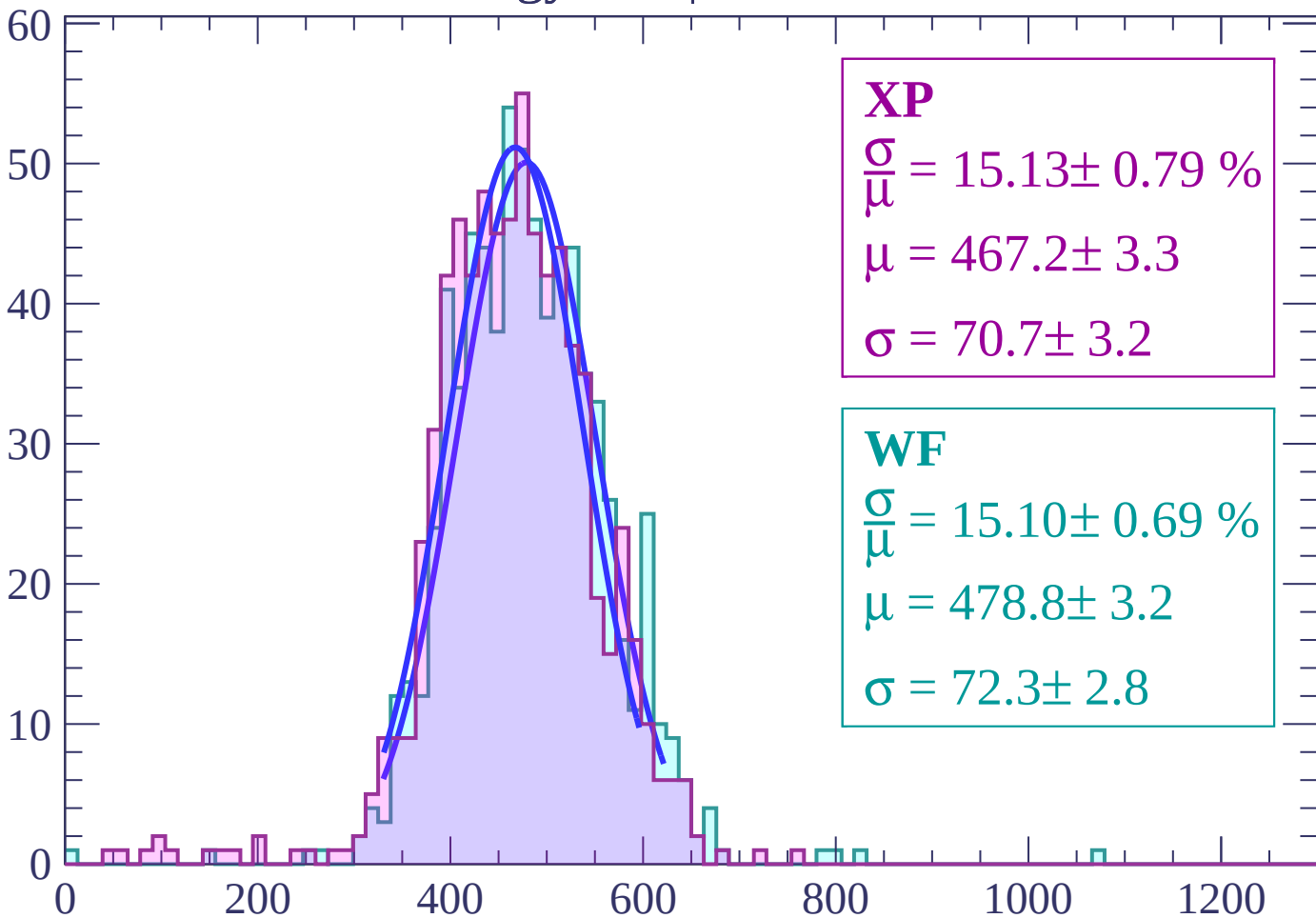
Energy loss | $14 < \theta < 16$

Count



Energy loss | $16 < \theta < 18$

Count



XP

$$\frac{\sigma}{\mu} = 15.13 \pm 0.79 \%$$

$$\mu = 467.2 \pm 3.3$$

$$\sigma = 70.7 \pm 3.2$$

WF

$$\frac{\sigma}{\mu} = 15.10 \pm 0.69 \%$$

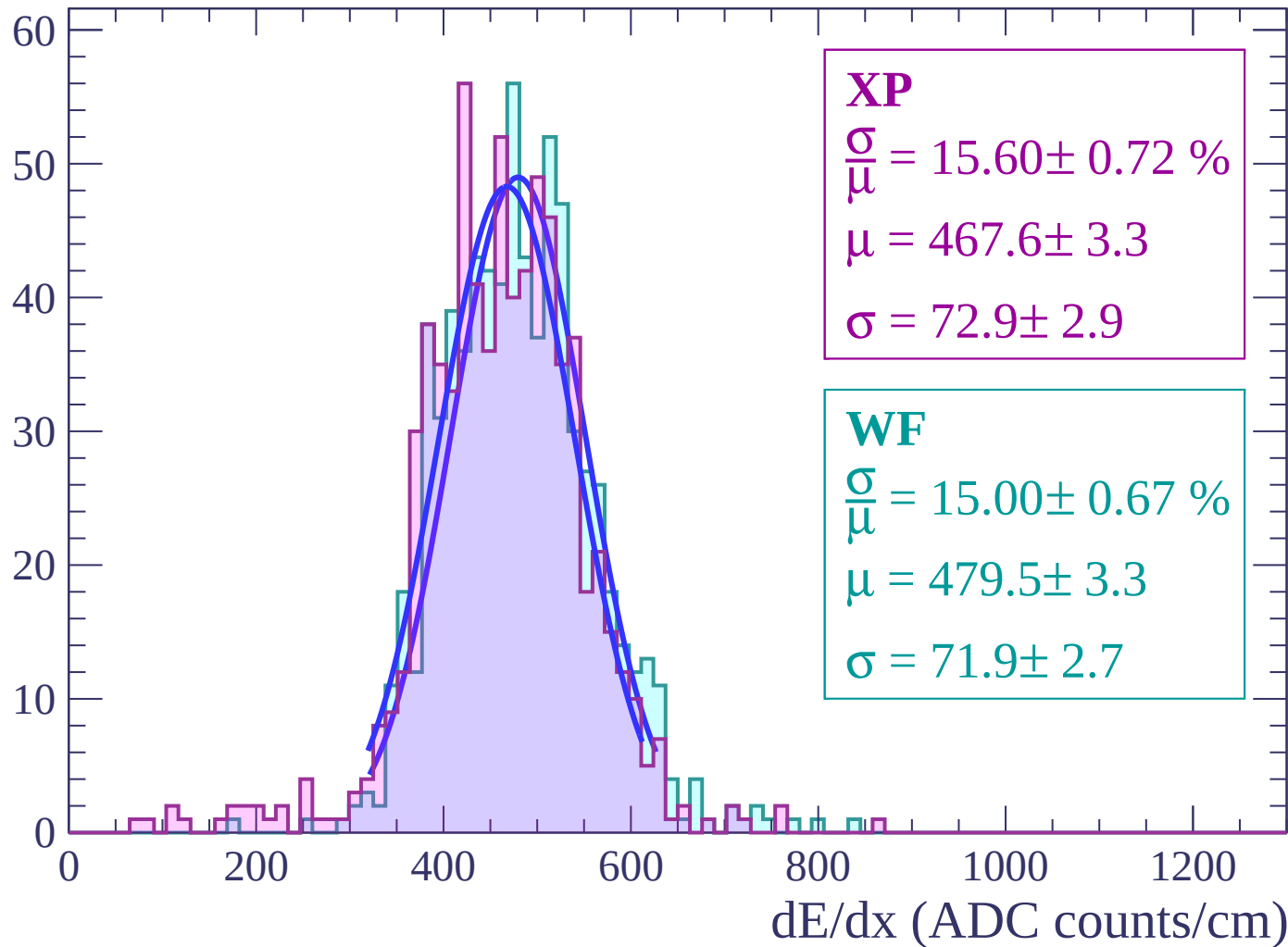
$$\mu = 478.8 \pm 3.2$$

$$\sigma = 72.3 \pm 2.8$$

dE/dx (ADC counts/cm)

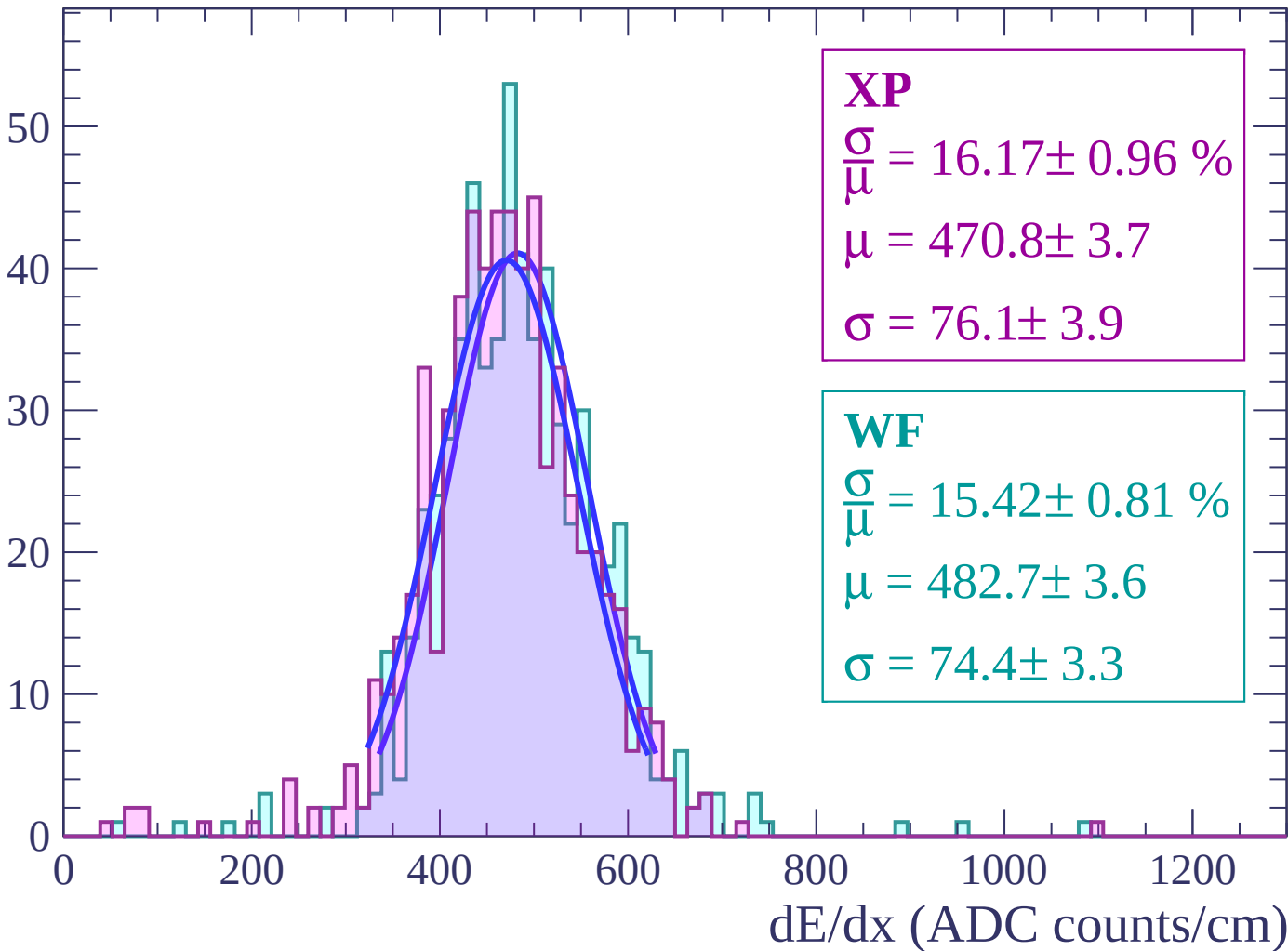
Energy loss | $18 < \theta < 20$

Count



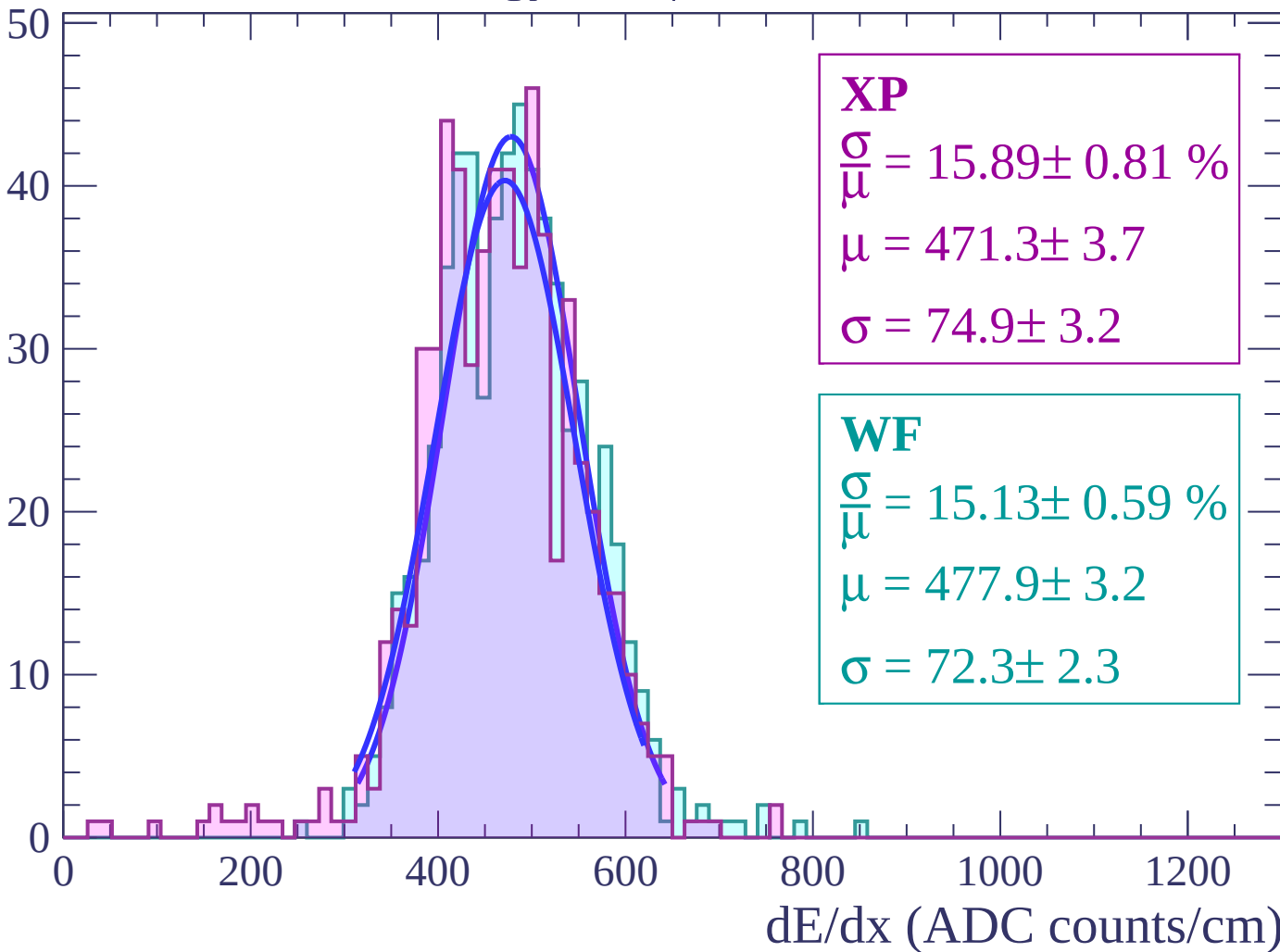
Energy loss | $20 < \theta < 22$

Count



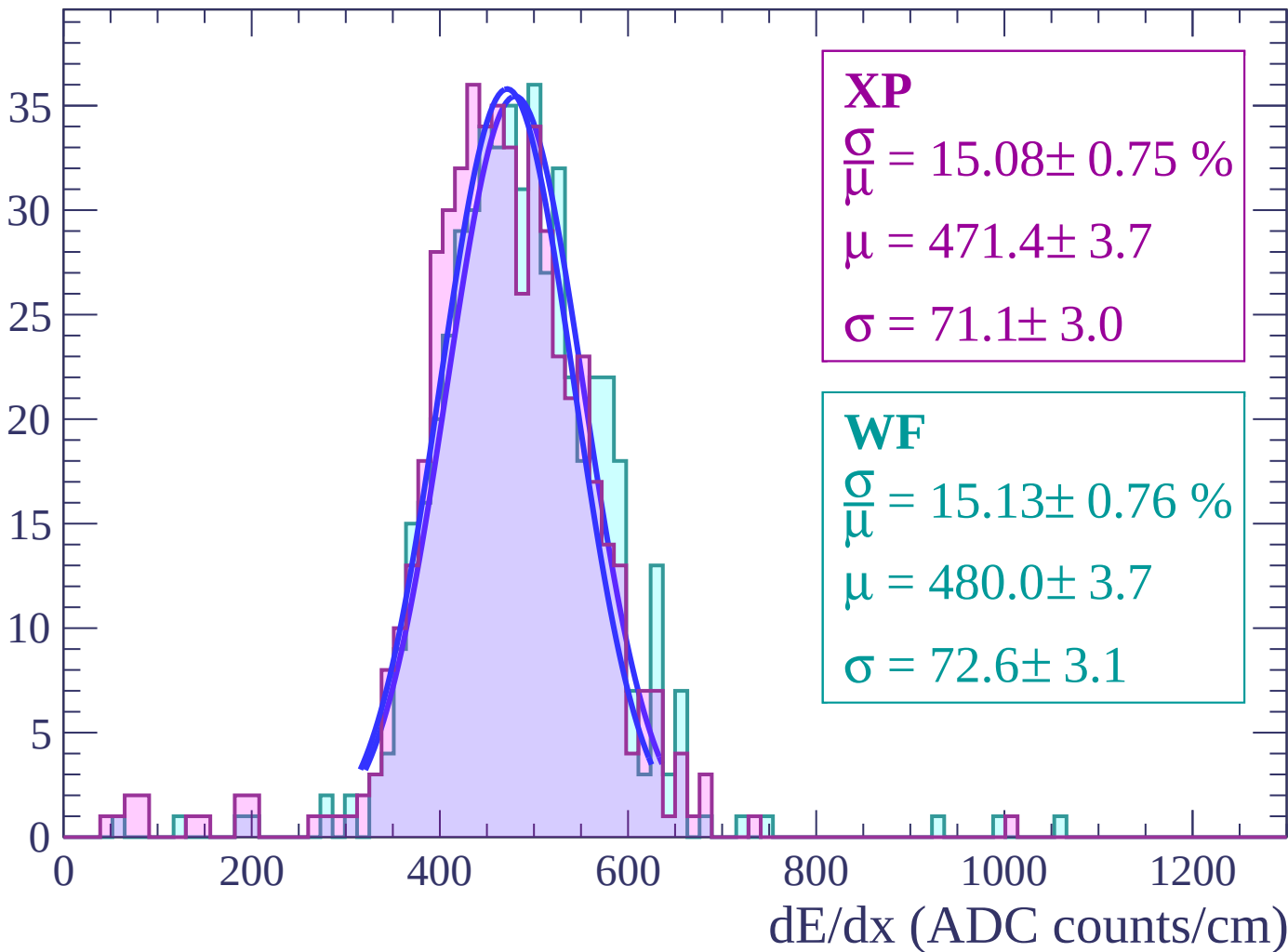
Energy loss | $22 < \theta < 24$

Count



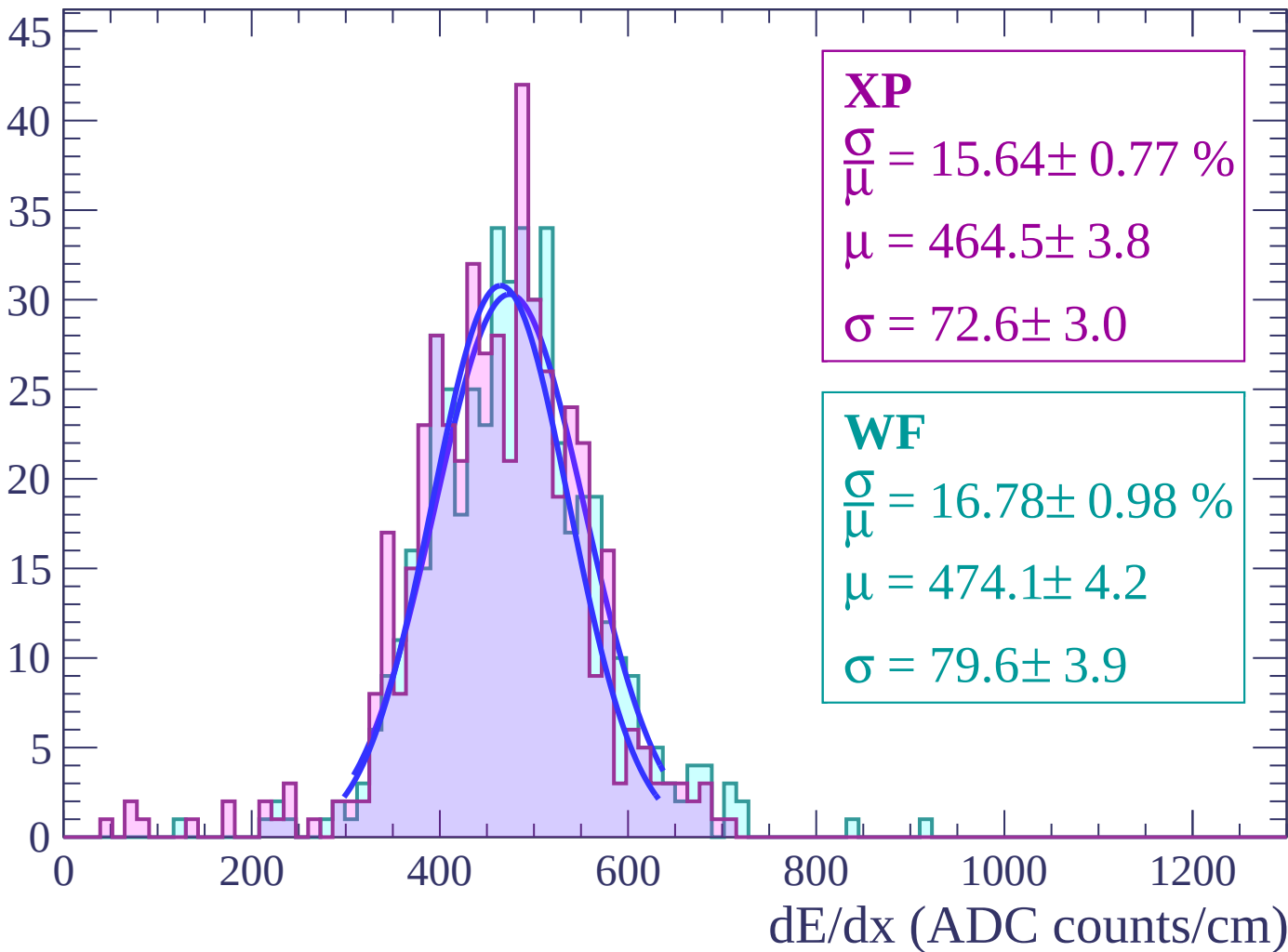
Energy loss | $24 < \theta < 26$

Count



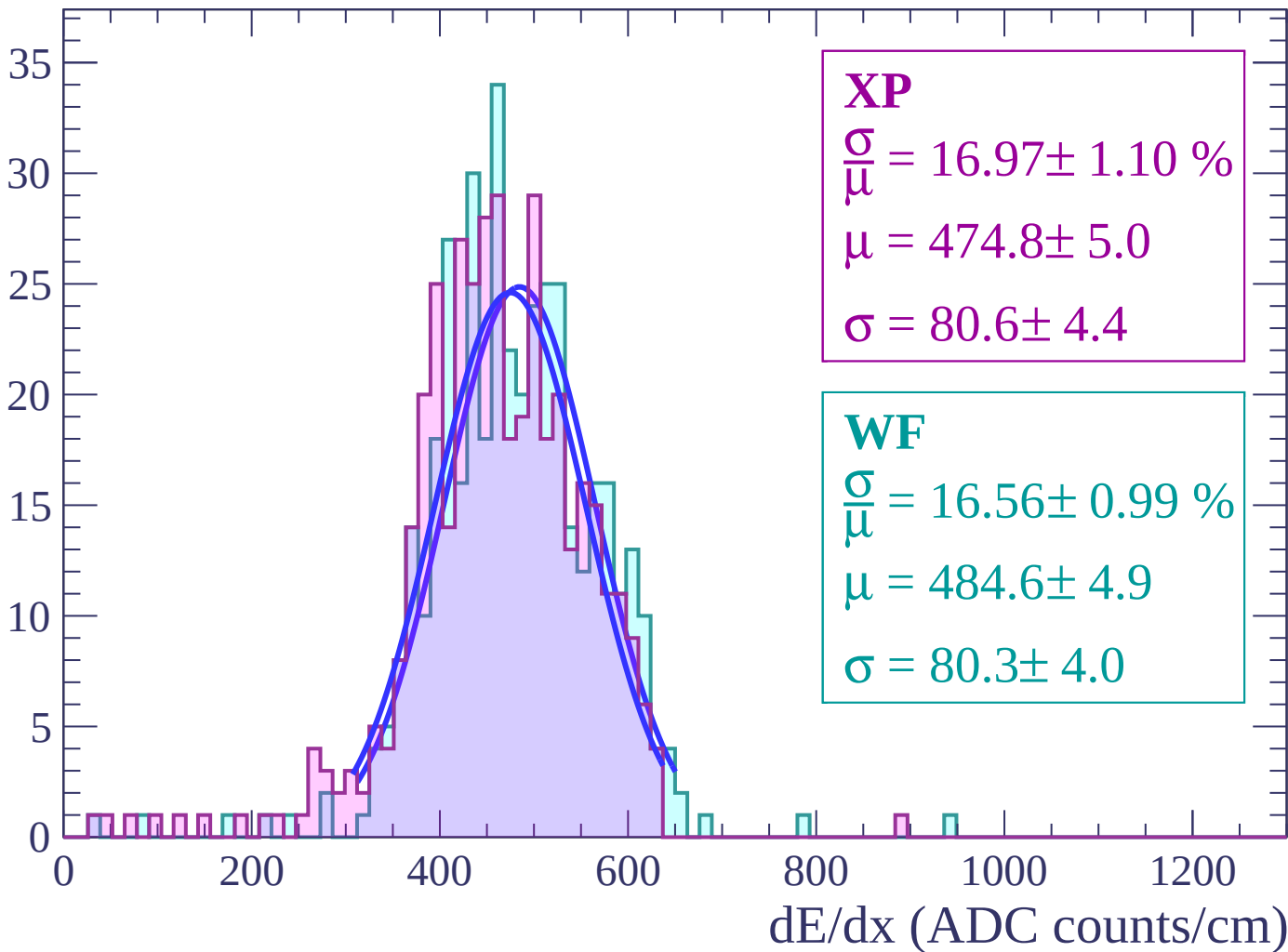
Energy loss | $26 < \theta < 28$

Count



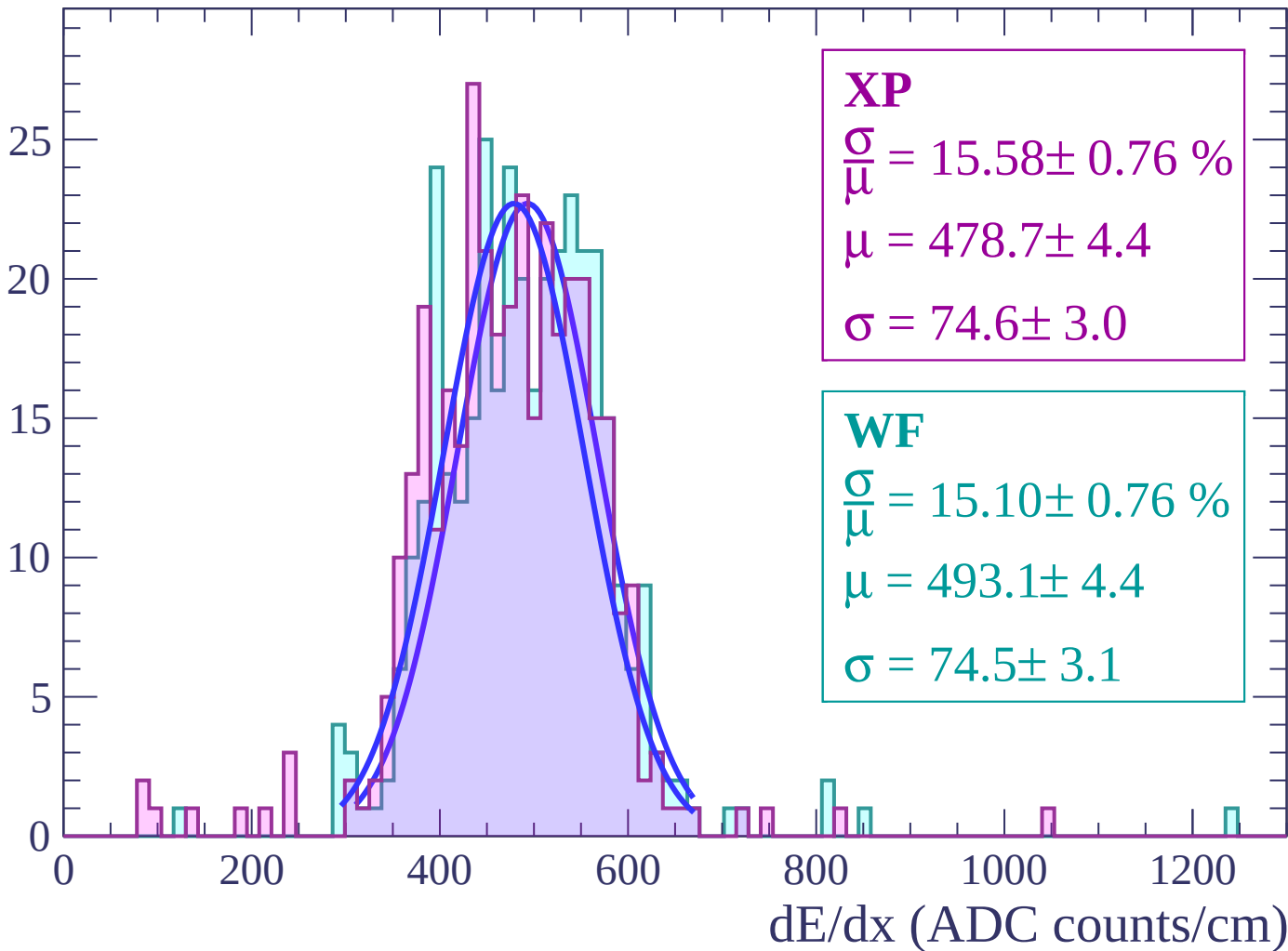
Energy loss | $28 < \theta < 30$

Count



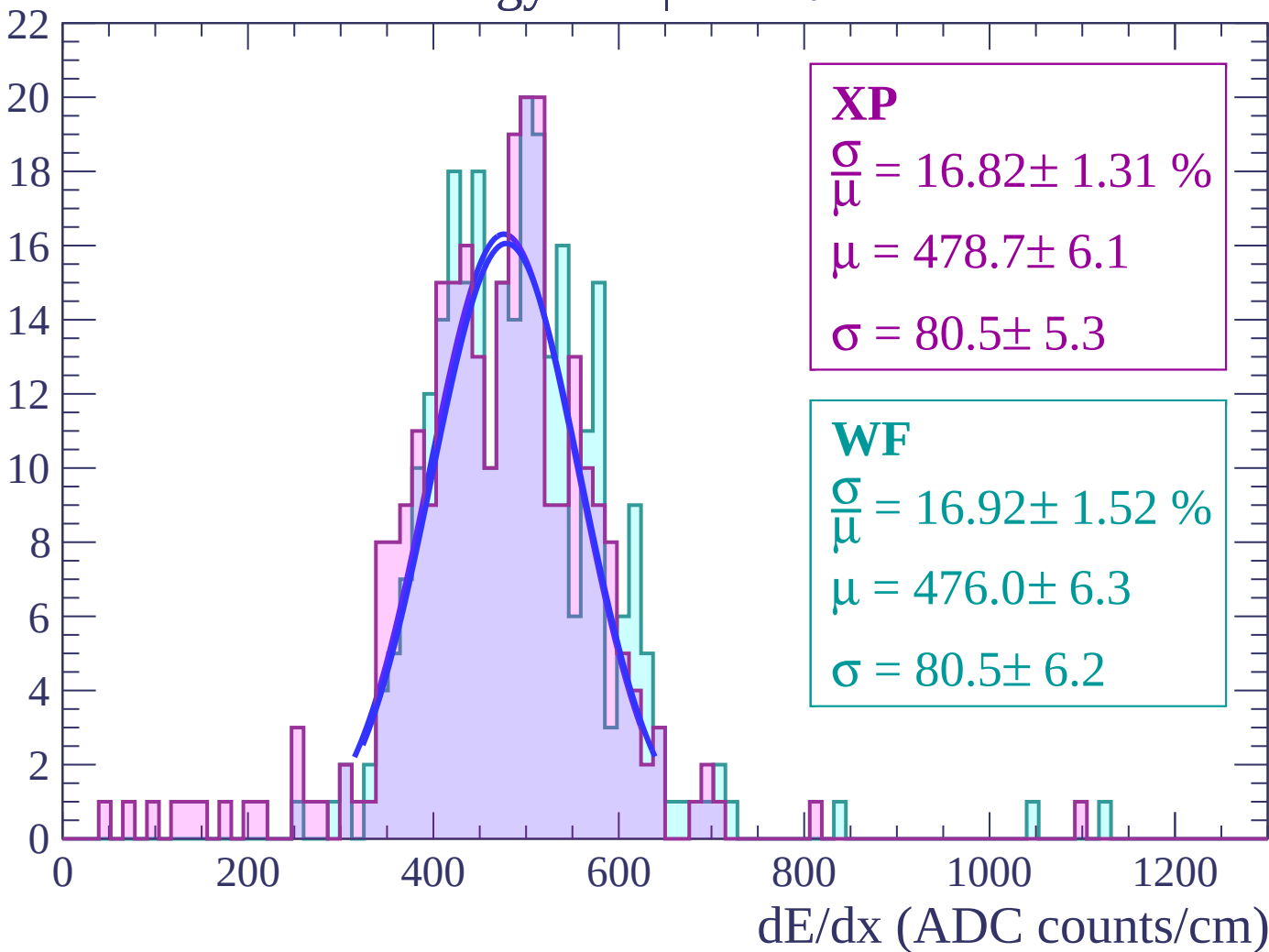
Energy loss | $30 < \theta < 32$

Count



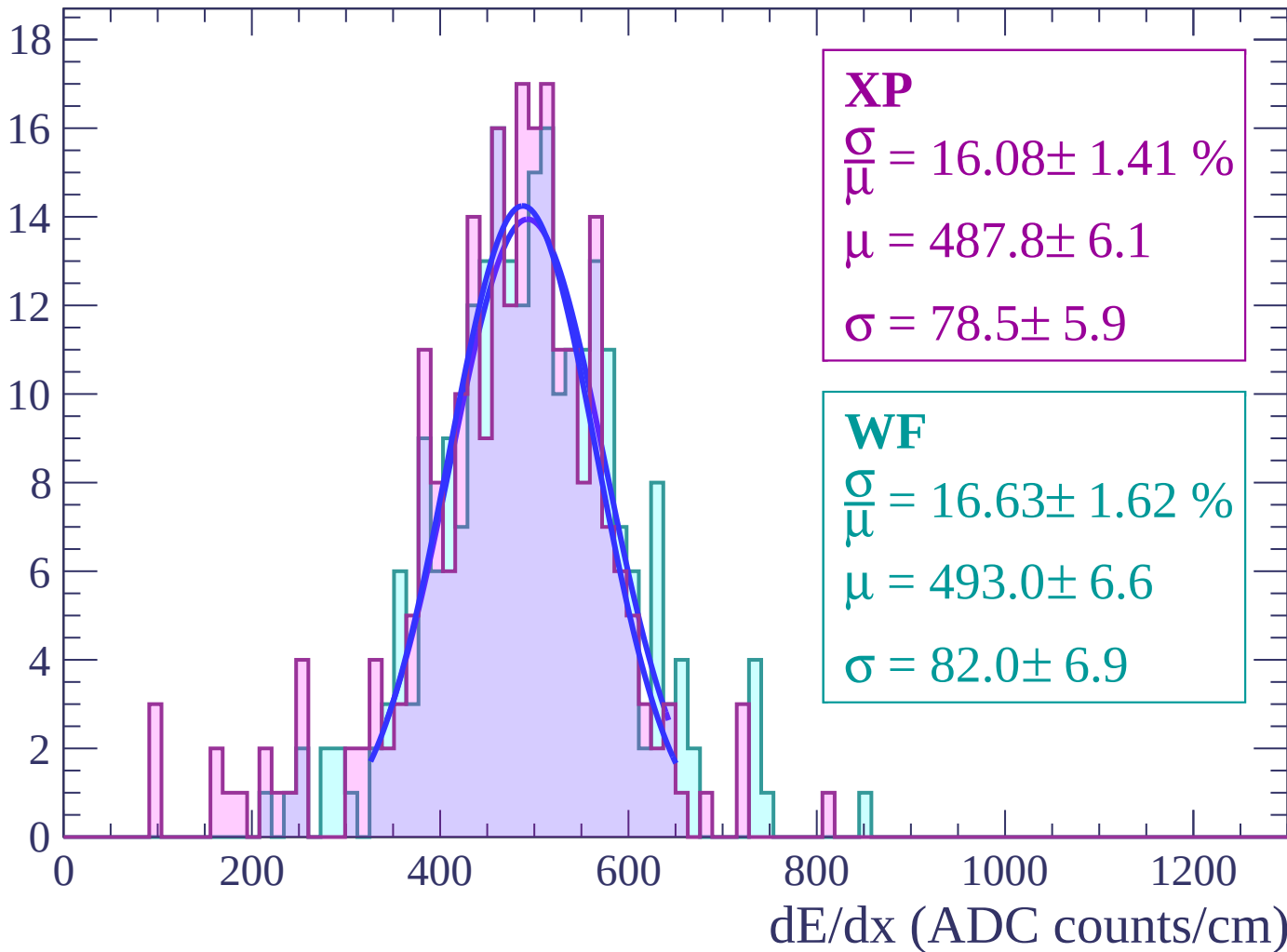
Energy loss | $32 < \theta < 34$

Count



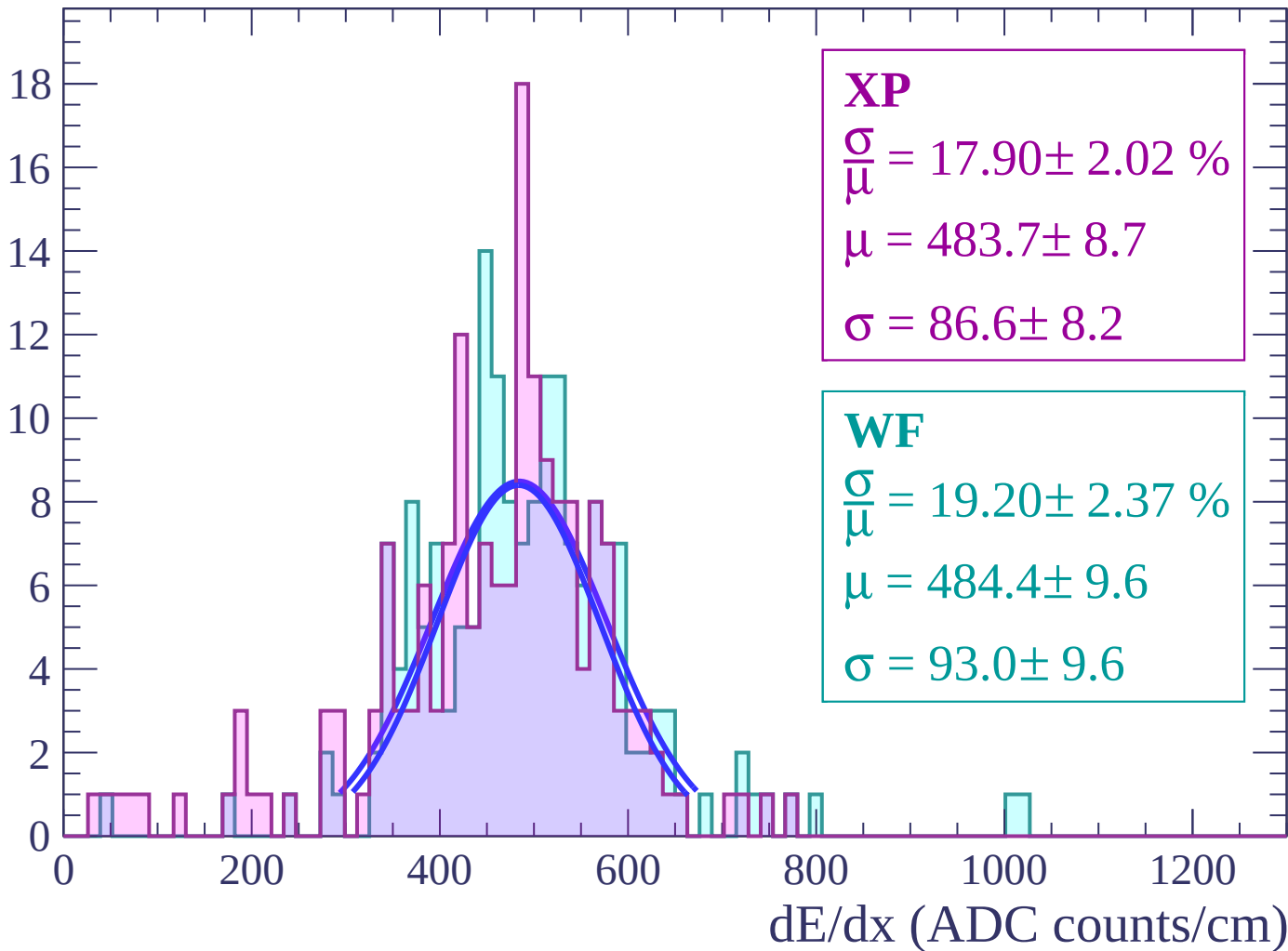
Energy loss | $34 < \theta < 36$

Count



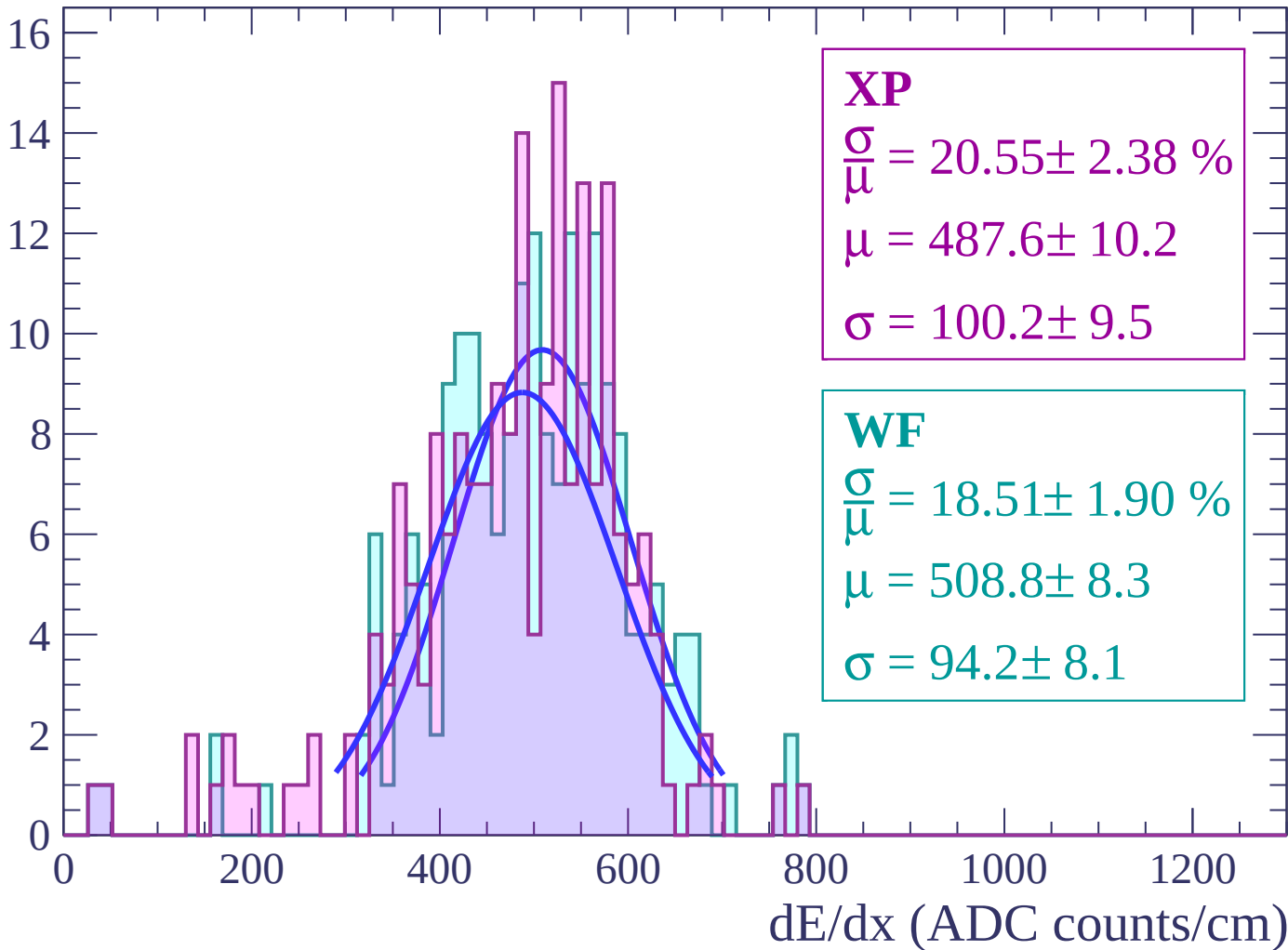
Energy loss | $36 < \theta < 38$

Count



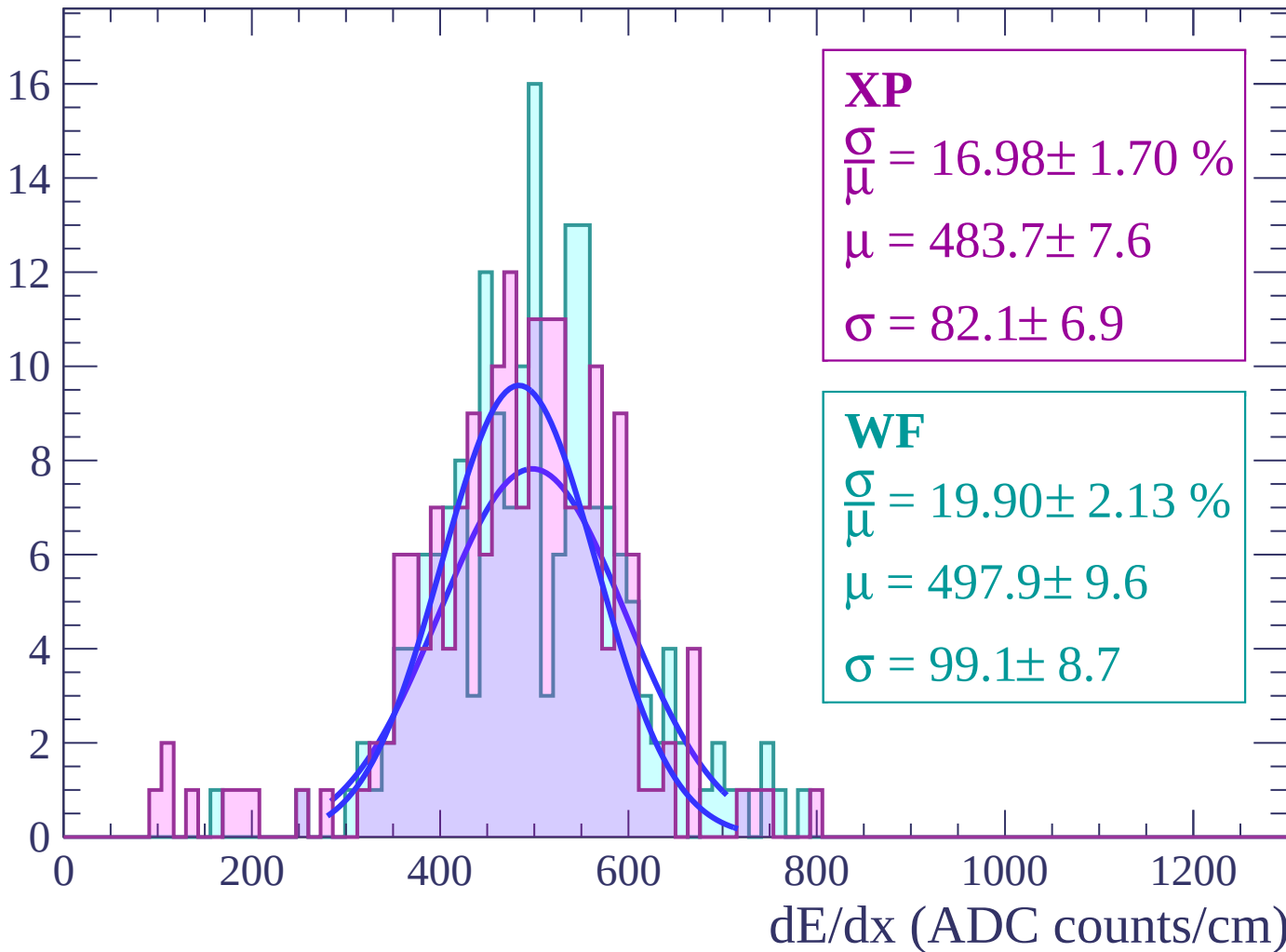
Energy loss | $38 < \theta < 40$

Count



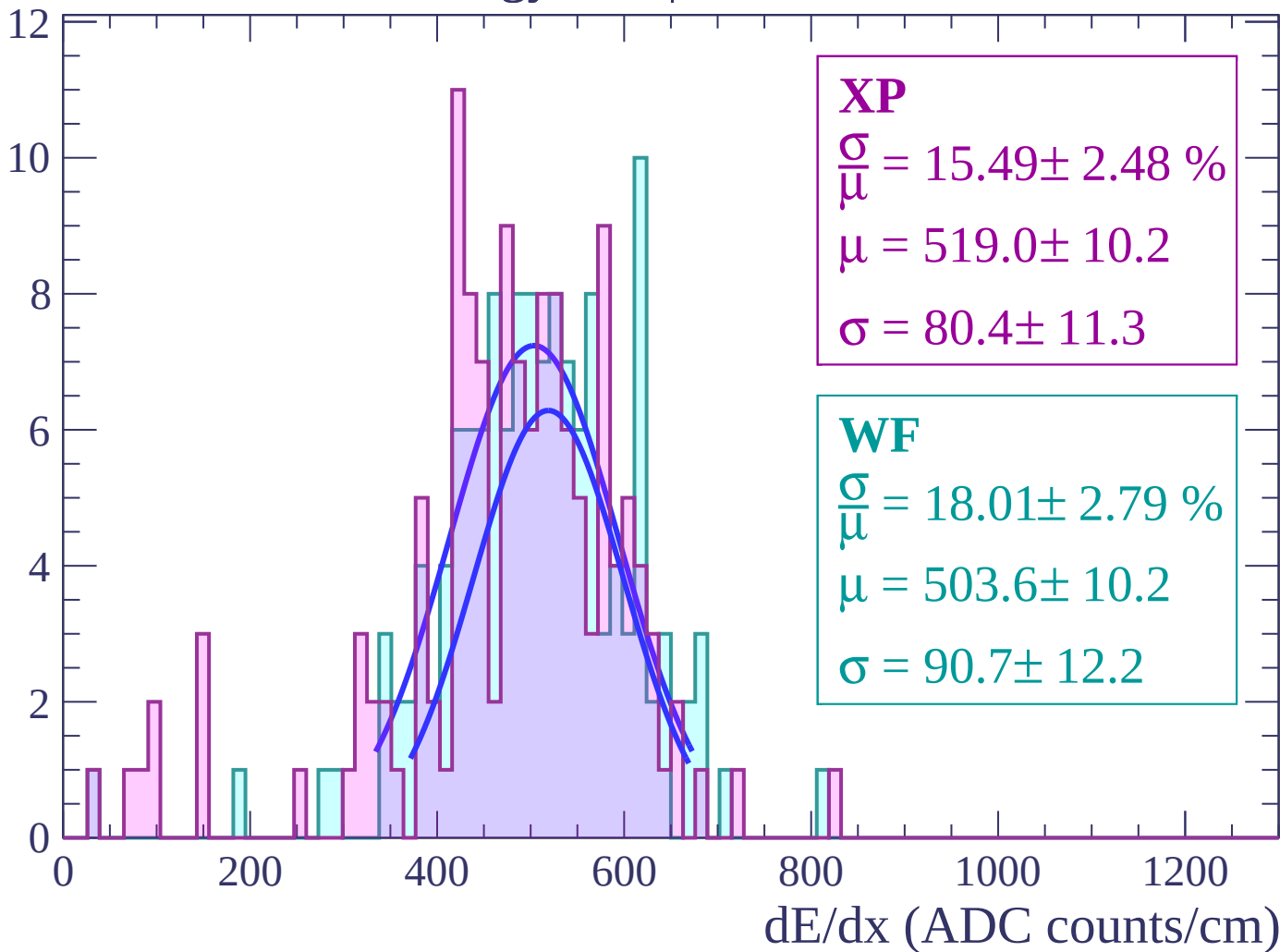
Energy loss | $40 < \theta < 42$

Count



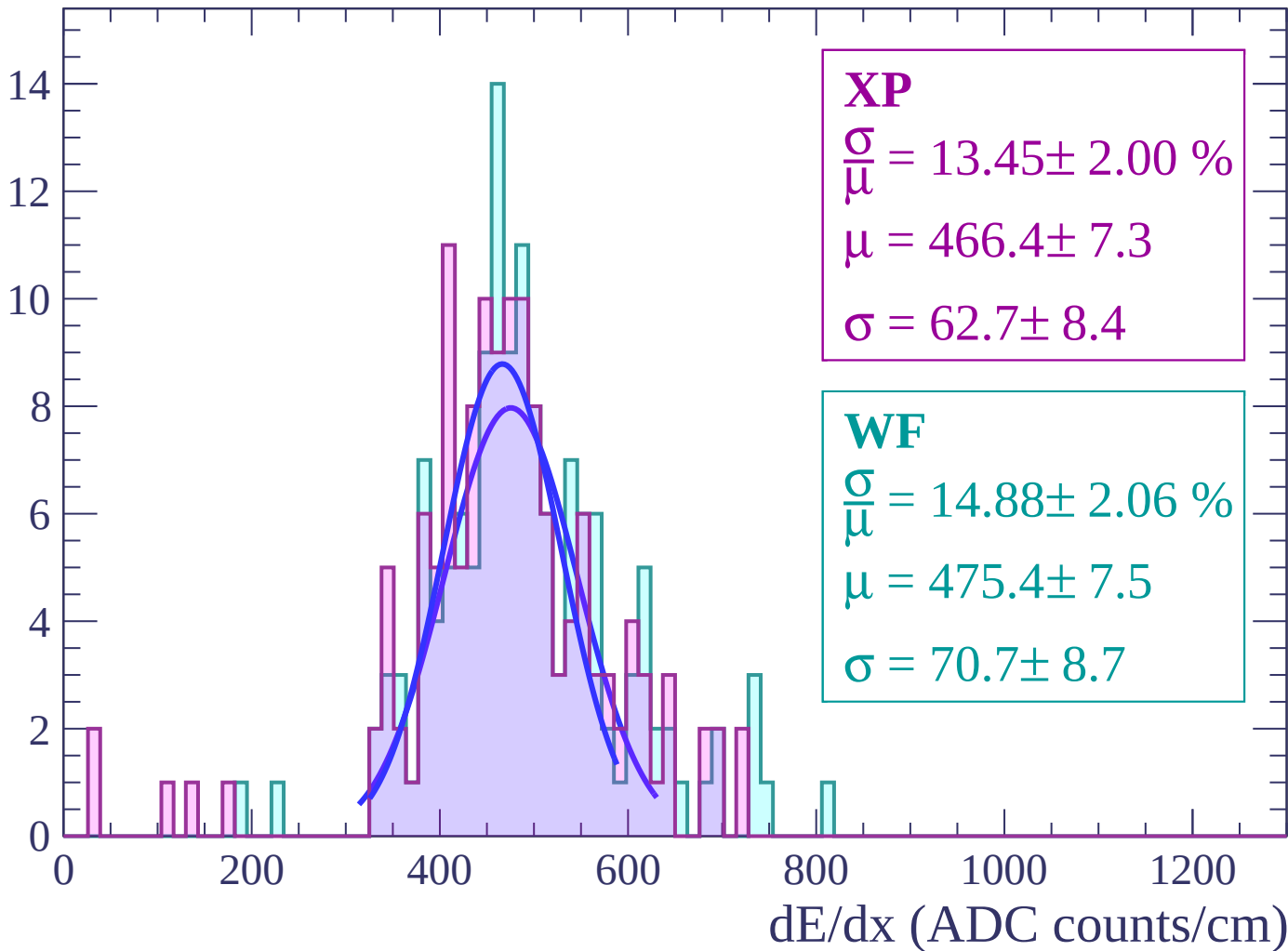
Energy loss | $42 < \theta < 44$

Count



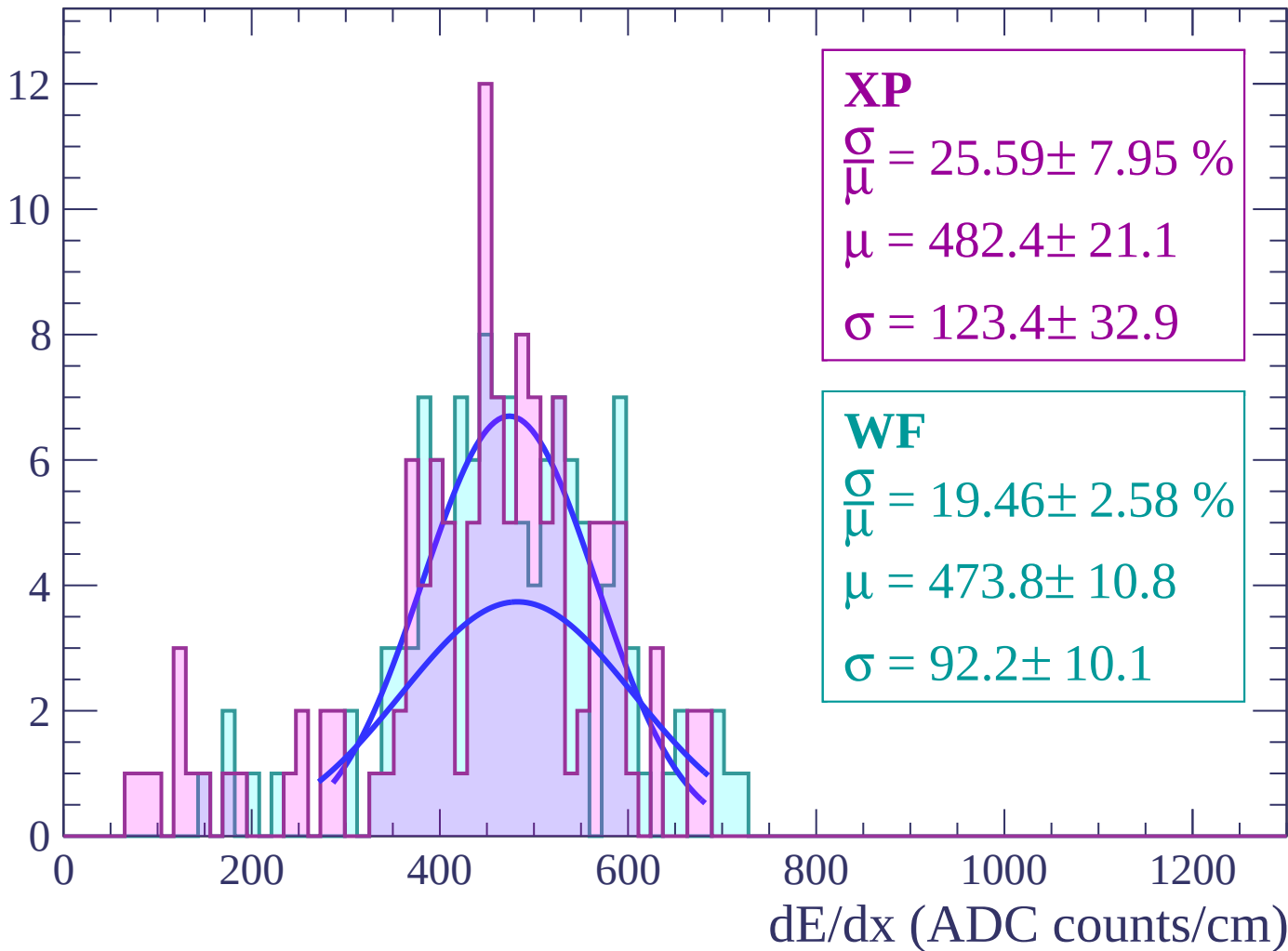
Energy loss | $44 < \theta < 46$

Count



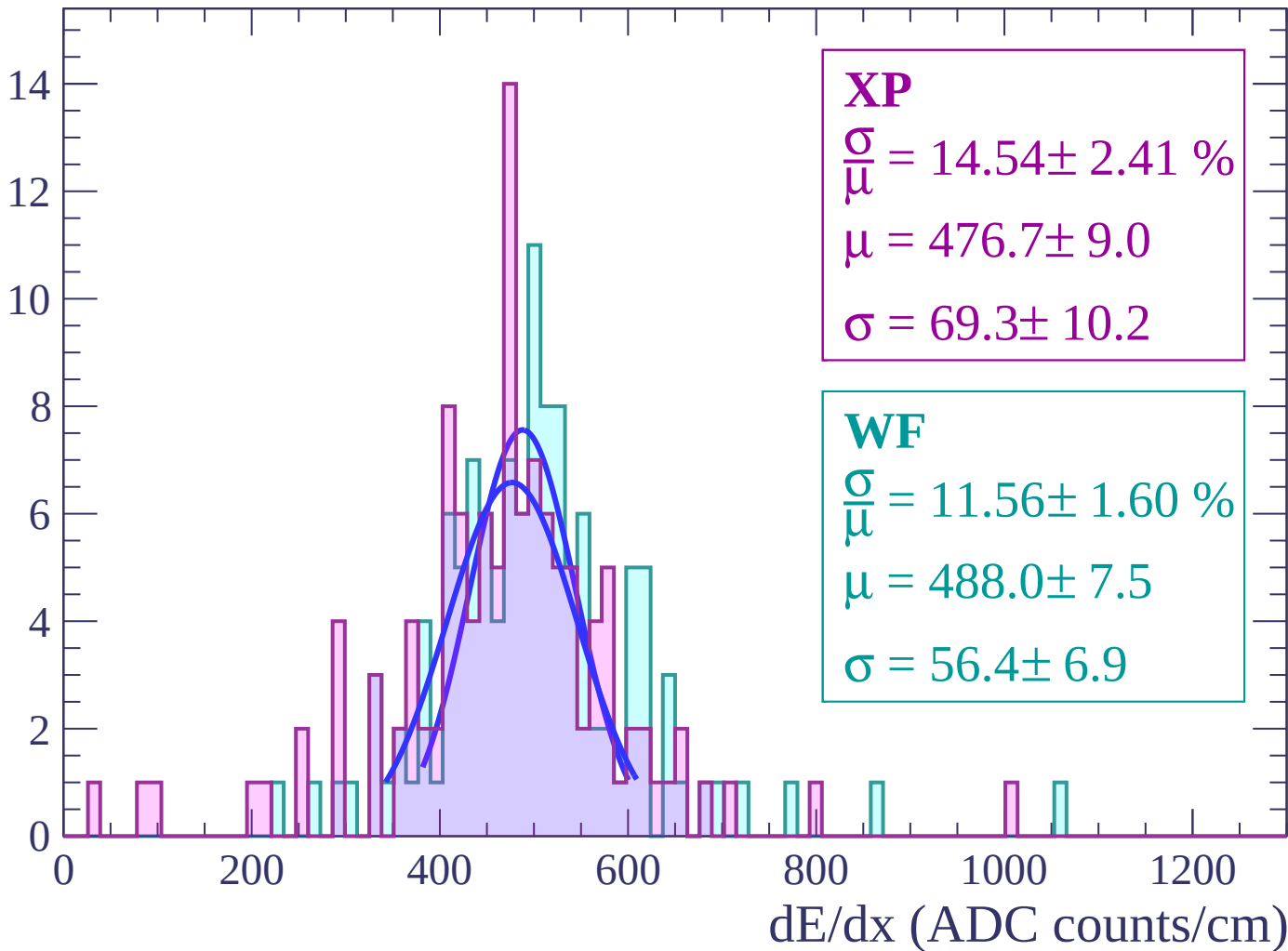
Energy loss | $46 < \theta < 48$

Count



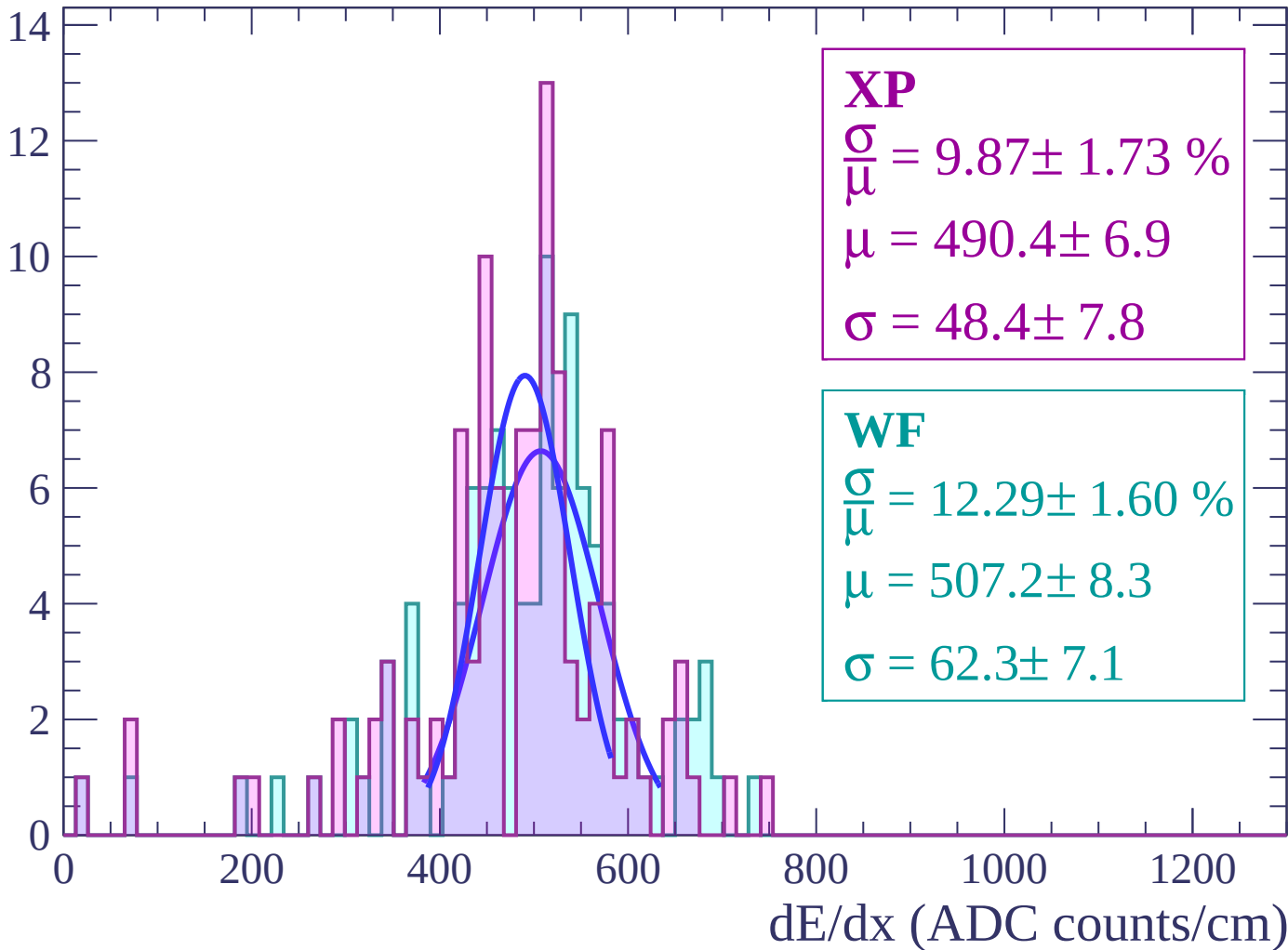
Energy loss | $48 < \theta < 50$

Count



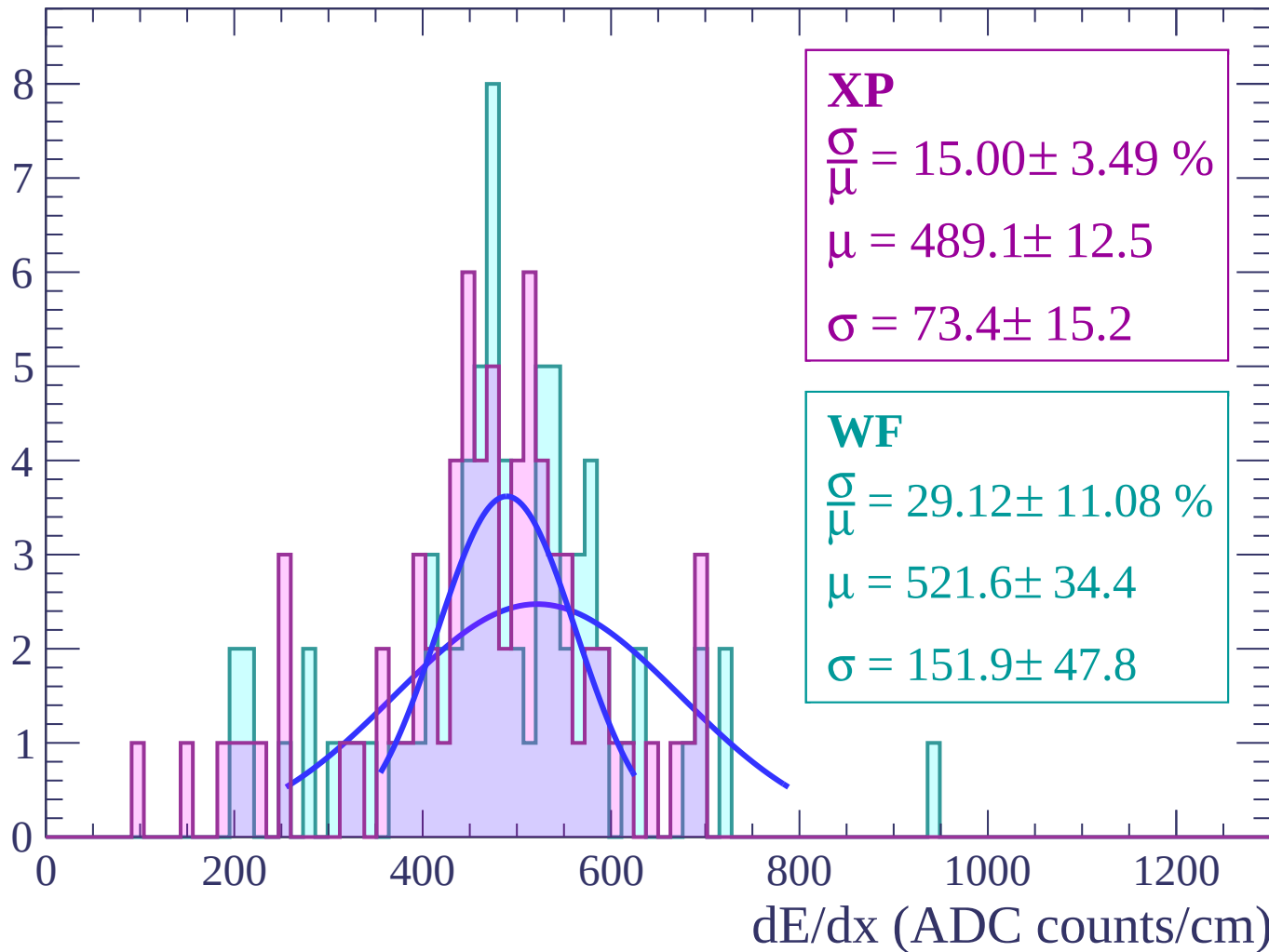
Energy loss | $50 < \theta < 52$

Count



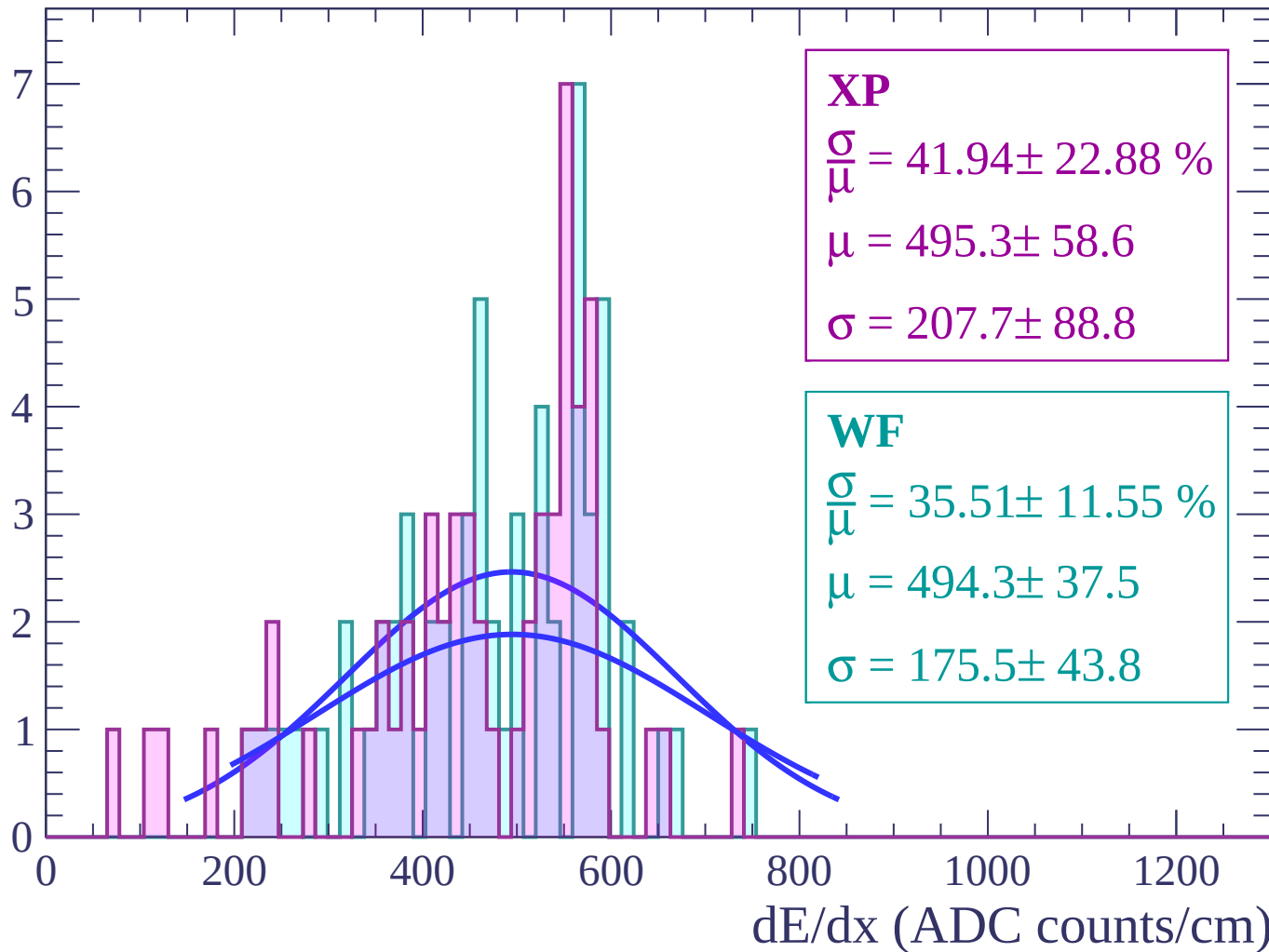
Energy loss | $52 < \theta < 54$

Count



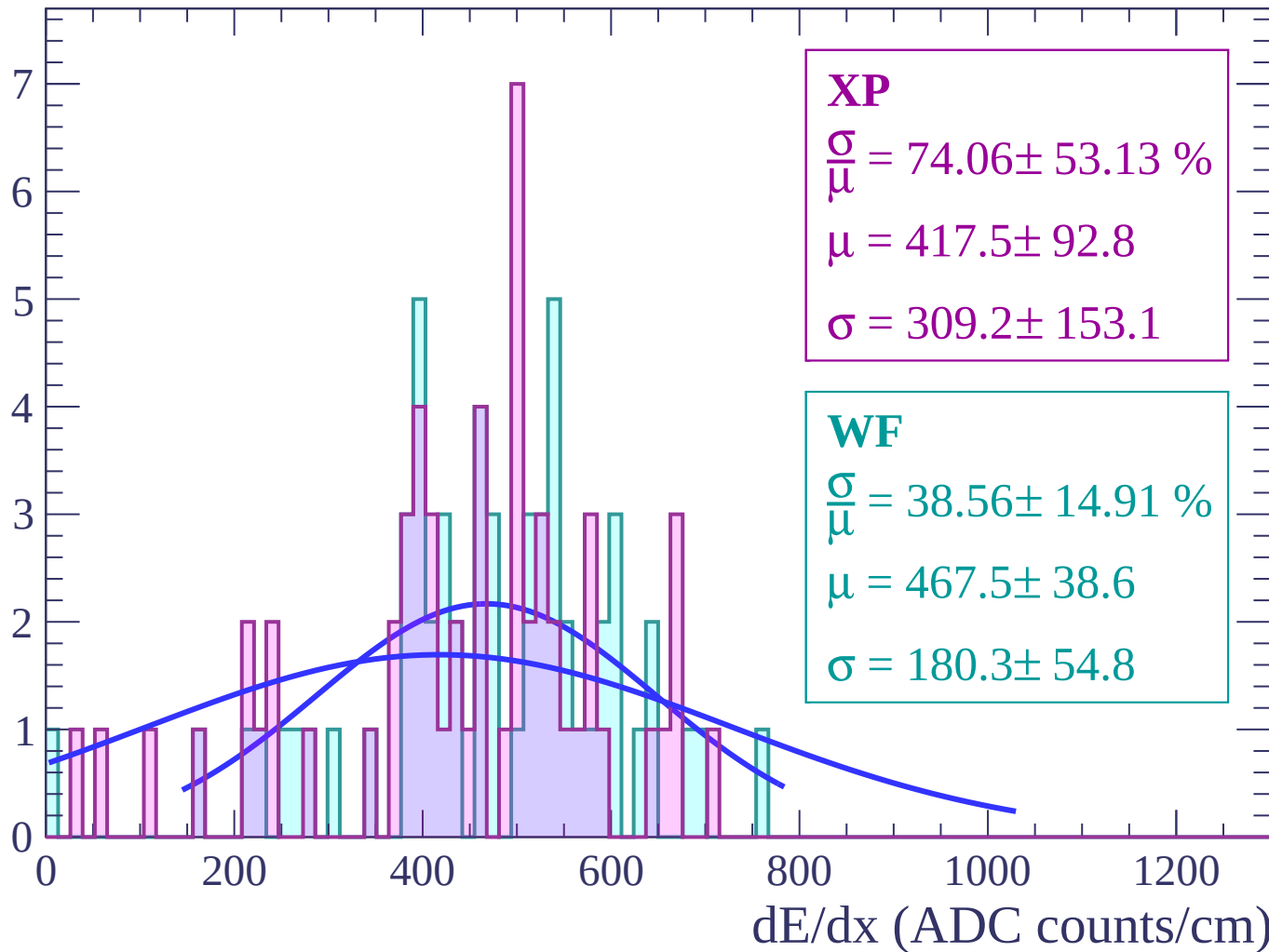
Energy loss | $54 < \theta < 56$

Count



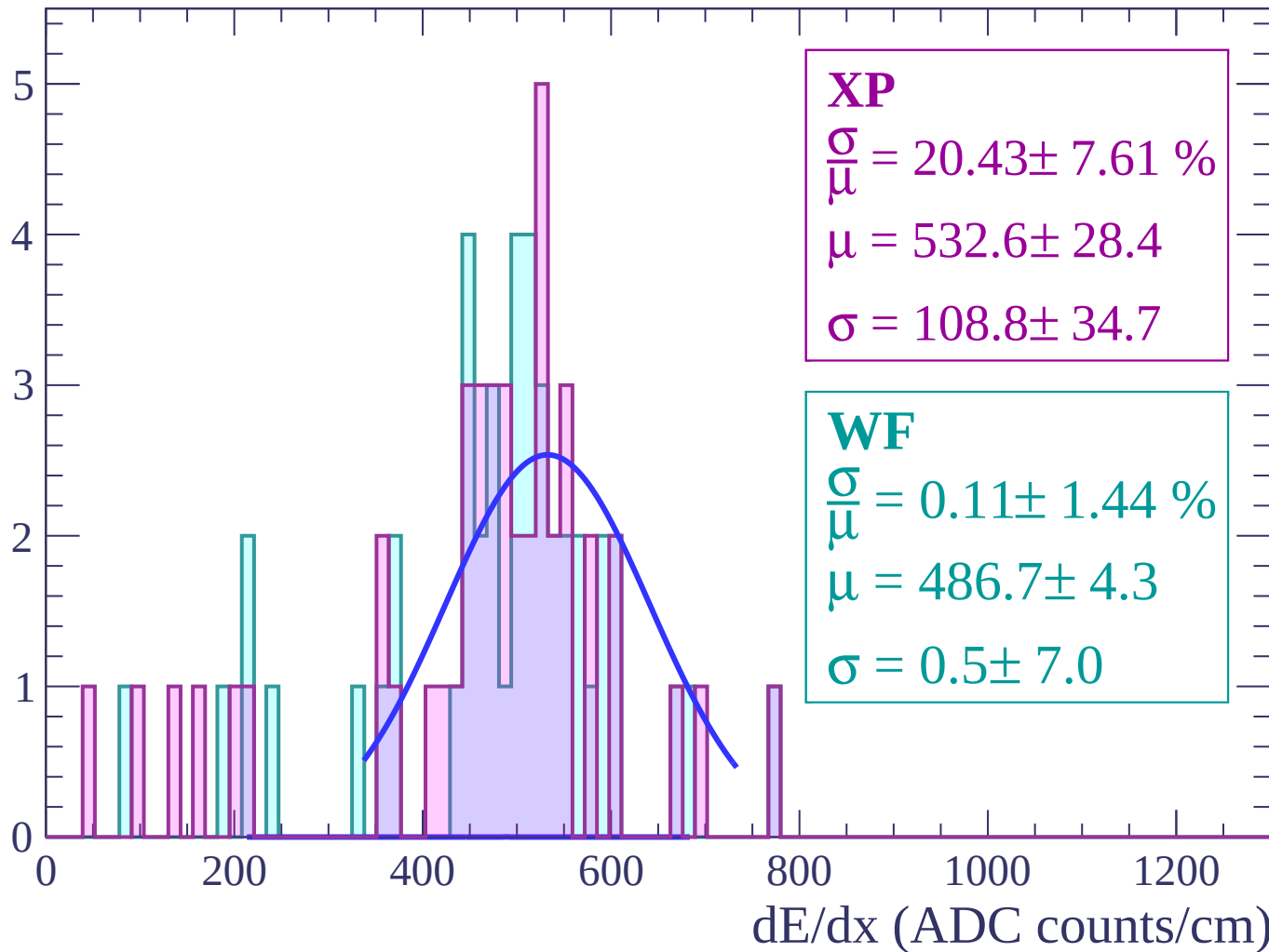
Energy loss | $56 < \theta < 58$

Count



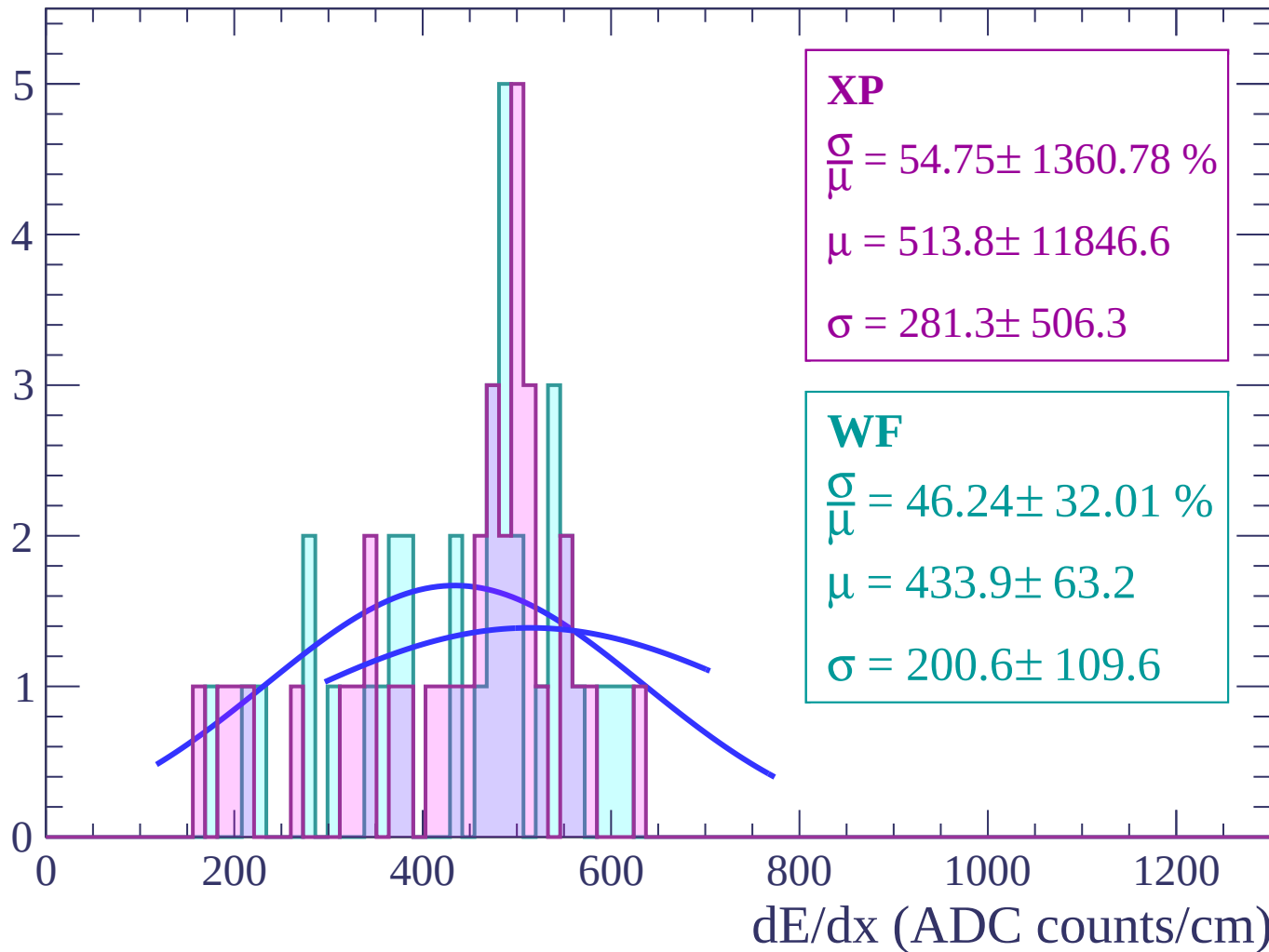
Energy loss | $58 < \theta < 60$

Count



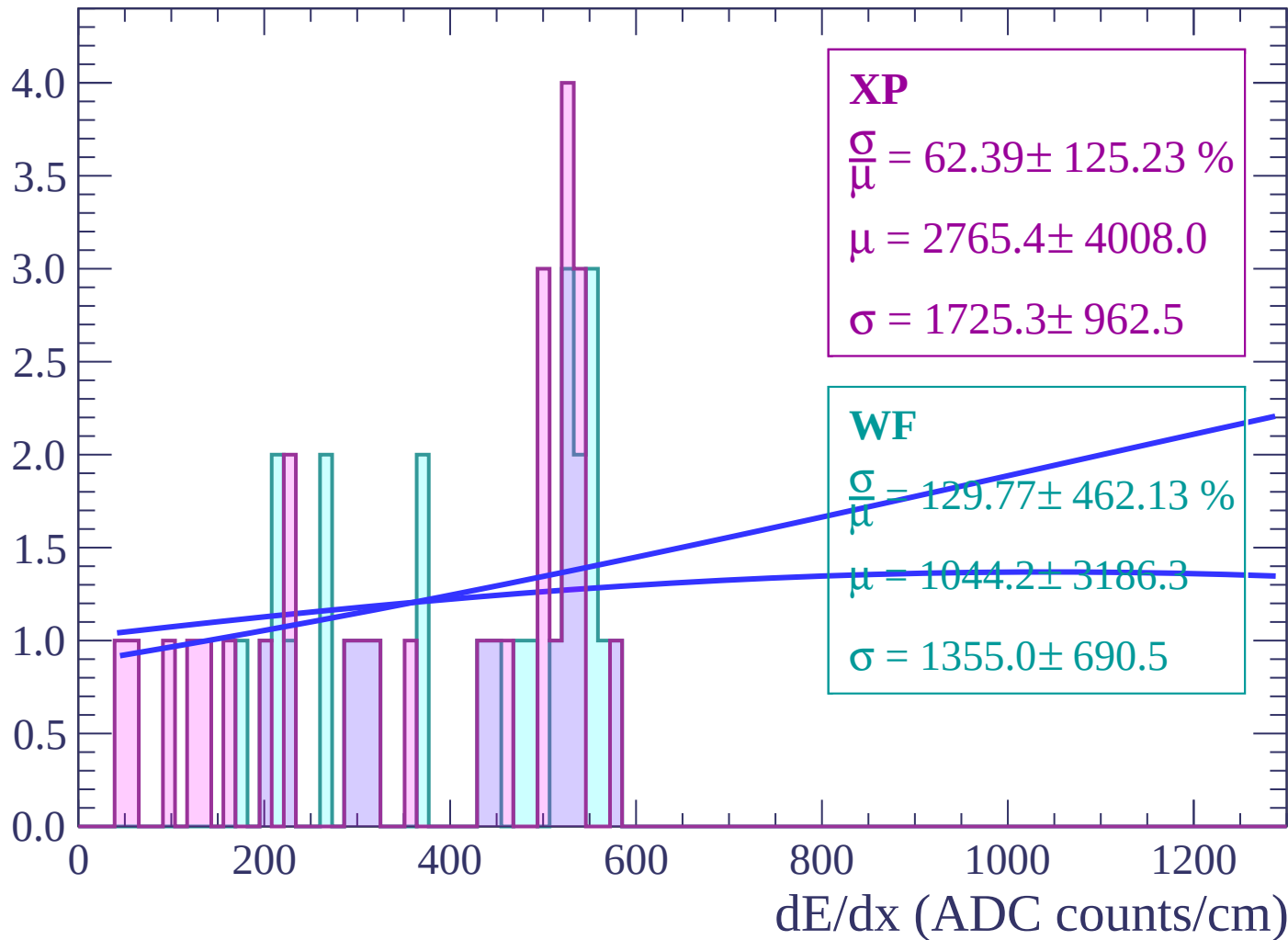
Energy loss | $60 < \theta < 62$

Count



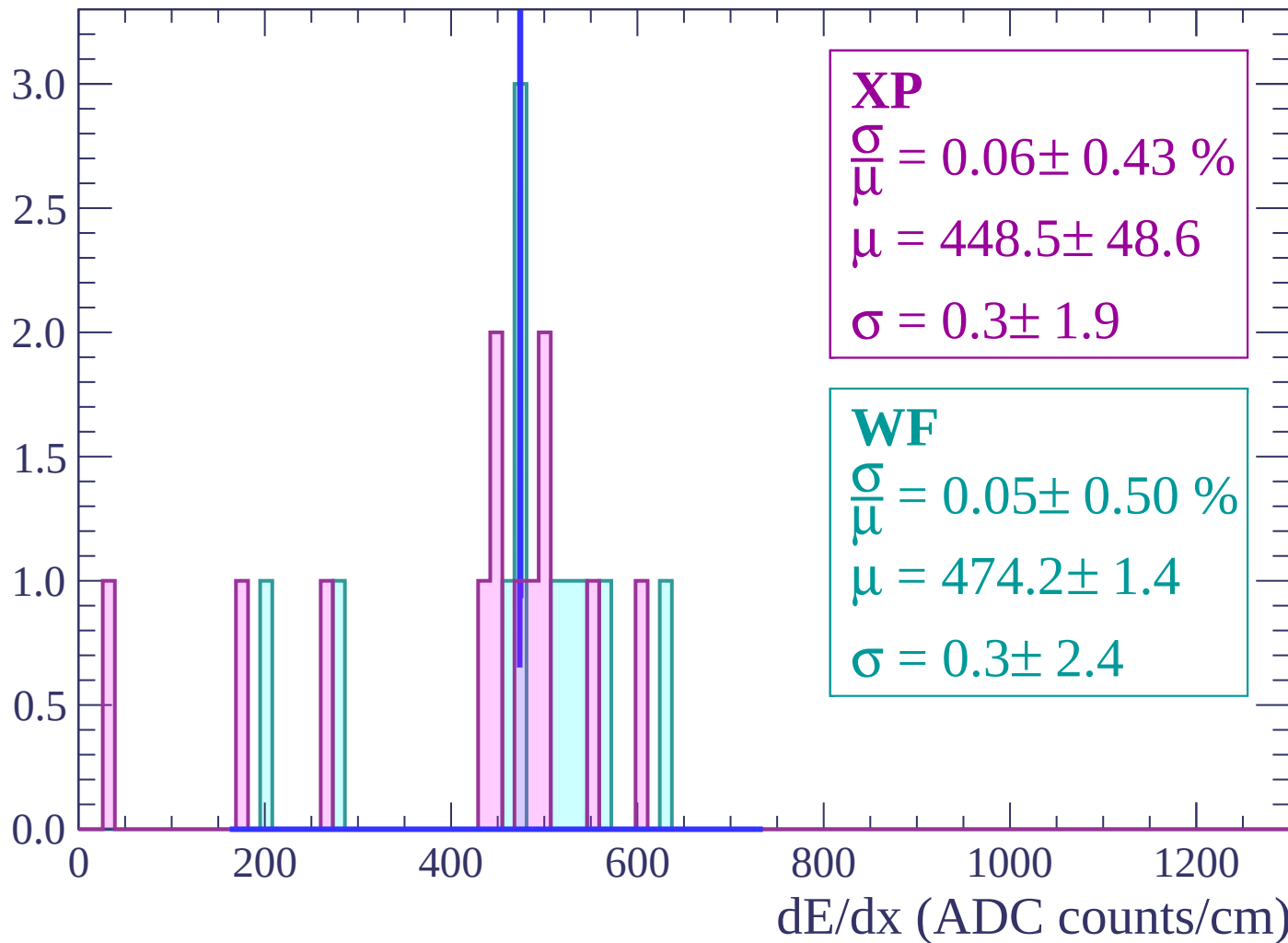
Energy loss | $62 < \theta < 64$

Count



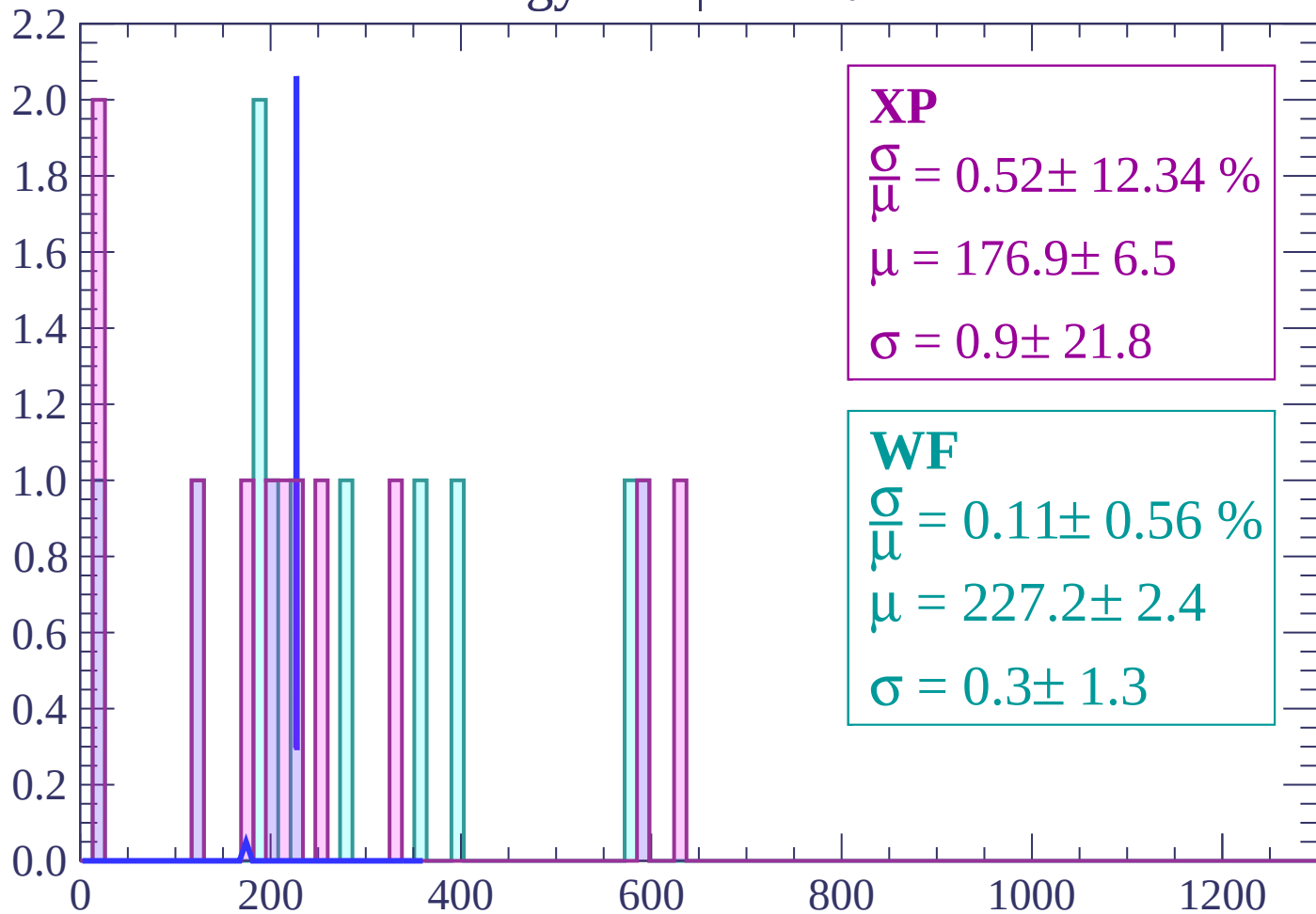
Energy loss | $64 < \theta < 66$

Count



Energy loss | $66 < \theta < 68$

Count



XP

$$\frac{\sigma}{\mu} = 0.52 \pm 12.34 \%$$

$$\mu = 176.9 \pm 6.5$$

$$\sigma = 0.9 \pm 21.8$$

WF

$$\frac{\sigma}{\mu} = 0.11 \pm 0.56 \%$$

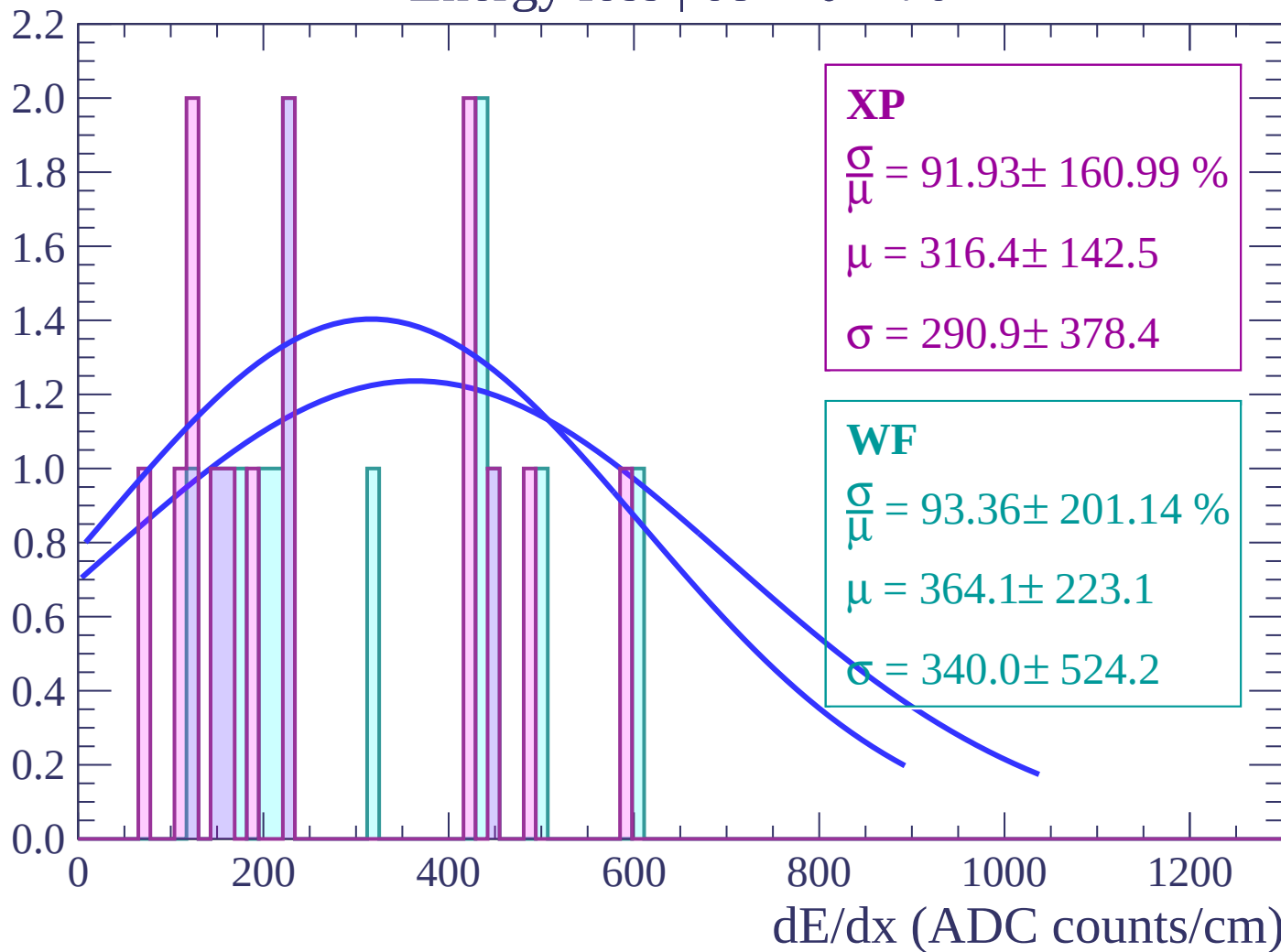
$$\mu = 227.2 \pm 2.4$$

$$\sigma = 0.3 \pm 1.3$$

dE/dx (ADC counts/cm)

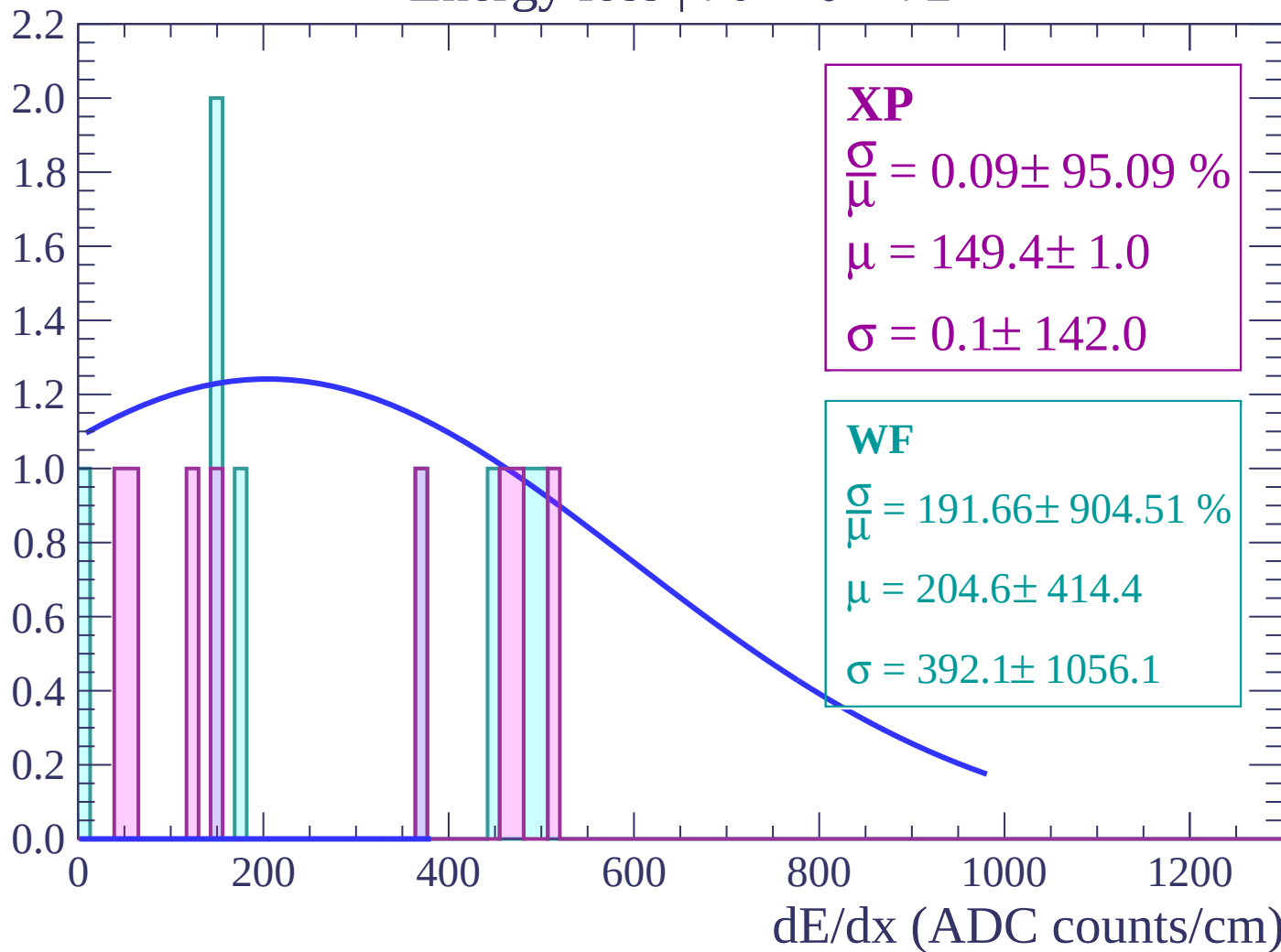
Energy loss | $68 < \theta < 70$

Count



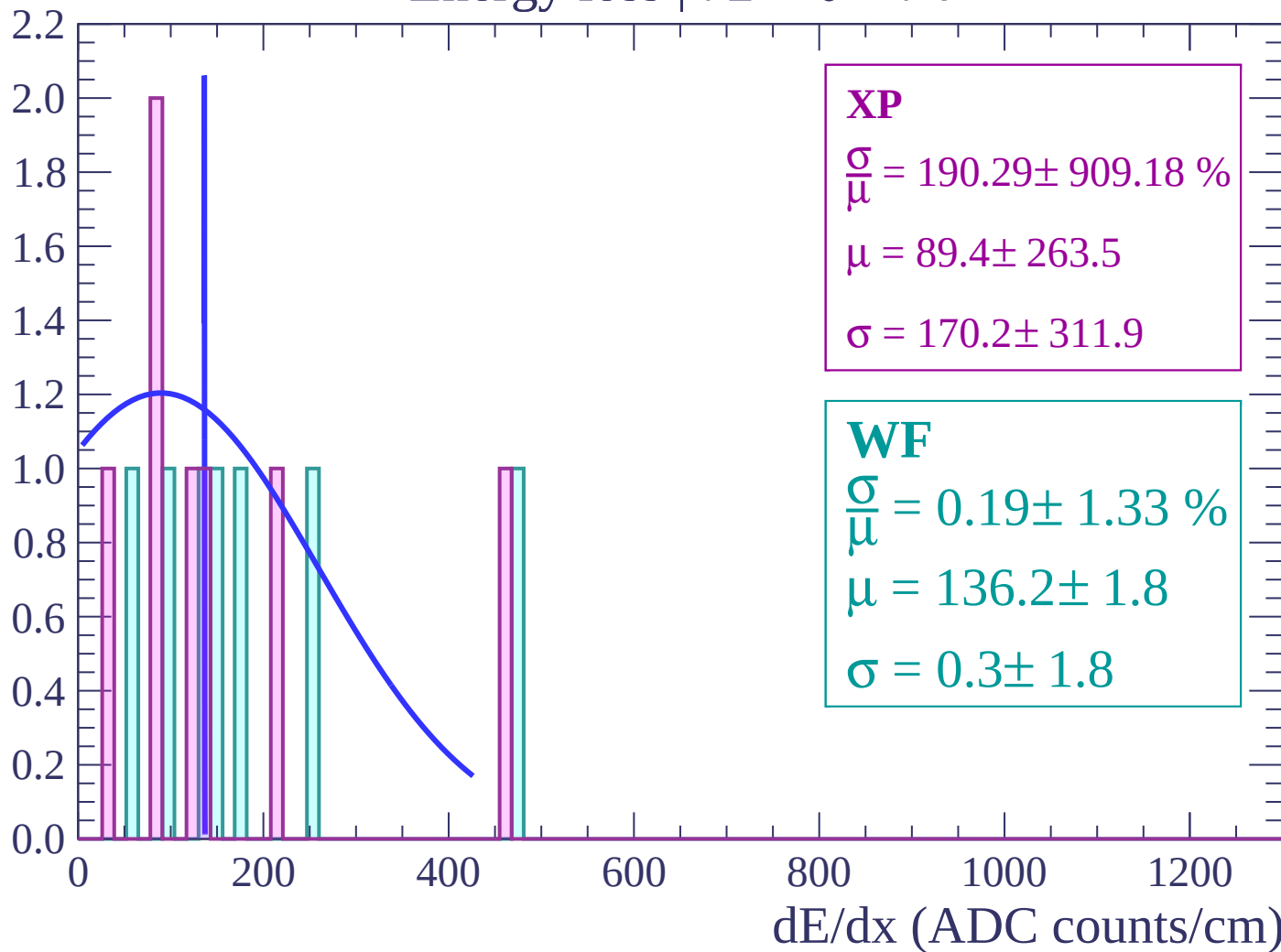
Energy loss | $70 < \theta < 72$

Count



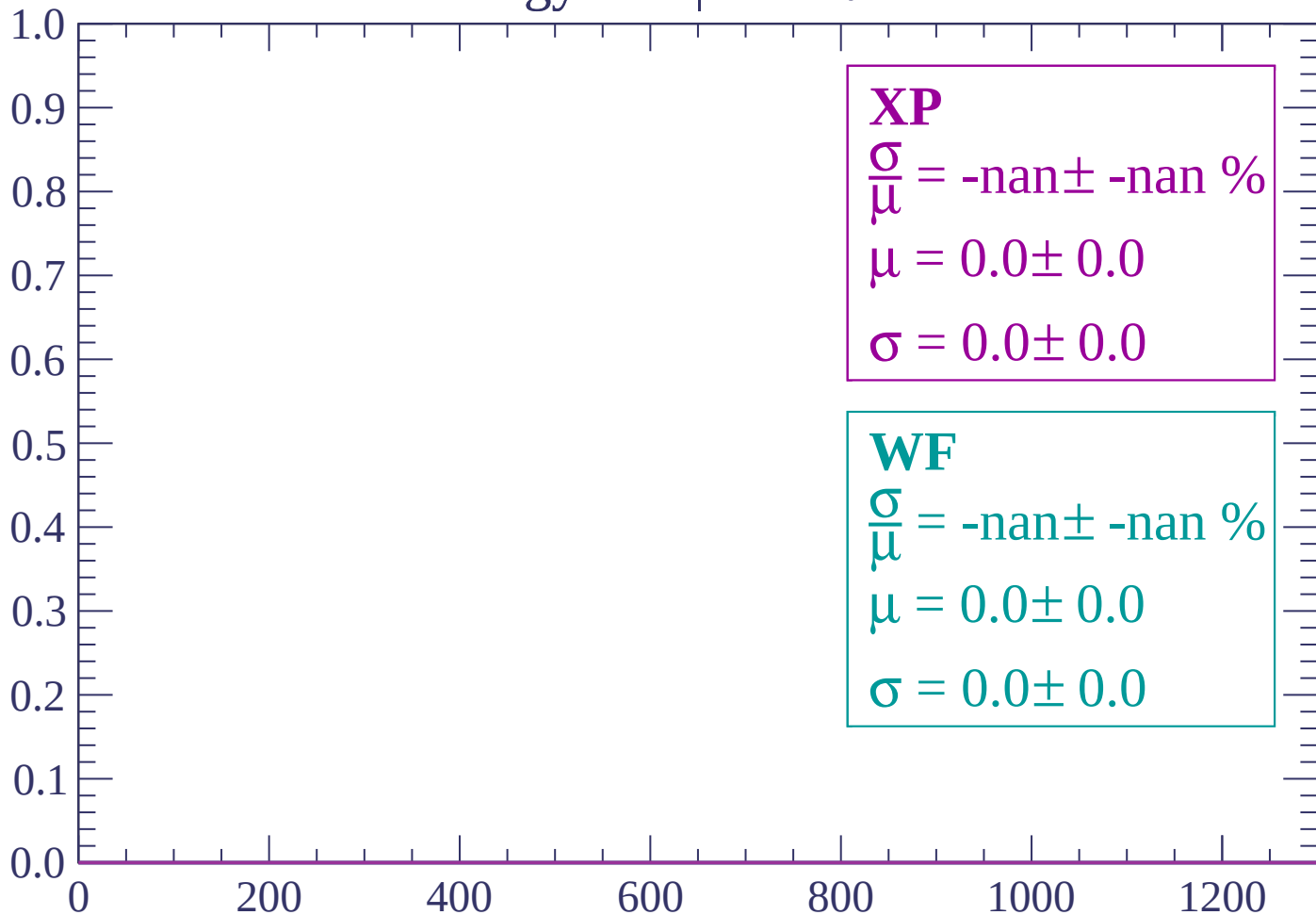
Energy loss | $72 < \theta < 74$

Count



Energy loss | $74 < \theta < 76$

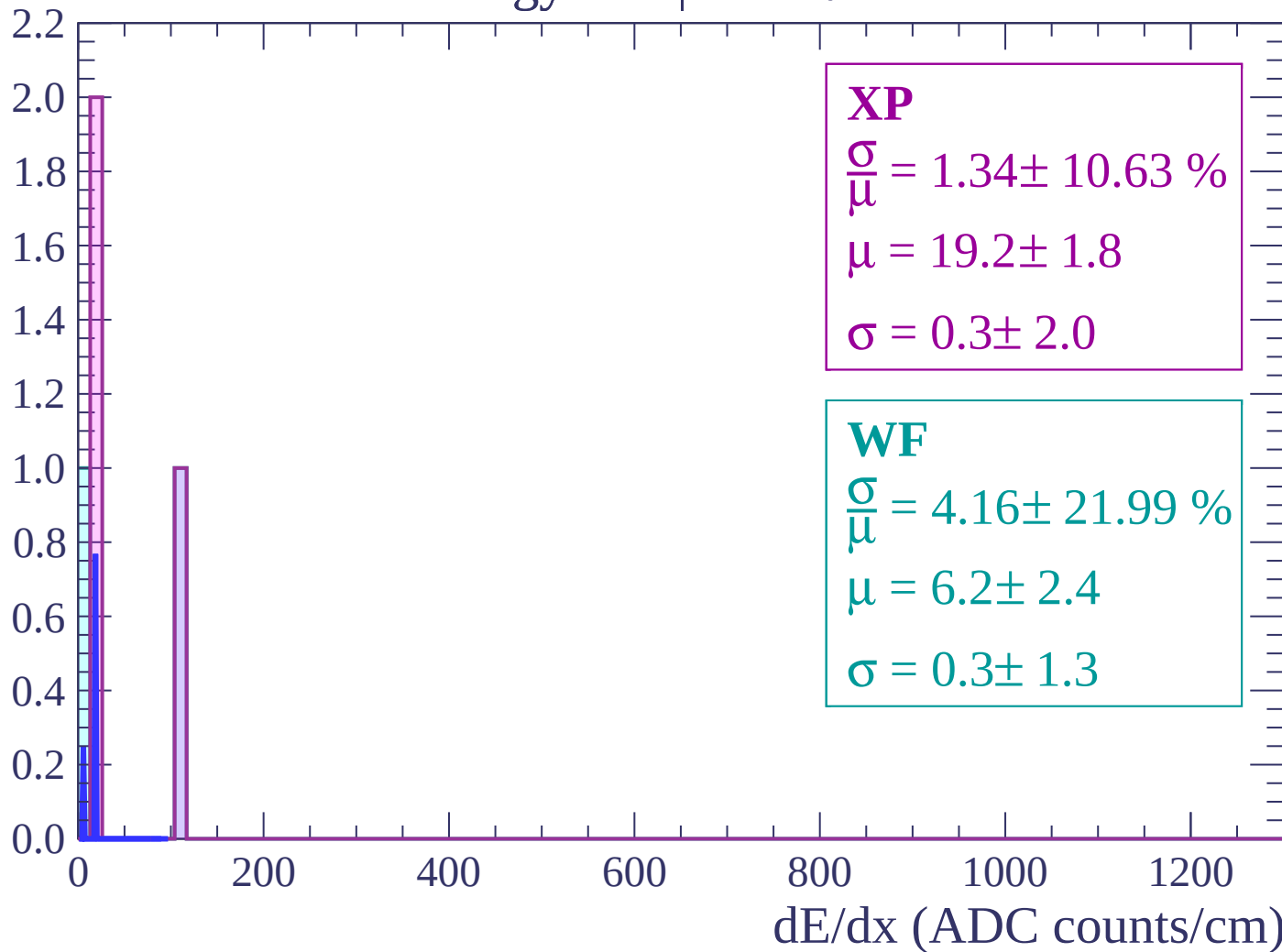
Count



dE/dx (ADC counts/cm)

Energy loss | $76 < \theta < 78$

Count



Energy loss | $78 < \theta < 80$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

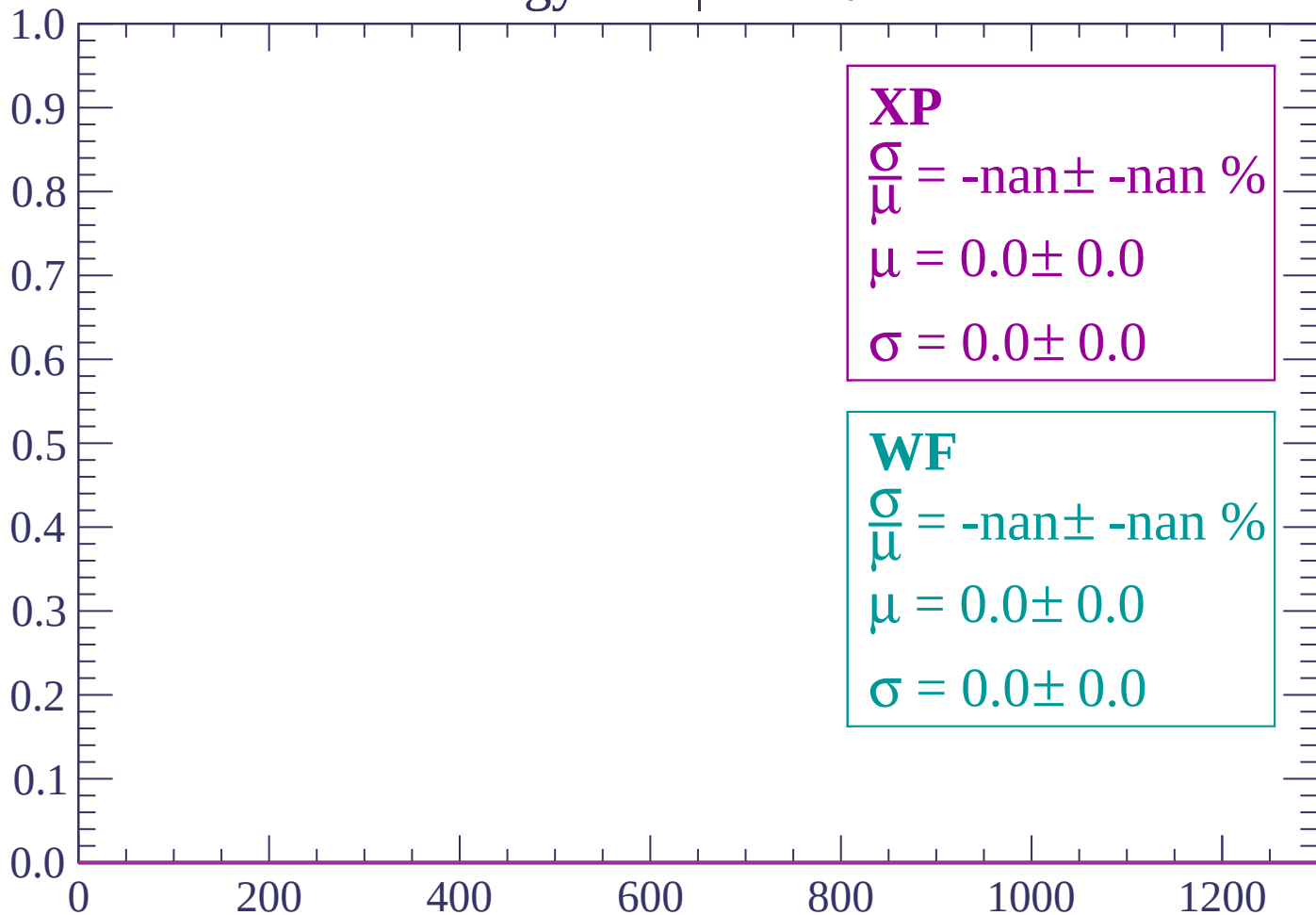
$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $80 < \theta < 82$

Count



dE/dx (ADC counts/cm)

Energy loss | $82 < \theta < 84$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $84 < \theta < 86$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

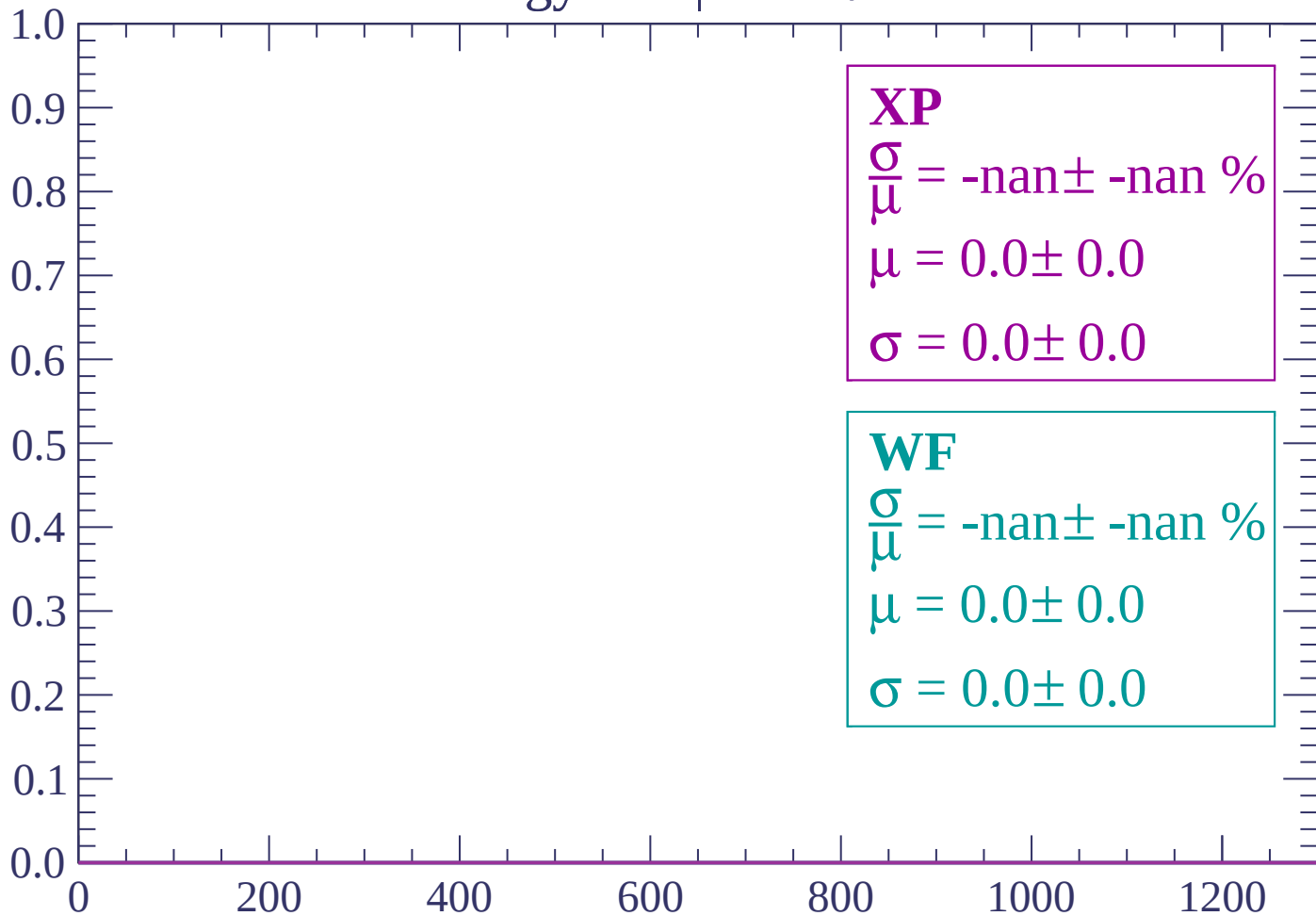
$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $86 < \theta < 88$

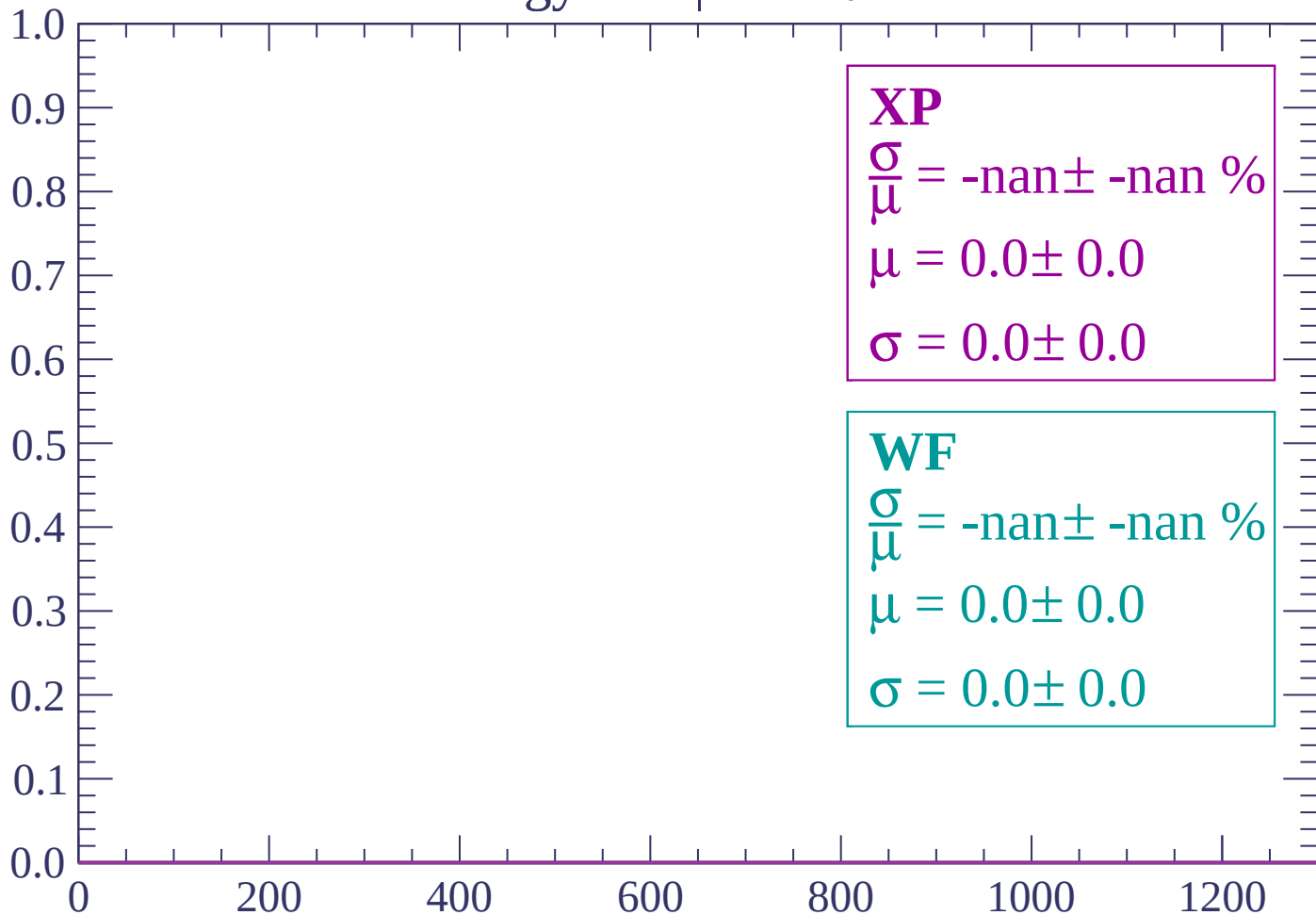
Count



dE/dx (ADC counts/cm)

Energy loss | $88 < \theta < 90$

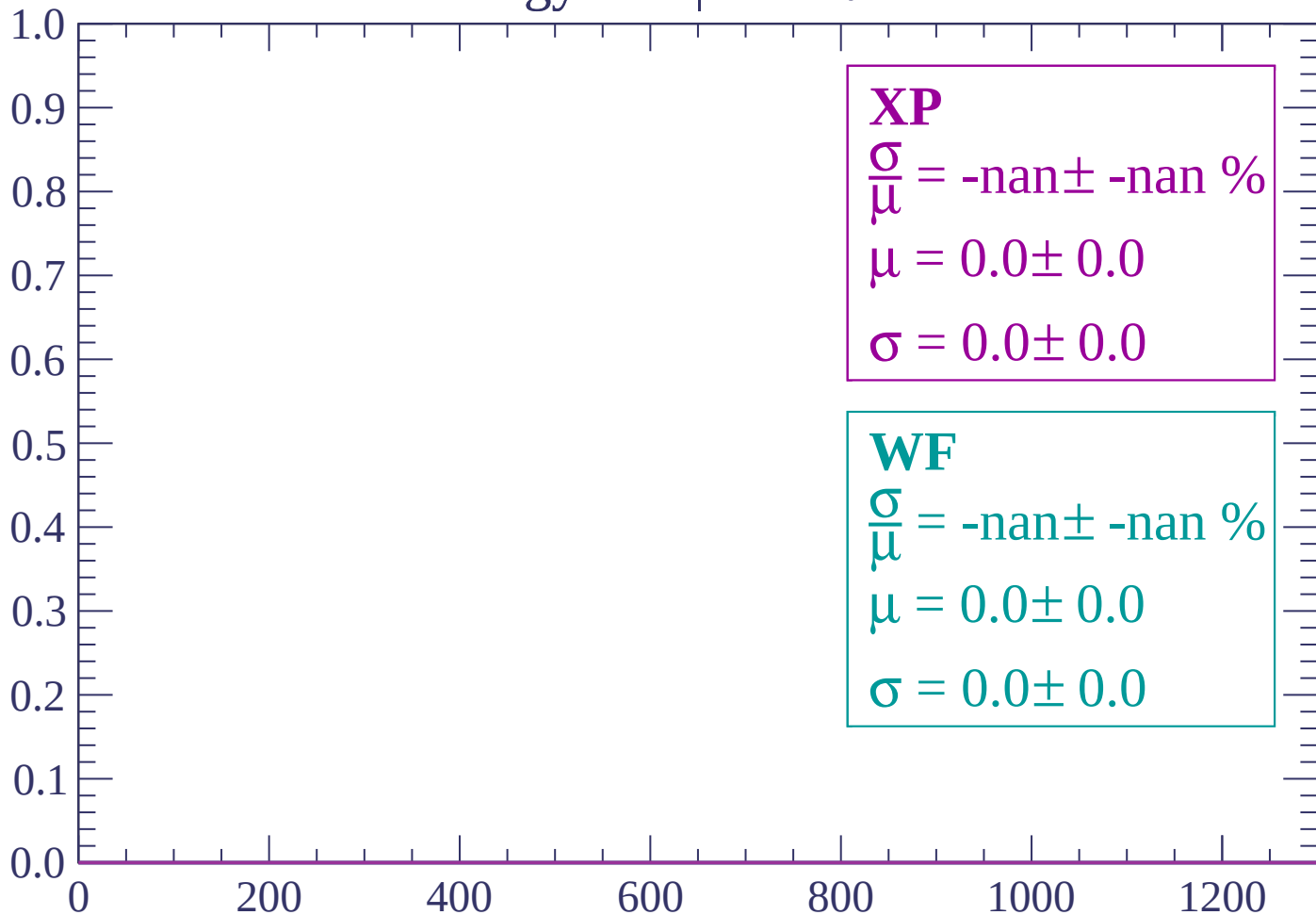
Count



dE/dx (ADC counts/cm)

Energy loss | $90 < \theta < 92$

Count



dE/dx (ADC counts/cm)

